

MAE 342 Space System Design

Final Design Report

Team Name: Magnetotail Systems Laboratory (MSL)

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Abstract

The MATRIX mission is a constellation of eight identical satellites studying Earth's magnetotail to answer critical questions about how energy is transported from the solar wind and the outer magnetotail into Earth's inner magnetosphere, with a specific focus on Bursty Bulk Flows (BBFs). NASA's current spacecraft fleet does not have good coverage of the magnetotail transition region located $6 - 12 R_E$ away downstream in the magnetotail, but this region is where BBFs arrive at Earth and deposit their energy so it is critical to better understand the mass and energy transport in this region. MATRIX fills this gap with a constellation architecture arranged into three groups of 2, 4, and 2 satellites, with apogees of $7 R_E$, $9 R_E$, and $12 R_E$ respectively. The satellites are spaced such that MATRIX provides continuous in-situ measurements of the magnetic field and plasma in this important magnetotail transition region.

Each MATRIX spacecraft is shaped like a hexagonal prism with solar panels covering one of the hexagonal faces. For the entirety of the MATRIX mission's lifetime, this solar panel face remains pointed toward the Sun to maximize power generation for the spacecraft. The three-instrument science payload suite consisting of a deployable magnetometer, electrostatic analyzer (ESA) and energetic particle detector (EPD) to measure magnetic fields and plasma is mounted radially along the rim of the spacecraft bus along with 12 hydrazine maneuvering thrusters that are used for attitude control as well as larger propulsive maneuvers. The center of the spacecraft bus contains a central fuel tank and smaller fuel tanks are located elsewhere within the hexagonal bus. The entire spacecraft is spin stabilized and rotates at 15 rpm. Through a combination of the rotation of the spacecraft and the fields of views of the instruments, each spacecraft in the MATRIX constellation continuously surveys the magnetotail. During nominal science operations each MATRIX spacecraft continuously gathers scientific data during its orbit. For a one-hour window centered on the orbit perigee, the science data along with housekeeping data on the health of spacecraft systems is transmitted down to the Near Space Network on Earth. Each MATRIX spacecraft has enough memory to store over a month's worth of data in case it cannot be transmitted for any reason. During every orbit, the collection and transmission of data along with any maneuvers all happen autonomously as the MATRIX spacecraft switches between different modes on its flight computer. Multi-nodal thermal simulations were also conducted to inform power requirements and determine the heating and cooling mechanisms on the spacecraft bus. MATRIX uses a combination of louvers and radiative insulation heaters for heat retention and dissipation.

This report presents the design of the MATRIX mission across eight subsystems: Guidance, Navigation, & Control (GNC); Attitude Determination & Control System (ADCS); Structures, Materials, & Mechanisms (SMM); Thermal (THM); Propulsion & Launch Vehicle (PLV); Power (POW); Communications, Command, & Data Handling (CCD); and Operations & Payload (OPL).



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List of Acronyms

- i. **ADCS:** Attitude Determination and Control System
- ii. **AKF:** Additive Kalman Filter
- iii. **AO:** Announcement of Opportunity
- iv. **AOP:** Argument of Periapsis
- v. **ARE:** Algebraic Riccati Equation
- vi. **ARW:** Angular Random Walk
- vii. **ASCENT:** Advanced Spacecraft Energetic Non-Toxic
- viii. **BAU:** Bus Avionics Unit
- ix. **BBF:** Bursty Bulk Flow
- x. **BOL:** Beginning of Life
- xi. **CAD:** Computer Aided Design
- xii. **CCD:** Communications Command and Data Handling
- xiii. **CCSDS:** Consultative Committee for Space Data Systems
- xiv. **CFDP:** CCSDS File Delivery Protocol
- xv. **CIDP:** Central Instrument Data Processor
- xvi. **CIM:** Communications Interface Module
- xvii. **Cmd:** Command
- xviii. **ConOps:** Concept of Operations
- xix. **COTS:** Consumer Off the Shelf
- xx. **CSDA:** Continuous Slowing Down Approximation
- xxi. **DOD:** Depth of Discharge
- xxii. **DOF:** Degree(s) of Freedom
- xxiii. **DPM:** Digital Processor Module
- xxiv. **bfDSN:** Deep Space Network
- xxv. **DTN:** Delay/Disruption Tolerant Networking
- xxvi. **EELV:** Evolved Expendable Launch Vehicle
- xxvii. **EIS+:** Energetic Ion Spectrometer (Modified)
- xxviii. **EKF:** Extended Kalman Filter
- xxix. **EOL:** End of Life



- xxx. **EP:** Electric Propulsion
- xxxi. **EPD:** Electrostatic Particle Detector
- xxxii. **ESA:** European Space Agency
- xxxiii. **ESPA:** EELV Secondary Payload Adapter
- xxxiv. **FEA:** Finite Element Analysis
- xxxv. **FGM:** Fluxgate Magnetometer
- xxxvi. **FOV:** Field of View
- xxxvii. **FDR:** Final Design Review
- xxxviii. **FPGA:** Field Programmable Gate Array
- xxxix. **GEO:** Geostationary Orbit
 - xl. **GGT:** Gravity Gradient Torque
 - xli. **GNC:** Guidance Navigation and Control
 - xl. **GOES-R:** Geostationary Operational Environmental Satellites -R
 - xl. **GPS:** Global Positioning System
 - xliv. **GPIM:** Green Propellant Infusion Mission
 - xl. **IDPU:** Instrument Data Processing Unit
 - xlvi. **IMAP:** Interstellar Mapping and Acceleration Probe
 - xlvi. **IMU:** Internal Measurement Unit
 - xlvi. **IR:** Infrared Radiation
 - xl. **ITR:** Initial Team Report
 - 1. **JHU APL:** Johns Hopkins University Applied Physics Laboratory
 - li. **JPL:** Jet Propulsion Laboratory
 - lii. **LQR:** Linear Quadratic Regulator
 - liii. **LSP:** Launch Service Provider
 - liv. **LV:** Launch Vehicle
 - lv. **MATRIX:** Magnetotail Transition Region In-situ eXplorer
 - lvi. **MEKF:** Multiplicative Extended Kalman Filter
 - lvii. **MEOP:** Maximum Expected Operating Pressure
 - lviii. **MIDEX:** Medium Explorer
 - lix. **MIB:** Minimum Impulse Bit
 - lx. **MLI:** Multi-Layer Insulation



- lxi. **MMS:** Magnetospheric Multiscale Mission
- lxii. **MSL:** Magnetotail Systems Laboratory
- lxiii. **NASA:** National Aeronautics and Space Administration
- lxiv. **NEN:** Near Earth Network
- lxv. **NIST:** National Institute of Standards and Technology
- lxvi. **NTC:** Negative Temperature Coefficient
- lxvii. **OpNav:** Optical Navigation
- lxviii. **OPL:** Operations and Payload
- lxix. **OS:** Operating System
- lxx. **OSAL:** Operating System Abstraction Layer
- lxxi. **PDR:** Preliminary Design Review
- lxxii. **PEU:** Power Electronics Unit
- lxxiii. **PFL:** Pump Fluid Loop
- lxxiv. **PLV:** Propulsion and Launch Vehicle
- lxxv. **PMD:** Power Management Device
- lxxvi. **PMS:** Propellant Management System
- lxxvii. **POSIX:** Portable Operating System Interface
- lxxviii. **POW:** Power
- lxxix. **PSD:** Power Spectral Density
- lxxx. **PSI:** Pounds per Square Inch
- lxxxi. **PWM:** Pulse Width Modulation
- lxxxii. **RAAN:** Right Ascension of Ascending Node
- lxxxiii. **Rad:** Radians
- lxxxiv. **RCS:** Reaction Control System
- lxxxv. **RFP:** Request for Proposal
- lxxxvi. **ROI:** Region of Interest
- lxxxvii. **RPM:** Revolutions per Minute
- lxxxviii. **RTD:** Resistance Temperature Detector
- lxxxix. **RTEMS:** Real-Time Executive for Multiprocessor Systems
- xc. **RTOS:** Real Time Operating System
- xc. **S/C:** Spacecraft



- xcii. **Sec:** Seconds
- xciii. **SEE(s):** Single Event Effect(s)
- xciv. **SEU:** Single Event Upset
- xcv. **SMAD:** Space Mission Analysis and Design
- xcvi. **SMEX:** Small Explorer
- xcvii. **SMM:** Structures Materials and Mechanisms
- xcviii. **SoC:** System-on-Chip
- xcix. **SOTA:** State of the Art
 - c. **SRP:** Solar Radiation Pressure
 - ci. **SRT:** Solar Radiation Torque
 - cii. **SSR:** Subsystem Report
 - ciii. **STK:** Systems Tool Kit (ANSYS)
 - civ. **SWaP:** Size, Weight, and Power
 - cv. **TBA:** To Be Announced
 - cvi. **TBC:** To Be Confirmed
 - cvii. **TBD:** To Be Determined
- cxviii. **TDRSS:** Tracking and Data Relay Satellite Systems
- cix. **THEMIS:** Time History of Events and Macroscale Interactions during Substorms
- cx. **THM:** Thermal
- cxii. **TID:** Total Ionizing Dose
- cxiii. **TRIDENT:** Tetrahedral Reconnection Investigation of Dynamic Energy in the Nightside Transition
- cxiv. **TT&C:** Telemetry, Tracking, and Command
- cxv. **VxWorks:** Virtual Works
- cxvi. **WBS:** Work Breakdown Structure



List of Symbols

[i.]

- r_p : Periapsis radius
- r_a : Apoapsis radius
- a : Semi-major axis
- e : Orbital eccentricity
- i : Orbital inclination
- ω : Argument of periapsis
- Ω : Right ascension of ascending node (RAAN)
- ν : True anomaly
- Δv : Change in velocity
- v : Velocity
- T : Orbital period
- R_E : Earth radius
- $\sim$: Order of magnitude / approximately
- ρ : Density
- m : Mass
- F : Force
- N : Thrust (Newtons)
- t : Time
- $\dot{m}$: Mass flow rate
- V : Volume
- P : Power
- f : Frequency
- θ : Angle
- α : Angle / coefficient (context-dependent)
- β : Angle / coefficient (context-dependent)
- γ : Ratio of specific heats (Propulsion)
- f_n : Natural frequency
- G : Acceleration response



- Q : Dynamic amplification factor (Structures)
- Q : Heat flux (Thermal)
- ζ : Damping ratio
- I : Moment of inertia
- k : Spring constant
- c : Damping constant
- P : Power
- I : Current
- R : Resistance



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1 Executive Summary

1.1 Mission Overview

This subsection presents the most recent version of the MATRIX (MAGnetotail Transition Region In-situ eXplorer) mission overview, carried forward from the Preliminary Design Report (PDR) and updated to reflect post-PDR subsystem refinement. The substantive mission architecture of eight spacecraft distributed across three coplanar, highly eccentric orbits with apogees of $7 R_E$, $9 R_E$, and $12 R_E$ and perigees of ~ 1000 km has not changed since PDR.

Earth's magnetosphere, formed by the interaction of the solar wind with Earth's magnetic field, stores and redistributes energy through the magnetotail. In the near-Earth transition region, approximately 6-12 Earth radii (R_E), energy and mass are transported earthward through bursty bulk flows (BBFs), which are plasma structures with mesoscale cross-tail extent on the order of one to a few R_E . While heritage missions such as THEMIS, MMS, and Cluster have successfully characterized microscale processes and broader magnetospheric structure, a key scientific gap remains in resolving how BBFs evolve across this larger transition-region corridor and how their associated magnetic and particle energy is redistributed as they move Earthward. The 2025 NAS Decadal Survey identifies this as a critical deficiency, noting that existing observations are insufficient to resolve the cross-tail structure of the transition region.

Within this broader 6-12 R_E region of interest (ROI), different subregions are associated with distinct BBF behaviors and particle energetics. Near the outer portion of the region, around 12 R_E , BBFs can be observed as they propagate earthward from the mid-tail plasma sheet, where they are generated at distances of order $\sim 17 R_E$. Closer in, around 9 R_E , BBFs encounter the increasing magnetic pressure of the near-Earth dipole field and begin to undergo braking, resulting in rapid deceleration, dipolarization, and flow diversion. This represents a key mesoscale interaction boundary where earthward-propagating flows convert kinetic energy into magnetic and particle energy and generate strong variability in fields and currents. Further inward, between 6.6-9 R_E , the consequences of this braking are observed in the inner magnetosphere, where energized particles, enhanced currents, and reconfigured magnetic fields contribute to ring current buildup and radiation belt dynamics. Sampling across these regions enables MATRIX to distinguish between the processes governing BBF evolution and braking and the downstream transport and deposition of energy into the inner magnetosphere. This motivates the central RFP-aligned question of how energy and mass transported by mesoscale phenomena (like BBFs) are redistributed and ultimately coupled into the inner magnetosphere despite flow braking in the transition region, and how this process can be resolved through coordinated multipoint, in-situ measurements.

MATRIX addresses this gap as a MIDEX-class, eight-spacecraft mission designed to obtain coordinated, in-situ measurements of magnetic field and plasma properties across this full transition-region corridor. The concept of operations (ConOps) distributes the spacecraft across three orbital groups with apogees at $7 R_E$, $9 R_E$, and $12 R_E$, so that each group samples a distinct but physically connected portion of the larger ROI. The $7 R_E$ apogee, two-spacecraft pair provides closely spaced measurements of inner-region flow braking and microscale structure; the $9 R_E$ apogee, four-spacecraft group resolves mesoscale structure and variability within the transition region; and the two spacecraft in the $12 R_E$ apogee orbit capture incoming BBFs and upstream conditions prior to further earthward propagation. Together, these measurements enable MATRIX to link upstream BBF formation, mesoscale evolution, and inner-region energy deposition within a unified



mission architecture.

Updates since PDR. The mission-level science motivation, ROI definition, and three-group orbital architecture remain as baselined at PDR. Post-PDR subsystem work has refined the implementation of this architecture without altering the mission overview itself: GNC and PLV have continued to mature the Δv budget and orbit-raising/lowering maneuver sequence used to deploy the three groups from the shared $9 R_E$ insertion orbit, ADCS has converged on a 23.5° ecliptic-plane stationkeeping strategy consistent across all three groups, and SMM, THM, POW, and CCD have refined their mass, volume, power, and data budgets against this fixed mission architecture.

Magnetotail Systems Laboratory (MSL) thus affirms that the mission, as described above, remains compliant with all mission-level requirements, constraints, and resource budgets, including the MIDEX-class \$350M (FY2025) cost cap (MISS-C-003), the 2,280 kg launch wet mass (MISS-C-005) and 14.16 m^3 launch volume (MISS-C-004) allocations, the 6-year minimum operational lifetime (MISS-F-004), and the spacecraft-level mass, volume, power, and data budgets maintained across all subsystems.

1.2 Mission Objectives

The MATRIX mission objectives focus on coordinated, concurrent measurements of bursty bulk flows (BBFs) across the full 6–12 R_E magnetotail transition region, with particular emphasis on resolving the mesoscale physics of flow braking and energy deposition within the 6.6–9 R_E sub-region [1]. These objectives are derived from the identified science gap in understanding how BBFs evolve across the transition-region plasma sheet, and how that energy is subsequently redistributed into the inner magnetosphere.

Primary Objectives:

- Characterize the **spatial and temporal structure of BBFs across the 6–12 R_E transition region** using coordinated, concurrent multi-point measurements.
- Resolve the **radial evolution and braking of BBFs within the 6.6–9 R_E region** by measuring flows at multiple radial distances.
- Quantify the **magnetic field and particle signatures associated with BBF braking** to determine how energy is deposited and redistributed within the transition region.

No major updates to the MATRIX mission objectives since the PDR.

1.3 Mission Requirements

MATRIX's mission-level requirements remain largely consistent with those defined at PDR, with no major changes to the overall requirement set. Table 1 presents a subset of the most critical mission-level requirements used to guide subsystem design and verification. Minor refinements have been made at the subsystem level to better capture implementation details and interface constraints, which can be seen in the [MATRIX Requirements Spreadsheet](#). These updates do not alter the fundamental mission specifications, and all subsystem requirements continue to trace directly to the same mission-level requirements.



ID	Requirement	Rationale	Verification Method
MISS-F-001	The mission must obtain in situ magnetic field and plasma measurements of bursty bulk flows within the magnetotail nightside transition region (6-12 Earth radii away)	The RFP requires measurements of critical magnetotail transition regions. The NAS 2024 Heliophysics Decadal Survey highlighted a need to resolve mesoscale cross-tail structures in the magnetotail's transition region	Satisfaction of trace-down requirements
MISS-F-002	The mission must make spatial measurements of the magnetotail transition region on a mesoscale (1-3 Earth radii)	The RFP requires multipoint in situ measurements at mesoscale	Satisfaction of trace-down requirements
MISS-F-003	The mission should make measurements at multiple spatial scales	RFP recommendation with strong scientific value by adding an additional perspective to the mission's science data	Satisfaction of trace-down requirements
MISS-C-003	The mission must cost at most \$350M FY2025	The cost of a MIDEX mission must not exceed \$350M FY2025 not including standard launch services as defined in the RFP	Cost budget breakdown
MISS-P-001	Spacecraft instruments should perform field and particle measurements with a resolution greater than or equal to the THEMIS mission measurements	A mission launching in 2033 should significantly improve upon measurements from the THEMIS mission launched in 2007	Reference instrument datasheet/paper
MISS-E-001	All mission components must be operational within the radiation environment of Earth's magnetotail (33 krad/year TBC) for the duration of the mission	The environment of the Van Allen belts and Earth orbits high above the Van Allen belts exposes the spacecraft electronics to harsh radiation environments	Selection of parts. Use datasheets or flight experience to verify radiation-hardened capabilities

Table 1: Mission Level Requirements (Subset)

1.4 Mission Constraints

1. **Cost:** The maximum cost for a MIDEX-class mission is \$350M (FY2025), with an additional \$50M available if a non-standard launch service is used, as outlined in the RFP [2]. This cost cap constrains the overall complexity of the spacecraft subsystems. Exceeding it introduces significant programmatic risk, including the possibility of descoping or even mission cancellation. Keeping costs lower provides more flexibility to accommodate design changes and implement risk-mitigation strategies. Since the PDR, this constraint has remained unchanged, but subsystem-level budget allocations have adjusted to ensure the total mission cost remains within the allowable limit.
2. **Timeline:** As a MIDEX-class mission, MATRIX must be delivered for launch by the end of 2032 and be fully operational by the end of 2033. Furthermore, MATRIX must operate for a minimum of 5 years. Since the PDR, these schedule constraints have remained constant; however, a more defined and detailed launch timeline has been established to support mission planning and subsystem integration. Schedule constraints strongly influence system design decisions, and timely delivery is critical to securing launch opportunities. Significant delays could result in the mission being deprioritised in favor of other scientific programs, reallocation of MATRIX funding, or reduced access to scheduled launch vehicles.
3. **Mass and Volume:** MATRIX must be designed to comply with the launch mass and volume requirements of its selected launch vehicle. Estimates based on NASA Launch Services Program performance tools [3] have since been refined since the PDR, and require the final constellation to satisfy a launch wet mass of at most 2880 kg and a launch volume of at most 14.16 m³. These constraints drive subsystem mass allocation and volumetric trades across



the spacecraft.

4. **Environmental Durability:** The spacecraft and its onboard electronics must survive continuous exposure to the intense radiation environment of the magnetotail without faulting. The spacecraft must also be able to function effectively in a low-temperature, vacuum environment, and withstand impact from space debris. This constraint remains consistent with the PDR.

1.5 Mission Drivers

1. **2-D Field and Particle Measurements for Mesoscale Science:** MATRIX seeks to measure both particles and magnetic fields to satisfy RFP requirements. This necessitates at least two scientific instruments, a magnetometer, and a particle detector. To achieve measurements on a mesoscale ($\sim 1 R_E$), the spacecraft in the MATRIX constellation must be spaced appropriately and have high precision in their GNC/ADCS systems. The transition from 3-D to 2-D field measurements carried through between the ITR and PDR remains unchanged in the FDR, and continues to guide the final mission design.
2. **High Temporal Sampling Rate:** The highly dynamic plasma, magnetic, and electric environments of the magnetotail require high-frequency measurements to capture transient phenomena, consistent with the approach established at PDR. The resulting data volume has driven the development of data handling and communication systems, including antenna sizing, transmitter power and instrument data collection modes.
3. **Orbit Geometry for Transition Region Access:** The MATRIX constellation must be able to access critical transition regions of the magnetotail with sufficient dwell time to take the required measurements. Since the PDR, the major orbit design has now been established as final, with trajectories selected to support a 2-D measurement geometry and a set of orbits with apogees spanning the 6–12, R_E range. Key parameters such as eccentricity and inclination have been defined to ensure adequate dwell time in the transition region while meeting mission trajectory and Δv requirements.
4. **Accurate Constellation Stationkeeping:** This driver remains unchanged from the PDR. Accurate 2D measurements require maintaining a well-defined spacecraft formation with coordinated orbital periods and controlled phasing. Spacecraft spacing must be preserved to resolve mesoscale structure, while orbital drift must be monitored and corrected through periodic maneuvers. These requirements drive factors like propulsion capability and navigation and control system design to ensure precise relative positioning and synchronised measurements across the constellation.
5. **Data Storage and Communication:** Multipoint and high cadence measurements require sufficient onboard storage and processing capability, with the resulting data volume driving the design of data handling and communication systems. Since the PDR, data prioritisation strategies, multiple collection modes, and downlink budget analysis have been developed to support reliable and efficient transmission to ground stations, driving requirements on electronics, antenna sizing and transmitter power.

1.6 Concept of Operations

The MATRIX mission consists of eight satellites in three highly eccentric orbits aligned with the ecliptic plane and with apogees of $7R_E$, $9R_E$, and $12R_E$ and perigees of ~ 1000 km. These orbits were selected to sample different regions within the magnetotail transition region from $6 - 12R_E$, enabling the mission to track the evolution of bursty bulk flows (BBFs) as they propagate earthward, undergo braking, and deposit energy into the inner magnetosphere.

Spacecraft are organized into three groups: (Group 1, Red) a pair of spacecraft with apogee radius $12 R_E$ spaced out at near opposite sides of the orbit, (Group 2, Purple) a train of four spacecraft with apogee radius $9 R_E$ separated at the mesoscale (a few Earth radii) to study the mesoscale structure and variability of BBFs, and (Group 3, Blue) a “buddy” pair of two spacecraft with apogee radius $7 R_E$ separated at the microscale (tens to hundreds of km) to resolve inner-magnetosphere dynamics and energy deposition. This architecture enables simultaneous sampling of BBF evolution across multiple spatial scales while maintaining continuous coverage of the transition region.

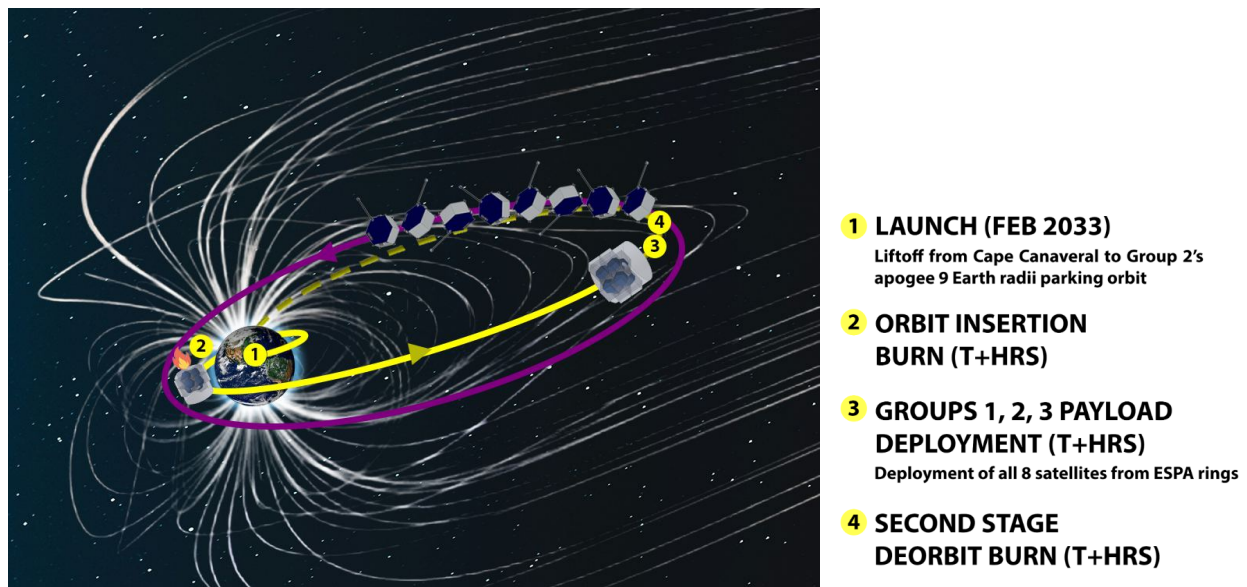
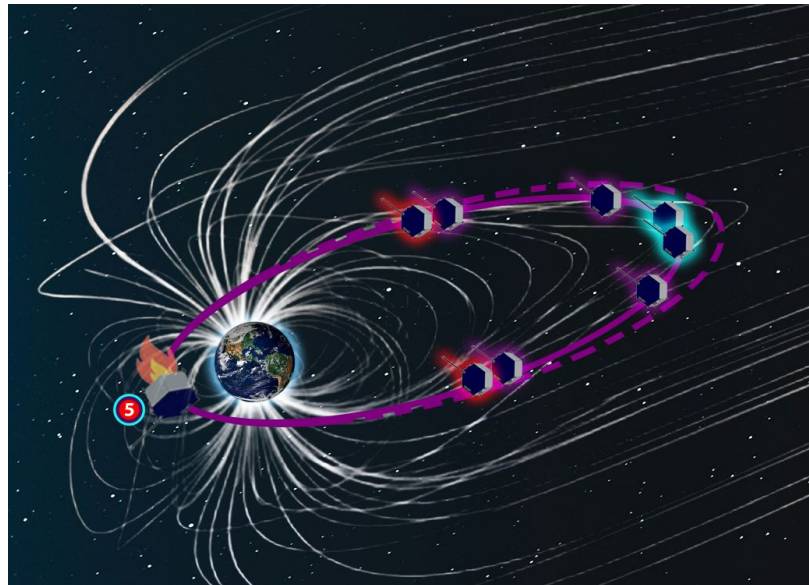


Figure 1: MATRIX Concept of Operations: Launch and Early Orbit

All eight satellites in the MATRIX constellation are launched as one payload from Cape Canaveral in February 2033. NASA's Launch Services Program deploys the entire constellation stack into the apogee radius $9R_E$ orbit of Group 2 (Figure 1). Immediately after deployment, each spacecraft powers on, detumbles and acquires a Sun-pointing state, and makes contact with the Near Space Network. In the meantime, the second stage of the launch vehicle deorbits and burns up in Earth's atmosphere.



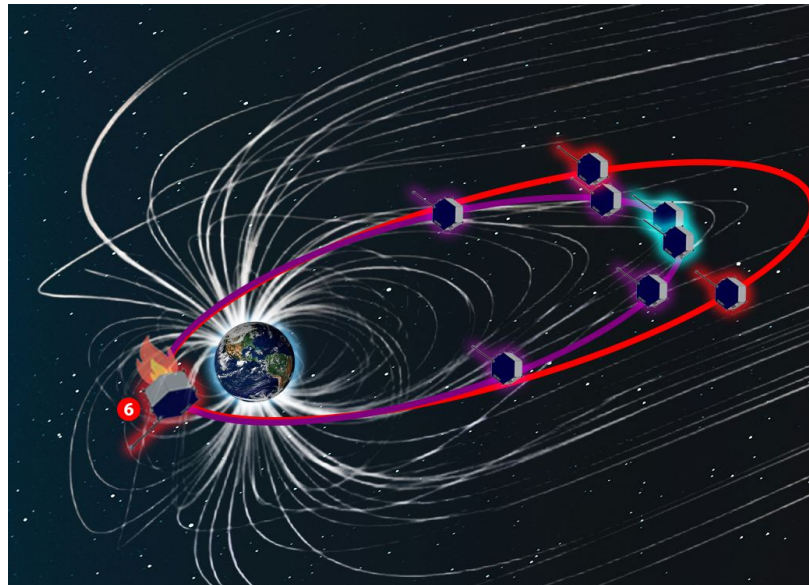
5 GROUPS 1, 2, 3 SPACING AND COMMISSIONING (T+MONTHS)

While in Group 2 orbit, slightly modify orbit apogees so the change in orbital period causes satellites to drift apart

Power on satellites, establish communications, and perform system checkouts

Figure 2: MATRIX Concept of Operations: Orbit Spacing & Commissioning

After all the satellites have successfully powered on, most of 2033 is spent spacing out the satellites in their three groups, shown in Figure 2. One at a time, the MATRIX spacecraft rotate while approaching perigee to achieve a prograde orientation and perform a very small thruster firing to raise the apogee of the orbit. Then they return to a sun-pointing orientation. This maneuver increases the orbital period slightly and allows the spacecraft to drift away from each other. Once each spacecraft is spaced correctly relative to the other spacecraft as described earlier in this section, they perform the same perigee maneuver but in the retrograde direction to return to a $9 R_E$ apogee and stop drifting apart. During this time in 2033, each spacecraft is also commissioned by turning on the payload instruments and performing checks of all spacecraft systems.

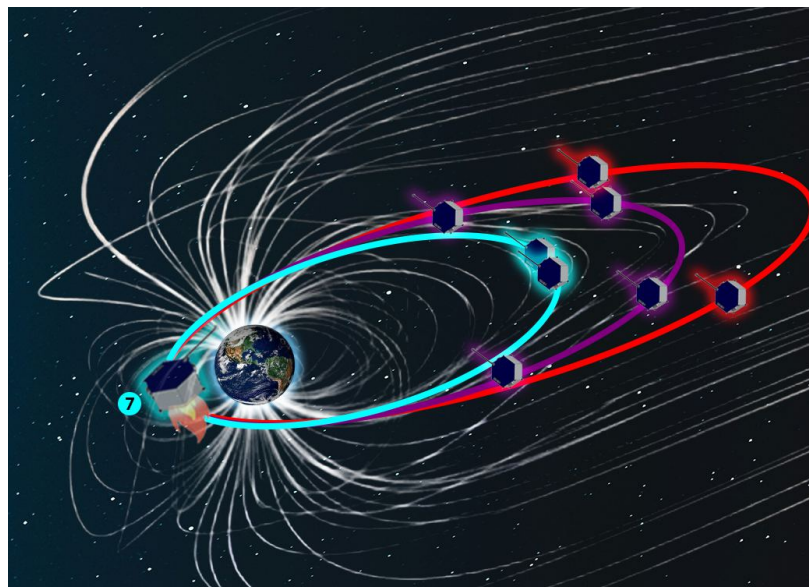


**6 GROUP 1 ORBIT RAISING
(NOV 2033)**

Raise apogee to a radius of 12 Earth radii

Figure 3: MATRIX Concept of Operations: Group 1 Orbit Raising

After all spacecraft have achieved the correct spacing by late 2033, the two Group 1 satellites perform a prograde orbit-raising maneuver at perigee following the same method as the orbit spacing maneuvers except with greater Δv to raise the orbit apogee all the way up to $12 R_E$.



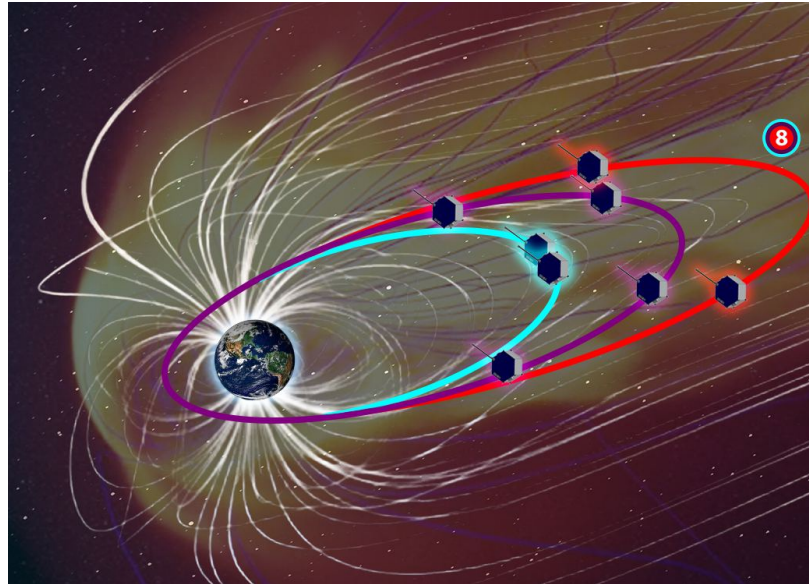
**7 GROUP 3 ORBIT LOWERING
(NOV 2033)**

Lower apogee to a radius of 7 Earth radii

Figure 4: MATRIX Concept of Operations: Group 3 Orbit Lowering

Following the orbit-raising of the Group 1 satellites, the two Group 3 satellites perform a retrograde burn at perigee to lower their apogee to $7 R_E$. The orbit raising and lowering burns for each of the four total satellites in Group 1 and Group 3 occur in direct succession to maintain the

correct constellation spacing (as soon as one of the spacecraft reaches perigee and performs the maneuver, the next spacecraft to reach perigee also performs the maneuver.)



**8 5-YEAR PRIME MISSION
(STARTS DEC 2033)**

Take in-situ measurements of Earth's magnetotail for approximately half of each year

Figure 5: MATRIX Concept of Operations: Nominal Science Operations

After all spacecraft have arrived at the correct orbit by December 2033, nominal science operations begin and continue for a period of at least five years. During this time, MATRIX spacecraft simply go around the Earth in their orbits with science instruments continuously on (except during maneuvers) and taking measurements of the magnetic field and plasma in the magnetotail. When each spacecraft passes perigee, the data from that orbit is downlinked to the Near Space Network and the spacecraft receives telemetry and, if necessary, commands from Earth. Every 28 days there is a stationkeeping burn performed at apogee to maintain inclination and perigee altitude.

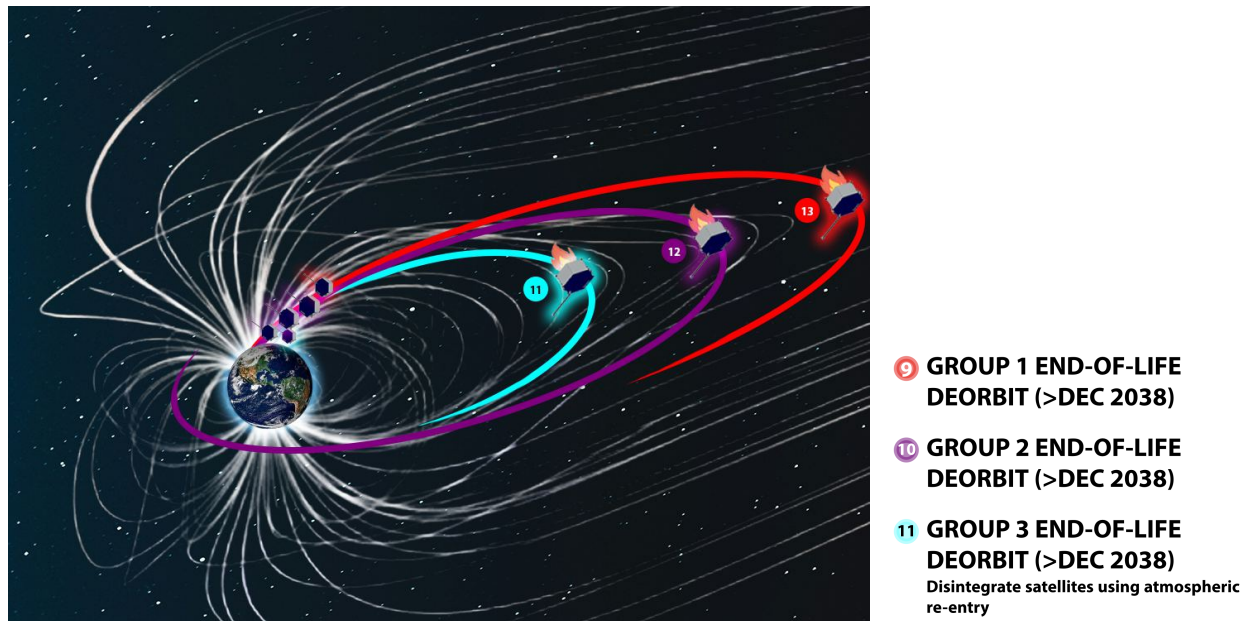


Figure 6: MATRIX Concept of Operations: End-of-Life

After five years of operations, in late 2038, the MATRIX mission comes to an end. If at this time all spacecraft are still healthy and contain sufficient fuel, the mission could be extended, but if not, each MATRIX spacecraft performs a retrograde burn while at apogee to lower the orbit perigee into Earth's atmosphere and disintegrate.

1.7 Key Design Decisions

The most important design decisions made in order to arrive at the final iteration of the MATRIX Concept of Operations came from orbit design work carried out by GNC.

In the early days of the MATRIX mission, the satellite groups were inclined relative to each other by as much as 10° . However, after looking at available launch options and calculating the Δv required to perform the inclination change burns, the Δv cost was too significant to be able to change inclination and have enough fuel leftover to stationkeep for at least 5 years of nominal science operations. As a result, the orbits became coplanar.

Another key design decision in the ConOps was launching directly into the Group 2 orbit with an apogee of $9 R_E$. After taking into account the mass budget of the entire MATRIX constellation, PLV analysis showed that NASA LSP was able to place the entire payload of satellites directly into the Group 2 orbit. GNC then confirmed that this configuration would result in the least amount of Δv required because Group 2 is the intermediate orbit and therefore minimizes the change in apogee required to achieve the orbits of the other two groups.



2 Compliance

2.1 Mission Requirements Compliance

Tables 2, 3, 4, and 5 show the verification and validation (V&V) details of the Mission Level Functional, Constraint, Performance, and Environment Requirements, respectively, to ensure that all these requirements are in compliance with the mission design. See the [MATRIX Requirements Spreadsheet](#) for the complete list of RFP, mission, and subsystem level requirements. The rows highlighted in yellow reflect the major changes or values confirmed since the PDR stage.

ID	Requirement	Status	Verification Details
MISS-F-001	The mission must obtain in situ magnetic field and plasma measurements of BBFs in magnetotail transition region	MET	Analysis performed in Sections 6.7.1, 4.1, and 4.7
MISS-F-002	The mission must make spatial measurements of the magnetotail transition region on a mesoscale (1-3 Earth radii)	MET	Analysis performed in Section 6.7.1
MISS-F-003	The mission should make measurements at multiple spatial scales	MET	Analysis performed in Section 6.7.1
MISS-F-004	The mission components must be operational for at least 6 years	MET	Analysis performed in Sections 8.7.1 and 4.7.4
MISS-F-005	The mission components must have an end-of-life plan for disposal	MET	Analysis performed in Section 6.7.3
MISS-F-006	The mission components shall collect and transmit science and telemetry data back to Earth at least once per orbit	MET	Analysis performed in Sections 5.7.10 and 6.7.1
MISS-F-007	The mission components shall operate with full autonomy in the magnetotail transition region and at least partial autonomy at orbit perigee	MET	Analysis performed in Sections 5.7.3 and 5.7.5
MISS-F-008	The mission components shall have propellant available to conduct mission operations for the duration of the mission	MET	Analysis performed in Sections 6.7.3, 11.7.3, and 11.7.4
MISS-F-009	The mission shall have power available to conduct mission operations for the duration of the mission	MET	Analysis performed in Section 10.7.3
MISS-F-010	Mission components shall have the ability to maneuver around the scenario of a potential collision in their nominal orbit	MET	Analysis performed in Sections 7.7.1 and 11.7.5
MISS-F-011	The nominal mission orbit must be designed in a way that avoids collisions between mission components	MET	Analysis performed in Section 6.7.1
MISS-F-012	The mission components must monitor temperature changes on the spacecraft throughout the mission and provide heating and heat dissipation mechanisms	MET	Analysis performed in Section 9.7.6
MISS-F-013	Each spacecraft should have the ability to receive uplinked data from Earth at any point in the orbit	MET	Analysis performed in Section 5.7.10

Table 2: Mission Level Functional Requirements



ID	Requirement	Status	Verification Details
MISS-C-001	The spacecraft must be delivered for launch by the end of 2032	MET	Analysis performed in Section 1.6
MISS-C-002	The mission components must be fully operational by the end of 2033	MET	Analysis performed in Section 1.6
MISS-C-003	The mission must cost at most \$350M FY2025	MET	Analysis performed in Section 2.5
MISS-C-004	The final constellation must have a launch volume of at most 14.16 m ³	MET	Analysis performed in Sections 11.1 and 8.7.2
MISS-C-005	The final constellation must have a launch wet mass of at most 2280 kg	MET	Analysis performed in Section 2.2
MISS-C-006	Each mission component shall have a volume of at most TBD at the beginning of mission operations	MET	Analysis performed in Section 8.7.2

Table 3: Mission Level Constraint Requirements

ID	Requirement	Status	Verification Details
MISS-P-001	Spacecraft instruments should perform field and particle measurements with a resolution greater than or equal to the THEMIS mission measurements	MET	Analysis performed in Sections 4.7.6 , 4.7.9 , and 4.7
MISS-P-002	Spacecraft instruments should have the dynamic range necessary to observe BBFs	MET	Analysis performed in Section 4.7
MISS-P-003	The mission components shall spend at most 20 hours (Group 1), 13 hours (Group 2), or 10 hours (Group 3) per orbital period charging/generating power	MET	Analysis performed in Sections 10.7.1 and 10.7.5

Table 4: Mission Level Performance Requirements

ID	Requirement	Status	Verification Details
MISS-E-001	All components must be operational within the radiation environment of Earth's magnetotail (33 krad/year) for the duration of the mission	MET	Analysis performed in Section 8.7.1
MISS-E-002	All mission components must withstand the Launch Vehicle load environment of 7g axial and 3g lateral during launch	MET	Analysis performed in Section 8.7.3
MISS-E-003	All mission components must remain operational when exposed to the full range of heat fluxes experienced on-orbit	MET	Analysis performed in Sections 9.7.7 and 9.7.1
MISS-E-004	All mission components must be operational within the microgravity environment for the duration of the mission	MET	Analysis performed in Section 8.7.3

Table 5: Mission Level Environmental Requirements



2.2 Mission Mass Compliance

The mass budget has been iteratively refined as the design of each subsystem matured. The table below presents the final mass budget, reflecting the percentage mass allocations, the actual mass allocations, and the final mass used by every subsystem. Recall that the maximum mass each spacecraft can have is 285 kg, as determined by the constraint mass of the separation mechanism with a 5% margin. With the propellant mass of 153.6 kg reserved, the allowable dry mass is therefore 131.4 kg.

Subsystem	Percentage Allocated (%)	Mass Allocated (kg)	Mass Used (kg)	Status
CCD	1.1%	1.39	1.22	Compliant
OPL	4.9%	6.48	5.70	Compliant
GNC	0.0%	0.0	0	Compliant
ADCS	4.6%	6.05	5.33	Compliant
POW	3.7%	4.93	4.34	Compliant
PLV	19.3%	25.36	22.32	Compliant
SMM	56.8%	74.58	65.63	Compliant
THM	9.6%	12.61	11.10	Compliant
Total	100.00%	131.40	115.6	

Table 6: Subsystem Mass Budget

As it can be observed above, all subsystems designs stayed within their allocated mass with some margin. The final design has a wet mass of 269.23 kg, which leaves the spacecraft with a mass margin of 5.53%. Thus, the MATRIX spacecraft is compliant with the mission's mass requirements and constraints.

2.3 Mission Volume Compliance

The volume budget has been refined as the design of each subsystem matured. The table below presents the final volume budget, reflecting the actual volume allocated to every subsystem. Recall that the maximum volume each spacecraft can have is of $42 \times 46 \times 56 \text{ in}^3$, or 1.77 m^3 as determined by the constraint volume of the ESPA ring.

Subsystem	Volume Allocated (cm ³)	Volume Used (cm ³)	Status
CCD	18736.3	5774.3	Compliant
OPL	87395.3	9056.3	Compliant
GNC	0.0	0.0	Compliant
ADCS	81691.6	813.2	Compliant
POW	66481.8	2835.9	Compliant
PLV	457103.4	457103.4	Compliant
SMM	891345.2	23668.3	Compliant
THM	170196.1	26414.2	Compliant
Total	1772949.2	525665.6	

Table 7: Subsystem Volume Budget

It should be noted here that the design process was facilitated by first giving volume assignments based on the mass percentages as a bottom-up approach. As such, there were minimal volume budget trades, however the inclusion of the propellant in the PLV subteam budget necessitated a trade with the SMM subteam. Overall, there are significant margins in the volume budget with



the possibility for further trades if necessary. A full mass and volume budget breakdown is shown in Appendix A.7.5.

2.4 Mission Power Compliance

Table 8 below shows the MATRIX mission power budget breakdown. In the last row, status of compliance is indicated for every mode in green. Within each mode, every subsystem utilizes less power than allocated, contributing to mission power compliance based on the positive percentage margin of power not being used out of the total allocated power for each subsystem respectively.

Subsystem	Day (W)		Eclipse (W)		Transit (W)		Stationkeeping (W)		Safe Mode (W)	
	Use	Alloc	Use	Alloc	Use	Alloc	Use	Alloc	Use	Alloc
ADCS/GNC	3.50	6.41	3.00	5.00	3.50	4.24	3.50	3.95	3.50	6.24
OPL/CCD	57.70	105.74	11.50	19.15	7.00	8.49	7.00	7.89	7.00	12.48
SMM/THM	1.00	1.83	46.30	77.11	1.00	1.21	46.30	52.21	46.30	82.54
POW/PLV	0.00	0.00	0.00	0.00	72.00	87.31	33.00	37.21	0.00	0.00
Total Subsystem Load	62.20	113.99	60.80	101.26	83.50	101.26	89.80	101.26	56.80	101.26
Battery Recharge	29.61	29.61	-	-	-	-	-	-	-	-
Total Power	91.81	143.60	60.80	101.26	83.50	101.26	89.80	101.26	56.80	101.26
Margin (%)	36%		40%		18%		11%		44%	
Compliance Verification	Yes		Yes		Yes		Yes		Yes	

Table 8: Compliance based on positive margin result from allocated power vs. subsystem power usage for all operational modes.

To see a more extensive breakdown of the power budget, please see Table 85 for the power demands of each instrument from all 8 subsystems.

The power budget was refined after the MATRIX preliminary design phase to include duty cycling for the spacecraft and its instruments. During this current final design phase, additional operational modes were annexed to the power budget to consider periods of time that more precisely defined the operational times of different instruments, such as Transit and Stationkeeping Modes. The duty cycling assisted in ensuring that the MATRIX power system would be compliant with the mission requirements. It did so by reducing the total power demands of all instruments on the spacecraft and accordingly reducing the solar array sizing. This allowed for the solar array to fit on the face of the spacecraft, avoid the demand of being deployable, and comply with overall mission mass and volume requirements.

Additional power compliance can be noted in Section 2.9 where the pointing of the solar array is noted to always be sun-facing. Because of this, the spacecraft is in the most optimal position to generate power from the solar arrays in the most consistent and efficient manner.

2.5 Mission Cost Compliance

The MATRIX mission lifecycle cost has been re-baselined since PDR using a bottom-up Bill of Materials (BOM) rollup augmented by SMAD-style programmatic and operations wraps [4], in compliance with MISS-C-003 (\$350M FY25 MIDEX cost cap). Table 9 consolidates the subsystem flight-hardware costs derived from the per-subsystem BOMs (full component-level tables provided in Appendix A.1.1). Where applicable, learning-curve recurrence discounts were applied



across the eight-spacecraft fleet using a 90% slope ($s = 0.9$), consistent with SMAD recommendations for production runs of under ten units [4].

Subsystem	Allocated (\$M)	BOM Total (\$M)	% Used	Compliance
GNC/ADCS	3.00	0.17	5.67	MET
SMM	35.00	24.02	68.62	MET
THM	15.00	0.85	5.69	MET
POW	5.00	0.62	12.49	MET
PLV	30.00	15.25	50.82	MET
OPL	75.00	70.67	94.22	MET
CCD	10.00	2.99	29.91	MET
Flight HW Total	173.00	114.57	66.23	MET

Table 9: MATRIX subsystem flight-hardware cost allocations versus BOM-based actuals. OPL value reflects learning-curve recurrence across eight spacecraft.

Adding the SMAD-aligned programmatic and operational wraps to the flight hardware total yields the full mission lifecycle cost shown in Table 10. Programmatic & Systems Engineering covers project management, mission assurance, and systems engineering activities; Ground & Operations covers ground segment development and the five-year nominal mission operations wrap. A 25% programmatic reserve (\$70M) has been held against the \$350M MIDEX cap, exceeding the 20% reserve baseline used at PDR and providing additional margin against TBC items in the BOM (notably propellant tank and Quick-Sep separation system pricing in SMM, which remain pending vendor quote confirmation).

Cost Element	Value (\$M FY25)
Flight Hardware (sum across subsystems)	173.00
Programmatic & Systems Engineering	15.00
Ground Segment & Mission Operations	15.00
Subtotal (Pre-Reserve)	203.00
Programmatic Reserve	70.00
Total Mission Lifecycle Cost	273.00
MIDEX Cost Cap (MISS-C-003)	350.00
Margin Against Cap	77.00 (22.0%)

Table 10: MATRIX mission lifecycle cost rollup against the \$350M FY25 MIDEX cost cap. Standard NASA-provided launch services are excluded from the cap per the RFP.

The total mission lifecycle cost of \$273M FY25 (including reserve) is \$77M below the \$350M FY25 MIDEX cap, leaving 22.0% margin against MISS-C-003. The MATRIX mission therefore complies with the cost constraint with substantial headroom for programmatic risk. The dominant cost driver remains the OPL payload allocation, which after learning-curve recurrence discount consumes 94.22% of its envelope; this is consistent with heritage magnetospheric missions in which payload instrument suites dominate flight hardware cost [4], [5]. All other subsystems retain margin in excess of 30% against their allocations, providing flexibility to absorb component-level cost growth between PDR and CDR.

The full per-subsystem Bill of Materials, including unit costs, quantities, and vendor sources, is documented in Appendix A.1.1, and OPL's analysis for subsystem budget allocation via SMAD is



presented in Section 4.7.3.

2.6 Thermal Compliance

Please refer to Section 9.7.6 for comments regarding the temperature-monitoring sensor placement and integration that anchors the thermal compliance budget, with the worst-case incoming flux characterization, multi-node equilibrium analysis, and supporting radiative/conductive insulation, heater, and louver trades documented throughout the remainder of Section 9. Together these analyses confirm that all spacecraft components remain within their allowable temperature ranges across hot- and cold-case orbital conditions, satisfying MISS-F-012, MISS-E-003, and all thermal performance requirements.

2.7 Data & Comms Compliance

Please refer to Section 5.7.2 and Section 5.7.6 for comments regarding the on-board data generation, processing, storage, and virtual-channel allocation that define the data-handling compliance budget, and to the link budget analysis in Section 5.7.10 for the corresponding RF compliance budget on both the downlink and uplink across the full operational range to the Near Earth Network. Supporting calculations for bandwidth allocation, flight software throughput, and link budget parameters are documented in Appendices A.2.1, A.3, and A.2.2. These analyses collectively verify that the MATRIX spacecraft meet MISS-F-006 and MISS-F-013.

2.8 Delta V Compliance

Please refer to Section 11.7.2 for the heritage-derived Δv context and Section 11.7.3 for the per-group Δv requirements that define the propulsion budget, with Section 11.7.4 demonstrating that the finalized five-tank hydrazine architecture provides approximately 1899 m/s of Δv capability against the maximum required 1129 m/s, leaving roughly 770 m/s of margin. The thruster selection supporting these maneuvers is documented in Section 11.7.5, and the in-depth orbit-transfer, stationkeeping, and decommission breakdown is provided in Appendix A.5.2. Together these analyses confirm compliance with MISS-F-008.

2.9 Pointing Budget & Scheduling Compliance

Please refer to Section 11.7.5.3 for comments regarding the compliance budgets related to thruster pointing, Sections 10.7.5 and 10.7.5.6 for comments regarding the compliance budgets related to solar array pointing, Section 5.7.8 for comments regarding the compliance budgets related to antenna pointing across the omnidirectional S-band link, and Sections 4.7.8 and 8.7.4 for comments regarding the compliance budgets related to magnetometer boom standoff and deployment that govern FGM pointing and magnetic cleanliness. Together with the spin-stabilized 15 rpm Sun-pointing architecture described in Section 1.6, these references confirm that all pointing budgets are satisfied across the mission operational modes.

3 Systems Integration

In the final design review stage, the MATRIX mission architecture has reached a mature state with well-defined interfaces between all subsystems. The overall mission goal and concept of operations with three groups of spacecraft in elliptical orbits remain unchanged as outlined in Section 1.6. The spin-stabilized architecture at 15 rpm with the spin axis along the Sun-Earth line has been retained from PDR and continues into the design consideration for all subteams. Since PDR, the **mass budget**, **power budget**, and **bill of materials** have all been updated to reflect final component selections, and a team-wide **parameter list** has been maintained to ensure consistency across subsystems.

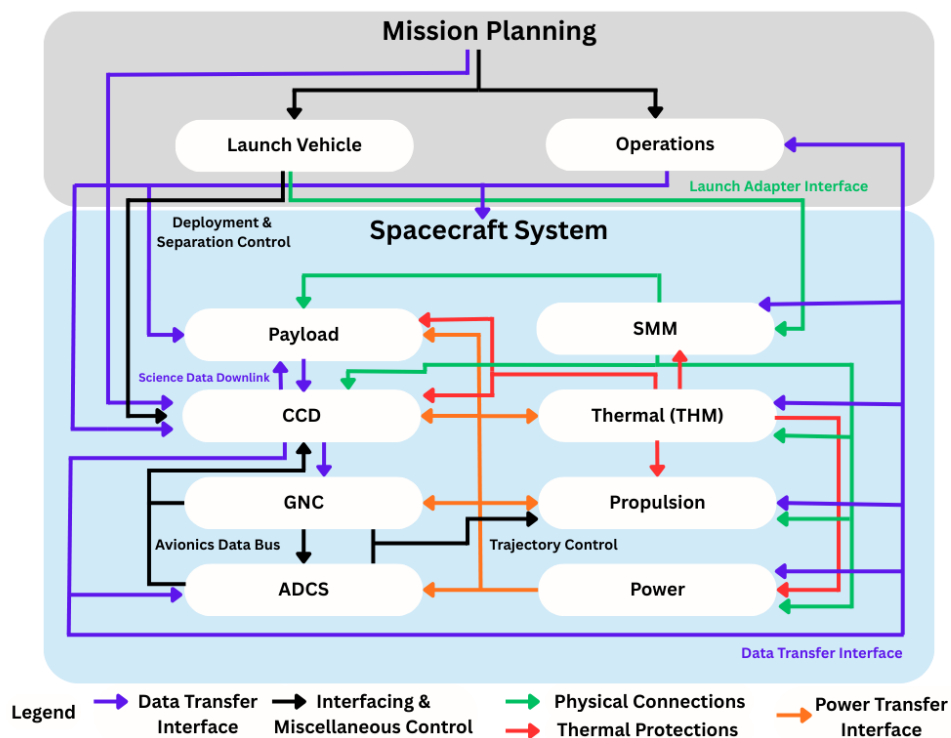


Figure 7: System Block Diagram

Figure 7 presents the current system block diagram (SBD), which shows the types of interactions between all 8 subsystems. Certain subsystems are critical dependencies for a large number of others. For example, SMM has distributed mass and volume budgets to the rest of the team, supporting the physical integration of all components - fuel tanks (PLV), science instruments (OPL), solar panels (POW) - within the spacecraft structure. The GNC, ADCS, and PLV teams are in close agreement on the feasibility of the updated ConOps orbits. The rest of this section details the more notable connections between subsystems.

- **GNC, PLV, and SMM:** The GNC subsystem drives the Δv budget, which in turn determines the propellant mass that PLV must provide. At FDR, GNC's STK simulations have confirmed the Δv costs per orbit group, enabling PLV to finalize hydrazine tank sizing and thruster selection. The resulting wet mass and tankage volume were passed to SMM, which



incorporated four custom ellipsoidal propellant.

- **POW and all subsystems:** The power subsystem presents a significant constraint for all other subsystems. Each subsystem's component selections feed into the power budget, and the total spacecraft power demand must remain within what the solar arrays can generate and the battery can store across the orbit. Power allocations have been agreed between POW and all subsystems, and duty cycling of high-power components such as the ESA and EPD during eclipse has been incorporated into the operational mode definitions managed by OPL.
- **POW and THM:** Eclipse survival is a shared challenge between POW and THM. During eclipse periods of up to 4.4 hours, the spacecraft must maintain component temperatures above survival limits using heaters, while simultaneously managing the battery to its depth of discharge floor. This creates a direct trade between heater power demand and battery capacity. POW and THM have coordinated on heater sizing and duty cycling to ensure both constraints are met within the available energy storage budget.
- **SMM and all subsystems:** The SMM subsystem provides the physical integration for the entire spacecraft. The structure accommodates all physical components from every other subsystem, and the mass budget is maintained by SMM through a bottom-up component spreadsheet. The total dry mass currently stands at 118.2 kg, with PLV and SMM components occupying the majority of the budget. Volume allocation has been managed through the CAD model, with trades in both budgets made as component selections were finalized.
- **OPL and SMM:** The fluxgate magnetometer science requirement imposes a minimum stand-off distance between the FGM sensor and the spacecraft center to limit contamination from spacecraft-generated magnetic fields. This has driven the inclusion of a deployable boom in the structural design and is a significant driver for the spin-stabilized architecture as conventional reaction wheel systems produce considerable magnetic field pollution.
- **CCD and OPL:** Science data generated by the three instruments must be stored onboard and downlinked during the approximately 1-hour perigee contact window using the omnidirectional S-band antenna. The daily science data volume is constrained to remain within CCD downlink and storage capacity, and OPL has sized instrument duty cycles accordingly.

It should be noted that at this stage there is still the opportunity for optimization and improvement between subteams. Feedback provided has suggested that PLV and SMM investigate ways to decrease the required propellant given a spacecraft structure with less mass, since total wet mass directly drives propellant budget through the rocket equation. The volume budget also requires finalization as component placements are confirmed in the CAD model. However, the overall systems integration shall be considered ready for further development of the MATRIX mission.



4 OPL Design

4.1 Subsystem Overview

The Operations & Payload (OPL) subsystem ensures the MATRIX mission executes feasibly across all lifecycle phases, with science return goals met within mission-level resource and programmatic constraints. OPL has two coupled roles: *Operations* defines the mission's end-to-end timeline and phase structure, spacecraft operational modes, executable procedures, resource and cost allocations, and risk analysis and mitigation, and *Payload*, which selects, characterizes, and operationally configures the science instruments, where three were selected: a fluxgate magnetometer (FGM), an electrostatic analyzer (ESA), and an energetic particle detector (EPD) to provide the concurrent magnetic-field and plasma measurements required to resolve bursty bulk flow (BBF) signatures across the 6-12 R_E magnetotail transition region [6]. The two roles are coupled because instrument duty cycles, data rates, and power profiles are themselves operational quantities: the same trade that selects an instrument also fixes the resource demand that operations must schedule against. Across the eight-spacecraft constellation, OPL further serves as the integration point for all subsystems, enforcing coherent mode transitions across vehicles, defining resource-feasible duty cycles for the POW and CCD subsystems, implementing fault-response policies that minimize science interruption, and maintaining a cost-aware doctrine consistent with the \$350M FY25 MIDEX cost cap [2], [4].

4.1.1 Operations Overview

The Operations side of OPL turns mission-level scientific and programmatic requirements into a flyable mission. It owns the mission timeline (when activities happen, from pre-launch fabrication through end-of-life disposal), the operational mode set (what configuration the spacecraft is in at any instant), the procedure set (how the spacecraft enters, executes, and exits each mode), the cost budget (how the \$350M FY25 MIDEX cap is allocated across subsystems), and the risk analysis (what can go wrong and how it is mitigated). Because MATRIX is an eight-spacecraft constellation rather than a single vehicle, Operations is also responsible for constellation-level concerns that requires multi-subsystem interfacing: coordinated deployment from a shared dispenser, mesoscale spacing maintenance, and time-synchronized science cadence across all eight spacecraft. Two mission-architecture decisions, fixed during PDR and retained at FDR, propagate into every operations product: sun-facing spin stabilization at 15 rpm with the spin axis along the Sun-Earth line, which makes science-instrument pointing inseparable from spin phase and removes the need for dedicated downlink-pointing maneuvers, and stacked co-launch into a 9 R_E parking orbit, which mandates a sequenced ~ 9 -month commissioning phase between launch and nominal science to allow for spacing. Together, the autonomous execution between intermittent perigee contacts, strict adherence to cost and resource constraints, and sun-pointing as the failure-safe attitude requirements motivate the analyses conducted below.

4.1.2 Payload Instrument Suite Overview

The MATRIX payload comprises three instruments selected to detect and measure the magnetic field and plasma signatures of bursty bulk flows (BBFs) in the nightside transition region of the magnetotail. All three instruments are based on flight-proven heritage from the THEMIS and MMS missions, leveraging mature designs to reduce technical risk for MATRIX as a MIDEX-class

mission:

- **Fluxgate Magnetometer (FGM):** The THEMIS FGM [7], providing 0.003 nT resolution, 4–128 Hz selectable sampling rate, and a $\pm 25,000$ nT dynamic range. The FGM is mounted on a 1 m boom to satisfy magnetic cleanliness requirements.
- **Electrostatic Analyzer (ESA):** The THEMIS ESA [8], providing ion measurements from 1.6 eV to 25 keV and electron measurements from 2 eV to 32 keV, with 4π sr coverage achieved by combining a 180° instantaneous field of view with the spacecraft's 4 s spin period.
- **Energetic Particle Detector (EPD):** A modified version of the MMS Energetic Ion Spectrometer [9], designated EIS+, extending electron energy detection to 1.5 MeV and rated for 200 krad cumulative radiation dose to support the 6-year mission lifetime.

The physical layout of these instruments on the spacecraft bus is shown in Figure 8.

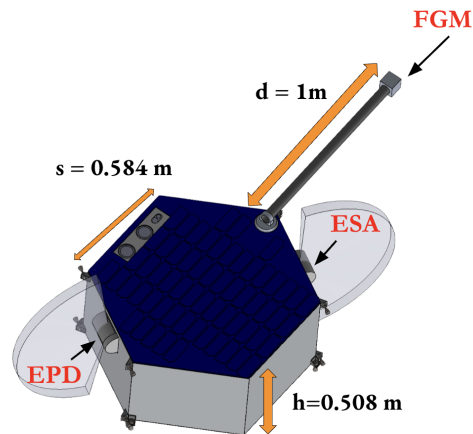


Figure 8: Layout of the MATRIX payload instruments on the spacecraft bus, showing the FGM mounted on a 1 m deployable boom and the accommodation of the ESA and EIS+ instruments on the main bus.



4.2 Subsystem Objectives

The OPL subsystem objectives flow directly from both the mission's primary science goal of characterizing the structure and evolution of bursty bulk flows across the 6-12 R_E magnetotail transition region, and from the programmatic constraints imposed by the MIDEX cost cap and the RFP-mandated timeline [2], [6]. The five OPL objectives are:

1. **Define and validate operational modes and procedures across all mission lifecycle phases**, from pre-launch integration through end-of-life disposal. This objective owns the conversion of mission-level timeline requirements (RFP-C-001 through RFP-F-005) into executable spacecraft behavior.
2. **Maximize science availability within power, data storage, downlink, and ΔV constraints**. The constellation must produce enough science return to satisfy the mesoscale-coverage and high-cadence measurement requirements (RFP-F-007 through RFP-F-011) while remaining inside the resource envelopes provided by POW, CCD, and PLV.
3. **Implement a risk-informed anomaly response doctrine** that protects spacecraft and constellation health. With eight spacecraft operating semi-autonomously between intermittent perigee contacts, detection-to-safe-mode transition must complete within $t_{\text{response}} \leq 600$ s without ground intervention.
4. **Maintain compliance with mission-level cost and schedule constraints**, holding lifecycle cost below \$350M FY25 (excluding standard launch services) and full operational capability achieved by end of 2033 (RFP-C-002).
5. **Select and operationally characterize the science payload** to meet MATRIX measurement requirements traced from the RFP scientific objectives (RFP-F-006 through RFP-F-011).

Objectives 1, 3, and 4 are owned by the Operations side, objective 5 is owned by the Payload side, while objective 2 sits at the coupling between them and drives both the duty-cycle analysis (Section 4.7.7) and the operational mode definitions.

4.3 Subsystem Requirements

The OPL subsystem carries five classes of requirements: functional (OPL-F), payload performance (OPL-P), operational performance (OPL-O), constraint requirements (OPL-C) and environmental requirements (OPL-E). The payload performance requirements drive instrument selection and accommodation, while the functional and operational requirements drive the mission timeline, spacecraft operational modes, and constellation-level science availability. The most significant updates since PDR are a renumbering of the magnetic cleanliness and boom-length requirements (formerly OPL-P-011 and OPL-P-012, now **OPL-P-010** and **OPL-P-011**) and the consolidation of the EPD species-specific requirements following retirement of the He^+ requirement; both updates are reflected in this section. The [MATRIX Requirements Spreadsheet](#) contains the full text of all OPL requirements.

Operational Functional Requirements (OPL-F-001-OPL-F-011) The operational functional requirements define what the OPL subsystem must *do* across the mission lifecycle, independent of how it does it. They are grouped into three sub-classes:



- **Mission execution and procedure definition (OPL-F-001 through OPL-F-005):** OPL-F-001 (executable nominal procedures for coordinated multi-spacecraft science) and OPL-F-002 (contingency and safe-mode procedures for off-nominal events) together require that every spacecraft state, nominal or off-nominal, has a defined entry trigger and action sequence. OPL-F-003 (operational procedures for failures during nominal operations) extends OPL-F-002 to in-flight anomalies specifically. OPL-F-004 (procedures for pre-launch activities) covers fabrication, integration, qualification, and launch-campaign sequencing. OPL-F-005 (launch, commissioning, and constellation-establishment sequence) defines the deploy-and-space sequence that takes the eight spacecraft from a stacked launch configuration into the final 7, 9, and 12 R_E orbit groups within the timeline imposed by RFP-C-001 and RFP-C-002.
- **Risk analysis and mitigation (OPL-F-006 through OPL-F-008):** OPL-F-006 (mission-level risk analysis), OPL-F-007 (consolidation of subsystem-level risks into a unified register), and OPL-F-008 (mitigation measures for identified risks) together place OPL in the role of constellation-level risk owner, responsible for capturing failure modes that cross subsystem boundaries (e.g., a CCD downlink fault that compromises constellation-level science availability).
- **Cost and compliance (OPL-F-009 through OPL-F-011):** OPL-F-009 (total mission cost analysis) and OPL-F-010 (subsystem cost allocation) together require that OPL maintain a defensible WBS-structured cost model demonstrating compliance with the MIDEX cap. OPL-F-011 (RFP compliance analysis) requires that the final design carries a complete requirements compliance matrix demonstrating that every RFP requirement has been satisfied by trace-down requirements within the subsystems.

Operations Constraint Requirements (OPL-C-006-OPL-C-008) The operations constraint requirements bound the OPL design in cost, schedule, and communications-feasibility space, and trace directly to mission-level RFP requirements.

- **OPL-C-006** (Mission lifecycle cost $\leq$ \$350M FY25): direct trace-down from RFP-C-003 (the \$350M FY25 MIDEX cost cap, excluding standard launch services). The cost model carries a 20% programmatic reserve on the pre-reserve total, following the SMAD principle that early-phase cost estimates should preserve a distinction between baseline estimate and programmatically responsible lifecycle estimate to absorb cost-growth risk as subsystem designs mature [4]. This constraint bounds payload selection complexity and operational complexity, and is verified through the WBS-structured cost budget developed in Section ??.
- **OPL-C-007** (Daily science data $\leq$ CCD downlink/storage capacity): ensures that operational duty cycles do not generate science volume in excess of what CCD can buffer onboard and downlink during the $\sim$ 1-hour perigee contact, traced up from RFP-F-013 (per-orbit communication of science data). This constraint is closed jointly with CCD through the per-orbit data-volume and link-budget analysis.
- **OPL-C-008** (Full operational capability by end of 2033): direct trace-down from RFP-C-002 (mission fully operational by end of 2033), fixing the December 2033 nominal-ops deadline that anchors the pre-launch schedule and the post-launch commissioning sequence. RFP-C-001 (mission begins operations by end of 2034) provides a secondary backstop should the December 2033 magnetotail-alignment window be missed.



Operational Performance Requirements (OPL-O-001-OPL-O-004) The operational performance requirements define measurable performance thresholds for the constellation as a whole rather than for individual spacecraft or instruments. They are derived from SMAD’s framing of mission effectiveness as a function of availability, reliability, and resource margin [4].

- **OPL-O-001** (Constellation science availability ≥ 0.90): fraction of mission lifetime over which sufficient spacecraft are healthy and within their nominal orbits to deliver mesoscale science. The threshold follows from the SMAD operational availability decomposition $A = 1 - (f_{\text{planned}} + f_{\text{env}} + f_{\text{anomaly}})$ with conservative Phase-A allocations of 5%, 3%, and 2% respectively for planned downtime, environment-driven protective operation, and anomaly recovery, yielding $A \geq 0.90$ [4].
- **OPL-O-002** (Data return efficiency ≥ 0.85): fraction of acquired science data successfully downlinked to the ground. The threshold follows from $\eta_{\text{data}} = (1 - \ell_{\text{contact}})(1 - \ell_{\text{link}})$ with conservative loss allocations of 10% for contact-window limitations and 5% for link and protocol overhead, yielding $\eta_{\text{data}} \geq 0.855 \approx 0.85$ [4].
- **OPL-O-003** (Anomaly response time ≤ 600 s): autonomous transition from fault detection to a safe, sun-pointing, power-positive state without ground intervention. The 600 s ceiling is set to bound fault-propagation risk in the magnetotail radiation environment, following the SMAD principle that survivability in radiation-intensive environments depends on rapid transition to safe configuration after critical fault detection [4].
- **OPL-O-004** (Operational design lifetime ≥ 6 years): 20% margin above the 5-year prime mission required by RFP-F-003. The margin factor follows SMAD’s recommendation to incorporate explicit lifetime margin against modeling uncertainty in propellant consumption, battery degradation, and cumulative radiation effects [4].

Payload Constraint Requirements (OPL-C-001-OPL-C-005) Requirements OPL-C-001 through OPL-C-005 define the mass, volume, power, and onboard processing allocations available to OPL. The mass, volume and power limits were derived from the mission-level subsystem budget allocations.

Payload Instrumentation Performance Requirements (OPL-P-001-OPL-P-011) The performance requirements for the payload instrumentation were derived by translating the mission’s primary science objective of characterizing mesoscale plasma dynamics within the magnetotail transition region into measurable physical quantities. Following the payload design methodology outlined in SMAD [10], this derivation proceeds through three steps: (1) identifying the relevant physical phenomena, (2) determining the physical quantities required to characterize those phenomena, and (3) translating these measurement needs into instrument performance requirements.

Mesoscale processes such as bursty bulk flows (BBFs) and dipolarization fronts involve coupled variations in magnetic field structure, thermal plasma distributions, and energetic particle populations. Accordingly, three primary instrument classes were selected: a fluxgate magnetometer (FGM) to measure vector magnetic fields, an electrostatic analyzer (ESA) to measure thermal ion and electron distributions, and an energetic particle detector (EPD) to measure suprathermal and energetic particle populations. Specifically, the payload instrumentation shall support:



- **Magnetic Field Sampling Rate (OPL-P-001):** The magnetometer shall support a magnetic field sampling rate ≥ 32 Hz during high-cadence science operations.
- **Magnetic Field Resolution (OPL-P-002):** The magnetometer shall resolve magnetic field variations of ≤ 0.1 nT.
- **Magnetic Field Measurement Range (OPL-P-003):** The magnetometer shall support a magnetic field measurement range of at least $\pm 25,000$ nT.
- **ESA Energy Range (OPL-P-004):** The ESA shall measure ion energy distributions across 100 eV to 25 keV and electron energy distributions across 5 eV to 30 keV.
- **ESA Field of View (OPL-P-005):** The ESA shall provide full field of view 4π sr measurements for ion and electron distributions.
- **Signal Processing/Aliasing (OPL-P-006):** The magnetometer measurement chain shall include an anti-alias low-pass filter with a cutoff frequency less than or equal to the Nyquist frequency associated with the selected sampling rate.
- **H⁺ Energy and Direction Measurements (OPL-P-007):** The energetic particle detector shall measure H⁺ energy and direction measurements from 25 keV to 150 keV.
- **O⁺ Energy and Direction Measurements (OPL-P-008):** The energetic particle detector shall measure O⁺ energy and direction measurements at energies of > 90 keV.
- **Electron Energy and Direction Measurements (OPL-P-009):** The energetic particle detector shall measure electron energy and direction measurements from 25 keV to 1 MeV.
- **Magnetic Cleanliness Requirement (OPL-P-010):** The spacecraft-induced magnetic field at the FGM sensor shall not exceed 5 nT.
- **Boom Length Requirement (OPL-P-011):** The magnetometer sensor shall be located at a minimum distance of 1.34 m from the spacecraft centre.

Derivations of the individual payload instrumentation requirements are included as part of the formal analysis for instrument selection and magnetic cleanliness in Sec 4.7.

Environmental Requirements (OPL-E-001-OPL-E-003) Environmental requirement OPL-E-001 is driven by the magnetotail plasma and radiation environment encountered between 6 and 12 R_E . The magnetosphere is characterized by low-density plasma, strong magnetic field coupling, and energetic particle populations. In such regimes, spacecraft charging and differential potential formation become significant due to increased Debye length and plasma sheath formation [11]. Operational procedures must therefore include safe-mode transitions and charging-aware operational constraints to mitigate arcing risk and electronics degradation.

Additionally, radiation exposure and eclipse cycles introduce operational duty-cycle constraints and potential science interruption. Operations must therefore define degraded modes that preserve spacecraft health while minimizing loss of mesoscale science coverage.



4.4 Subsystem Constraints

The OPL subsystem operates within four categories of constraint, each anchored to a mission-level RFP requirement or to a cross-subsystem interface agreement.

Resource Constraints. Mass (OPL-C-001), volume (OPL-C-002), peak power (OPL-C-003), average power (OPL-C-004), and onboard processing (OPL-C-005) constraints are imposed by the overall spacecraft architecture and ensure compatibility with SMM and POW capabilities.

Data and Communications Constraints. OPL-C-007 binds payload data generation to the storage and downlink capacity defined by CCD. The OPL operations plan must schedule science acquisition duty cycles such that per-orbit science volume can be both buffered onboard and downlinked during the ~ 1 -hour perigee contact, ensuring traceability to RFP-F-013 (per-orbit communication of science data, equivalent to MISS-F-017 in the prior numbering scheme).

Cost and Schedule Constraints. OPL-C-006 enforces lifecycle cost compliance with the \$350M FY25 MIDEX cap (traced up from RFP-F-006-equivalent). OPL-C-008 enforces the end-of-2033 operational capability deadline (traced up from RFP-C-002). Together these constraints bound payload selection complexity (more capable instruments cost more and take longer to qualify) and operational complexity (more elaborate procedures require more ground-segment infrastructure). RFP-C-001 (mission begins operations by end of 2034, TBC) provides a secondary backstop should a launch slip force the December 2033 magnetotail-alignment window to be missed.

Integration and Interface Constraints. The magnetic cleanliness requirement (OPL-P-010) and its derived boom-length requirement (OPL-P-011) impose direct structural constraints on SMM, requiring that the FGM be mounted at ≥ 1.34 m from the spacecraft centre. Payload data rates and CCSDS protocol compatibility impose interface requirements on CCD. The sun-facing spin-stabilization architecture imposed by ADCS in turn constrains payload pointing geometry to instruments compatible with spin-modulated sampling.

4.5 Subsystem Drivers

Science Measurement Drivers. The primary OPL driver is the requirement to detect and characterize BBFs in the 6-12 R_E transition region, traced up from RFP-F-006 (particle and magnetic-field measurements in the magnetotail), RFP-F-007 (high-resolution, multipoint, in situ measurements), RFP-F-008 (mesoscale observations), and RFP-F-009 (measurements of critical magnetotail regions). This drives the selection of the FGM, ESA, and EPD and the performance requirements OPL-P-001 through OPL-P-009.

Measurement Quality Drivers. The need to resolve sub-nT magnetic perturbations associated with dipolarization fronts drives both the magnetometer resolution requirement (OPL-P-002) and the cleanliness/boom requirements (OPL-P-010, OPL-P-011), which directly impact spacecraft structural layout and impose the 1.34 m minimum sensor standoff.



Constellation Architecture Drivers. The transition to a coplanar multi-apogee architecture at 7, 9, and 12 R_E drives the deployment and commissioning sequence (OPL-F-005), the mission timeline (Section 4.7.1), and the per-orbit duty-cycle analysis (Section 4.7.7). Maximizing radial coverage replaces the prior tetrahedral-formation driver and reflects a more scientifically targeted alignment with RFP-F-008 (mesoscale observations) and RFP-F-009 (critical-region measurements).

Operations and Data Drivers. The operational-performance requirements OPL-O-001 (science availability ≥ 0.90), OPL-O-002 (data return efficiency ≥ 0.85), and OPL-O-003 (anomaly response ≤ 600 s) drive the operational mode definitions and the contingency procedure design. RFP-F-011 (high-cadence concurrent measurements during opportune events) drives the burst-mode trigger logic described in Section 4.7.7.

Resource and Programmatic Drivers. The MIDEX cost cap (RFP-F-006-equivalent, traced down to OPL-C-006) and the 2033/2034 operational deadlines (RFP-C-001, RFP-C-002, RFP-F-003, traced down to OPL-C-008) together drive payload selection toward heritage instruments (THEMIS FGM, THEMIS ESA, modified MMS EIS) rather than custom-developed hardware, and drive the mission timeline toward an aggressive but executable back-planned schedule from a February 2033 launch.

4.6 Subsystem Design Approach

Operations At PDR, the Operations work focused on bounding mission feasibility at the architectural level: the lifecycle was decomposed into the six standard NASA phases, modes were sketched as preliminary state assignments per phase, a first-order data-volume estimate was produced to test downlink feasibility, a parametric WBS cost model was constructed to test MIDEX-cap compliance, and OPL-to-subsystem interfaces were captured in a five-category block diagram [4]. This established mission feasibility but deliberately deferred resource-resolved closure to FDR, when subsystem designs would be mature enough to support a bottom-up rollup. In this final design iteration, the Operations work focused on the following:

- Converting the PDR phase decomposition into a fully back-planned mission timeline anchored to the GNC-selected February 2033 launch window and the December 2033 magnetotail alignment, closing OPL-F-004, OPL-F-005, and OPL-C-008.
- Formalizing the canonical nine-mode behavioral set with executable procedures, including the High-Rate Science (Burst) mode and its onboard $|\partial B_z / \partial t|$ trigger, closing OPL-F-001 through OPL-F-003 and OPL-O-003.
- Re-baselining the PDR top-down parametric cost model into a bottom-up Bill-of-Materials rollup with a 90% learning-curve discount across the eight-spacecraft constellation, closing OPL-F-009, OPL-F-010, and OPL-C-006.
- Advancing the PDR interface block diagram into a quantitative analysis of the multi-subsystem integration tensions that emerge when those interfaces are exercised simultaneously, supporting OPL-F-007 and OPL-F-008.
- Advancing the risk register from the PDR initial likelihood-impact matrix to a post-mitigation matrix with explicit residual-risk reclassifications, closing OPL-F-006 through OPL-F-008.



The PDR data-volume thread is not carried forward as a standalone Operations analysis: it has migrated into the payload duty-cycle analysis (Section 4.7.7) and into CCD's link-budget work, where OPL-O-001 and OPL-O-002 are jointly closed.

Payload The payload design approach follows the structured process for defining space payloads described in *Space Mission Analysis and Design (SMAD)* [10]. In this framework, payload definition begins by translating mission science objectives into measurable physical quantities, followed by development of a payload operations concept and identification of candidate instruments capable of meeting the derived performance requirements.

In previous design iterations, subject trades were completed to identify the key mesoscale phenomena within the magnetotail transition region, the payload performance requirements were derived from the science objectives, and the instrument suite was selected through trade studies. In this final design iteration, the payload work focused on the following:

- Finalizing the payload operational modes and the FGM cadence states within Science mode, including the onboard high-cadence trigger condition derived from the dipolarization front selection criteria.
- Re-evaluating the magnetometer dynamic range requirement (OPL-P-003) using updated trajectory data from GNC, with the worst-case near-Earth field magnitude at perigee derived from the Group 3 orbit.
- Computing the orbit-averaged payload power for each MATRIX orbit group using the per-group ROI fractions provided by GNC, closing OPL-C-004.
- Completing the verification of all OPL performance and constraint requirements at FDR maturity.

4.7 Formal Analysis

The formal analysis below presents the work that closes the OPL subsystem requirements at FDR maturity, organized to mirror the two-role structure of the subsystem. The first four analyses are Operations-side products: the mission timeline (Section 4.7.1), the spacecraft phase, mode, and procedure definitions (Section 4.7.2), the BOM-based cost rollup (Section ??), the risk analysis and mitigation register (Section 4.7.4), and the cross-subsystem interfacing and engineering-conflict mitigation analysis (Section ??). The remaining analyses are Payload-side products: magnetometer selection and requirement derivation (Section 4.7.6), payload operational modes and duty-cycle analysis (Section 4.7.7), magnetic cleanliness and boom-length derivation (Section 4.7.8), ESA selection (Section 4.7.9), and EPD selection (Section 4.7.10). Each analysis identifies the team members who conducted the work and the principal updates since PDR, and concludes by tying its results back to the OPL requirements being closed.

4.7.1 Mission Timeline: From Pre-Launch to End-of-Life

Analysis Conducted by: Sonal

Updates since PDR: The PDR timeline gave the launch and nominal-ops dates and a phase-by-phase narrative, but did not back-plan the seven-year pre-launch period into specific activities. For FDR, this period has been decomposed into nine concurrent activities across three threads (design & fabrication, integration &



qualification, ground segment, launch campaign), and two MATRIX-specific verifications (Helmholtz-coil cleanliness mapping and constellation time-synchronization testing) have been explicitly scheduled to close OPL-P-010 and constellation operational availability, respectively.

The MATRIX mission timeline is structured to satisfy the RFP requirements of delivering all eight spacecraft for launch by the end of 2032, achieving full operational capability by the end of 2033 (RFP-C-002), and operating from end-of-2033 through end-of-2038 as a five-year prime science mission (RFP-F-003). Table 11 summarizes the end-to-end lifecycle from pre-launch through end-of-life disposal, organized by phase and labeled with the corresponding ConOps step numbers. The timeline is back-planned from the February 2033 launch window selected by GNC analysis, which provides approximately nine months of post-launch operations before the nominal science start date of December 2033, when all three orbit-group apogees are optimally aligned with the magnetotail during the northern winter solstice [12]. The two timeline drivers are therefore (i) the 2032 delivery deadline imposed by RFP-C-002, which fixes the upstream pre-launch schedule, and (ii) the December 2033 magnetotail-alignment window, which fixes the launch date and bounds the duration available for deployment, spacing, and commissioning between launch and nominal science.

Phase	Steps	Key Activities
Pre-Launch	(0)	Spacecraft fabrication, integration, and environmental testing; stacking of all eight spacecraft into shared dispenser configuration; end-to-end communications and timing-synchronization validation; propellant loading and launch site integration; mission readiness and operational validation for autonomous early-phase execution.
Launch	(1)	All eight spacecraft launched in stacked configuration (February 2033); injected into transfer trajectory targeting $9 R_E$ highly elliptical parking orbit.
Orbit Insertion & Stack Deployment	(2-4)	Orbit insertion burn into $9 R_E$ orbit; stack separates from launch vehicle; upper stage performs disposal maneuver for debris compliance.
Sequential Deployment & Spacing	(5-7)	Eight spacecraft deployed in three groups from dispenser rings; staggered separation imparts differential ΔV (cm/s scale) to ensure along-track drift and eliminate collision risk; initial detumble, attitude stabilization, and communications checkout begin.
Orbit Transfer, Commissioning & Readiness	(8-10)	Group 1 $\rightarrow 12 R_E$ (apogee raise); Group 2 spacing maneuvers at $9 R_E$ (temporary orbit drift); Group 3 $\rightarrow 7 R_E$ (apogee lower); subsystem commissioning; payload activation, calibration, cross-calibration, and time synchronization.
Nominal Science Operations	(11)	Multi-point BBF observations: $12 R_E$ (initiation), $9 R_E$ (evolution), $7 R_E$ (braking and deposition); includes Nominal and Burst Modes, routine perigee downlink, and annual station-keeping; ≥ 5 -year mission duration per RFP-F-003.
End-of-Life Disposal	(12-14)	Controlled, staggered perigee-lowering burns; atmospheric re-entry and burnup; spacecraft passivation prior to disposal, satisfying RFP-F-005.

Table 11: MATRIX mission operational phases, derived top-down from RFP delivery, lifetime, and end-of-life compliance requirements (RFP-C-001, RFP-C-002, RFP-F-003, RFP-F-005). Step numbers correspond to the labels in the ConOps diagram.

Pre-Launch (2026-2032). The pre-launch period spans roughly seven years and is the most schedule-driven portion of the lifecycle: it must deliver eight identical flight-qualified spacecraft, validate a shared launch and ground-segment architecture, and ship to Cape Canaveral by January 2033 in



time for an early-February launch.

Table 12 decomposes this period into nine activities organized across three concurrent threads, all back-planned from the launch date: flight hardware development, ground-segment readiness, and launch-campaign integration.

Detailed design (Phase C) runs from January 2026 through June 2028, producing finalized drawings, specifications, and procurement plans for all eight spacecraft and freezing instrument interface definitions with the THEMIS-FGM, THEMIS-ESA, and EIS+ vendors [4], [7], [8], [9]. Subsystem fabrication and qualification (Phase D) follows from July 2028 through December 2030, during which long-lead components (thrusters, batteries, magnetometer sensors) are procured early to smooth the parts-flow profile and expose interface problems before integration begins, consistent with low-cost spacecraft development practice [4]. Spacecraft integration and test of all eight units runs from January 2031 through March 2032; serial integration of the eight nearly-identical buses realizes the recurring-cost reduction from the aerospace learning curve and exposes any unit-to-unit variability before environmental qualification begins.

Environmental qualification (TVAC, vibration, EMI) runs from April through September 2032 and validates survivability through the launch and on-orbit environments. Two MATRIX-specific verifications are scheduled in parallel with environmental qualification: Helmholtz-coil magnetic-cleanliness mapping (May-July 2032) closes verification of the spacecraft-induced-field requirement OPL-P-010 (the FGM-at-sensor field must remain ≤ 5 nT, which sets the magnetometer boom length [13]); and constellation-level time-synchronization testing (September-October 2032) validates that the eight spacecraft can maintain the coordinated cadence required for mesoscale BBF reconstruction. Near Earth Network end-to-end ground-segment testing (August-October 2032) confirms that the CCSDS store-and-forward downlink architecture, command uplink, and ranging service all function across the eight-vehicle constellation [14].

Propellant loading and ESPA-class stack integration (November-December 2032) prepare the integrated stack for shipment, after which the stack ships to Cape Canaveral, undergoes launch-vehicle integration, and passes the Operational Readiness Review (January 2033) which is the final go/no-go gate before launch. The schedule retains approximately one month of slack at each major milestone, however, loss of more than that on any critical-path activity (integration & test, environmental qualification, or LV integration) would slip the February launch and forfeit the December magnetotail-alignment window, requiring at minimum a six-month delay to the next viable launch opportunity and risking violation of RFP-C-002.



Activity	Dates	Duration	Thread
Detailed design (Phase C)	Jan 2026 - Jun 2028	30 mo	Design & fabrication
Subsystem fabrication & qualification (Phase D)	Jul 2028 - Dec 2030	30 mo	Design & fabrication
Spacecraft integration & test (all 8 units)	Jan 2031 - Mar 2032	15 mo	Integration & qualification
Environmental test (TVAC, vibration, EMI)	Apr - Sep 2032	6 mo	Integration & qualification
Helmholtz-coil magnetic-cleanliness verification	May - Jul 2032	3 mo	Integration & qualification
NEN / ground-segment end-to-end test	Aug - Oct 2032	3 mo	Ground segment
Constellation time-synchronization check	Sep - Oct 2032	2 mo	Ground segment
Propellant load & ESPA stack integration	Nov - Dec 2032	2 mo	Launch campaign
Ship to CCSFS, LV integration, ORR	Jan 2033	1 mo	Launch campaign
Launch	Feb 2033	-	Launch campaign

Table 12: MATRIX pre-launch and launch-campaign timeline, decomposed across three concurrent threads. Critical-path activities are spacecraft integration & test and environmental qualification; MATRIX-specific verifications (Helmholtz-coil magnetic-cleanliness mapping and constellation time-synchronization) are explicitly scheduled to close verification of OPL-P-010 and constellation-level operational availability respectively.

Launch and Early-Orbit Operations (T+0 to T+9 months). The February 2033 launch from Cape Canaveral injects the stacked eight-spacecraft configuration into a $9 R_E$ highly elliptical parking orbit [12]; upper-stage disposal is performed immediately following separation to satisfy orbital-debris mitigation requirements (RFP-F-005). Deployment from the shared dispenser uses a sequenced, velocity-biased separation strategy in which each release imparts a small differential ΔV on the order of cm/s to induce along-track drift, with ground tracking verifying safe separation margins before each subsequent release [12]. After full deployment, all eight spacecraft remain in the $9 R_E$ parking orbit while spacing maneuvers establish the final constellation geometry: Group 2 (four spacecraft at $9 R_E$) spaced 4 hours apart in along-track separation; Group 1 (two spacecraft) spaced 12 hours apart and then raised to $12 R_E$; Group 3 (two spacecraft) maintained at ~ 7 km separation for microscale measurements and then lowered to $7 R_E$. This spacing and orbit-transfer process requires approximately 2-8 months depending on the ΔV allocated to temporary-orbit maneuvers [12], and is followed by subsystem commissioning, payload activation, calibration, cross-calibration, and constellation time synchronization.

Nominal Operations (Late 2033 - Late 2038). Nominal science operations begin once all eight spacecraft have reached their final orbits, completed commissioning, and verified constellation readiness, targeting a December 2033 start to exploit peak magnetotail alignment with full operational readiness confirmed by early 2034 at the latest [2], [12]. Because Earth's orbital motion drifts the magnetotail out of alignment with the spacecraft apogees on a seasonal basis, active science data collection is available for approximately half of each calendar year; no station-keeping is required between annual realignment opportunities [12]. The per-orbit operations cycle, described in detail in Section 4.7.2, is repeated continuously across the five-year prime mission to satisfy RFP-F-003.



End-of-Life Disposal (After December 2038). End-of-life disposal begins no earlier than December 2038. Each spacecraft executes a perigee-lowering burn timed near apogee to reduce perigee altitude from approximately 1000 km to below approximately 300 km, after which atmospheric drag drives natural orbital decay and re-entry. Spacecraft are expected to burn up completely during re-entry; no controlled ocean impact is required. Prior to the disposal burn, each spacecraft is passivated (batteries discharged, residual propellant vented, transmitters shut down) to eliminate stored-energy sources and prevent fragmentation, in compliance with the post-mission disposal requirement RFP-F-005 and NASA orbital-debris mitigation guidelines [11]. Disposal is executed in a staggered sequence by group, with tracking maintained throughout to verify compliance.

4.7.2 Definition of Spacecraft Phases, Modes, & Procedures

Analysis Conducted by: Sonal

Updates since PDR: The PDR defined the canonical mode set and described procedures qualitatively in prose. For FDR, each procedure has been formalized into a structured table (Table 14) specifying entry trigger, key actions, and dominant resource or interface constraint, making each procedure verifiable against a specific subsystem-level limit. The phase-mode-procedure hierarchy is now also explicitly framed as the analytical mechanism by which OPL-F-001 through OPL-F-005 are closed.

The mission timeline above is realized in flight through three coupled behavioral abstractions, following standard heliophysics-mission practice [4], [11]: phases, modes, and procedures. *Phases* are the temporally ordered stages tabulated in Table 11, and answer the question of *when* the spacecraft is doing something.

Modes, summarized in Table 13, are the behavioral configurations that govern bus and payload state at any instant; they answer the question of *what configuration* the spacecraft is in. The mode set follows the canonical start-up, commissioning, nominal science, high-rate science (burst), downlink, eclipse, maneuver, safe, and end-of-life structure recommended for spin-stabilized magnetospheric missions [4].

Procedures, summarized in Table 14, are the sequential actions that map between modes and accomplish each phase's objectives; they answer the question of *how* the spacecraft enters, executes, and exits a mode. The phase-mode-procedure hierarchy ensures that mission-level requirements (delivery date, lifetime, debris compliance) decompose cleanly through behavioral configurations into executable command sequences with bounded resource demand, and is the analytical mechanism by which OPL satisfies OPL-F-001 through OPL-F-005 (executable nominal procedures, contingency procedures, pre-launch procedures, and the launch/commissioning/establishment sequence).

Two architecture-level decisions bound the operational design space and propagate into every procedure. First, MATRIX employs sun-facing spin stabilization at 15 rpm with the spin axis aligned to the Sun-Earth line [15]. This single choice has mission-wide operational implications: science instrument pointing is inseparable from spin phase (the THEMIS ESA's 360° azimuthal field of view is swept through the remaining solid angle once per 4 s rotation [8]), power generation is continuous whenever sun-pointing is held, and a body-mounted omnidirectional S-band antenna can be used with no dedicated downlink-pointing maneuvers [14], [16]. Second, all eight spacecraft are co-launched in a stacked dispenser configuration into the $9 R_E$ parking orbit, requiring a sequenced commissioning and constellation-establishment phase of approximately nine



months before nominal science can begin [12]. Together, these decisions impose three operational invariants: procedures must execute autonomously between intermittent perigee ground contacts (~ 1 h per orbit), must respect the resource envelopes provided by POW, THM, and CCD, and must preserve sun-pointing as the failure-safe attitude across all contingency transitions.

Mode	Spacecraft Behavior	Payload Configuration
Start-Up	System boot, detumble, establish power and comms	Payload OFF
Commissioning	Subsystem checkout and calibration	ESA/EPD OFF; FGM low cadence for calibration
Nominal (Survey)	Routine science ops, stable spin, periodic perigee downlink	ESA ON, EPD ON; FGM survey cadence (4 Hz)
Burst / High-Rate Science	Event-triggered ops, prioritized data storage and downlink	ESA ON, EPD ON; FGM high cadence (up to 128 Hz)
Downlink	Omnidirectional S-band active; CCSDS data transmission near perigee	ESA, EPD OFF; FGM continues survey cadence
Maneuver	Spin-down to thrust-vector pointing; thruster burn; re-acquire 15 rpm spin	Payload OFF
Eclipse	Power-constrained ops; non-essential heaters cycled; battery managed to 30% DoD floor	ESA/EPD OFF; FGM low-cadence survey only
Safe Mode	Fault protection; sun-pointing attitude re-acquired; beacon active; await ground up-link	Payload OFF
End-of-Life	Apogee perigee-lowering burn and passivation	Payload OFF

Table 13: MATRIX spacecraft operational modes: bus and payload behavioral configuration at any instant. Modes follow the canonical SMAD heliophysics structure [4] and span the full mission lifecycle.

The most common procedure is the per-orbit nominal science cycle, which is the pattern executed for the bulk of the ≥ 5 -year prime mission and which directly delivers the science measurements required by RFP-F-006 through RFP-F-009. On entering the region of interest at $r > 6 R_E$, the FGM transitions to its survey cadence and the ESA and EPD activate at full cadence; an onboard $|\partial B_z / \partial t|$ trigger consistent with established dipolarization-front signatures [17], [18] elevates the FGM to its 128 Hz burst cadence and reprioritizes the CCD buffer (satisfying the high-cadence concurrent-measurement requirement RFP-F-011). The spacecraft exits the ROI approximately 30 minutes before perigee, queues buffered science for downlink, and executes a ~ 1 h CCSDS store-and-forward contact with the Near Earth Network during which command and ranging uplinks are also performed [14], satisfying the per-orbit science-data communication requirement RFP-F-013. Eclipse intervals up to 4.356 h are absorbed by powering the ESA and EPD off, holding the FGM at low cadence, and managing the battery to its depth of discharge (DoD) floor [12], [19]. The resulting per-orbit pointing schedule allocates 79.7%, 66.1%, and 47.8% of orbit period above $6 R_E$ for orbit Groups 1, 2, and 3 respectively, with $\sim 3\%$ allocated to perigee downlink and the remainder shared between eclipse survival and ADCS housekeeping [12].

Off-nominal and lifecycle procedures complete the framework. The start-up procedure executes autonomously following dispenser separation: bus power-on, magnetorquer detumble to $< 2^\circ/\text{s}$, sun acquisition, spin-up to 15 rpm, and first NEN beacon contact for initial orbit determination,



all within approximately six hours of separation [12], [15]. Commissioning spans the subsequent nine months and includes magnetometer boom deploy with in-flight magnetic-cleanliness calibration [7], ESA and EPD calibration sweeps, and the orbit-raising and orbit-lowering maneuvers that establish the three-group constellation. The maneuver procedure powers the payload off, aligns the thrust vector via ADCS, fires the 1 N main or 100 mN attitude-control thrusters, re-acquires spin, and resumes the prior mode [15]. The safe-mode procedure, triggered by onboard fault detection within $t_{\text{response}} \leq 600$ s (OPL-O-003), powers the payload off, holds sun-pointing attitude, enables the beacon, and awaits ground diagnosis and recovery uplink; given MATRIX's highly eccentric orbits and intermittent contact windows, the spacecraft must autonomously maintain safe-mode stability between contacts. Finally, the end-of-life procedure executes the perigee-lowering burn and passivation sequence described in Section 4.7.1, in compliance with RFP-F-005 and NASA orbital-debris mitigation guidelines [11].



Procedure (Mode)	Entry Trigger	Key Actions	Constraint
Start-Up (Start-Up)	Separation from dispenser at $T + 0$	Bus power-on and boot ($T + 0$ to $T + 30$ s); magnetorquer detumble to $< 2^\circ/\text{s}$ ($T + 30$ s to $T + 2$ h); sun acquisition and 15 rpm spin-up ($T + 2$ h to $T + 4$ h); first NEN beacon contact and initial orbit-determination solution ($T + 4$ h to $T + 6$ h)	ADCS detumble authority; NEN visibility
Commissioning (Commissioning)	Stable sun-pointing attained ($\sim T + 6$ h)	Bus health checkout (POW, THM, CCD, ADCS) in week 1; magnetometer boom deploy and in-flight cleanliness calibration in week 2; ESA/EPD power-up and calibration sweeps in weeks 3-6; orbit raising/lowering and group spacing in months 2-9	PLV ΔV ; SMM deployment reliability
Nominal Science (Nominal/Survey)	ROI entry ($r > 6 R_E$) and commissioning complete	Science ON entering ROI: FGM at 4 Hz survey cadence, ESA and EPD at full cadence; burst trigger on $ \partial B_z/\partial t $ threshold elevates FGM to 128 Hz with prioritized buffer; exit ROI to survey cadence and prep downlink queue ~ 30 min before perigee; routine housekeeping telemetry	POW science-mode load; CCD storage rate
High-Rate Science (Burst)	Onboard $ \partial B_z/\partial t $ threshold consistent with BBF signature	FGM elevated to 128 Hz; prioritized buffer allocation; ESA/EPD retain full cadence	POW peak load; CCD priority queue depth
Downlink (Downlink)	Perigee entry; ground station in view (~ 1 h/orbit)	Activate omnidirectional S-band; CCSDS store-and-forward downlink; command uplink and ranging for orbit determination; post-perigee cruise out and eclipse survival if in shadow before returning to ROI entry	CCD link budget; ground contact window
Eclipse (Eclipse)	Earth-shadow entry (up to 4.356 h)	ESA/EPD powered off at shadow entry; FGM held at low-cadence survey only; non-essential heaters cycled; battery managed to 60% DoD floor	POW battery capacity; THM survival
Maneuver (Maneuver)	Scheduled ΔV event (orbit transfer, spacing, station-keeping, disposal)	Payload off; ADCS spins spacecraft down from 15 rpm to thrust-vector pointing; main 1 N or 100 mN ACS thrusters fire; re-acquire 15 rpm spin and resume prior mode	PLV propellant; ADCS pointing accuracy
Safe Mode (Safe)	Onboard fault detection, $t_{\text{response}} \leq 600$ s	Fault detected, payload powered off; sun-pointing attitude re-acquired; beacon enabled and ground uplink awaited between contact windows	FDIR autonomy; ground latency
End-of-Life (EoL)	Completion of ≥ 5 -yr prime mission ($\geq$ Dec 2038)	Apogee perigee-lowering burn to < 300 km; passivation (battery discharge, propellant vent, transmitter shutdown); atmospheric re-entry and burnup	Orbital-debris compliance; PLV residual ΔV

Table 14: MATRIX operational procedures: mapping from spacecraft modes to executable action sequences. Each procedure identifies the entry trigger, key actions, and dominant resource or interface constraint that bounds execution.

4.7.3 Mission Costs & Subsystem Allocations

Analysis Conducted by: Sonal.

Updates since PDR: The PDR cost model produced a \$306M mid-cycle estimate (\$255M pre-reserve plus 20% reserve) using a mix of top-down WBS allocations and partial BOM feedback, and was expressed at the per-subsystem level. For FDR, this model has been fully re-baselined as a bottom-up Bill-of-Materials rollup driven by component-level pricing from each subsystem, with a 90% learning-curve slope applied



to the OPL payload allocation across the eight-spacecraft constellation. The result re-identifies OPL itself, rather than PLV, as the binding cost driver at 5.78% margin against its allocation.

The MATRIX mission cost budget enforces compliance with the \$350M FY25 MIDEX cost cap (RFP-C-003, traced down to OPL-C-006) and discharges the total-cost-analysis and subsystem-cost-allocation obligations imposed by OPL-F-009 and OPL-F-010. The budget methodology follows SMAD guidance, employing a combination of analogy, parametric, and grassroots estimation methods [4] calibrated against heritage magnetospheric missions (THEMIS, MMS, Cluster) [5], [20], [21]. The mission uses the standard NASA Launch Services Program launch vehicle provided under the MIDEX AO; because standard launch services are excluded from the MIDEX cost cap, the full \$350M FY25 cap applies to all hardware, wrap, and programmatic costs excluding launch [2].

Since PDR, the budget has been re-baselined from a parametric WBS estimate to a bottom-up Bill of Materials (BOM) rollup informed by vendor quotes, heritage instrument pricing, and learning-curve recurrence modeling for the eight-spacecraft constellation. Each subsystem provided component-level pricing for flight hardware, which was aggregated into the subsystem allocations shown in Table 15. Programmatic and operations wraps were retained at SMAD-consistent fractions (\$15M Programmatic & SE; \$15M Ground & Ops over a five-year nominal mission). For the OPL payload allocation (\$75M), a 90% learning-curve slope was applied across the eight units consistent with SMAD guidance for production runs of fewer than ten spacecraft [4], yielding a recurrence-discounted payload cost of \$70.67M against a budget of \$75M.

Subsystem	Budget (\$M)	BOM Total (\$M)	% Used	% Margin	Compliance
GNC/ADCS	3.00	0.17	5.67	94.33	MET
SMM	35.00	24.02	68.62	31.38	MET
THM	15.00	0.85	5.69	94.31	MET
POW	5.00	0.62	12.49	87.51	MET
PLV	30.00	15.25	50.82	49.18	MET
OPL	75.00	70.67	94.22	5.78	MET
CCD	10.00	2.99	29.91	70.09	MET
Flight HW Total	173.00	114.57	66.23	33.77	MET

Table 15: Subsystem flight hardware budget allocations and BOM-based compliance. OPL value reflects 90% learning-curve recurrence discount across eight spacecraft.

The OPL subsystem itself carries the largest single allocation because the \$75M envelope absorbs all flight payload instruments (FGM, ESA, EPD) across the eight-spacecraft constellation. As shown above, the OPL subsystem retains only a 5.78% margin against its allocation and is therefore the binding cost driver of the mission; all other subsystems retain margins greater than 30%. A full mission-level rollup against the MIDEX cap, including programmatic and operations wraps and reserve, was presented earlier in Section 2.5.

4.7.4 Risk Analysis & Mitigation

Analysis conducted by: Sawooly Li (mission-level risk identification, development of likelihood-impact risk matrices, and coordination of cross-subsystem mitigation strategies).

Updates since PDR: Final risk matrix established; ensured mitigation strategies implemented across



subsystems and remaining risks reevaluated based on updated design maturity and validated analyses.

In the PDR, the OPL team collaborated closely with all subsystems to identify and evaluate the most critical risks across the mission. These risks were organised into an initial likelihood-impact risk matrix, shown in Table 16, which captured the preliminary mission risk. Risk severity is represented by color, with red indicating high risk, orange medium-high risk, yellow medium risk, light green low-medium risk, and green low risk.

	Negligible	Minor	Moderate	Significant	Severe
Very Likely	Orbital precession causing loss of science coverage				
Likely		Radiation effects on instruments			Solar array failure
Possible			Space weather disrupts comms Timeline delays	Incorrect orbital spacing Insufficient autonomous anomaly response Power budget exceedance during peak operations	Multi-satellite launch failure Δv budget exceedance MIDEX cost cap exceedance
Unlikely			Time synchronisation error	Downlink bottleneck or data backlog Battery failure	Comms and hardware failure
Very Unlikely				Magnetic torquers corrupt payload measurements	Payload instrumental failure Launch vehicle explosion Maneuver execution error

Table 16: Initial Risk Matrix

Through continued discussion across subsystems, design iteration, and targeted identification of key failures by OPL, these risks were revisited and refined during the final design cycle. This process enabled the team to implement mitigation strategies and reduce uncertainty across subsystems, with the goal of downgrading all medium-high risks to medium or below. The resulting mitigation efforts and updated risk classifications can be seen in the updated risk matrix in Table 17 and are described below. The addressed risks are bolded for ease of identification.



	Negligible	Minor	Moderate	Significant	Severe
Very Likely	Orbital precession causing loss of science coverage				
Likely		Radiation effects on instruments			
Possible			Space weather disrupts comms		
Unlikely			Time synchronisation error Timeline delays	Power budget exceedance during peak operations Incorrect orbital spacing Insufficient autonomous anomaly response	Comms and hardware failure
Very Unlikely				Magnetic torquers corrupt payload measurements Downlink bottleneck or data backlog	Payload instrumental failure Launch vehicle explosion Maneuver execution error MIDEX cost cap exceedance Multi-satellite launch failure Δv budget exceedance EPD-Comms Failure Solar Array Failure

Table 17: Risk Matrix After Mitigation

1. **Solar Array Failure:** This risk was initially classified as **high** due to uncertainty in solar array integration, driven by limited available spacecraft surface area and the requirement for array area comparable to a spacecraft face. The impact was significant, as insufficient power generation would compromise mission operations. At the PDR stage, this risk was elevated primarily due to incomplete design maturity rather than a fundamental feasibility issue. Since then, solar cell selection, array sizing, and integration have been fully defined, including resolution of structural mounting and compatibility with spacecraft geometry. In addition, the power system architecture, including battery sizing and storage capacity, has been



finalised, ensuring that both continuous and peak power requirements can be met across all mission modes. Consequently, the likelihood of this risk has been **reclassified as very unlikely**, reducing the overall risk to **medium**.

2. **Incorrect Orbital Spacing:** This risk was initially classified as **medium-high** due to the complexity of multi-satellite deployment, orbit raising, and the sensitivity of spacing to timing and Δv uncertainties. The impact is significant, as maintaining mesoscale spacing $\sim 1 R_E$ is required for coordinated BBF measurements. Since then, the constellation geometry has been fully defined, including finalisation of Group 3 spacing. In addition, orbit structures and stationkeeping/correction strategies have been finalised, reducing uncertainty in long-term spacing maintenance. These have been validated using high-fidelity STK simulations, providing confidence in the predicted orbital behaviour. As a result, the likelihood of this risk has been **reclassified as very unlikely**, reducing the overall risk to **medium**.
3. **Insufficient Autonomous Anomaly Response:** This risk was initially classified as medium-high due to limited onboard fault management capability and reliance on ground intervention. The impact is significant, as delayed anomaly response could lead to loss of functionality or degraded mission performance. Since then, operational modes (including safe mode and standby configurations) have been defined, and baseline fault protection behaviors have been incorporated into the ConOps, enabling autonomous transition to stable configurations during off-nominal events. In addition, the IDPU has been verified to provide an additional layer of fault protection, as the main flight computer is capable of processing instrument data independently, while the dedicated processor introduces redundancy within the data handling architecture. Consequently, the likelihood of this risk has been **reclassified as very unlikely**, and the overall risk is therefore reduced to **medium**.
4. **Power Budget Exceedance During Peak Operations:** This risk was initially classified as medium-high due to concurrent subsystem demands during high-load operational phases, with the potential to exceed available power. Since then, mission modes, such as eclipse and nominal, have been formalised with defined duty cycles, and subsystem power requirements have been refined and verified within an updated power budget that includes margin. In addition, worst-case orbit-averaged power analyses have been conducted to ensure compliance under peak operational conditions. As a result, the likelihood of this risk has been **reclassified as unlikely**, and the overall risk is therefore reduced to **medium**.
5. **Multi-Satellite Launch and Deployment Failure:** This risk was initially classified as **medium-high** due to the complexity of deploying multiple spacecraft from a shared launch configuration, with severe impact on mission capability if unsuccessful. Since then, heritage separation mechanisms, specifically QwkSep-24, have been confirmed, and structural validation against launch loads has been completed by the SMM team. Deployment sequencing strategies have also been defined to minimize collision risk and ensure reliable separation. The use of flight proven hardware and validated deployment architecture therefore **downgrades this risk to medium**.
6. **Δv Budget Exceedance:** The possibility of exceeding the Δv budget is classified as possible due to cumulative maneuver demands across the mission. The impact is severe, as exceeding the Δv budget would reduce mission lifetime or prevent completion of required operations. Since then, all stationkeeping requirements, orbit geometries, and correction strategies have



been finalised, removing earlier uncertainties in Δv estimation. These have been validated using high-fidelity STK simulations, ensuring consistency between predicted and required maneuver budgets. As a result, the likelihood of this risk has been **reclassified as very unlikely**, and the overall risk is therefore reduced to **medium**.

7. **MIDEX Cost Cap Exceedance:** The possibility of exceeding the MIDEX cost cap is classified as possible but severe due to uncertainty in subsystem costs and the complexity of a multi-spacecraft mission. This risk was initially classified as **medium-high** due to uncertainty in subsystem costs and overall mission complexity, with particular concern around evolving propellant estimates across design cycles. Since then, a complete bill of materials (BOM) has been constructed and finalised, with margins applied at the subsystem level. While the OPL subsystem retains a relatively tight margin (5%), all other subsystems maintain margins of at least 30%, providing system-level reserves that enable cost reallocation if required. Propellant requirements have also been fully defined and modelled, removing a major source of uncertainty. As a result, the likelihood of this risk has been **reclassified as very unlikely**, and the overall risk is therefore reduced to **medium**.
8. **Comms and Hardware Failure:** This risk was initially classified as **medium-high** due to the severe impact of losing communication capability, despite relatively low likelihood from heritage hardware. Since then, the communication system design has been refined, including validated link budgets with margin, elimination of strict antenna pointing requirements, and implementation of onboard data buffering and prioritisation. These measures improve resilience to temporary downlink interruptions and reduce operational sensitivity. However, some uncertainty remains in the end-to-end power chain, particularly regarding potential thermal losses in the RF power amplification stage. While these losses are expected to be accounted for within component datasheet specifications, they introduce residual uncertainty in whether the required transmit power can be consistently achieved under all conditions. As a result, this risk has **not been fully downgraded** and remains classified as **medium-high**, making it the only risk that has not been reduced to medium or below. However, the refined data budget and prioritisation scheme successfully reduced the likelihood of a downlink bottleneck or data backlog from **unlikely** to **very unlikely**.
9. **EPD-Communication Failure:** This risk was not included in the initial risk matrix at PDR, but was identified as an area requiring further consideration following subsystem presentations. It is classified as **severe but very unlikely**, as failure of the modified EPD would prevent collection of key science data and therefore compromise primary mission objectives. However, the likelihood is very low. Although the modified EPD configuration has not been previously flown in this exact form, it is based on a heritage design that has been fully functional and flight-tested. In addition, the mission timeline allocates sufficient time for comprehensive end-to-end integration and testing, ensuring compatibility with the flight computer and communications systems prior to launch. As a result, the likelihood of this risk is assessed as **very unlikely**, and it is retained as a **medium** risk due to its high impact.

Thus, across the final design cycle, all previously identified medium-high risks have been reduced through a combination of subsystem maturation, validated analyses, and refined operational planning. Remaining risks are either low-likelihood or bounded by implemented mitigation strategies.



4.7.5 Subsystem Interfacing & Engineering Conflict Mitigation

Analysis Conducted by: Sonal

Updates since PDR: The PDR identified two integration tensions (per-orbit resource contention and constellation-level integration risk) and discussed cross-subsystem coupling. For FDR, this analysis has been expanded to four explicit tensions, adding cleanliness-driven structural constraints on SMM (OPL-P-010, OPL-P-011) and architecture-level constraints from the ADCS sun-facing spin and PLV propellant selection, with each tension tied to a specific OPL mitigation mechanism.

OPL functions as the mission-level integration point across the eight subsystems, translating science objectives and mode definitions into resource demands that must be jointly satisfied. The subsystem interface diagram (referenced in Section 4.1) identifies the data, power, structural/mechanical, and control couplings between OPL and each of the other subsystems; this analysis focuses on the multi-subsystem integration tensions that emerge when those interfaces are exercised simultaneously, and on how OPL's mode resolved load cases mitigate them.

Per-Orbit Resource Contention. The most acute integration tension is per-orbit resource allocation: within each orbit, the spacecraft must simultaneously balance science collection, power generation, communications, and attitude correction within finite pools of time, power, and pointing. The conflict is sharpest among OPL, POW, and CCD: the FGM, ESA, and EIS+ draw peak power during science mode, and the science data they generate must be buffered onboard and downlinked within the ~ 1 h perigee contact. Without careful scheduling, either POW battery margins are violated or CCD storage overflows, with science data lost in either case. OPL resolves this by publishing the mode-resolved load cases summarized in Tables 13 and 14, which both POW (for solar-array and battery sizing) and CCD (for downlink rate and buffer depth) use as design inputs [4]. The duty-cycle analysis in Section 4.7.7 provides the quantitative orbit-averaged loads that underpin these allocations.

Constellation-Level Integration Risk. The eight-spacecraft architecture introduces failure modes that single-spacecraft missions do not face. Constellation science availability (OPL-O-001, $A \geq 0.90$) depends not on any single spacecraft's health but on the coordinated performance of all eight: a single CCD hardware fault preventing downlink, or a PLV anomaly that misplaces a spacecraft during orbit-raising, can degrade constellation geometry and reduce science return below threshold even if seven spacecraft remain fully healthy. OPL therefore maintains a constellation-level view of interdependencies through the risk register (Section 4.7.4), flags emerging conflicts between subsystem design assumptions, and defines contingency operational modes (degraded constellation operation with < 8 healthy vehicles) that preserve science return in degraded configurations [4].

Cleanliness-Driven Structural Constraints. The magnetic-cleanliness requirement OPL-P-010 (≤ 5 nT at the FGM sensor) and its derived boom-length requirement OPL-P-011 (≥ 1.34 m sensor standoff) impose structural constraints on SMM that go beyond conventional payload accommodation. Magnetically noisy components (magnetorquers, reaction wheels, current loops) must be kept far from the sensor or magnetically compensated, and the boom mounting geometry must be compatible with the derived standoff distance. OPL coordinates with SMM on instrument placement and with ADCS on magnetorquer current profiles to mitigate this risk; the current 2 m



boom design provides >159% margin above the derived minimum standoff, absorbing residual uncertainty in the spacecraft dipole estimate.

Architecture-Level Constraints from ADCS and PLV. The sun-facing spin stabilization architecture imposed by ADCS makes science instrument pointing inseparable from spin phase; OPL accommodates this by selecting the THEMIS ESA (whose 360° azimuthal field of view sweeps 4π sr per spin) and by event-triggering the FGM burst cadence onboard rather than relying on ground-commanded mode transitions [8], [15]. The PLV propellant selection (hydrazine) determines the achievable ΔV envelope and constrains the maneuver schedule; OPL must keep the cumulative maneuver demand (deployment, spacing, transfer, station-keeping, disposal) inside PLV's lifetime allocation [22]. Any increase in orbit-maintenance demand propagates directly to propellant mass and spacecraft wet mass, with knock-on effects on the cost budget and SMM mass allocation.

4.7.6 Magnetometer Selection

Analysis conducted by: Maia (Derived the magnetometer performance requirements through research on the magnetotail literature, and conducted a heritage instrument trade study selecting the THEMIS FGM as the MATRIX magnetometer. Updates since PDR: Refined the OPL-P-002 resolution derivation to derive directly from literature rather than inheriting directly from THEMIS, updated the OPL-P-003 derivation using GNC's perigee data, and removed the hardware range-switching framing in favor of the continuous-range THEMIS architecture as a trade study selection criterion.

The magnetometer selection analysis derives the four magnetometer performance requirements (OPL-P-001 through OPL-P-003 and OPL-P-006) from the mesoscale science objectives and selects a heritage instrument capable of meeting them. The derivation proceeds from the physical phenomena (dipolarization fronts within BBFs), through the measurement quantities required to characterize them, to the instrument performance specifications that follow.

Sampling Rate (OPL-P-001). The fastest mesoscale magnetic structure associated with bursty bulk flows is the dipolarization front (DF). Dipolarization fronts are characterized by a sharp increase in the northward magnetic field component B_z , which typically occurs over a timescale of approximately 4 seconds on average [17]. This sets the fastest physical timescale the magnetometer must resolve:

$$T_{DF} \approx 4 \text{ s}, \quad f_{DF} = \frac{1}{T_{DF}} = 0.25 \text{ Hz}. \quad (1)$$

Adequate temporal resolution of the DF ramp shape requires sampling well above the Nyquist limit of f_{DF} . Following standard signal-processing practice and consistent with heritage magnetospheric mission cadence design [7], 128 samples across the 4 s ramp is adopted as the design target. The corresponding required sampling rate is:

$$f_s = \frac{N}{T_{DF}} = \frac{128}{4 \text{ s}} = 32 \text{ Hz}. \quad (2)$$

This establishes **OPL-P-001**: the magnetometer shall support $f_s \geq 32$ Hz during high-cadence science operations.



Anti-Aliasing (OPL-P-006). The dipolarization front signal of interest sits at $f_{DF} = 0.25$ Hz, well below the Nyquist limit of the 32 Hz sampling rate. However, the magnetometer also sees higher-frequency content from the broader magnetotail environment and from spacecraft-induced electromagnetic interference. If any of this content is present in the analog signal at the moment of sampling, it aliases into the sampled band and contaminates the science measurement [10]. The Nyquist frequency for the high-cadence sampling rate is:

$$f_N = \frac{f_s}{2} = 16 \text{ Hz.} \quad (3)$$

The magnetometer measurement chain must therefore include an analog low-pass filter with cut-off at or below f_N , placed before the analog-to-digital converter, to remove signal content above 16 Hz prior to sampling. This is standard practice for fluxgate magnetometers [7]. This establishes **OPL-P-006**: the magnetometer measurement chain shall include an anti-alias low-pass filter with cutoff frequency $\leq f_N$ associated with the selected sampling rate.

Resolution (OPL-P-002). The magnetometer must also provide sufficient amplitude resolution to detect the magnetic field perturbations associated with mesoscale magnetotail dynamics. Fu et al. [17] characterize dipolarization fronts as magnetic field jumps satisfying $\Delta B_z \geq 4$ nT in their study of DF events.

Resolving the shape of a ~ 4 nT perturbation requires the magnetometer to measure the field finely enough that real changes can be distinguished from rounding error. If the measurement resolution is comparable to the signal amplitude, apparent changes between consecutive samples become indistinguishable from rounding effects rather than physical variations. To reliably capture the structure of small-amplitude magnetic perturbations, the measurement resolution must therefore be substantially finer than the signal of interest. Heritage magnetospheric missions adopt a design margin of approximately an order of magnitude between the smallest resolved signal and the instrument resolution [7]. Applying this criterion to the ~ 4 nT dipolarization front perturbation yields:

$$\Delta B_{\text{res}} \leq 0.4 \text{ nT.} \quad (4)$$

The resolution requirement is set at ≤ 0.1 nT, consistent with heritage FGM design [7] and providing additional margin to capture finer-scale variations within mesoscale magnetotail structures. This establishes **OPL-P-002**: the magnetometer shall resolve magnetic field variations of ≤ 0.1 nT.

Dynamic Range (OPL-P-003). The magnetometer must additionally provide sufficient dynamic range to measure the magnetic fields encountered along the spacecraft orbit without saturation. While the primary science observations occur within the magnetotail transition region where field strengths are on the order of tens of nanotesla, the spacecraft also passes through significantly stronger near-Earth magnetic environments at low altitude near perigee.

To estimate the maximum magnetic field magnitude expected along the orbit, the perigee altitudes and inclination provided by the GNC subsystem were used. The lowest perigee in the constellation is the Group 3 orbit, with $r_{\text{perigee}} = 1.142 R_E$ corresponding to an altitude of approximately 905 km, and an estimated inclination of 23.5° . Using these parameters, the Earth's magnetic field magnitude was estimated using the NOAA World Magnetic Model (WMM) magnetic field calculator [23], yielding a worst-case field magnitude of approximately 24,235 nT at the highest-latitude



point of perigee passage. This value represents the approximate upper bound of the magnetic field strength that the spacecraft may encounter during the orbit. To ensure that the magnetometer remains operational and does not saturate in this environment, the instrument dynamic range must exceed this value [24].

Heritage missions provide guidance on the appropriate range margin. The THEMIS fluxgate magnetometer is designed with a continuous measurement range of $\pm 25,000$ nT, accommodating near-Earth magnetic field strengths while maintaining sensitivity to weak magnetotail fields [7]. Adopting a similar dynamic range ensures that the instrument can operate across both weak-field magnetotail environments and stronger near-Earth magnetic environments encountered near perigee. This establishes **OPL-P-003**: the magnetometer shall support a magnetic field measurement range of at least $\pm 25,000$ nT.

A further consideration is whether the dynamic range is provided as a single continuous range or split across multiple hardware-switched configurations. Some heritage magnetospheric magnetometers such as MMS achieved wide dynamic range by switching between discrete low-field and high-field modes [25], which introduces operational complexity through commanded transitions. A continuous-range design, in which the full $\pm 25,000$ nT range is available at all times without commanded mode switching, avoids this complexity. The MATRIX worst-case field of 24,235 nT lies within this continuous range, eliminating the need for hardware range switching to satisfy OPL-P-003. The preference for a continuous-range instrument is revisited as a selection criterion in the magnetometer trade study.

Magnetometer Trade Study A trade study was conducted to identify the magnetometer instrument best suited to meet the payload magnetic field measurement requirements derived above.

Table 18: Magnetometer instrument trade study comparing candidate fluxgate magnetometers.

Parameter	THEMIS FGM	MMS DFG/AFG*	GTOSat	MAVEN
Highest Sampling Rate	128 Hz	128 Hz	32 Hz	32 Hz
Field Resolution	0.003 nT	$\sim 0.005\text{--}0.01$ nT	< 0.1 nT	0.015 nT
Dynamic Range	$\pm 25,000$ nT	$\pm 8200 / \pm 10500$ nT	$\pm 4096 / \pm 65536$ nT	$\pm 512 / \pm 2048 / \pm 65536$ nT
Anti-Alias Filtering	Yes	Yes	Yes	Yes
Power Consumption	0.8 W	1.65 W	0.85 W	0.6 W
Mass	~ 0.38 kg	~ 0.51 kg	~ 0.44 kg	Not isolated

*For MMS, the magnetic field investigation includes two triaxial fluxgate magnetometers: the Analog Fluxgate Magnetometer (AFG) and the Digital Fluxgate Magnetometer (DFG). The mass and power values shown here correspond to that dual-fluxgate system only and do not include the separate search-coil magnetometer (SCM). MATRIX only requires a fluxgate magnetometer architecture because its science objectives target mesoscale plasma dynamics rather than the higher-frequency microscale measurements emphasized by MMS for which an additional SCM was required.

Sources: THEMIS [7], MMS [25], GTOSat [26], MAVEN [27].

First, the candidate instruments were compared against the dynamic range requirement necessary to avoid sensor saturation within the expected magnetic field environment. The MMS AFG/DFG



system provides a maximum dynamic range of approximately ± 8200 – ± 10500 nT, which is significantly lower than the dynamic range provided by the other candidate instruments and does not satisfy requirement OPL-P-003. The MMS architecture was therefore eliminated from further consideration.

The remaining candidates (THEMIS FGM, GTOsSat, and MAVEN) all satisfy the magnetic field resolution, sampling cadence, and dynamic range requirements (OPL-P-001 through OPL-P-003), as well as the anti-alias filtering requirement (OPL-P-006), necessary to observe mesoscale magnetic field structures associated with bursty bulk flows and dipolarization fronts. However, the GTOsSat magnetometer currently has a technology readiness level (TRL) of approximately 6, as it has not yet flown on an operational mission [26]. In contrast, both the THEMIS and MAVEN fluxgate magnetometers have extensive flight heritage and are considered TRL 9 instruments [7], [27]. For a MIDEX-class mission with strict reliability and schedule constraints, the lower TRL of the GTOsSat instrument represents additional technical and programmatic risk. It was therefore deprioritized for the current design cycle in favor of higher-TRL instruments.

A final comparison was therefore conducted between the THEMIS and MAVEN fluxgate magnetometers. Both instruments provide sufficient dynamic range and measurement resolution to meet the derived payload requirements. However, the THEMIS fluxgate magnetometer provides higher field resolution (~ 0.003 nT) compared with MAVEN (~ 0.015 nT), which improves sensitivity to small-amplitude magnetic field perturbations and increases robustness to measurement noise. The THEMIS instrument also supports a higher maximum sampling rate (128 Hz), providing additional operational margin should future design cycles require higher temporal resolution.

The two instruments also differ in how they implement their dynamic range. The MAVEN magnetometer uses three hardware-switched ranges (± 512 , ± 2048 , $\pm 65,536$ nT) selected based on the local field environment [27]. While this provides flexibility, range switching introduces operational complexity as the on-board computer must switch between low and high-field ranges. The THEMIS FGM, by contrast, provides a single continuous $\pm 25,000$ nT range, which fully accommodates the MATRIX worst-case field of 24,235 nT without requiring range switching, simplifying payload mode management.

Based on these considerations, the THEMIS fluxgate magnetometer provides the most suitable heritage architecture for the MATRIX payload in the current design cycle.

4.7.7 Payload Operational Modes and Duty Cycle Analysis

Analysis Conducted by: Maia (Defined the payload-level operational modes and the FGM cadence states within Science mode, specified the onboard high-cadence trigger condition, and derived per-orbit duty cycles and orbit-averaged payload power for each MATRIX orbit group, closing the OPL-C-004 average power constraint. Updates since PDR: New analysis for FDR with modes refined, trigger defined and duty cycle analysis.

Payload Operational Modes. For the purposes of payload duty cycle analysis, the spacecraft operational modes are abstracted into two payload-level operational modes that distinguish whether and how the science instruments are actively collecting data: **Science mode**, active when the spacecraft is within the magnetotail transition region of interest ($r > 6 R_E$, the ROI), and **Non-Science mode**, active outside the ROI. The instrument states in each mode are summarized in



Table 19. This two-mode abstraction applies during nominal operations (Nominal Science Mode); during off-nominal spacecraft modes (Safe, Maneuver, Standby/Idle, Start-Up), all payload instruments including the FGM are powered OFF.

Table 19: MATRIX payload operational modes during nominal operations.

Instrument / Parameter	Science Mode	Non-Science Mode
ESA	ON	OFF
EIS+ (EPD)	ON	OFF
FGM	ON (low or high cadence, event-triggered)	ON (low cadence)

The ESA and EIS+ are powered ON only within the ROI to capture concurrent bulk plasma and energetic particle measurements during BBF events, and are powered OFF outside the ROI to conserve spacecraft power and reduce data volume. The FGM remains ON across both nominal-operations modes to provide contextual magnetic field measurements and support sensor offset calibration during low-field portions of the orbit [24]. The selected THEMIS FGM provides a continuous $\pm 25,000$ nT measurement range that accommodates both the weak magnetotail fields encountered within the ROI and the stronger near-Earth fields encountered at low altitude near perigee, without requiring the on-board computer to switch between low- and high-field ranges.

Magnetometer High-Cadence Trigger. Within Science mode, the FGM operates in one of two cadence states: a low-cadence survey rate ($f_s = 4$ Hz) for continuous monitoring, and a high-cadence rate ($f_s \geq 32$ Hz, per OPL-P-001) for targeted observations of dipolarization front (DF) events. The transition from low cadence to high cadence is event-triggered onboard by the IDPU.

The IDPU continuously monitors the 4 Hz B_z survey stream against the DF selection criteria reported by Fu et al. [17]: $\Delta B_z \geq 4$ nT within a window of $t_{\text{duration}} < 8$ s. When the IDPU detects a B_z increase satisfying these conditions, the high-cadence state is triggered and the FGM transitions to its higher sampling rate.

High-Cadence Duty Cycle. The fraction of time the FGM operates in high cadence is determined by the occurrence rate of dipolarization front (DF) events and the duration of each high-cadence interval. Fu et al. [17] report a maximum DF occurrence rate of approximately one event per 3.9 hours of magnetotail observation time. This rate is interpreted as representative of the MATRIX region of interest (ROI), such that, on average, one DF event occurs every 3.9 hours while the spacecraft is within the ROI.

Each detected DF triggers a high-cadence interval of approximately 60 s, sized to capture both the < 8 s DF ramp and the trailing bursty bulk flow (BBF), which persists for tens of seconds [6]. Thus, for every 3.9-hour interval of ROI time, the instrument operates in high cadence for approximately 60 s. The resulting high-cadence duty cycle is:

$$f_{\text{HC}} = \frac{60 \text{ s}}{3.9 \text{ h} \times 3600 \text{ s/h}} = \frac{60}{14,040} \approx 0.43\%. \quad (5)$$

The high-cadence state is therefore expected to occupy less than 1% of nominal science operations. The cadence state does not affect FGM power consumption: the THEMIS FGM analog-to-digital



converter (ADC) operates at a fixed internal sampling rate of 32,768 Hz regardless of output cadence, with output rates produced by averaging different numbers of ADC samples per output value [7]. The cadence state therefore primarily affects the FGM data rate, with the high-cadence state producing $32\times$ more data per second than the low-cadence state. This supports compliance with OPL-C-007, which requires that nominal science data generation per spacecraft remain within the downlink and storage capacity allocated by CCD; the corresponding link and storage budgets are accounted for in the CCD subsystem's analysis.

Payload Per-Orbit Duty Cycles and Orbit-Averaged Power. The per-orbit duty cycle of each instrument is determined by the fraction of orbit time spent within the ROI. The MATRIX constellation comprises three orbit groups with distinct apogees and periods, and correspondingly distinct ROI coverage. The FGM operates continuously across both modes (duty cycle 1.0). The ESA and EIS+ are powered ON within the ROI and OFF otherwise, so their per-orbit duty cycle equals the ROI fraction directly. The orbit-averaged payload power for each group is:

$$P_{\text{avg}} = P_{\text{FGM}} + f_{\text{ROI}} \cdot (P_{\text{ESA}} + P_{\text{EIS+}}). \quad (6)$$

With per-instrument powers $P_{\text{FGM}} = 0.8 \text{ W}$, $P_{\text{ESA}} = 1.7 \text{ W}$, and $P_{\text{EIS+}} = 2.0 \text{ W}$, and where f_{ROI} is the ROI fraction for each group, taken from GNC analysis. Per-group results are summarized in Table 20.

Table 20: Per-group orbit characteristics from GNC analysis, instrument duty cycles, and resulting orbit-averaged payload power.

Group	Apogee (R_E)	Period (h)	f_{ROI}	f_{FGM}	$f_{\text{ESA, EIS+}}$	P_{avg} (W)
Group 1	12.09	24	79.73%	100%	79.73%	3.75
Group 2	8.96	16	66.06%	100%	66.06%	3.24
Group 3	7.20	12	47.82%	100%	47.82%	2.57

Group 1 is the worst case for orbit-averaged payload power. The longer 24 h orbit period combined with the highest ROI fraction (79.73%) gives the largest fraction of orbit time with all three instruments active. Group 1 is therefore used for OPL-C-004 average power as an upper bound. This Group 1 value of 3.75 W satisfies the maximum average power consumption of 4.0 W thus verifying the power constraint requirement OPL-C-004.

4.7.8 Magnetic Cleanliness

Analysis Conducted by: Maia (Derived the magnetic cleanliness requirement, estimated the MATRIX spacecraft dipole moment based on the THEMIS magnetic cleanliness program, and computed the minimum FGM boom standoff distance requirement. Updates since PDR: No major changes beyond re-wording and requirement re-numbering.

While instrument sensitivity determines the ability to resolve small magnetic perturbations, measurement accuracy is also constrained by spacecraft-generated magnetic fields. Having established that the fluxgate magnetometer must resolve magnetic field variations of $\leq 0.1 \text{ nT}$ (OPL-P-002), it is also necessary to ensure that strength of the spacecraft-generated magnetic field at the sensor is accounted for, and controlled via the magnetometer boom length.



To address this, a dedicated magnetic cleanliness requirement is introduced, following the approach of the THEMIS magnetic cleanliness program [13]. THEMIS provides direct heritage for this additional requirement: the absolute DC spacecraft-induced field at the FGM sensor was required not to exceed 5 nT.

- **OPL-P-010:** The spacecraft-induced magnetic field at the FGM sensor shall not exceed 5 nT.

THEMIS verified this requirement through magnetic mapping of the spacecraft in a Helmholtz coil facility at JPL [13]. These measurements were used to estimate the effective spacecraft magnetic dipole moment, which provides a characterization of the spacecraft-generated magnetic field. Table 21 summarises the key instrument-level dipole moment contributions measured by Ludlam et al., reported as means across the five THEMIS flight models.

Table 21: Selected instrument-level dipole magnetic moments for the THEMIS FM1 flight model, based on Table 1 of Ludlam et al. [13]. FM1 is selected as a conservative representative case. The EFI spin-plane boom contribution is computed as the sum of SPB1-SPB4. The spacecraft-level total corresponds to the measured integrated dipole moment reported in Table 3 of [13].

Subsystem	Dipole Moment (mA m ²)
IDPU / ESA / SCM	6.8
EFI Spin Plane Boom motors (SPB1-SPB4)	20.5
Solid State Telescope 1 (SST1)	31.2
Solid State Telescope 2 (SST2)	36.2
FM1 spacecraft total (measured)	116.43

Note: The listed subsystems represent the dominant contributors to the spacecraft magnetic moment. The total spacecraft dipole moment also includes additional contributions from other subsystems and structural elements, is reported and sourced from Table 3 of [13].

The MATRIX observatory carries a simplified instrument suite relative to THEMIS. To obtain a conservative estimate of the MATRIX spacecraft dipole moment, the THEMIS FM1 integrated spacecraft dipole moment is adopted as the heritage baseline. Contributions from subsystems known to be absent from MATRIX were subtracted using the corresponding FM1 instrument-level dipole moments reported by Ludlam et al. [13]. Specifically, MATRIX does not include the electric field instrument spin-plane boom motors and carries only one SST (equivalent to the EPD) rather than two. The combined THEMIS IDPU/ESA/SCM contribution was not subtracted, as the individual magnetic contributions of the IDPU, ESA, and SCM are not separately reported. Although MATRIX does not include a dedicated IDPU or SCM, it does include an ESA, and therefore removing the full combined contribution would not be justified. Retaining this term ensures that the resulting estimate remains conservative. The resulting estimates are summarised in Table 22.

Note: The combined IDPU/ESA/SCM contribution is retained because subcomponent-level moments are not separately reported by Ludlam et al. [13], and MATRIX carries an ESA.

Boom Length Derivation The spacecraft-induced magnetic field is approximated as that of a magnetic dipole. The field along the dipole axis is given by [28]:

$$B(r) = \frac{\mu_0}{4\pi} \frac{M}{r^3} \quad (7)$$



Table 22: Estimated MATRIX spacecraft dipole moment derived by subtracting subsystems known to be absent from the THEMIS FM1 heritage baseline. Instrument-level subtractions use FM1 values from Table 1 of Ludlam et al. [13]. The spacecraft-level baseline of 116.43 mA m^2 is taken from Table 3 of Ludlam et al. [13].

Adjustment	$\Delta M \text{ (mA m}^2\text{)}$	Justification
THEMIS FM1 model total	+116.43	Conservative THEMIS reference case
EFI boom motors removed	-20.5	MATRIX carries no electric field instrument
Second SST removed	-36.2	MATRIX carries one SST instead of two
MATRIX estimated total	≈ 59.7	

where M is the total spacecraft dipole moment and r is the distance from the spacecraft magnetic moment centroid to the FGM sensor, approximated as the geometric centre of the spacecraft bus.

To account for uncertainty in the estimated dipole moment, a factor of $2\times$ margin is applied to M . Enforcing the magnetic cleanliness requirement defined in OPL-P-011, $B(r) \leq B_{\text{req}}$, and solving for the minimum required radial distance gives:

$$r_{\min} = \left(\frac{\mu_0}{4\pi} \frac{2M}{B_{\text{req}}} \right)^{1/3} \quad (8)$$

where the factor of 2 represents the applied margin on M . Substituting $M = 59.7 \text{ mA m}^2 = 5.97 \times 10^{-2} \text{ A m}^2$ and $B_{\text{req}} = 5 \text{ nT} = 5 \times 10^{-9} \text{ T}$ gives:

$$r_{\min} = \left(\frac{10^{-7} \cdot 2(5.97 \times 10^{-2})}{5 \times 10^{-9}} \right)^{1/3} \approx 1.34 \text{ m} \quad (9)$$

The FGM sensor shall therefore be located at a minimum distance of 1.34 m from the spacecraft centre. This leads to the derived requirement:

- **OPL-P-011:** The FGM sensor shall be located at a minimum distance of 1.34 m from the spacecraft centre.

The MATRIX magnetometer boom is mounted at the upper edge of the spacecraft. The spacecraft bus is hexagonal with side length $s = 23 \text{ in} = 0.584 \text{ m}$ and height $h = 20 \text{ in} = 0.508 \text{ m}$. The distance from the centre of the bus to the boom attachment point is therefore:

$$d_{\text{attach}} = \sqrt{\left(\frac{\sqrt{3}}{2}s \right)^2 + \left(\frac{h}{2} \right)^2} = \sqrt{(0.505)^2 + (0.254)^2} \approx 0.566 \text{ m} \quad (10)$$

Given this attachment location the minimum required boom length to satisfy OPL-P-011 and OPL-P-012 with no additional margin is:

$$L_{\text{boom,min}} = r_{\min} - d_{\text{attach}} = 1.34 - 0.566 \approx 0.77 \text{ m} \quad (11)$$



4.7.9 ESA Selection

Analysis Conducted by: Maia (Derived the ESA requirements based on research on the magnetotail literature, and selected the ESA for MATRIX through a trade study. Updates since PDR: No major changes.

BBF events are fundamentally high-speed earthward flow phenomena. The most common criteria for the identification of a BBF requires the flow to exceed 100 km s^{-1} throughout the interval and reach peak values of at least 250 km s^{-1} [6]. Bulk velocity is defined as the first moment of the particle velocity distribution function:

$$\mathbf{V} = \frac{1}{n} \int \mathbf{v} f(\mathbf{v}) d^3v, \quad (12)$$

and therefore measurement of the underlying ion and electron energy distributions using an electrostatic analyzer (ESA) is required to identify a BBF event. Electrostatic analyzers (ESAs) are the standard instrument class used in space plasma missions to measure these distributions. An ESA operates by applying a voltage across a set of electrodes to generate an electrostatic field that deflects incoming charged particles onto curved trajectories. Based on the applied voltage, only particles with a specific energy-per-charge ratio (E/q) are transmitted through the analyzer to the detector. By stepping the applied voltage, the ESA samples discrete energy-per-charge intervals and, in combination with angular sampling, reconstructs the local three-dimensional velocity distribution, from which plasma moments such as density, bulk velocity, and temperature are derived [29].

Energy Range Requirement

In the absence of BBF activity, the background ion population has a bulk velocity close to zero and is concentrated in the 0.5-4 keV range [30]. During BBF events, energy transport is carried by ions in the 10-20 keV range and mass transport by a broader population spanning 1-15 keV [30]. A lower bound of 100 eV and an upper bound of 25 keV for the ion distribution energy range requirement are adopted to capture both populations, with margin on either side to account for the tails of the distributions.

For electrons, electron energy ranges from 6 eV to 25 keV during BBF events [31], so a lower bound of 5 eV is adopted for the requirement. During BBF events, electron number density ramps up and electron energies extend beyond 25 keV into energetic electron bursts (EEBs) [32], with this higher-energy population addressed by the MATRIX EPD. This leads to the energy range requirement.

Field of View Requirement

Another marker of BBFs is that the ion and electron distributions are observed to be highly anisotropic [30], meaning the distribution of velocities is asymmetric, with a preferred flow direction rather than being evenly spread across all directions. To detect this behaviour, the local flow at the spacecraft must be resolved into its V_x , V_y , and V_z components, leading to the field of view requirement.

ESA Instrumentation Requirements

- **OPL-P-004:** The ESA shall measure ion energy distributions across 100 eV to 25 keV and electron energy distributions across 5 eV to 30 keV.
- **OPL-P-005:** The ESA shall provide full 4π sr field-of-view coverage for ion and electron measurements.



Instrument Selection

Several ESA geometries exist, including spherical, cylindrical, and top-hat analyzers. Among these, the top-hat ESA provides 360° azimuthal field of view and is the standard configuration for space plasma missions due to its wide angular coverage and compatibility with position-sensitive detectors [29].

When combined with spacecraft spin, this azimuthal coverage is swept through the remaining angular directions, enabling reconstruction of the full 4π sr velocity distribution. Each MATRIX spacecraft has a spin rate of 15 rpm (4 s period), so one complete three-dimensional distribution is obtained locally per rotation. Accordingly, a top-hat ESA is selected as the baseline instrument, as it satisfies the field-of-view requirement (OPL-P-005).

Table 23: ESA instrument trade study comparing candidate electrostatic analyzers.

Parameter	THEMIS ESA	MMS FPI (DIS+DES)
Instrument Type	Top-hat (dual ion+electron)	Top-hat (DIS + DES)
Ion Energy Range	1.6 eV - 25 keV	10 eV - 30 keV
Electron Energy Range	2 eV - 32 keV	10 eV - 30 keV
Instantaneous FOV	$180^\circ \times 6^\circ$	$180^\circ \times 11.25^\circ$
FOV per Spin	4π sr	4π sr [†]
Mass (combined)	2.96 kg	12.08 kg
Power	1.7 W	11.0 W
Volume	5,470 cm ³	~9,000 cm ³ *
Species	Protons+Electrons	Protons+Electrons

Sources: [8]; [33]

*Volume estimated from DES photograph since full instrument dimensions not reported. [33];

[†]Achieved via 8 sensors per species without relying on spacecraft spin.

Both candidate instruments are top-hat electrostatic analyzers providing full 4π sr coverage per spin and ion and electron energy ranges that overlap with the ESA requirements (OPL-P-004, OPL-P-005). However, the two instruments differ substantially in mass, power, and volume. The MMS FPI DIS+DES pair draws 11.0 W per spacecraft. Combined with the selected magnetometer (0.8 W) and EPD (2.0 W), the total OPL peak power would reach 13.8 W: nearly three times the 5 W peak power constraint (OPL-C-003). The MMS FPI is therefore eliminated from consideration on power grounds alone, independent of its mass and volume, which would similarly exceed the OPL subsystem budget.

The THEMIS ESA draws 1.7 W per spacecraft. Combined with the selected magnetometer (0.8 W) and EPD (2.0 W), the total OPL peak power is 4.5 W, satisfying OPL-C-003 with margin. The combined OPL mass of 2.96 kg (ESA) + 0.38 kg (magnetometer) + 2.2 kg (EPD) = 5.54 kg similarly remains within the OPL mass allocation.

Regarding radiation tolerance, the THEMIS spacecraft have operated continuously in the near-Earth magnetotail environment for over a decade, reaching their 10-year operational mark in February 2017 and remaining operational thereafter [34]. The MATRIX orbital environment exposes the payload to approximately the same magnetotail radiation environment over a 5-year mission lifetime, satisfying OPL-E-002. The THEMIS ESA's demonstrated longevity in this environment provides strong heritage validation of its radiation tolerance without requiring additional qualification.



Based on these considerations, the THEMIS ESA is selected as the MATRIX electrostatic analyzer. It satisfies all derived ESA payload requirements (OPL-P-004, OPL-P-005), meets the power constraint (OPL-C-003) and radiation tolerance requirement (OPL-E-002), and its compact dual ion-electron package provides extensive plasma sheet flight heritage directly applicable to the MATRIX mission [8].

4.7.10 EPD Selection

Analysis Conducted by: William (trade study, instrument selection, datasheet creation)

Changes since PDR: Discussion of SMM effects of repurposing EPD from MMS¹

The MATRIX Energetic Particle Detector (EPD) is necessary to provide scientific data on the energy, composition, and direction of ions and electrons at higher energies than what the ESA instrument can measure. This is accomplished by using a different instrument architecture, namely a time-of-flight system rather than an electrostatic analyzer architecture which allows for measurements of higher energy particles at the expense of energy resolution.

During BBFs, electron energies increase beyond 25 keV and there is evidence for increases of up to ~ 300 keV from observations by the European Space Agency's Cluster mission [36]. Bursts of electrons with energies up to 1 MeV have also been observed, but farther out in the magnetotail at distances of $\sim 20 - 30 R_{Earth}$. Both of these reported energy limits were observed at radial distances farther from Earth observed exactly in the magnetotail transition region that MATRIX is exploring but provide a good starting point for the selection of an energetic particle detector.

For ions, MATRIX is primarily interested in protons (H^+), which are the dominant component of BBFs and other space plasmas. These protons primarily reside in energies between 10 keV and 150 keV [37]. The ESA instrument is capable of measuring ions up to 30 keV in energy so the EPD is only required to measure ions down to 25 keV in energy to create a little overlap with the ESA. In addition to protons, during geomagnetic storms the plasma in the magnetotail can become dominated by oxygen (O^+) ions with energies greater than 90 keV even though there is not an abundance of O^+ in the outer magnetotail [38]. This leads some authors to suggest that the ionosphere is the dominant contributor of O^+ . MATRIX's contribution to the current gap in mesoscale observations of the magnetotail transition region could provide the missing piece of information to answer where the O^+ in magnetotail plasma originates.

Table 24 summarizes the results of a trade study of heritage magnetotail and space plasma mission energetic particle detectors to determine the best EPD for MATRIX.

¹All except discussion of effects of repurposing EPD is copied over from the MATRIX OPL PDR Subsystem Report [35].



	Features	Proton (H+) Energy	Oxygen (O+) En- ergy	Electron e- Energy
Requirement		25 keV to 150 keV	>90 keV	25 keV to 1 MeV
THEMIS SST [39]	Assumes H+	30 to 300 keV Maximum: 20 keV to 6 MeV	–	30 to 100 keV Maximum: 25 keV to 1 MeV
MMS EIS [9]	Energy, Angle, Composition	20 keV to >1 MeV	45 keV to >1 MeV	25 keV to 1 MeV
CLUSTER II HIA + RAPID [40]	HIA for ions. RAPID for elec- trons	5 eV to 32 keV	–	20 keV to 400 keV
IS$\odot$IS EPI-Lo [41]	–	20 keV to 15 MeV	320 keV to 15 MeV	25 keV to 1 MeV
IMAP CoDICE [42]	–	30 keV to 5 MeV	0.5 keV to 80 keV, 480 keV to 80 MeV	–

Table 24: Energetic Particle Detector Trade Study

As seen in Table 24, the MMS Energetic Ion Spectrometer, satisfies the energy ranges of H+, O+, and e- particles typically seen in BBFs. Crucially, the EIS instrument's 20 keV and 25 keV minimum energy requirement for ions and electrons overlaps with the 30 keV maximum energy requirements for the ESA instrument, allowing for calibration and continuity across the entire range of energies that MATRIX measures.

Although MMS's EIS instrument fits the energy ranges very well, it is not perfect. It would be preferable if its upper limit of electron energy detection extended beyond the 1 MeV maximum observed in previous literature, so that if there is new science to be discovered in regards to high energy electrons, the mission would be capable of that. The EIS instrument is also only certified for a 100 krad radiation dose, but the environment that MATRIX operates in receives 33 krad/year so the EIS instrument would only be certified for a 3-year mission instead of 6 years as required by **MISS-F-004** and **OPL-E-002**.

Therefore, rather than using the EIS instrument outright on MATRIX, we propose using a hypothetical modified EIS instrument which we will call EIS+. EIS+ would be an updated version of EIS with an upper bound of electron energy measurements of 1.5 MeV and a more advanced flight computer capable of withstanding a 200 krad radiation dose. The increased radiation hardening will come from more advanced electronics components and improved circuit design rather than by adding thicker aluminum walls in the instrument to block ionizing radiation. This allows the mass of the instrument to remain comparable to the original EIS instrument on MMS. Finally, the increased energy range of EIS+ compared to EIS does not necessarily translate into an increase in mass and instead the increased energy range can be achieved by using higher voltages (and lower currents to maintain equal power) on the electrodes in the instrument to bend particles of higher energy. A great example of this tradeoff is found in the Solar Wind Around Pluto (SWAP) and Solar Wind And Pickup Ion (SWAPI) instruments which both measure the solar wind. SWAP weighs 3.29 kg, has a volume of 0.011 m^3 , and has an energy range of 35 eV to 7.5 keV [43]. SWAPI weighs 3.01 kg, has a volume of 0.011 m^3 , and has an energy range of 89 eV to 21.5 keV [44].

The parameters of the proposed EIS+ instrument that are relevant to other MATRIX systems are summarized in Table 25.



Parameter	Value	Relevant Subteam
H+ Detection Energy	20 keV to > 1 MeV	OPL
O+ Detection Energy	45 keV to > 1 MeV	OPL
e- Detection Energy	25 keV to 1.5 MeV	OPL
Total Mass	2.2 kg	SMM
Volume	23.3 x 16.9 x 12.8 cm	SMM
Field of View	160° x 12°	SMM
Operational Power	2.0 W	POW
Data Rate	0.67 kbps	CCD
Radiation Protection	200 krad	SMM

Table 25: MATRIX Energetic Ion Spectrometer (EIS+) Parameters [9]



5 CCD Design

5.1 Subsystem Overview

The broad goals and responsibilities of the CCD subsystem are to design methods of onboard computation, data storage, command execution, and ground communication.² For MATRIX, CCD defined a data budget by compiling hardware from other subsystems and determining the rate of data generation and total data generated per orbit. This information was then used to define the specific methods of data processing, identify hardware for flight computers and payload data processors that were compatible with subsystem requirements, identify communications hardware including a radio, RF power amplifier, and antenna, and create a link budget for downlink and uplink of data to each MATRIX spacecraft.

5.2 Subsystem Objectives

The objectives of the CCD subsystem are to³:

1. Enable the processing and communication to Earth of scientific data collected by science instruments while in the magnetotail transition region.
2. Operate the spacecraft efficiently by taking in data from all subsystems and using prioritization to processing and/or store the data on the spacecraft.
3. Provide enough computational power to support scientific instruments as well as tasks from other subsystems such as GNC, ADCS, or THM.
4. Provide robustness in the design of mission components so that single point failures are minimized.

5.3 Subsystem Requirements

The CCD subsystem requirements remain entirely consistent with those established at the PDR stage, but since then, minor refinements have been made to specific numerical values following the revision and finalization of subsystem budgets. Table 26 below shows the most important requirements to the CCD subsystem and the full set of requirements can be found in the CCD tab of the [MATRIX Requirements Spreadsheet](#).

²This sentence was copied over from the MATRIX CCD PDR Subsystem Report [14].

³List copied from CCD PDR Subsystem Report [14]



ID	Requirement	Rationale	Verification/Validation Method	Trace Up ID
CCD-P-002	The communication technologies selected by the CCD subsystem must have a signal-to-noise ratio of at least 6.6 dB	Robustness to error within a margin of 3 dB (a factor of 2) is good practice for the transmission and reception of communications between spacecraft and Earth ground stations. According to JPL, an additional 3.6 dB is required [45].	Validation of sufficient link budget Eb/N0 parameter	MISS-F-007
CCD-P-004	The communication technologies selected by the CCD subsystem shall accommodate an uplink rate of at least 9.6 kb/s	The Near Earth Network can uplink UTDF tracking data at 9.6 kb/s	Link Budget	MISS-F-013
CCD-P-005	The communication technologies selected by the CCD subsystem shall have a downlink rate of at least 253.394 kb/s	The data rate required to communicate all scientific data on each orbit. Derived from data generation rate of instruments and other sensors onboard the spacecraft	Link budget calculation	MISS-F-006
CCD-C-003	The CCD subsystem shall have a peak power consumption of at most 47 W of the total generated power when communicating with Earth	Compliance with mission architecture power budget	Subsystem power budget	MISS-F-009
CCD-C-004	The CCD subsystem shall have an average power consumption of at most 7 W	Compliance with mission architecture power budget	Subsystem power budget	MISS-F-009
CCD-E-003	All CCD components shall remain operational for the full 6-year mission in a total ionizing dose environment of 198 krad.	The expected radiation environment corresponds to approximately 33 krad/year, resulting in a total mission dose of 198 krad over 6 years [46].	Cross-reference datasheets of selected components	MISS-E-001

Table 26: CCD Subsystem Requirements

5.4 Subsystem Constraints

This section summarizes the CCD subsystem constraints. There have been no significant changes to the subsystem-level or mission-level constraint definitions since the last design cycle; however, specific numerical values within the power and mass budgets have been finalised. In particular, revised estimates for the flight computer and antenna power amplifier have resulted in updated CCD power allocations. The overarching constraints remain unchanged, as they continue to sufficiently bound and motivate the subsystem design.

- **Mass and Power Constraints:** The CCD subsystem must remain within an allocated mass limit of 32.5 kg and comply with updated power allocations derived from subsystem-level trades. While the average power consumption target remains at 7 W, the finalised power budget allocates 53.2 W to the antenna power amplifier. These limits derive from mission-level constraints MISS-C-005 and MISS-F-009.
- **Environmental and Radiation Constraints:** CCD electronics must remain functional within the environmental conditions encountered in Earth's magnetotail, including operation in vacuum and exposure to a cumulative radiation dose of 198 krad over a 6-year mission duration (CCD-E-003)[5]. This requirement traces to MISS-E-001 and protects against radiation-



related anomalies in the MATRIX flight computer.

- **Reliability and Fault Protection Constraints:** The CCD subsystem must maintain reliable operation throughout the mission lifetime to ensure continuous command, telemetry, and science data return capabilities. This requirement traces up to mission-level requirements MISS-F-006 and MISS-F-007. This constraint requires the CCD architecture to incorporate reliability measures such as fault-monitoring in memory systems, and protective logic to ensure that critical command and communication functions remain available even during anomalies.

5.5 Subsystem Drivers

Below is a summary of the CCD subsystem's drivers. These are unchanged since the PDR Subsystem Report because they continued to fully motivate the subsystem design in this design cycle and no new drivers were needed to complete the design. ⁴.

- **Science Data Processing Throughput:** The CCD subsystem must be able to continuously process instrument measurements, including time-tagging, packetisation, buffering, and preparation for transmission to Earth. This supports **Mission Drivers 1, 2, and 5** as described in the MATRIX PDR [47] which drive continuous in-situ field and particle measurements of the magnetotail and the transmission of that data to Earth.
- **Onboard Data Storage and Data Management:** The CCD subsystem must provide sufficient non-volatile storage and data management capabilities to store data for downlink if it is unable to be downlinked on any given orbit. This is derived from **MISS-F-006** which requires that telemetry and collected science data be transmitted back to Earth at least once per orbit
- **Reliable Autonomous Operation and Communication Support:** The CCD subsystem must maintain reliable operation while supporting spacecraft autonomy during periods without ground contact, effectively forming the operational backbone of the spacecraft. This driver derives primarily from **MISS-F-004**, which requires mission components to remain operational for at least six years, and **MISS-F-007** which requires the spacecraft to operate autonomously in the magnetotail transition region.
- **Minimize Single Point Failures:** Although reliability was not a mission driver described in the MATRIX PDR, this driver derives from **MISS-F-004** which requires the mission to be operational for at least 6 years and therefore requires robust design to prevent an anomaly like an antenna malfunction from ending the mission.
- **Minimize Power Utilization:** Throughout the design of the MATRIX mission, power has been one of the most constrained resource and the antenna requires significant power so minimizing power consumption is a significant driver of the CCD subsystem. Minimizing power consumption allows more power to be distributed to other systems and also minimizes thermal losses that must be radiated away.

⁴These drivers have been mostly copied over from the previous CCD PDR subsystem report [14], but have been rewritten for conciseness and updated with correct mission level numbers.



5.6 Subsystem Design Approach

The CCD subsystem design approach follows a logical flow starting with the data budget, which drives design decisions for selection of communications equipment and flight computer hardware and software. Since the PDR report, most of the design work for CCD has been finalizing trades with other subsystems like providing final mass for a rescaled Bus Avionics Unit (BAU) and receiving an updated number of thermal sensors from THM. Additional work has also been done to clearly define when spacecraft communicate with Earth and explain the omnidirectionality of the communication antennas.

5.7 Formal Analysis

5.7.1 Data Budget

Analysis conducted by: William (quantitatively defining the data collected by various subsystems on the spacecraft based on data rate and time spent collecting data) *Updates since PDR:* Trade with THM on the exact number of thermal sensors, considering margin in the occurrence of BBFs.

The data budget considers the rate of data generation from all systems on a MATRIX spacecraft and the time during which that data is generated in order to determine the total amount of data generated during one orbit. This is the amount of data that must be transmitted to the ground and directly impacts the link budget which is discussed in Section 5.7.10.

Table 27 shows the different sources of data on MATRIX. Since each MATRIX spacecraft rotates at 15 rpm or 4 seconds per rotation, many of the sources of housekeeping data are taken on a once per rotation basis. The primary sources of data come from the science instruments from OPL which also produce data at a rate proportional to the spin rate of the spacecraft.

Item	Data Rate (bps)	Total Data Volume (bits, 1 orbit)
Magnetometer	6 (Survey Mode), 3072 (Burst Mode)	14310180
ESA	5632	481704285
EPD	670	57305020
ADCS Silicon Sensing DMU41 IMU	812	2923200
TE Connectivity RTD Platinum Temperature Sensor	88	4704144
Other Housekeeping Data	250	21382470
Total (with margin of 1.5x)		912220174
Required Downlink Rate	253 kbps	

Table 27: MATRIX Spacecraft Data Budget

5.7.2 Onboard Computer Hardware Trade

Analysis conducted by: Sawooly Li (trade study on flight computer hardware, including computational performance, storage capacity, and power requirements).

Updates since PDR: Processor selection corrected following reanalysis; trade study repeated using updated processor assumptions.

MATRIX's onboard computer trade study began with an evaluation of three candidate systems with heritage in magnetospheric science missions: the THEMIS Command and Data Handling (C&DH) computer [48], the MMS Central Instrument Data Processor (CIDP) [49], and the RTG4-LEON3 processing architecture proposed for MagCon [26]. These systems were selected due to

their demonstrated or anticipated capability to support high-rate plasma and field measurements relevant to the MATRIX mission.

Table 28: Performance Metrics of Candidate Onboard Computers

	THEMIS C&DH Computer [48]	MMS Central Instrument Data Processor (CIDP) [49]	MagCon RTG4-LEON3 Processing Unit [26]
Mission Heritage	THEMIS	MMS	Proposed for MagCon (RTG4 FPGA heritage in multiple space missions)
Primary Processor	RH5208 ColdFire (ATK Digital Processor Module) [50]	RAD750 [51]	LEON3 soft-core implemented in RTG4 FPGA [52]
Computational Capability	Up to 159 MIPS (16 MIPS)	200–400 MIPS	70–200 MIPS
System Architecture	Centralised	Centralised	Semi-centralised
Dimensions (L × W × H)	21.3 × 15 × 11.9 cm	25 × 25 × 15 cm	17 × 17 × 3 cm
Volume	3800 cm ³	9375 cm ³	870 cm ³
Non-Volatile Storage	4 GB solid-state recorder	96 GB mass memory module	4–16 GB external NAND flash
Volatile Storage	64 MB SDRAM with EDAC (triple-voting)	1 GB SDRAM	512 MB–1 GB SDRAM
Fault Tolerance	Radiation hardening + EDAC memory + watchdog (cFS HS) + IDPU isolation layer	Radiation-hardened processor with ECC memory and redundant instrument processors	RTG4 FPGA supporting triple modular redundancy (TMR), configuration scrubbing, and duplicated processing logic
Average Power Consumption	< 7 W	10–14 W	8–15 W

Initial analysis of computational requirements was performed using SMAD-based [53] estimates for command handling, telemetry processing, fault monitoring, and science data processing. The total required processing throughput was found to be approximately 1.41 MIPS, which was updated to a better estimate of 1.49 MIPS in the PDR. In an earlier iteration, the THEMIS flight computer was incorrectly identified as being based on the RAD6000 processor [54], but this has since been corrected to the RH5208 ColdFire processor within the ATK Digital Processor Module [50]. Even with this modification, all candidate systems significantly exceed the baseline computational capacity, demonstrating that the flight system selection is not driven by raw computational performance.

Both the THEMIS C&DH and MMS CIDP are based on predominantly centralised data handling architectures, in which a primary onboard processor performs spacecraft command, telemetry, and data management functions [48] [49]. In contrast, the MagCon RTG4-LEON3 concept adopts a semi-centralised architecture with distributed processing elements explicitly integrated into the system design [26]. Given that the required processing capability is significantly lower than that provided by all candidate systems, a centralised architecture was preferred for MATRIX to minimise system complexity and integration risk without sacrificing performance.

Power consumption was also a key design constraint as all onboard electronics must operate within a limited budget. The THEMIS C&DH computer operates at less than 7 W [50] compared to estimated ranges of 8–15 W [51] [52] for higher-performance alternatives, and as all systems provide sufficient fault tolerance through radiation-hardened components or redundancy mecha-



nisms, the Themis computer was chosen for MATRIX, providing a balanced solution that satisfies performance requirements while adhering to strict power constraints and minimising overall system risk.

5.7.3 Flight Software

Analysis conducted by: Sonal (flight software framework trade study, mapping of cFE services to MATRIX functions, throughput derivation against the corrected RH5208 ColdFire baseline, and definition of the ten flight computer operational modes).

Updates since PDR: Throughput derivation re-baselined against the corrected RH5208 ColdFire processor (the PDR analysis still partly referenced the RAD6000 figures from the original CCD report); per-mode flight computer configuration table consolidated into Appendix A.4 and an accompanying mode-transition diagram added; explicit linkage between cFS services and the IDPU interface defined in Section 5.7.5.

The MATRIX flight computer baseline established in Section 5.7.2 is the THEMIS Command and Data Handling system, built around the ATK Digital Processor Module's RH5208 ColdFire processor running at 16 MHz [55]. The present analysis investigates the software layer that executes atop this hardware: specifically, the candidate flight software (FSW) frameworks capable of implementing MATRIX's onboard data prioritisation, scheduling, and subsystem coordination strategy as defined in CCD requirements CCD-F-002, CCD-F-007, CCD-F-011, CCD-F-014, and CCD-F-016. The analysis builds directly on the IDPU functions defined in Section 5.7.5 (which determine what data the DPM receives) and on the RS-422 channel mapping defined in Section 5.7.11 (which determines how that data physically arrives).

Flight Software Framework Trade Study Three candidate FSW frameworks were evaluated: NASA's Core Flight System (cFS), the Real-Time Executive for Multiprocessor Systems (RTEMS) with a custom application layer, and Wind River VxWorks with a custom application layer. All three provide a preemptive priority-based real-time scheduler suitable for the RH5208 ColdFire processor; the differentiator is what mission-level infrastructure each provides without additional development effort.

RTEMS is an open-source POSIX-compliant RTOS with strong NASA and ESA science mission heritage, having flown on the Mars Reconnaissance Orbiter, Fermi Gamma-ray Space Telescope, the Magnetospheric Multiscale mission, and Parker Solar Probe, and used by ESA on Herschel, Planck, BepiColombo, and ExoMars [56], [57]. Critically, RTEMS is one of the supported RTOS platforms underneath the cFS Operating System Abstraction Layer (OSAL), meaning it is not strictly an alternative to cFS but can serve as its substrate [56]. VxWorks has equally extensive spaceflight heritage, powering Mars Pathfinder (1997), Curiosity (2012), InSight (2018), and dozens of other NASA JPL missions [58], [59]. However, VxWorks is a proprietary commercial product, carrying licensing costs and restricting source-level visibility. Crucially, neither RTEMS nor VxWorks provides an inter-application messaging bus, built-in health monitoring, or telemetry routing out of the box - all of which would require substantial bespoke software development, increasing both cost and verification burden.

NASA's Core Flight System (cFS) provides all of these as mature, flight-proven applications [60], [61]. Originally developed at NASA Goddard Space Flight Center, the cFE Core Flight Executive was first flown on the Lunar Reconnaissance Orbiter (LRO) in 2009, and the full cFS application



suite first flew on the Global Precipitation Measurement (GPM) spacecraft in 2014 [62]. Since open-sourcing under the NASA Open Source Agreement in 2015, cFS has been used on over 40 missions [60]. Table 29 summarizes the trade and downselection.

Table 29: Flight Software Framework Trade Study

Criterion	NASA cFS	RTEMS + custom	VxWorks + custom	Notes
Priority scheduling	✓	✓	✓	All support preemptive RTOS [56], [58]
Inter-app messaging bus	✓	×	×	cFE Software Bus built-in [60]
Fault detection (built-in)	✓	×	×	cFS HS and LC apps [60]
Telemetry & cmd routing	✓	×	×	cFE Event and Executive Services [60]
Open-source	✓	✓	×	VxWorks proprietary [58]
> 10 missions heritage	✓	✓	✓	cFS: 40+ NASA [60]; RTEMS: MRO, MMS [56]; VxWorks: Curiosity, InSight [58]
Verdict	Selected	Viable; high dev. cost	High cost; proprietary	

Given MATRIX’s modest 1.49 MIPS processing requirement (derived in Appendix A.3), the custom development and verification cost of building equivalent messaging, health monitoring, and telemetry infrastructure on top of RTEMS or VxWorks is unjustified. NASA’s cFS is therefore selected. cFS provides all required capabilities in a single open-source, flight-heritage framework. Its OSAL ensures portability to the ColdFire target without processor-specific source changes [60], and the cFS Scheduler (SCH) and Data Storage (DS) apps directly implement the tiered science data prioritisation strategy defined in Section 5.7.6.

How cFS Implements MATRIX’s Data Strategy cFS runs on top of the ColdFire RTOS and provides mission-level services via its Core Flight Executive (cFE). Table 30 maps each cFE service to its MATRIX function. The cFE Software Bus is the key enabler: every subsystem application publishes CCSDS-formatted packets by topic, and subscribing applications (data storage, downlink manager, fault monitor) receive them without direct coupling. For example, when GNC detects that the spacecraft has entered the science zone ($> 6 R_{\oplus}$), it publishes a single mode-change message that all subscribed applications (instruments, scheduler, SSR manager) reconfigure against simultaneously. This satisfies CCD-F-002, CCD-F-011, and CCD-F-014.

Table 30: cFE Services and Their MATRIX Application

cFE service	Function	MATRIX application
Executive Services (ES)	App lifecycle, watchdog timer	Safe-mode entry, processor reset
Software Bus (SB)	Publish-subscribe messaging	Science data routing, mode changes
Scheduler (SCH)	Major/minor frame task timing	4 Hz ADCS, 1 Hz HK, 0.1 Hz THM
Event Services (EVS)	Async fault/status messages	BBF trigger relay, anomaly logging
Table Services (TBL)	In-flight reconfigurable tables	Priority thresholds, limit values
Health & Safety (HS)	CPU monitoring, watchdog	Hung-task detection, processor reset
Limit Checker (LC)	Telemetry watch-points	Power, thermal, attitude interlocks
Data Storage (DS)	Priority-queued SSR writes	Tiered science/HK buffering



Throughput and Fault Protection Because the IDPU offloads all instrument-specific processing from the DPM (Section 5.7.5), the DPM only ever receives CCSDS packets, not raw samples. This significantly reduces the DPM software load. The total DPM FSW load is approximately 1.49 MIPS (TBC) against ~ 16 MIPS available on the RH5208 ColdFire, giving a $7.5\times$ margin (full derivation in Appendix A.3). Applying SMAD's 70% utilization guideline [53], usable capacity is $0.7 \times 16 = 11.2$ MIPS, which still leaves a factor of ~ 7.5 in headroom, which is comfortably above the 50% on-orbit spare reserve recommended for post-launch anomaly correction.

Fault protection is implemented in three complementary layers and is the principal mechanism for satisfying **CCD-F-016**:

1. *IDPU payload-level isolation.* An instrument fault is contained at the MA61C IDPU and cannot propagate to the DPM, because the IDPU validates packet integrity (opcode and length checks) before forwarding any data over the IDPU High Rate Data RS-422 channel (Section 5.7.5).
2. *cFS Health & Safety watchdog.* The cFS HS application monitors the DPM watchdog timer and detects hung tasks, triggering a controlled safe-mode entry on timeout [53], [60].
3. *ATK DPM hardware EDAC.* The ATK DPM's triple-voting EDAC on the 64 MB SDRAM corrects single-bit upsets in software memory and flags multi-bit upsets to the HS application [55].

The Limit Checker (LC) application provides an additional cross-subsystem layer: it subscribes to housekeeping telemetry on the cFE Software Bus and triggers safe-mode or mode-change responses when power, thermal, or attitude telemetry crosses configurable thresholds. Because LC limits live in cFS Table Services, they are reconfigurable on-orbit without a flight software re-upload.

Operational Modes The flight software supports ten operational modes spanning the full mission lifecycle: Start-Up, Commissioning, Nominal (Survey), Burst/High-Rate Science, Downlink, Maneuver, Eclipse, Safe Mode, Standby/Idle, and End-of-Life. Each mode is implemented as a distinct cFS Scheduler configuration with its own task cadence, IDPU command, and SSR write priority. The full mode table and accompanying transition diagram are provided in Appendix A.4. The most operationally significant transition is Survey $\rightarrow$ Burst: as defined in Section 5.7.5, this transition is initiated by the IDPU itself when the FGM dipolarization-front criterion ($\Delta B_z \geq 4$ nT in < 8 s) is satisfied, and the cFS Software Bus then propagates a *BBF-Detected* message that simultaneously raises the SSR write queue to Tier 3 priority, increases the SCH frame rate, and arms the Limit Checker on the SSR fill level. This event-driven design satisfies **CCD-F-014** and **CCD-F-015** by minimizing latency between the physical event and the onboard response.

5.7.4 Bus Avionics Unit Design and Instrument Data Processing Unit Analysis

Analysis conducted by: Maia (All analysis tasks in this subsection including BAU architecture design, MA61C-SmallSat selection and IDPU integration and figure design.)

Updates since PDR: Refined BAU structural mass to account for aluminium chassis extension; defined power and data path separation.



Payload Data Handling Unit In order to downlink the data collected by the science instruments on the payload to the ground, the instruments must first transfer their data to the spacecraft's data handling system. In the traditional model, each instrument or instrument suite is served by a dedicated payload processor, referred to as an Instrument Data Processing Unit (IDPU) [63], [64]. The IDPU sits in the data path between the instruments and the spacecraft's main computer, receiving raw sensor data over dedicated digital interfaces, performing analog-to-digital conversion and instrument-specific processing, and routing the processed data to the spacecraft's main computer. In the distributed model, raw unprocessed sensor data is passed directly to the spacecraft computer, which takes on all instrument-related processing [64]. While this reduces payload-side hardware complexity, it concentrates processing load on the main computer. The use of a dedicated IDPU additionally enables payload-level fault protection and isolation, satisfying requirement **CCD-F-016**. This is critical for the MATRIX mission, in particular during nominal science operations in the ROI when the payload generates high-rate data from multiple instruments simultaneously. Under these conditions, an uncontained instrument fault could propagate to the main command system and disrupt nominal spacecraft operations [10].

This design decision is also supported by heritage missions. THEMIS used a single central IDPU per spacecraft to serve all five science instruments [65]. The IDPU handled all data processing of the raw sensor data, and triggered the various science acquisition modes. MMS used a two-tier payload processing architecture: a Central Instrument Data Processor (CIDP) coordinating the majority of instruments, and a dedicated IDPU (FPI-IDPU) specifically supporting its 8 electrostatic analyzers [66]. MMS's two-IDPU architecture was a consequence of instrument suite scale - each MMS spacecraft housed 25 individual sensors. By contrast, MATRIX features a much more compact design similar to the THEMIS scale of 3 science instruments per spacecraft, thus a single central IDPU selected as more suitable.

MATRIX Bus Avionics Unit Design In the THEMIS architecture, the IDPU was implemented as a physically separate unit requiring multiple boards and a dedicated enclosure, imposing additional mass (+5kg), volume, and harness complexity [65]. Relative to the THEMIS implementation, a modernisation is proposed: rather than housing the IDPU in a fully separate chassis, the IDPU functions are consolidated onto a dedicated payload interface board integrated as an additional board within the BAU stack. This consolidation is motivated by advances in miniaturised payload data processing unit design since the launch of THEMIS in 2007. Modern payload IDPU's feature a 'System-on-chip' (SoC) architecture which leverages high density field-programmable gate arrays (FPGAs) whereby all the IDPU functions are integrated on a single chip [67]. Specifically, LEON SPARC processors have accumulated broad flight heritage across NASA and ESA science missions as instrument and spacecraft data processing units, some notable examples include:

Table 31: Flight heritage examples of the LEON SPARC processor family in space missions

Mission	Agency	Launch	Processor	Ref.
PROBA-V	ESA	2013	LEON2	[68]
CHEOPS	ESA	2019	LEON3-FT	[69]
Artemis I	NASA	2022	LEON3-FT	[70]
JUICE (RPWI)	ESA	2023	LEON3-FT	[71]

The current generation of the LEON processor family, the LEON3-FT, is therefore baselined as the

MATRIX IDPU processor. The LEON3-FT is procured via the SPiN MA61C-SmallSat; a dedicated board-level product which packages the LEON3-FT processor alongside an interface adaptation capability [72]. The MA61C-SmallSat automatically adapts between communication protocol standards, supporting the OPL subsystem’s mixture of heritage instruments including RS-422 communication with the instruments [73]. The MA61C-SmallSat PCB measures 105×105 mm and weighs 150 g [72]. As the THEMIS BAU structure cross-section is 213×150 mm [73], the MA61C-SmallSat PCB fits within the chassis width and depth without modification; only the chassis height requires an incremental increase of approximately 35 mm to accommodate the additional card layer, yielding modified BAU dimensions of $213 \times 150 \times \sim 154$ mm. The chassis extension also contributes additional structural mass. The THEMIS BAU had aluminium (Al 6061) housing [74], with a density of 2700 kg/m^3 . Assuming a 2.5 mm wall thickness typical of small spacecraft enclosures [75], the additional perimeter walls of the 35 mm chassis extension contribute a volume of $6.35 \times 10^{-5} \text{ m}^3$ (perimeter $0.726 \text{ m} \times$ extension height $0.035 \text{ m} \times$ wall thickness 0.0025 m), giving an added structural mass of approximately 0.17 kg and a total BAU mass increase of ~ 0.32 kg over the heritage design.

Therefore the chosen IDPU integrates as an additional module within the BAU card stack with minimal structural modification, supporting the CCD subsystem mass and volume requirements **CCD-C-001**, **CCD-C-002**. The MA61C is also rated to 300 krad TID [76], exceeding the mission radiation requirement **CCD-C-003** of 165 krad. The key parameters are summarized in Table 32.

Table 32: MATRIX Bus Avionics Unit (BAU) Key Parameters

Parameter	Value	Ref.
Mass (THEMIS BAU)	2.8 kg	[73]
Mass (IDPU board)	+0.15 kg	[76]
Mass (chassis extension)	+ ~ 0.17 kg	[74], [75]
Mass (Total for MATRIX)	~ 3.12 kg	
Dimensions (THEMIS)	$213 \times 150 \times 119$ mm	[73]
Dimensions (MATRIX)	$213 \times 150 \times \sim 154$ mm	
Power (THEMIS BAU)	6.4 W	[73]
Power (IDPU board)	+1.45 W	[72]
Power (Total for MATRIX)	~ 7.85 W	
MATRIX IDPU radiation rating	300 krad TID	[76]

The MATRIX BAU retains the three modules from the THEMIS BAU architecture [73] without modification. The Digital Processor Module (DPM) hosts the RH5208 ColdFire flight computer, running the spacecraft flight software and managing the IDPU interface via RS-422. The Communications Interface Module (CIM) handles uplink command decoding and downlink telemetry formatting, interfacing directly with the S-band radio. The Power Electronics Unit (PEU) processes solar array power, controls battery charging, and distributes power across all spacecraft subsystems. A fourth module, the IDPU, is added to the card stack as described above.

Figure 9 shows how the science instruments interface with the BAU along two physically and functionally separate paths. The data path is routed through the IDPU: each instrument transmits raw sensor data to the IDPU over RS-422, where it is processed and forwarded to the DPM for storage and downlink. The power path is routed directly from the PEU: each instrument receives a dedicated switched voltage line, independent of the data system. This separation underpins the payload-level fault isolation strategy and satisfies requirement **CCD-F-016**. Electrical faults (e.g. overcurrent or short-circuiting) are contained by the PEU’s overcurrent protection mechanisms

[73], while instrument data errors are contained within the IDPU and cannot propagate to the DPM that hosts the main flight computer.

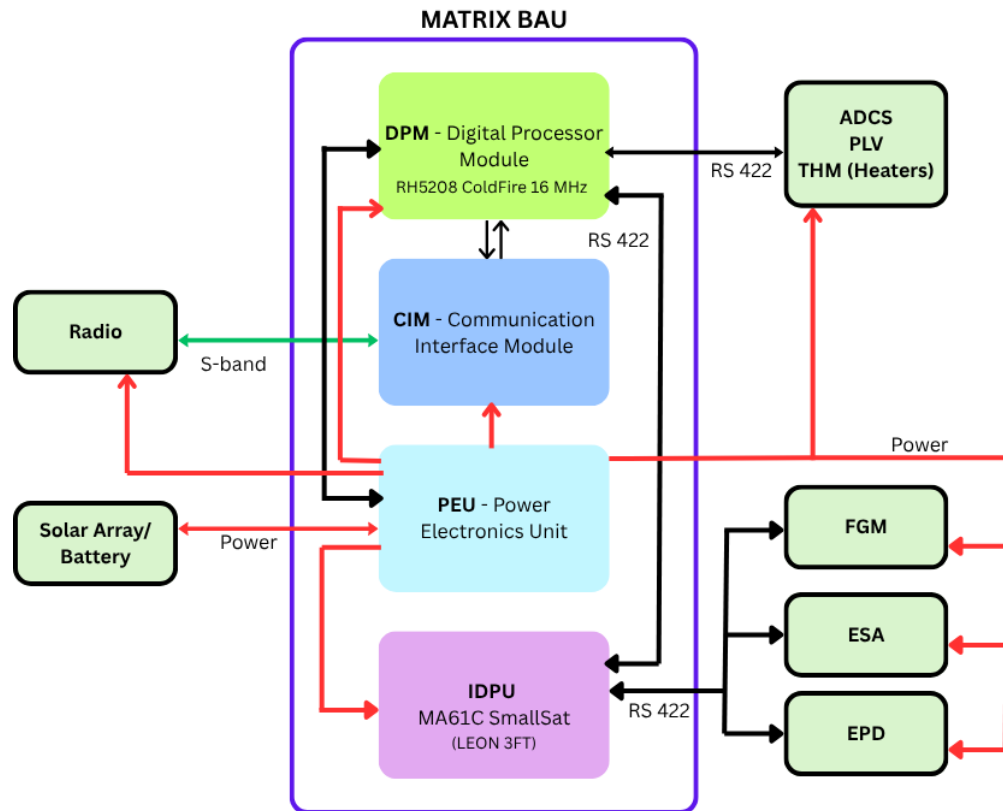


Figure 9: Functional architecture of the MATRIX BAU, showing the four-module card stack (DPM, CIM, PEU, IDPU) with data interfaces (black, RS-422), power distribution (red), and the S-band RF interface (green).



5.7.5 IDPU-Instrument Interface

Analysis conducted by: Maia (Defined the per-instrument IDPU processing functions addressing remaining CCD requirements.)

New analysis since PDR

The IDPU analysis in Section 5.7.4 established the IDPU as a dedicated payload interface module within the BAU. This analysis defines specifically how the IDPU is intended to interact with each of the science instruments for the MATRIX mission. In doing so, this analysis addresses three CCD requirements directly:

- **CCD-F-008** (onboard computational capabilities): defining the per-instrument computational functions performed by the IDPU formalises closure by specifying not just that computation occurs on the IDPU but what computation runs.
- **CCD-F-014** (trigger and indicate operational mode changes): specifying the BBF detection algorithm in concrete terms defines the actual trigger condition.
- **CCD-F-015** (autonomy in nominal operations): establishing that BBF detection and high-cadence mode initiation execute on the IDPU without ground intervention defines the level of autonomy in nominal science operations.

As discussed in Sec. 5.7.4, THEMIS housed separate hardware boards for each of its five instrument classes (EPD, FGM, SCM, EFI, and ESA) [77]. MATRIX houses only three of these instrument classes (FGM, ESA, and EPD) and consolidates all per-instrument processing into firmware modules running on the LEON3-FT processor on the MA61C-SmallSat, leveraging the SoC architecture established in Section 5.7.4. This consolidation pattern is flight-proven, notably on the CHEOPS mission which leveraged the same LEON3-FT processor [78]. The key per-instrument processing functions performed by the IDPU for the MATRIX mission are detailed below.

FGM The magnetometer takes a magnetic field sample which it transmits to the IDPU via the RS-422 data channel at the selected sampling rate of 4 Hz in low-cadence and 128 Hz in high cadence mode. With respect to the FGM - IDPU interface, there are three key functions:

- **High-cadence trigger:** The IDPU continuously monitors the 4 Hz B_z low-cadence data stream against the dipolarization-front selection criteria reported by Fu et al. [17]: a B_z increase of $\Delta B_z \geq 4$ nT within a window of $t_{\text{duration}} < 8$ s. When this condition is satisfied, the IDPU autonomously commands the FGM to transition to its 128 Hz high-cadence mode (raising the FGM data rate to 3072 bps), and raises a high-cadence flag on the IDPU High Rate Data RS-422 channel for the DPM to propagate the mode change to the rest of the spacecraft.
- **Data compression:** In low-cadence mode, the IDPU applies a sine-wave least-squares fit to the samples acquired per 4 s spin period. This compresses the raw data into a 24-bit packet per spin [77] which is then passed on to the DPM.

ESA The ESA generates ion and electron count data binned by energy and incoming direction, transmitted to the IDPU via RS-422 [8]. With respect to the ESA - IDPU interface, there are two key functions:



- **Onboard moment computation:** The IDPU accumulates incoming counts to build the full distribution function over each spin period, from which it computes the standard plasma moments (number density, bulk velocity, pressure) onboard [8]. This significantly reduces the downlink volume relative to transmitting full distributions.
- **Data compression:** The IDPU applies the pseudo-square-root lossy compression to ESA data, reducing the transmitted volume by a factor of 2 [79]. This compression scheme is chosen for compatibility with the selected ESA (THEMIS).

EIS+ MATRIX's energetic particle detector (EIS+) measures time-stamped data of detected high-energy particles, transmitted to the IDPU via RS-422 at a rate of 670 bps [9].

- **Species classification:** The IDPU classifies incoming particle events by species (H^+ , electrons) based on energy levels and categorizes the data into packets before transmission to the DPM [77].

Additionally, across all incoming data streams from all instruments, the IDPU validates data packets through standard error detection protocols (opcodes, length checks) [80]. Corrupted packets are stopped at the IDPU before they can be forwarded to the DPM. All functions are summarised in Table 33 along with the corresponding CCD requirements satisfied.

Table 33: Summary of IDPU per-instrument processing functions and corresponding CCD requirements.

Instrument	Function	Description	Requirement
FGM	High-cadence trigger	Autonomous BBF detection via $\Delta B_z \geq 4$ nT in < 8 s window; commands FGM to 128 Hz and flags DPM	CCD-F-014, CCD-F-015
FGM	Data compression	Sine-wave least-squares spin fit, packed to 24-bit packet per spin	CCD-F-008
ESA	Onboard moment computation	Plasma moments computed per spin	CCD-F-008
ESA	Data compression	Pseudo-square-root lossy compression (factor of 2 reduction)	CCD-F-008
EIS+	Species classification	H^+ / electron classification by energy; packetised for DPM	CCD-F-008
All	Fault protection	Packet integrity validation (opcode, length checks); corrupted packets rejected at IDPU	CCD-F-016

5.7.6 Definition of Transmission Protocol

Analysis conducted by: Sawooly Li (trade study on transmission architectures, including CCSDS, CFDP, and DTN, and definition of downlink prioritisation and virtual channel allocation).

Updates since PDR: No major architectural changes; downlink data rates and bandwidth allocation refined based on finalised data budget, with minor updates to uplink and downlink assumptions.

Firstly, a trade study was conducted between three candidate transmission architectures: CCSDS, CFDP, and DTN-based networking. CCSDS provides a standardized and low-complexity approach for packetised data transmission over a direct store-and-forward architecture, as demonstrated in heritage missions such as THEMIS [65]. CFDP extends this by enabling reliable file-level transfer by dividing data into smaller segments and retransmitting any lost or corrupted portions,



but remains fundamentally point-to-point [81]. In contrast, DTN introduces a networking architecture capable of multi-hop communication, at the cost of increased system complexity [81].

For MATRIX, communication is limited to scheduled spacecraft-to-ground contacts, with no requirement for inter-satellite relay. Data are buffered onboard and downlinked once per orbit. In this context, DTN introduces unnecessary architectural complexity, while CFDP provides limited additional benefit beyond existing onboard buffering and controlled downlink. Therefore, a CCSDS-based transmission architecture is selected as the baseline, providing a flight-proven solution that aligns well with MATRIX's chosen software, as will be further analysed in the Flight Software subsection.

Following selection of CCSDS, data transmission is organized using virtual channels, which enable multiple data streams to be shared over a single physical link while maintaining prioritisation. This allows critical telemetry to be transmitted independently of science data, preventing contention within the downlink. MATRIX adopts a channel structure separating engineering and science data, as summarized in Table 34.

VC ID	Data Type	Priority	Bandwidth Allocation (%)
0	Critical Engineering Telemetry (Faults, Alerts)	Highest	5
1	Event Metadata (BBF flags / metadata)	High	5
2	Stored Science (Nominal + BBF Data)	Medium	85
3	Nominal Engineering Telemetry (Routine HK)	Low	5
4	Fill Data (Idle / Padding)	Lowest	-

Table 34: Virtual Channel Allocation and Prioritization

Given the mission orbit, science data acquisition occurs primarily within the magnetotail transition region, which does not coincide with ground station visibility. As a result, it is assumed that no real-time science transmission is required, and all science data, including high-rate BBF-triggered measurements, are treated as stored data for later downlink.

Data prioritisation within the downlink is driven by both data volume and mission criticality. Based on the downlink calculations from the last CCD report [14], a derived total data per orbit of approximately 1.16×10^9 bits and a required downlink rate of 3.21×10^5 bps over a 3600 s contact window bandwidth must be allocated carefully to ensure complete data return each orbit. House-keeping telemetry, including ADCS, thermal, and power data (on the order of 10^2 – 10^3 bps [14]), represents only a small fraction of the total data volume but is assigned the highest priority, as it is essential for monitoring spacecraft health. Accordingly, a small but guaranteed portion of the downlink capacity is reserved for critical and nominal engineering data. The detailed justification for bandwidth allocation can be seen in Appendix A.2.1.

The majority of the available bandwidth is allocated to science data, which dominates the total data volume due to continuous measurements and burst-mode BBF events. Event metadata, such as BBF triggers, are assigned a higher-priority channel to ensure that scientifically significant events are identified quickly, while the full high-rate datasets are downlinked later as part of the stored science stream. This introduces a trade between maximizing science return and maintaining reliable telemetry: allocating too much bandwidth to engineering reduces science throughput, while allocating too little risks loss of spacecraft health visibility.

In addition to the downlink architecture, the uplink transmission protocol is defined to support



reliable command and control of the spacecraft. Given that uplink traffic consists only of telecommands rather than bulk data transfer, a low-rate CCSDS telecommand protocol is adopted, consistent with heritage missions such as THEMIS [15]. Commands are generated on the ground, encoded into telecommand transfer frames, and transmitted during scheduled communication windows. These frames are received by the spacecraft radio, decoded by the onboard computer, and either executed immediately or stored for later execution.

5.7.7 Frequency Band and Communications Network Selection

Analysis conducted by: William Li (selection of communications band for the small spacecraft architecture of each MATRIX spacecraft, identifying a communications network).

Updates since PDR: Refined explanation of use of the Near Space Network, reconsidered use of the TDRS fleet, updated link budgets throughout.

This section describes the trade studies carried out to select a frequency band and communications network for MATRIX spacecraft to communicate to. The options considered were chosen out of a small set of typical options for a mission with a “standard” orbit that doesn’t have exotic communications requirements.

According to NASA, small satellites typically communicate over four different frequency bands: UHF, S, X, and Ka [82].

	UHF	S	X	Ka
Frequency (GHz)	0.3-1	2-4	8-12	27-40

Table 35: Typical Satellite Communication Downlink Frequency Bands

The free space path loss of a radio wave is defined as, $L_s = (\frac{4\pi Rf}{c})^2$ and is proportional to the square of the frequency, therefore it is desirable to minimize the communications frequency. Frequencies in the Ka range are ~ 7 times the frequencies in the S range, meaning that Ka-band losses are an astounding ~ 49 times greater compared to S-band. As explained in Chapter 9 of NASA’s State-of-the-Art of Small Spacecraft Technology Report, higher frequency bands are capable of higher data rates but they require higher power or higher gain antennas which also require greater pointing accuracy [82]. The UHF band appears to be the most desirable in terms of minimizing power and losses, but it is actually susceptible to interference if there is not perfect line of sight between the satellite and the ground station. Therefore, the **S band is chosen for MATRIX’s communications frequency** because it is a good middle ground between reliability and minimizing burdensome power and pointing constraints for the spacecraft.⁵

The communications networks considered were the Near Earth Network (NEN), the Deep Space Network (DSN), and the Tracking and Data Relay Satellite System (TDRSS). Parameters from datasheets for each network were used in a link budget calculation to determine the signal-to-noise ratio of each option. The target signal-to-noise ratio is > 6.6 dB (derived from requirement CCD-P-002), which is slightly larger than a factor of 4, providing robust margins and ensuring that data from each MATRIX is received clearly at the ground.

⁵Paragraph copied from MATRIX CCD Subsystem Report with minor modifications to wording



	TDRSS [83]	NEN [84]	DSN [85]
Frequency f (MHz)	2200 MHz	2200 MHz	2295 MHz
Slant Range R (km)	1173.536	11090.587	11090.587
Receiving Antenna Gain G_r (dBi)	37	45.56	63.34
Data Rate R_b (bps)	$2.434 \cdot 10^5$	$2.434 \cdot 10^5$	$2.434 \cdot 10^5$

Table 36: Link Budget Parameters

The slant range value for the NEN and DSN are based on an orbit perigee altitude of 1000 km and a 5° minimum elevation angle [84], [86] while the TDRSS slant range is calculated based on a distance of 34,786 km between GEO where TDRS satellites reside and the 1000 km perigee of MATRIX satellites. Additional constants in the link budget are listed in Table 37.

	Value
Transmit Power P_{tx} (dBm)	42.55 (18W)
Transmit Antenna Gain G_{tx} (dBi)	0 (worst-case)
Implementation Losses L_{tx} and L_{rx} (dB)	10 (typical)
Receiver Noise Figure NF (dB)	1.014 (cryogenic low-noise amplifier)

Table 37: Link Budget Constants

Based on these parameters, the signal to noise ratio (SNR) achievable with each network selection is shown in Table 38 and the equations for the link budget calculation are in Appendix A.2.2.

	TDRSS	NSN	DSN
Signal-to-Noise Ratio E_b/N_0 (db)	-2.59	7.16	27.76

Table 38: Link Budget Network Signal-to-Noise Ratio

After calculating a link budget for all three communications network options, both the NSN and DSN fulfill the minimum 6.6 dB SNR requirement of CCD-P-002.⁶ The DSN has better signal-to-noise performance but both fulfill the requirement and the NSN has a wider selection of ground stations available to use than the DSN, which avoids adding strict pointing requirements for MATRIX. Furthermore, NASA is commercializing the network which will bring prices down and result in more accessible stations compared to the DSN. The TDRSS network, due to its increased distance from the MATRIX spacecraft except for a small portion of time when the orbits intersect, does not have enough signal-to-noise ratio to support the downlink of all data on every orbit.

The Near Space Network is the nominal ground station network used for communication with MATRIX. However, it is good to know that in the case of off-nominal situations where communication with the NEN is not possible (severe weather on Earth attenuating signal gain, loss of attitude control, etc.) the DSN could be used as a backup.

5.7.8 Antenna Trade Study

Analysis conducted by: William Li (trade study and downselection of space-rated communications antennas).

⁶This sentence copied from MATRIX CCD Subsystem Report



Updates since PDR: Refined explanation for when each of the four antennas is used, refined explanation of pointing requirements for communication with Earth.

After the data is processed onboard the spacecraft in the BAU, it must be transmitted to Earth. The goal of the antenna trade study is to identify a suitable antenna to downlink spacecraft data to Earth and receive uplinked commands and verify that it is compatible with the MATRIX link budget. Table 39 shows the achievable signal-to-noise ratios achievable with a broad set of different space-rated antennas of different architectures such as patch antennas, horn antennas, and phased array antennas.⁷

	Frequency (MHz)	Gain $G_{tx_{dB_i}}$ (dBi)	Power $P_{tx_{dB_m}}$	Signal-to-Noise Ratio (dB)
Space Inventor S-Band Patch Antenna PATCH2X2-S-R [87]	2200	7	33.0103 dBm (2W)	6.13
EnduroSat S-Band Patch Antenna Commercial [88]	2200	7	36.0206 dBm (4 W)	3.14
Texas Instruments & NASA S-Band Phased Array Antenna [89]	2217.5	12	51.139 dBm (130 W)	23.19
Anywaves S-Band TT&C Patch Antenna [90]	2200	4.5	46.0206 dBm (40 W)	10.64

Table 39: Link Budget Analysis: Antenna Signal-to-Noise Ratio

The Anywaves S-Band TT&C patch antenna was selected as MATRIX's primary communications antenna because of its balance between gain and power input. The Anywaves antenna has a moderate power consumption of 40 W, adequate signal-to-noise ratio according to Table 39, and a low mass of only 136 g. The Space Inventor and EnduroSat antennas have higher gains than the Anywaves antenna but the Anywaves antenna operates at a higher power and even though it has a lower gain, it has sufficient signal-to-noise ratio while the other two patch antennas do not fulfill the CCD-P-002 requirement. The Texas Instruments phased array antenna has a superb signal-to-noise ratio but requires 130 W of power. The benefit of a phased array antenna is that it can point itself which removes CCD pointing requirements from the ADCS team. However, when considering the full spacecraft power budget, it is not appropriate to give that much power to just one subsystem especially since depending on the time of the year, power may be simultaneously required by THM.⁸

Due to the low mass and small size of the Anywaves, 4 are used on each MATRIX spacecraft and CCD still remains within mass and volume compliance. As seen in Figure 10, there are two on each hexagonal face of the spacecraft. On each of the hexagonal faces, one of the two antennas is redundant and exists as a backup. Those antennas are completely turned off and draw no power unless the primary antenna breaks. Of the remaining two primary antennas, both are used during a downlink window, depending on which one has a better view of Earth. As a MATRIX spacecraft descends towards perigee, one face will naturally be pointing more at the Earth and therefore able to transmit data with higher gain. At some point in the perigee pass, the geometry becomes reversed and the opposite of the spacecraft now has a clearer view of Earth. When this occurs, the

⁷Table 39 and this paragraph copied from MATRIX CCD Subsystem Report

⁸Paragraph partially copied from MATRIX CCD Subsystem Report

original antenna is switched off and the other primary antenna on the other side of the spacecraft begins transmitting, therefore saving power because only 1 antenna is on at any given moment.

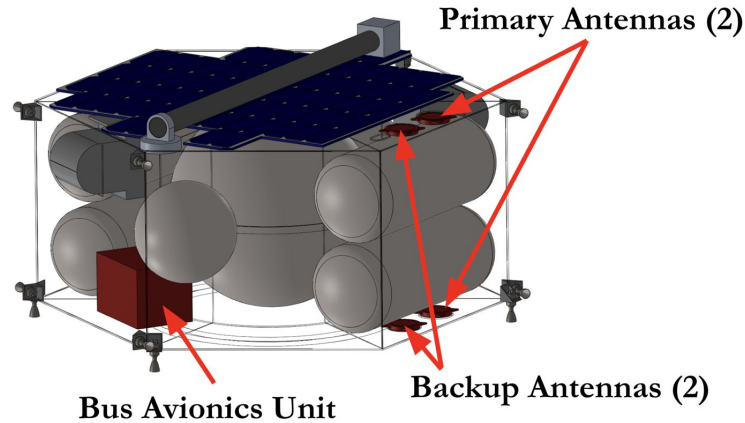


Figure 10: Location of Antennas on MATRIX Spacecraft Bus

5.7.9 Radio and Power Amplifier Selection

Analysis conducted by: William (curation of radio and power amplifier specifications and downselection)

To accompany the Anywaves S-Band antenna and complete the communications pathway, a RF power amplifier and S-band radio are required. The radio actually generates the signal that the antenna transmits, and the power amplifier increases the power of the signal that the antenna transmits, which increases signal-to-noise ratio.

To conduct a trade study for S-band radios, a list of viable satellite radios was obtained from NASA's 2024 State-of-the-Art of Small Spacecraft Technology Report [91]. The available options were downselected based on flight heritage, support of S-band frequency channels for both uplink and downlink, and transmit power.

	Transmit Power (dBm)	Maximum Data Rate (kbps)	Power Required (W)	Idle Power (W)
Rincon Research AS-TROSDR	Unknown	5	30	4
SkyLabs NANOLink-boost Gen2	37	4000	17	1.9
IQ Technologies SLINK-PHY	30	20000	13	4.5
Endurosat S-band Transceiver II	125	33	12	1.2

Table 40: S-Band Radio Trade Study

The SkyLabs NANOLink-boost Gen 2 radio was selected for MATRIX on the basis of its high transmit power of 37 dBm, along with verification of its ability to simultaneously receive data at 2025 MHz and transmit at 2200 MHz [92]. The radio also has strong weight savings, weighing only 250g.



To perform the power amplifier trade study, a broad online search for commercially available space-rated power amplifiers was conducted. The most important parameters when looking at radios were the amount of power after amplification, which had to be ~ 40 W in order to satisfy the link budget, as well as the amount of power consumed which is ideally minimized to be considerate of the MATRIX spacecraft power budget.

	Transmit Power (dBm)	Transmit Power Consumption (W)	
NuPower 12B01A-01	42.55	42	
NuPowerTM 15D05A-C01-S01	45.18	75.6	
Erzia ERZ-HPA-0200-0400-30 105	35	7	Erzia ERZ-HPA-0200-0400-43 43

Table 41: Power Amplifier Trade Study

The NuPower 12B01A-01 RF Power Amplifier was selected for MATRIX because of its high transmit power to power consumption ratio. Very few space-rated power amplifiers were found that were able to transmit power in the > 40 dBm range while also consuming only tens of watts of input power instead of hundreds of watts. As shown in Table 41, the ERZ-HPA-0200-0400-30 has a low power consumption of 7 W but as a result can only amplify to 35 dBm. On the other hand, the ERZ-HPA-0200-0400-43 can amplify signals to 43 dBm but as a result consumes 105 W of power to do so.

5.7.10 Link Budget

Analysis conducted by: William (quantitatively defining the data collected by various subsystems on the spacecraft, updating link budget, defining uplink values in link budget)

The final link budget for the MATRIX mission is shown in Table 42 below. The transmit power is also limited to 18 W instead of the 40 W that the Anywaves antenna is capable of because the power amplifier outputs 18 W to the antenna. A gain of 0 dBi is used to account for the worst case scenario when the spacecraft must communicate nearly perpendicular to Earth and the Anywaves antennas have reduced gain. Although the Anywaves S-Band Patch Antenna is omnidirectional, it is not a perfect omnidirectional emitter where the gain is the same in every direction. Rather, there is a gain pattern, shown in Figure 11 where the highest gain, 6.5 dB, occurs when the antenna is perfectly pointed at a receiver, and this decreases to 0 dB when the antenna is pointed $\pm 70^\circ$ away from the receiver. MATRIX uses the blue line in Figure 11 which corresponds to a co-polarization angle of 0° . Since 0 dB was used in the link budget, this means that as long as the MATRIX spacecraft is pointed within $\pm 70^\circ$ of Earth, it is able to downlink data with sufficient signal-to-noise ratio. Additionally, a slant angle of 5° was used in the link budget calculation, which allows MATRIX to communicate with Near Space Network stations even if they are extremely low on the horizon. Due to the abundance of radio dishes in the Near Space Network and this combination of conservative factors in the link budget, it is highly unlikely that there will not be a NSN station available on any given perigee pass.

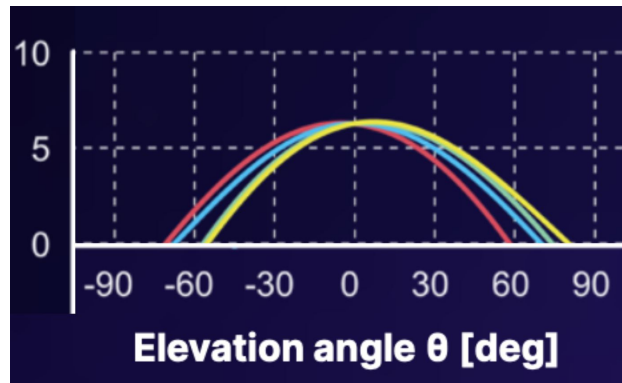


Figure 11: Gain Pattern of Anywaves S-Band Patch Antenna [93]

Parameter	Value
Frequency	2200 MHz
Slant Range	11090.587 km
Transmit Power	42.55 dBm (18 W)
Transmit Antenna Gain	0 dBi
Near Space Network Receiving Gain	45.86 dBi
Miscellaneous Losses (Cable, Implementation, Polarization)	20 dB
Data Rate	253394.4928 bps
Receiver Noise Figure	1.014 dB
Signal-to-Noise Ratio (minimum 6.6dB)	7.16 dB

Table 42: MATRIX Link Budget (Downlink)

This analysis shows that MATRIX has adequate signal-to-noise ratio to downlink all data to the Near Space Network on every orbit. The data rate used in the link budget also includes a 1.5x margin in case backlogged data needs to be downlinked. In the event that a spacecraft is unable to downlink its data for any reason, the ~ 115 MB of data from that orbit is stored in the 4GB of Solid State Drive (SSD) storage that each MATRIX spacecraft carries. This is sufficient to store more than a month of backlogged data. The link budget accounts for the transmission of backlogged data by allowing half of a backlogged orbit's data to be downlinked while still maintaining adequate signal-to-noise ratio. It then takes 2 orbits to downlink 1 orbit of backlogged data.

CCD also considered the link budget for the uplink of telemetry and any other commands to MATRIX. Nominally, each MATRIX spacecraft receives a telemetry uplink from the Near Space Network once per orbit after passing perigee. The onboard computer then propagates the orbit of the spacecraft during that orbit and the spacecraft receives updated telemetry on its next perigee pass. In addition to this routine telemetry uplink, ADCS and GNC expressed a desire to uplink telemetry data to MATRIX at other points in the orbit. Every 28 days there is a stationkeeping maneuver that is performed at perigee and it is best to receive an updated set of telemetry from the Near Space Network before the maneuver to make the maneuver as accurate as possible. Table 43 shows the link budget for uplinking telemetry data in the Universal Tracking Data Format (UTDF) from the Near Space Network to a MATRIX spacecraft at maximum apogee radius of 12 Earth radii (altitude 11 Earth radii.) UTDF data can be uplinked at either 110 baud (bit/s) or 9600 bps [94]. For the link budget calculation, higher data rates lower the signal-to-noise ratio so the

higher data rate is used to verify a worst-case scenario. In S-band communications, 2200 MHz is used as a typical downlink frequency and uplink occurs at a separate 2025 MHz frequency and this is reflected in Table 43.⁹

Parameter	Value
Frequency	2025 MHz
Slant Range	70158 km (11 R_E distance at apogee)
Transmit Power	63.9794 dBm (2500 W) [95]
Transmit Antenna Gain	45.86 dBi
Near Space Network Receiving Gain	0.0 dBi
Miscellaneous Losses (Cable, Implementation, Polarization)	20 dB
Data Rate	9600 bps
Receiver Noise Figure	1.014 dB
Signal-to-Noise Ratio (minimum 6.6dB)	27.50 dB

Table 43: MATRIX Link Budget (Uplink)

This analysis confirms that data is able to be uplinked to the MATRIX spacecraft with adequate signal-to-noise ratio even at an apogee radius of $12R_E$, which is a result of the high transmit power that the Earth-based Near Space Network is capable of.

Figure 12 below shows the communications windows for MATRIX. As described in Section 5.7.8, each MATRIX spacecraft communicates to the Near Space Network for a one-hour window centered around perigee, which is derived from the heritage MMS mission [96]

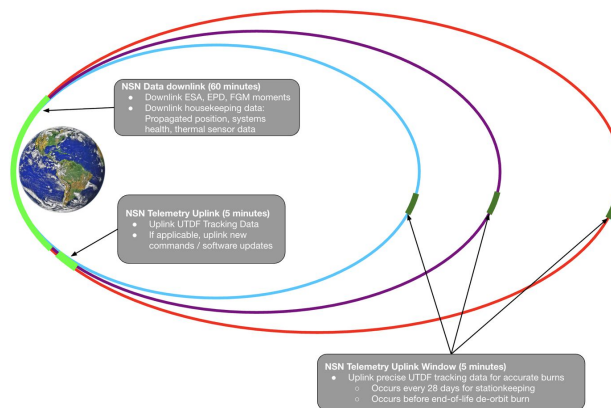


Figure 12: MATRIX Communications Windows

5.7.11 Interfacing Onboard Each Spacecraft & Data Flow

Analysis conducted by: Sonal (defined the full RS-422 channel mapping between subsystems and the DPM, verified MA61C-SmallSat to ATK DPM wiring/protocol compatibility, generated the integrated data flow diagram).

Updates since PDR: Expanded subsystem interfacing analysis to explicitly address wiring and protocol compatibility across all RS-422 channels (the principal gap flagged in PDR feedback). Added per-channel

⁹Paragraph partially copied from MATRIX CCD PDR Subsystem Report



link-rate margin analysis, MA61C-to-DPM electrical/protocol compatibility verification, and explicit treatment of the spare-channel and unused-interface upgrade paths.

The ATK Digital Processor Module (DPM) serves as the single centralized command and data hub for each MATRIX spacecraft [55]. Every subsystem connects to the DPM via a dedicated RS-422 point-to-point serial link, forming the centralized star topology described in SMAD [53]: a failure on any one interface does not propagate to the others, and all spacecraft-level command routing, telemetry aggregation, and data management are concentrated in a single, well-defined processor. The complete data flow architecture is shown in Figure 13, which illustrates the centralized star topology, the two-tier IDPU-to-DPM science data path defined in Section 5.7.4, and the full communications chain from the DPM through to the NEN ground station.

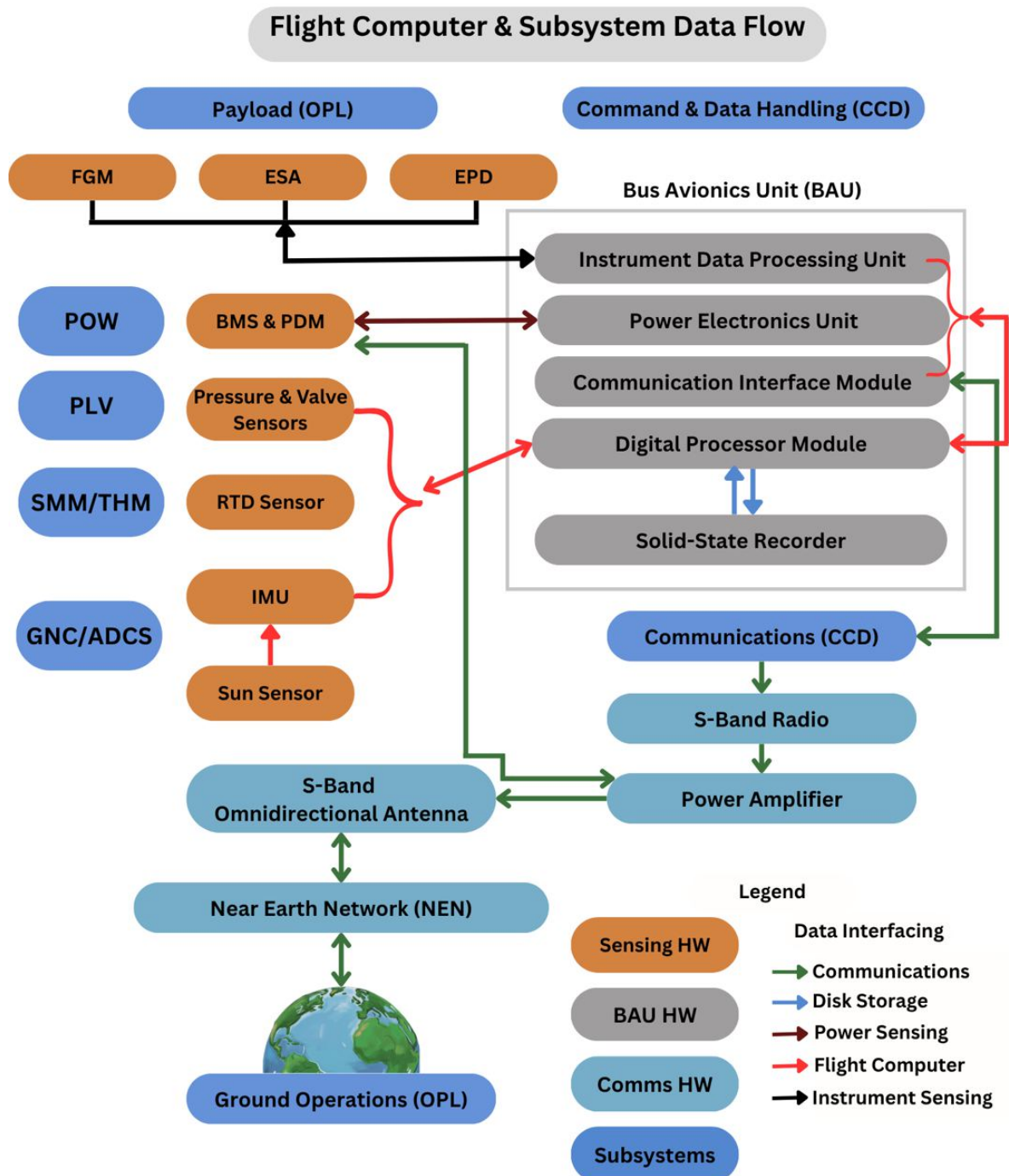


Figure 13: MATRIX flight computer and subsystem data flow. The IDPU sits between all science instruments and the DPM, handling analog-to-digital conversion, instrument-specific processing, and BBF burst-mode triggering before any data reaches the flight computer.



Two-Tier Onboard Data Architecture The MATRIX data handling architecture is two-tiered. The first tier is the IDPU (the SPiN MA61C-SmallSat, see Section 5.7.4), which intercepts all raw instrument data before it reaches the second tier, the DPM. The IDPU performs analog-to-digital conversion, applies instrument-specific calibration, and detects BBF events using onboard threshold logic, offloading these tasks entirely from the flight computer and providing hardware-level payload fault isolation. When a BBF is detected, the IDPU raises a burst-mode trigger over its IDPU High Rate Data RS-422 channel; the DPM's cFS IDPU Interface application receives the trigger, publishes a *BBF-Detected* message on the cFE Software Bus, simultaneously reconfigures the task scheduler, switches the SSR write queue to Tier 3 priority, and logs the event. The IDPU continues recording at high rate until the DPM sends a mode command to return to survey. This event-driven design means burst capture is initiated by the instruments themselves rather than by DPM polling, minimising latency between the physical event and the onboard response.

The second tier, the ATK DPM, receives pre-processed, CCSDS-formatted science packets from the IDPU and handles all spacecraft-level functions: command execution, housekeeping aggregation, data prioritisation, SSR management, downlink scheduling, and autonomous fault response via the cFS Health & Safety and Limit Checker applications.

RS-422 Channel Allocation and Wiring PDR feedback identified that the cross-subsystem wiring and protocol compatibility had not been explicitly verified at the channel level. This subsection closes that gap by mapping every subsystem interface to a specific DPM RS-422 channel, comparing the actual data load against the channel ceiling, and verifying electrical and protocol compatibility on the IDPU side. Table 44 lists all subsystem interfaces with the DPM. The IDPU occupies three of the DPM's available RS-422 channels (IDPU High Rate Data, IDPU Timing, and IDPU Cmd), matching directly the interface defined in the THEMIS IDPU-to-BAU Interface Control Document [97], [98]. Spacecraft subsystems POW, PLV, ADCS, GNC, THM, and SMM each occupy dedicated RS-422 links so that no subsystem shares a bus with another; this preserves the single-point-failure containment property of the star topology. The communications chain is the one directional path that exits the spacecraft: the DPM passes formatted CCSDS telemetry frames to the S-Band radio, which drives the power amplifier and omnidirectional antenna toward the NEN ground station, while received telecommand frames travel the reverse path back to the DPM for validation and routing [95], [99].



Table 44: MATRIX Flight Computer Subsystem Interface Summary (TBC)

Node	DPM channel	Rate	Direction	Data / function
IDPU (MA61C)	IDPU High Rate Data	Async	IDPU → DPM	Pre-processed FGM/ESA/EPD science packets
	IDPU Timing	1 Hz	DPM → IDPU	Spacecraft clock synchronisation
	IDPU Cmd	Event	DPM → IDPU	Survey/Burst mode commands
ADCS	Sun Sensor TM	4 Hz	ADCS → DPM	Sun sensor attitude data
	Thruster Cmd	2 Hz	DPM → ADCS	Spin-axis control commands
POW	Ctrl/Status	1 Hz	POW → DPM	Battery SoC, bus voltage
		Event	DPM → POW	Load-shed commands
PLV	RS-422	Event	DPM → PLV	Thruster fire commands, burn sequence
THM	Ctrl/Status	0.1 Hz	THM → DPM	Component temperatures
		Event	DPM → THM	Heater on/off commands
GNC	cFE SW Bus (int.)	1 Hz	GNC → DPM	Orbital position, ROI flag
SMM	RS-422	One-time	SMM → DPM	Deployment confirmation flags
Radio / comms	Downlink TM I/F	Sched.	DPM → Radio	CCSDS TM frames [99]
	Uplink TC	On-rcv	Radio → DPM	Ground TC frames
External SSR	Backplane I/F	Cont.	DPM ↔ SSR	Science/HK data read/write

MA61C-to-DPM Electrical and Protocol Compatibility A central concern raised in PDR feedback was whether the MA61C-SmallSat IDPU and the heritage ATK DPM are wiring-compatible without a custom protocol adaptor. The MA61C provides five RS-422 ports rated to 10 Mbps each, plus SpaceWire ($\times 2$ at 200 Mbps), MIL-STD-1553B, CAN, SPI, and I2C [100]. The THEMIS IDPU-to-BAU Interface Control Document [97] defines three RS-422 signal pairs between the IDPU position and the DPM: a high-rate science telemetry link, a low-rate timing/sync link, and an event-driven command link. These three signal definitions map one-to-one onto three of the MA61C's RS-422 ports, confirming a native electrical match with no protocol adaptor required. Table 45 shows the explicit port assignment along with the achievable ceiling and the actual MATRIX load on each link.

Table 45: MA61C-SmallSat to ATK DPM Interface Mapping and Link Margins

Signal	ATK DPM channel	MA61C port	Ceiling	Actual load
Science data (pre-processed)	IDPU High Rate Data	RS-422 #1	10 Mbps	~9.1 kbps
Spacecraft clock sync	IDPU Timing	RS-422 #2	10 Mbps	Low rate
Survey/Burst mode commands	IDPU Cmd	RS-422 #3	10 Mbps	Event-driven
Spare / future use	–	RS-422 #4–5	10 Mbps	Available

The peak MATRIX science data rate observed at the IDPU output is 9118.2 bps (Appendix A.3), which is approximately three orders of magnitude below the 10 Mbps RS-422 ceiling on the High Rate Data channel. The link is therefore not bandwidth-constrained, and substantial margin remains for future payload upgrades, packet overhead, and error-correction framing. The IDPU Timing and IDPU Cmd channels carry traffic that is several orders of magnitude below the same 10 Mbps ceiling, so they are similarly margin-rich. The two unused RS-422 ports (#4 and #5), as well as the unused SpaceWire and MIL-STD-1553B interfaces on the MA61C, represent explicit upgrade paths: a future high-rate instrument (e.g. an imager or wave instrument) requiring SpaceWire-class throughput could be added without re-baselining the BAU.



Wiring of Non-IDPU Subsystems The remaining subsystem-to-DPM links are all native RS-422 differential pairs, matching the standard interface used on the THEMIS BAU [73]. Because every subsystem is on a dedicated point-to-point link rather than a shared bus (such as MIL-STD-1553B or CAN), there is no contention or arbitration overhead, and the cFS IDPU Interface, ADCS, POW, THM, GNC, and SMM applications each receive a dedicated I/O service from the cFS Scheduler. The data rates on these links (1–4 Hz nominal, event-driven for commands) are far below the 10 Mbps RS-422 ceiling, so the wiring is bandwidth-trivial; the relevant design constraint is harness mass and connector count rather than electrical performance. With three IDPU channels and six external subsystem channels, the BAU harness terminates nine RS-422 differential pairs at the DPM plus the radio interface and the SSR backplane, all of which fit within the cable cutouts of the heritage THEMIS BAU enclosure without modification.

Implications for V&V The explicit channel-by-channel mapping above closes the wiring-compatibility gap flagged in PDR feedback for requirements **CCD-F-011** (provide data to communications subsystem) and **CCD-F-014** (communicate priority/mode changes), since both requirements now have a verified physical path from source to destination. End-to-end electrical verification (continuity, bit-error rate, and timing jitter on each RS-422 pair) remains a hardware-in-the-loop test deferred to the integration phase, but the present analysis confirms that no protocol adaptor, signal converter, or harness re-design is required between the selected IDPU and the heritage DPM.



6 GNC Design

6.1 Subsystem Overview

The guidance, navigation, and control (GNC) subsystem is responsible for defining orbits, maneuvers, and timing operations so that MATRIX best fulfills its mission-level objectives and scientific requirements. We must also devise a subsystem architecture that allows for state estimation and determination to be able to execute maneuver operations during the mission. We are responsible for interfacing with other subsystems and making sure our design choices are feasible. For example, we use eclipse times to size batteries for the power subsystem, and our station-keeping maneuvers are constrained by the Δv budget determined by the propulsion subsystem. These goals and responsibilities remain unchanged from earlier design iterations.

6.2 Subsystem Objectives

The subsystem objectives are largely unchanged from the objectives outlined in the PDR, remaining closely aligned with higher mission-level objectives.

GNC Subsystem Objectives

1. Design a set of preferred flight paths which meet primary mission requirements with preferred characteristics for transfer trajectories, station-keeping maneuvers, and orbital communications. Analyze these flight paths for total ΔV costs.
 - *Derived from Primary Mission Objectives 1 and 2*
2. Determine orbital state of MATRIX satellites in order to properly execute orbital transfer, station-keeping, and end-of-life maneuvers, accounting for deviations from planned flight paths.
 - *Derived from Primary Mission Objective 1*
3. Create and maintain proper spacing between satellites in each orbit group, allowing for measurements from the micro to macro scale.
 - *Derived from Primary Mission Objectives 2 and 3*
4. Develop control systems that correct for orbit deviations and maintain trajectories in line with planned flight paths, using attitude information from ADCS subsystem and propulsion from PLV.
 - *Derived from Primary Mission Objectives 1 and 2*

6.3 Subsystem Requirements

The GNC requirements that are most important for the MATRIX mission and pertinent to this design cycle are discussed below. The full list of requirements can be found here under the "GNC" tab: [GNC Subsystem Requirements List](#). The notable requirements listed below are grouped by a specific objective it satisfies and includes mission-level rationale. The rationale for these requirements provide key sources, calculations, and research done to arrive at these values.

1. Functional



- **GNC-F-002:** The GNC subsystem must identify orbits and trajectories including orbital radii between 6-12 Earth Radii (R_E).
 - Derived from Objective **MISS-F-001**
 - To properly study the magnetotail transition region, the spacecraft should be physically in that area, influencing GNC trajectory planning.
- **GNC-F-003 - GNC-F-007:** The GNC subsystem must be able to perform orbit maneuvers, correction maneuvers, stabilize during maneuvers, and stationkeep.
 - Derived from Objectives **MISS-F-007, MISS-F-017**
 - These requirements are crucial in ensuring that the satellites can be properly located and be scientifically useful during the duration of the mission. Without proper course correction, the mission execution is not sustainable.

2. Environment

- **GNC-E-001 - GNC-E-003:** The GNC subsystem must be able to operate within the space environment of its orbit
 - Derived from Objectives **MISS-E-001, MISS-E-004**
 - These requirements are fundamental in making sure the GNC components function properly during its entire mission. The spacecraft will be in the microgravity space facing large temperature changes, radiation pressure, and vibration.

3. Performance

- **GNC-P-002:** The GNC subsystem shall stationkeep so that perigee altitude does not change by more than 100 km (TBC) for any satellite.
 - Derived from Objectives **MISS-F-012, MISS-F-016**
 - This requirement is important for mission longevity in maintaining measurements in the ROI as well as preventing orbit degradation.
 - Heritage missions, MMS, THEMIS, and Cluster II, observed the magnetotail using highly elliptical orbits. These orbits require stationkeeping at perigee to avoid effects of atmospheric drag on the spacecraft, particularly from heating of the Thermosphere [4]. Drag here is a highly important factor when scaled to a large apogee of many earth radii, leading to a decaying orbit and potential crash into Earth's surface. This, in addition to other forces such as gravity gradients, vibrations, and coriolis force will impact the stability of the spacecraft's nominal orbit [4]. With this requirement, a box is defined at perigee for triggering autonomous or ground stationkeeping.
- **GNC-P-003:** The GNC subsystem shall ensure that satellites are spaced apart appropriately for mesoscale measurements, having at least 1 R_E of space between satellites in the same group.
 - Derived from Objectives **RFP-F-008, MISS-F-003**



- This requirement is fundamental for fulfilling the requirements of the scientific mission as stated in the RFP. This requirement differs from heritage missions which studied the ROI at a microscale, where satellites were spaced less than $1 R_E$ apart, (MMS [101], Cluster II [102]) or macroscale, where satellites were spaced $10 R_E$ apart (THEMIS [103]).
- For this particular ROI, mesoscale measurements require the GNC subsystem to maintain satellites in Groups 1 and 2 about $1-3 R_E$ apart [104]. To achieve this, satellites in the same measurement structure may differ in epoch while on the same orbit, inclination, apogee, or other orbital parameters. Additionally, STK analysis was done to show possible spacing strategies.
- **GNC-P-005:** The GNC subsystem shall communicate with ground control below by spending at least 1 hour below 3000 km (TBC) altitude.
 - Derived from Objectives **MISS-F-013**, **MISS-F-017**
 - This requirement is significant for orbit design, as it will determine the extent to which MATRIX satellites can provide constant data to ground stations, as well as perform state updates to ensure the continuation of the mission.
 - This values for this performance requirement come from the CCD team and their analysis of different communication options such as GPS and OpNav [105][106].

6.4 Subsystem Constraints

The GNC requirements that are most important for the MATRIX mission and pertinent to this design cycle are discussed below. The full list of requirements can be found here under the “GNC” tab: [GNC Subsystem Requirements List](#). The notable requirements listed below are grouped by a specific objective they satisfy and include a mission-level rationale. Note that the Performance requirements and numerical values listed are subject to change, but are better informed and researched compared to the first subteam report. The rationale for these requirements provides key sources, calculations, and research done to arrive at these values.

1. Functional

- **GNC-F-002:** The GNC subsystem must identify orbits and trajectories including orbital radii between 6-12 Earth Radii (R_E).
 - Derived from Objective **MISS-F-001**
 - To properly study the magnetotail transition region, the spacecraft should be physically in that area, influencing GNC trajectory planning.
- **GNC-F-003 - GNC-F-007:** The GNC subsystem must be able to perform orbit maneuvers, correction maneuvers, stabilize during maneuvers, and stationkeep.
 - Derived from Objectives **MISS-F-007**, **MISS-F-017**
 - These requirements are crucial in ensuring that the satellites can be properly located and be scientifically useful during the duration of the mission. Without proper course correction, the mission execution is not sustainable.



2. Environment

- **GNC-E-001 - GNC-E-003:** The GNC subsystem must be able to operate within the space environment of its orbit

Note: This requirement is unchanged from the previous design cycle.

- Derived from Objectives **MISS-E-001, MISS-E-004**
- These requirements are fundamental in making sure the GNC components function properly during its entire mission. The spacecraft will be in the microgravity space facing large temperature changes, radiation pressure, and vibration.

3. Performance

- **GNC-P-002:** The GNC subsystem shall stationkeep so that perigee altitude does not change by more than 100 km for any satellite.
 - Derived from Objectives **MISS-F-012, MISS-F-016**
 - This requirement is important for mission longevity in maintaining measurements in the ROI as well as preventing orbit degradation.
 - Heritage missions, MMS, THEMIS, and Cluster II, observed the magnetotail using highly elliptical orbits. These orbits require stationkeeping at perigee to avoid effects of atmospheric drag on the spacecraft, particularly from heating of the Thermosphere [4]. Drag here is a highly important factor when scaled to a large apogee of many earth radii, leading to a decaying orbit and potential crash into Earth's surface. This, in addition to other forces such as gravity gradients, vibrations, and coriolis force will impact the stability of the spacecraft's nominal orbit [4]. With this requirement, a box is defined at perigee for triggering autonomous or ground stationkeeping.
- **GNC-P-003:** The GNC subsystem shall ensure that satellites are spaced apart appropriately for mesoscale measurements, having at least $1 R_E$ of space between satellites in the same group.

Note: This requirement is unchanged from the previous design cycle, but included due to the analysis on spacing discussed later. ADDITIONALLY, this requirement only applies to satellites in Groups 1 and 2.

- Derived from Objectives **RFP-F-008, MISS-F-003**
- This requirement is fundamental for fulfilling the requirements of the scientific mission as stated in the RFP. This requirement differs from heritage missions which studied the ROI at a microscale, where satellites were spaced less than $1 R_E$ apart, (MMS [101], Cluster II [102]) or macroscale, where satellites were spaced $10 R_E$ apart (THEMIS [103]).
- For this particular ROI, mesoscale measurements require the GNC subsystem to maintain satellites in Groups 1 and 2 about $1-3 R_E$ apart [104]. To achieve this, satellites in the same measurement structure may differ in epoch while on the same orbit, inclination, apogee, or other orbital parameters.



- **GNC-P-005:** The GNC subsystem shall spend at least 1 hour below 3000 km altitude above the Earth's surface to enable scientific downlinking.
 - Derived from Objectives **MISS-F-013, MISS-F-017**
 - This requirement is significant for orbit design, as it will determine the extent to which MATRIX satellites can provide constant scientific downlinking to ground stations, as well as perform state updates to ensure the continuation of the mission.
 - This performance requirement comes from the CCD team and their specific communication system.
- 1. **Subsystem Mass and Volume:** The GNC subsystem must fit within specified mass and volume restrictions in order to accommodate launch vehicle requirements and needs from all other subsystems. These mass and volume budgets are largely set by Structures Materials and Mechanisms (SMM), but are subject to change.
 - Derived from Mission Constraint 3
- 2. **Power and Computation Allocation:** The GNC subsystem must be able to operate with its allotted power budget, maintaining operations even when in regions of eclipse. This power budget also extends to computational constraints, making sure that onboard processing power is not dominated by the GNC subsystem.
 - Derived from Mission Constraint 4
- 3. **Radiation Exposure:** GNC subsystem hardware must be able to withstand the harsh radiation environment of the magnetotail and continue to provide accurate data in order to meet navigation and control requirements.
 - Derived from Mission Constraint 4
- 4. **Thermal Exposure:** On-board GNC hardware must be able to withstand a wide range of temperatures as it moves in and out of eclipse zones. Hardware will be selected and modified based on expected temperature ranges and analysis from THM.
 - Derived from Mission Constraint 4
- 5. **Subsystem Cost:** MIDEX-class budget cap of \$350M requires GNC subsystem to remain under 4.2% (TBC) of total budget expenditure, as set by Operations and Payload (OPL).
 - Derived from Mission Constraint 1
- 6. **Timeline:** The GNC subsystem must design trajectories and have operating software/hardware in the given mission timeline. Mission orbits must be reasonable given a 5 year timeline and account for decay and course correction maneuvers during that time.
 - Derived from Mission Constraint 2
- 7. **ΔV :** The GNC subsystem must ensure that mission maneuvers are planned with the intention of limiting the cost of ΔV . Current estimates from PLV give a hard ceiling for the ΔV budget per satellite at 1000 m/s.
 - Derived from Mission Constraints 1 & 3



8. **Network Communication:** To account for CCD communications and ground tracking, GNC subsystem must ensure the orbit structure allows for consistent communication with ground stations. This is most important at or near perigee, allowing for uplink and downlink of scientific data along with position determination updates.

- Derived from Mission Constraint 2

6.5 Subsystem Drivers (6 Points)

The determination of the GNC subsystem drivers was heavily derived from the mission-level drivers. The following lists these drivers as well as the mission-level drivers that they are derived from.

1. **2D Field and Particle Measurements for Mesoscale Science:** The GNC subsystem's orbital design is driven by the requirement to take 2D measurements at a mesoscale. This determines satellite spacing between orbits (1-3 R_E [104]) as well as formations satellites must have with each other to satisfy this criteria as stated in the RFP.
 - Derived from Mission Driver 1
2. **Orbit Geometry for Transition Region Coverage:** The GNC subsystem's orbital design is also driven by the desire to access and cover certain critical regions in the magnetotail, as stated in requirement GNC-F-001.
 - Derived from Mission Driver 3
3. **High Temporal Sampling Rate:** The GNC subsystem must design effective orbits which optimize communication and time collecting data in the ROI given the high variability of space weather events and desire for frequent data sampling.
 - Derived from Mission Driver 2
4. **Accurate Constellation Stationkeeping:** The GNC subsystem must maintain the constellation spacing of satellites discussed in the Mission-Level ConOps Report. This drives rate or parameters for stationkeeping and orbit maintenance.
 - Derived from Mission Driver 4
5. **Data Storage and Communication:** The GNC subsystem's orbit design is driven by the need to frequently send data back to ground stations given the large amounts of measurements taken in the ROI, as determined by the Communications subsystem, and receive state position updates from ground tracking. Accurate communication timing and spacing influences the GNC subsystem's choice of orbits.
 - Derived from Mission Driver 5
6. **Delta-V:** The GNC subsystem's orbit design is driven by Δv budget provided by the Propulsion subsystem. While this budget is tied into the mass and volume constraint discussed in 6.4, it also drives orbit design knowing that certain ConOps of larger Δv are not possible.
 - Derived from Mission Drivers 1 and 2



6.6 Subsystem Design Approach (10 Points)

In the previous design cycle, the PDR, we expanded on analyzing a set of nominal orbit structures along with the maneuvers and ΔV costs associated with each orbit. In-depth analysis of spacing maneuvers and launch timelines was conducted, as well as orbit maintenance costs for our main science orbits. This design approach was conducted using information through heritage mission structure and SMAD [4].

In this design cycle, we iterated on the results from our PDR, conducting a more robust delta-v that accounted for attitude maneuvers, as well as a more refined spacing analysis. As a result, this design cycle was informed by SMAD, heritage structure, and results from our first subteam report and divided into five key tasks:

1. **Orbit Finalization:** Complete final analysis on a set of flight paths that comply with mission and subsystem level requirements, maximizing our coverage of the inner magnetotail.
 - Related to Requirements: **GNC-F-002, GNC-P-001, GNC-P-004**
 - The analysis will be reviewed for orbit characteristics, including eclipse times, time in ROI, and time around perigee, for uplink/downlink execution.
2. **Finalizing Spacing, Timing, and Launch:**
 - Related to Requirements: **GNC-P-003**
 - Here, we intend to review a plan for satellite spacing that benefits both ROI coverage and measurement resolution. Of particular importance is the spacing of the two Group 3 satellites, which was neglected in order to focus on Groups 1 and 2 in the last design cycle.
3. **Delta-V Analysis:** Create a full estimates for total ΔV values for all major and minor maneuvers executed over mission life, including attitude control. This will also be discussed in the ADCS subsystem section.
 - Related to Requirements: **GNC-F-003, GNC-C-006, GNC-P-002**
 - Our focus in this section is on incorporating attitude costs into the ΔV budget, as well as recalculating other operations with updated thruster and fuel tank specifications from PLV.
4. **Maneuver Pointing Accuracy:**
 - Related to Requirements: **GNC-F-001, GNC-F-005, GNC-F-007**
 - This builds upon the ΔV analysis done in our first design cycle, but incorporates further, more in-depth analysis. Our focus in this section is on station-keeping costs and timing orbit transfers to achieve proper spacing, as we gained accurate estimates for transfer from parking to mission orbits.
5. **Instrumentation and State Estimation:** Determining the specific method for state estimation and determination, in order to properly execute maneuvers and scientific data collection.
 - Related to Requirements: **GNC-F-001, GNC-F-005, GNC-F-007**



- Based on initial hardware analysis, we intend to make more specific recommendations on exact types of hardware (i.e. required OpNav cameras, etc) and see how state estimation will be performed.

6.7 Formal Analysis (60 Points)

Describe the results of your design approach. These could be in terms of the data you collected or generated, any simulation results, and any mathematical or analytical results. Include appropriate analysis for these results, describing what the data represents and comment on pertinent trends in the data. Your analysis may reinforce your comments on compliance to design objectives, requirements, and constraints from the previous subsection.

6.7.1 Orbit Finalization

Analysis conducted by: Ethan Spain (ROI and perigee analysis), Josephine Kwok (eclipse times)

In this section, we intend to perform final orbit analysis, building on results from the PDR and initial design cycle.

Mission Orbits

From the PDR, the nominal orbit parameters remain unchanged, with perigee and apogee selected to fulfill requirements for both ROI coverage and science spacing (i.e., maintaining mesoscale spacing while also obtaining macro- and microscale measurements). ROI and perigee coverage calculations were done the same way as in our first design report, using first principles related to Kepler's equations [107]. This process was included in A.5.1 since it is unchanged from the PDR.

Group #	$r_{apogee}(R_E)$	$r_{perigee}(R_E)$	Semi-Major Axis (km)	T (hrs)	% T Beyond $6R_E$	Time Below $1.08R_E$ (min)
1	12.09	1.1588	42241.08	24	79.73	59.94
2	8.96	1.150	32235.98	16	66.06	62.45
3	7.20	1.142	26610.21	12	47.82	64.96

Table 46: Summary of Mission Orbits and Important Characteristics

Table 46 shows the results of this analysis, with apogee and perigee values determined based on desired periods and their resulting time in the ROI. Time below $1.08 R_E$ is necessary as the CCD team needs to remain within communication range in order to execute uplink/downlink near perigee. Overall, we can see that all three groups spend around the same time near perigee, but differ widely in time spent in the ROI. From these results, we would expect most of our scientific data to be at the meso and macro scales, as measurements from Group 1 and Group 2 are dominated by these scales, while Group 3 primarily collects microscale data.

Figure 14 shows physically how much of each orbit near perigee is within the $1.08 R_E$ zone for the CCD requirement. As we can see, the highly elliptical nature of the orbit limits the time spent in the coverage region.

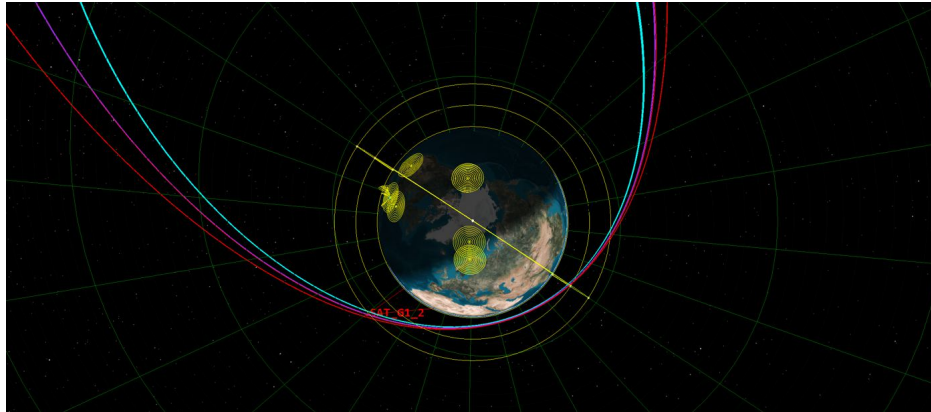


Figure 14: Snapshot of Mission Orbits in Uplink/Downlink Range

Below, 15 shows the nominal main mission orbits for each group, with Groups 1, 2, and 3 in red, purple, and blue respectively. This gives a sense of scale for each of the three groups, as well as the differences between them.

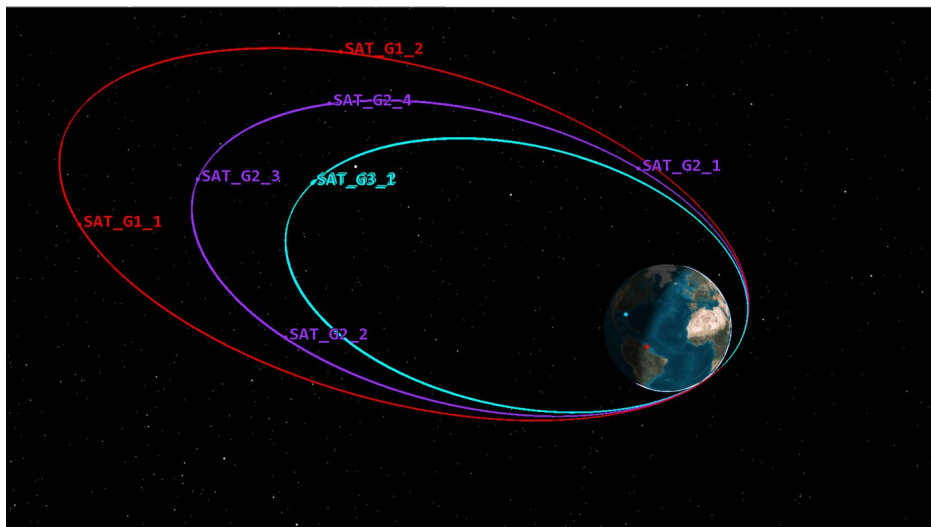


Figure 15: Snapshot of Mission Orbits via STK

Eclipse Times

The following table shows the minimum and maximum eclipse times for a satellite in each orbit group. Note that stationkeeping will be performed when apogee is most closely aligned with the magnetotail, and that is when the eclipse will be longest. When apogee is closest to the Earth, combined with the drifting orbital parameters, the eclipse period will be shortest.



Group #	Max Umbra (s)	Min Umbra (s)	Max Penumbra (s)	Min Penumbra (s)	Max Eclipse (hr)	Min Eclipse (hr)
1	13883.000	1468.427	1800.000	12.743	4.356	0.411
2	10805.000	1361.297	1048.000	13.808	3.293	0.382
3	9058.000	1273.979	715.000	14.952	2.715	0.358

Table 47: Eclipse Durations (1 Year Propagation via STK)

From Table 84, we see that the maximum eclipse time is 4.356 hours. Since all satellites will be designed the same way, the batteries will be sized accordingly. Expanding on the PDR, propagation was extended to a full 5 years after the satellite reached its mission orbit, but the maximum eclipse time remained the same at 4.356. As a result of this extended analysis, we now know what to expect over the entire mission timeline in terms of the maximum power needed during eclipse.

6.7.2 Spacing, Timing, and Launch

Analysis conducted by: Josephine Kwok (design of trade study, simulation, calculations, writing of section).

This section details all timing and spacing operations between initial launch and when all satellites become fully operational. We intend to launch in early 2033 and become fully operational in late 2033. We also plan to first spread the satellite groups in the parking orbit before the raising and lowering maneuvers. This section for Groups 1 and 2 is largely unchanged from the PDR, but contains more in-depth analysis for Group 3's spacing.

Launch Window

The launch window for MATRIX was first defined by mission requirements MISS-C-001 and MISS-C-002: The spacecraft must be delivered for launch by the end of 2032, and the mission components must be fully operational by the end of 2033, respectively. Given that we want to start operations when our apogees are aligned with the magnetotail, and that the magnetotail is largest during the winter season, we wish to start operations in December 2033 to best align with the magnetotail during the winter solstice. Table 48 shows the desired orbital values to begin operations for each group.

Group #	$r_p(R_E)$	$r_a(R_E)$	Inclination (deg)	RAAN (deg)	Arg of Periapsis (deg)
1	1.157	12.089	23.5	0	250
2	1.149	8.960	23.5	0	250
3	1.141	7.203	23.5	0	250

Table 48: Desired Orbits for 12/01/2033 Operations

For an appropriate window for launch, we wish to reference heritage missions, natural orbit decay, and the positioning of the launch site relative to the desired parking orbit. THEMIS performed orbit maneuvers before nominal mission operations from February 2007 to July 2008 [108]. However, their largest orbit was much higher than ours at roughly $30 R_E$, so our launch operations will fit inside the mission constraints. For reasons of maximum time to properly space the satellites and doing nominal checks, we anticipate to launch in February 2033, giving us 9 months after launch to become fully operational.



Group #	RAAN (deg)	Arg of Periapsis (deg)
1	-45.154	78.670
2	-71.606	124.161
3	-103.860	181.088

Table 49: Annual Drift of Orbital Parameters via STK

Looking at Table 49, we note the natural drift of orbital elements for each orbit group. The values for Group 2 are useful for determining the orbital parameters of our parking orbit since we are launch to Group 2's apogee and perigee. For minimum Δv usage, we wish to choose a RAAN and argument of periapsis (AOP) so that the orbit will naturally precess to the parameters listed in Table 48 by December. Knowing that the drift rate for these two parameters is linear, we are able to accurately calculate drift for 9 months, then work backwards to find our parking orbit parameters.

Group #	$r_p(R_E)$	$r_a(R_E)$	Inclination (deg)	RAAN (deg)	Arg of Periapsis (deg)
Parking	1.149	8.960	23.5	53.705	156.879

Table 50: Parking Orbit Parameters for February Launch

For a launch scheduled in early February, we aim to launch from Cape Canaveral Space Station near midnight so that the launch site best aligns with the perigee of the orbit described in Table 50. The day of launch is variable, but RAAN and AOP would be recalculated to ensure proper precession of the orbit.

Satellite Spacing

The mission orbits specified in Table 46 become resonant with the proper satellite spacing. We wish to space the 8 satellites so that there is constant coverage of the ROI, and that satellites in different groups will line up at apogee with the other groups for diverse scientific data.

Similar to what was mentioned in the PDR, satellite spacing works as follows: 2 satellites in Group 1 are 12 hours apart, 4 satellites in Group 2 are each 4 hours apart, and the remaining 2 satellites in Group 3 are spaced closely together for microscale measurements. The design of the spacing and orbits satisfy mission requirements MISS-F-002 and MISS-F-003 regarding measurements at multiple spatial scales.

To have satellites aligned at apogee for all 3 orbit groups, we first space satellites in the parking orbit, followed by orbit raising and lowering. This order ensures precise alignment of satellites between groups since we do not know the exact timing of orbit raising and lowering, or how many burns each satellite would perform to orbit raise and lower. For these reasons, we will perform all satellite spacing first in the parking orbit.

To space our satellites, we plan to perform a burn which causes the satellite to enter a temporary orbit with a slightly larger apogee. This temporary orbit has a slightly larger period, creating temporal separation. The satellites will remain in the temporary orbit until they are separated by the correct amount of hours, where it will then perform a burn at perigee to return back to Group 2 orbit.

The largest separation between the earliest and latest satellite in the orbits will be 12 hours. Table 51 gives examples of different temporary orbits that could be used for spacing.

Temp Apogee (km)	Δr_a (km)	ΔT (sec)	Total Time for Separation (months)	Δv per Satellite (m/s)
57373.519	229.564	320.000	3	4.633
57313.919	169.965	240.000	4	3.475
57278.146	134.192	192.000	5	2.780
57254.292	110.337	160.000	6	2.317
57237.251	93.296	137.143	7	1.986
57224.468	80.513	120.000	8	1.738

Table 51: Potential Separation Orbits

Given the February 2033 launch and the end-of-year starting operations, and assuming that orbit raising and lowering will take roughly a week, we are able to use less Δv and spend 8 months to properly space all satellites. In the event where orbit raising and lowering take longer or launch is delayed, the temporary orbit can be changed at the cost of Δv , though this trade off of a couple m/s for months of separation time is most costly for time compared to Δv . Figure 16 shows how the satellites will be spaced in the parking orbit, as well as satellite group assignments.

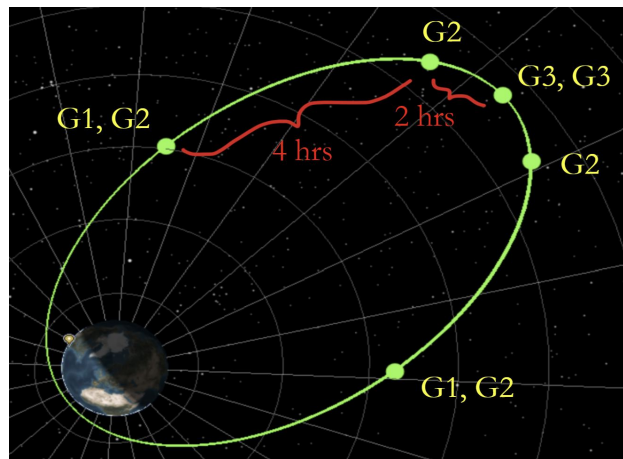


Figure 16: Parking Orbit Satellite Spacing via STK

To achieve the constant microscale spacing of the Group 3 satellites, they must be launched within a short time differential of each other. The MMS mission maintained its satellites in a small tetrahedral structure for microscale measurements. For this configuration, satellites were 10-160 km apart near Earth and 30-400 km apart in the magnetotail [5]. Since MMS also used HEO orbits with similar distances, we plan to maintain a similar range for microscale spacing in Group 3. Figure shows the distances at perigee and apogee for the satellite pair when spaced by different times.

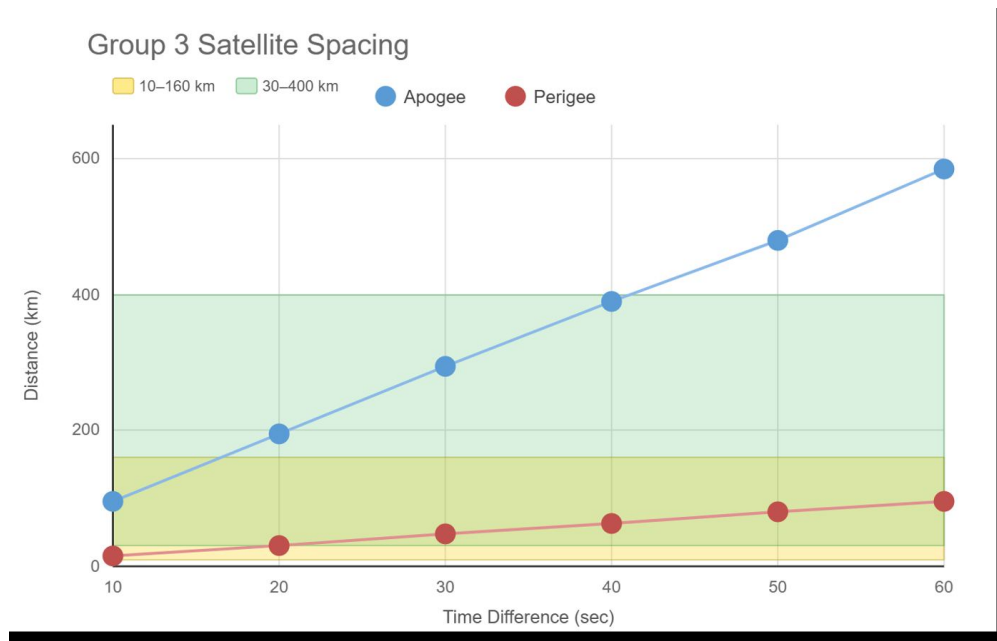


Figure 17: Group 3 Spacing Compared to MMS Micro-spacing

Based on heritage precedent, we will aim to launch the satellites in Group 3 30 seconds apart from each other. This will keep them roughly 292 km apart near apogee and 46 km apart near perigee. Specific maneuvers to account for drift from this separation distance is not planned, but each satellite will perform the same stationkeeping maneuvers which may also be an opportunity to reset satellite distances as well. Additionally, given the large Δv margin in Table 94, there is also the opportunity to perform one-off correction maneuvers if needed.

6.7.3 Maneuver Analysis

Analysis conducted by: Ethan Spain (design of trade study, simulation, calculations, writing of section)

In this section, we intend to provide a full breakdown of ΔV costs during the course of our 5-year active mission timeline, including attitude costs. We intend to keep the total cost for each satellite under 1200 m/s, per requirement GNC-C-006. Analysis for orbit transfers, stationkeeping, and decommissioning remain largely unchanged from PDR, as the same process was used with slightly updated variables via PLV (i.e., using STK to model the maneuvers), which resulted in minute changes in ΔV values. The addition of the ADCS budget is included here for completeness, but is discussed more in-depth in the formal analysis section within the ADCS chapter.

Group #	Orbit Transfer	Stationkeeping	ADCS	Decommission	Total Cost	Percent Margin
1	150.382	347.815	138.008	54.801	692.992	52.664%
2	0	449.100	180.996	71.001	703.082	49.264%
3	137.666	628.449	251.831	109.262	1129.193	21.256%

Table 52: Total ΔV for entire mission by group (in meters per second)



Table 52 provides a summary of the entirety of ΔV costs for the mission, verified through STK. As we can see, Groups 1 and 2 have relatively similar total costs, but Group 3's cost is over 400 m/s higher. This is mostly due to increased stationkeeping costs and attitude losses, caused by Group 3's tighter spacing and smaller orbit. From this analysis, it is possible that Group 1 and 2 satellites, with their larger ΔV margins, could be considered for an extended mission after 5 years of science operations. However, Group 3, with its tighter margin and greater chance of collision with LEO satellites, would likely need to be decommissioned. To see an in-depth breakdown of ΔV costs, which is largely unchanged from the PDR, see section A.5.2.

6.7.4 Maneuver Pointing Accuracy

Analysis conducted by: Joseph Norwood (trade studies, writing of section).

In this section, we analyze the impacts of attitude error on the Δv cost of the various maneuvers to formally calculate the necessary attitude accuracy required during GNC maneuvers.

Analysis started from the nominal orbits and their maneuvers in STK to set the baseline Δv 's of the two most Δv intensive maneuvers, orbit raising and orbit lowering. These maneuvers were chosen to be the focus of the pointing accuracy examination, since they are both the single maneuver that requires the most Δv . The nominal Δv 's differed slightly from those used in those in section 6.7.3, because the STK Attitude Control in Astrogator was changed to Euler Angles for this analysis. After the optimal attitude was determined, the satellite's orientation was misaligned by a chosen degree simultaneously in every Euler angle, representing a worst-case combined pointing error, and the required Δv was reanalyzed. The results of this analysis for the orbit raising and lowering maneuvers can be seen in Appendix A.5.3.

Examining the results, we see that the increasing attitude error does have a proportional, non-linear relationship with Δv and Fuel Use. The percentages for the ΔFuel are also similar for both maneuvers. Assuming similar sensitivity in all maneuvers, we can propagate this ΔFuel percentage to our whole Δv budget of 1000 m/s to get a rough estimate of the total fuel waste due to attitude error and compare this to the initial propellant mass of 94 kg. The results are seen in Table 53.

Table 53: Estimated Total Wasted Fuel

Error	Total Wasted Fuel (kg)	% Total Fuel Wasted
0 degrees	0.0000	0.00%
0.1 degrees	0.0060	0.01%
1 degree	0.5947	0.63%
2 degrees	2.3817	2.53%
3 degrees	5.3696	5.71%
4 degrees	9.5739	10.19%
5 degrees	15.018	15.98%
10 degrees	62.0244	65.98%

Examining the effects of attitude error on the total fuel wasted, we can see that around an accuracy error of 3 degrees the total wasted fuel during orbital maneuvers already exceeds 5%. Given our fuel budget only allocates a few percent of total propellant mass to losses and uncertainty, we can



choose a GNC maneuver pointing accuracy of 2 degrees, which increases control leniency without compromising fuel efficiency.

6.7.5 Instrumentation and State Estimation

Analysis conducted by: Joseph Norwood (analysis, writing of section).

Analysis of the instrumentation necessary for GNC state estimation began with a general trade study on the feasibility of different position determination methods, specifically Ground Tracking, GPS, and Optical Navigation (OpNav). GPS, defined as position determination based on communication with a specialized satellite constellation, has a typical operation range of 3000 to 36000 km [105]. This distance is significantly less than the Group 1 apogee of 77104 km meaning it is not viable for continued state position estimation. Examining OpNav, position determination based on images of celestial bodies, we found there are no OpNav systems capable of operating at 15 rpm. OpNav cameras already struggle to deal with smear just from the use of electronic shutter on a non-spin stabilized satellite [106]. This weakness indicates that added smearing due to a fast spin-stabilized rotation would make the system not viable. The only remaining option is Ground Tracking, which relies on satellite ground communication to calculate and update the satellite's position. Since our satellite will already have a 1 hour uplink/downlink period each orbit for the science downlink, our ground tracking system would use the doppler effect on the scientific data to determine the line-of-sight velocity of the satellite, range measurements using a call and response signal, and ground antenna angles to calculate the satellite's state and relay it back for onboard update [109]. Since our payload and communication systems don't have stringent pointing requirements, the coarse propagation of our course once per orbit is sufficient for our mission. However, the GNC maneuvers require an accurate position estimate for efficient maneuvers and do not always occur at perigee. Our satellite has a powerful S-band antenna chosen for CCD, which is capable of communication at any point in our orbit. This allows us to recompute and receive an accurate position estimate right before perigee maintenance at the apogee and inclination maintenance at the line of nodes.

As specified by the CCD subteam, our satellite will be using the Near Earth Network on the S-band, which has radiometric tracking services. Using the tone ranging and 1-way doppler of these tracking services, we get an accuracy of < 5 meters (1σ) for radial range and ≤ 30 mm/s (1σ) for line-of-sight velocity [110]. The state will then be propagated using orbit estimation until the next time the ground station updates its position.



7 ADCS Design

7.1 Subsystem Overview

The primary goal of the attitude determination and control systems (ADCS) subsystem is to provide accurate pointing maneuvers for each satellite, enabling MATRIX to best fulfill its mission-level objectives and scientific requirements. We are responsible for interfacing with other subsystems and making sure our design choices are feasible. For example, our choice to use spin stabilization impacts the physical design of the satellites, determined by the SMM subsystem, and antenna selection determined by the CCD subsystem. This subsystem consists of 2 sun sensors, an IMU, an MEKF for dynamics updates and attitude determination, and a PD controller. None of these choices has changed since the PDR, but additional analyses have been conducted.

7.2 Subsystem Objectives

Below are the subsystem objectives for ADCS, which remain unchanged from our PDR.

ADCS Subsystem Objectives

1. Provide accurate pointing and slewing for the successful implementation of GNC maneuvers.
 - *Derived from Primary Mission Objectives 1 and 2*
2. Provide stabilization and pointing for the communication of CCD systems and the data collection of the scientific payload.
 - *Derived from Primary Mission Objectives 1 and 3*
3. Develop a system capable of successful stabilization, jitter reduction and momentum dumping for precise, reliable control.
 - *Derived from Primary Mission Objectives 1 and 2*

7.3 Subsystem Requirements

The ADCS requirements that are most important for the MATRIX mission and pertinent to this design cycle are discussed below. The full list of requirements can be found here under the "ADCS" tab: [ADCS Subsystem Requirements List](#). The notable requirements listed below are grouped by a specific objective they satisfy and include mission-level rationale. The rationale for these requirements provides key sources, calculations, and research done to arrive at these values.

1. Functional

- **ADC F-009 - ADC-F-011:** The ADC subsystem shall be able to provide pointing capabilities and control attitudes for the purposes of data measurement, communication, and thruster maneuvers.
 - Derived from Objective **MISS-F-014, MISS-F-017**
 - This requirement relates to the satellite's ability to correct and reorient itself for a variety of mission reasons, including thruster maneuvers, sensor alignment, and



data collection. These are the most crucial and overarching requirements for the ADC subsystem design philosophy.

- **ADC F-008:** The ADCS shall be capable of executing pointing commands from ground control.
 - Derived from Objective **MISS-F-017**
 - In cases where manual input is needed to conduct pointing commands and attitude control, the ADC subsystem should be able to override on-board executions with requests from ground control teams.

2. Performance

- **ADC-F-002:** The ADC shall obtain a 3-sigma average pointing accuracy of at least 2 degrees for the mission architecture.
 - Derived from Objectives **MISS-F-013-015**
 - Pointing accuracy is a crucial component of the ADC subsystem, determining the spacecraft's ability to accurately orient with targets, completing tasks such as communicating with ground stations, reorienting for maneuvers, and knowing attitude for data collection.
 - Values for this requirement are set via table 11-8 in SMAD, outlining the effect of control accuracy on sensor selection [4]. Due to the purpose of our mission and extensive use of magnetometers, we determined that having pointing accuracy of $\sim 1^\circ$ fits with both the needs of our mission and the accuracy outlined in heritage missions [20].
- **ADC-F-004:** The ADC subsystem shall reduce the maximum drift rate to at most 1° per hour.
 - Derived from Objectives **MISS-F-010-012, MISS-F-014**
 - The ADC system must be able to compensate for low-frequency vehicle motion, which can build up over time and lead to accuracy loss in the ADC and GNC subsystems.
 - Values for this requirement are set via specifications in SMAD [4]. THEMIS satellites achieved much higher drift rate control, lowering this value to $<0.1^\circ$ per hour and maintaining spin rates of 16 ± 2 rpm [20]. In the future, these values may be refined to be more stringent in line with heritage missions, but for now the requirement is rooted directly in recommendations from SMAD.
- **ADC-F-005:** The ADC subsystem will maintain a momentum storage capacity of ~ 14 Nms.
 - Derived from Objectives **MISS-F-010-12, MISS-F-014**
 - A buildup of torque beyond margins within the ADC subsystem can be catastrophic, as reaction systems are no longer able effectively account for torque disturbances and the system loses attitude control.



- Numerical values for this requirement were determined through the analysis of microtorque build up, which is explained extensively in section ???. An analysis of worst-case Gravity Gradient Torque and Solar Radiation Torque was conducted based on nominal orbits determined through the GNC subsystem. A per-period torque increase of 1.157 Nms for Group 1 satellites. Depending on mission parameters, momentum dumping can be done on different schedules. Larger GEO satellites like GOES-R require daily momentum dumping, while smaller satellites can get by on a weekly basis [111]. For our purposes, we are targeting momentum dumping every two weeks.
- An adoption of a fluid nutation damper, like those seen on THEMIS and MMS, would allow for an automatic fulfillment of this requirement [112] [113].

3. Environment

- **ADC-E-001 - ADC-E-003:** The ADCS subsystem must be able to operate within the space environment of its orbit
 - Derived from Objectives **MISS-E-001, MISS-E-004**
 - These requirements are fundamental in making sure the ADCS components function properly during its entire mission. The spacecraft will be in the microgravity space facing large temperature changes, radiation pressure, and vibration.

7.4 Subsystem Constraints

The determination of the ADCS subsystem constraints was heavily derived from the mission-level constraints. The following lists these constraints as well as the mission-level constraints that they are derived from. These remain largely unchanged from the PDR.

1. **Subsystem Mass and Volume:** The ADCS subsystem must fit within specified mass and volume restrictions in order to accommodate launch vehicle requirements and needs from all other subsystems. These mass and volume budgets are largely set by SMM, but are subject to change.
 - Derived from Mission Constraint 3
2. **Power and Computation Allocation:** The ADCS subsystem must be able to operate with its allotted power budget, maintaining operations even when in regions of eclipse. This power budget also extends to computational constraints, making sure that onboard processing power is not dominated by the ADCS subsystem.
 - Derived from Mission Constraint 4
3. **Radiation Exposure:** The ADCS subsystem hardware must be able to withstand the harsh radiation environment of the magnetotail and continue to provide accurate data in order to meet attitude determination and control requirements.
 - Derived from Mission Constraint 5
4. **Thermal Exposure:** On-board ADCS hardware must be able to withstand a wide range of



temperatures as it moves in and out of eclipse zones. Hardware will be selected and modified based on expected temperature ranges and analysis from THM.

- Derived from Mission Constraint 5
5. **Subsystem Cost:** MIDEX-class budget cap of \$350M requires ADCS subsystem to remain under 12.9% (TBC) of total budget expenditure, as set by OPL.
 - Derived from Mission Constraint 1
 6. **Timeline:** The ADCS subsystem must have operating software/hardware in the given mission timeline and be operational for at least 5 years as well, as described in the mission overview.
 - Derived from Mission Constraint 2
 7. **Pointing Accuracy:** The ADCS subsystem must maintain a pointing accuracy of sufficient fidelity to enable accurate scientific measurements, ground station communication, maneuver capabilities and solar panel alignment.
 - Derived from Mission Constraint 2
 8. **Stability:** The ADCS subsystem must maintain satellite stability, reducing jitter, to avoid degradation of scientific measurements.
 - Derived from Mission Constraint 2

7.5 Subsystem Drivers

The determination of the ADCS subsystem drivers was heavily derived from the mission-level constraints. The following lists these constraints as well as the mission-level constraint that they are derived from.

1. **2D Field and Particle Measurements Accuracy:** The ADCS subsystem is driven by the objective to collect accurate data in conjunction with other satellites for 2D measurements. Given particular orbit designs in the ROI, ADCS must be able to validate this positioning at all points in orbit and orient itself for maneuvers. This influences sensor selection and accuracy.
 - Derived from Mission Drivers 1 and 3
2. **High Temporal Sampling Rate:** The ADCS subsystem must enable slews, formation reconfigurations, and rapid reorientation to satisfy time-critical observation windows.
 - Derived from Mission Driver 2
3. **Accurate Constellation Stationkeeping:** Similarly, the ADCS subsystem must enable slews, formation reconfigurations, and rapid reorientation to multi-satellite coordination in the tetrahedral configuration.
 - Derived from Mission Driver 4
4. **Data Sampling and Communication:** The ADCS subsystem must be capable of frequently sending data back to ground stations, given the large amounts of measurements taken in



the ROI, as determined by CCD. Accurate communication timing, spacing, and pointing influence the ADCS subsystem complexity.

- Derived from Mission Driver 5

7.6 Subsystem Design Approach

In the ITR, we analyzed the ADCS subsystem starting from the control modes and heritage missions before examining the effects of disturbance torques on spin stabilization and conducting preliminary hardware trade studies. In the PDR, we iterated these results and strengthened the analysis on the thrusters, nutation dampers, attitude estimation instruments and control frequency. The past design approaches were guided by heritage mission structure and SMAD guidelines [4].

In this final design cycle, we reexamined all the previous analyses while further refining weak points in the analysis. As a result, this design cycle was informed by SMAD, heritage structure, and results from our previous reports and is divided into eight key tasks:

1. **Control Modes:** Determine the required attitude control and estimation accuracies.
 - Related to Requirements: **ADC-F-001, ADC-F-002, ADC-F-006**
 - For the Final Design Report, we must reexamine the control modes and how they have changed since the initial subteam report to guide the final analysis of the ADCS subsystem.
2. **Heritage Mission Precedent:** Understand the ADCS systems and corresponding design choices for magnetosphere Heritage missions.
 - Related to Requirements: **ADC-F-001, ADC-F-002, ADC-F-006**
 - Besides SMAD, the Heritage Mission Analysis was an important driving factor throughout the design of the ADCS subsystem. This analysis was reexamined with new knowledge of the need for magnetic cleanliness.
3. **Micro-torque Disturbances:** Get more accurate estimates for micro-torque disturbances and the corresponding affects on the spin-stabilization.
 - Related to Requirements: **ADC-F-004, ADC-F-005**
 - With the satellite dimensions and properties solidified, this examination was reiterated through one last time.
4. **Nutation and Damper Selection:** Select proper hardware to reduce nutation and create a stable system.
 - Related to Requirements: **ADC-F-002, ADC-F-003, ADC-F-004, ADC-F-005**
 - This analysis reiterates the previous examinations based on the final disturbance torque values and performs additional calculations examining the specific sizing required of the chosen nutation damper.
5. **Thruster Analysis:** Determine and analyze a thruster orientation capable of accurate, reliable slewing maneuvers and determine the ADCS thruster requirements for successful attitude control.



- Related to Requirements: **ADC-F-002, ADC-F-006**
- All previous analyses of the thrusters and their orientation were reexamined. This analysis successfully solidifies the thruster requirements from an ADCS perspective.
- 6. **Attitude Delta-V Budget:** Determine the ΔV required for attitude maintenance and pointing.
 - Related to Requirements: **ADC-C-006**
 - This analysis examines the ΔV requirements by looking at the different attitude maneuvers required throughout the mission.
- 7. **Instrumentation:** Determine and analyze the specific instruments required for attitude estimation and determination.
 - Related to Requirements: **ADC-F-001**
 - This analysis combines the previous analysis done in the initial subteam report and the PDR to successfully analyze and explain instrument selection.
- 8. **Control Algorithms:** Determining the specific control algorithms used for attitude control and analyze their computational load.
 - Related to Requirements: **ADC-F-002, ADC-F-003, ADC-F-004, ADC-C-005**
 - This combines all previous analysis on the control algorithms for attitude estimation and control, while further deepening said analysis.

7.7 Formal Analysis

7.7.1 Control Modes

Analysis conducted by: Joseph Norwood (analysis, writing of section).

Before designing this subsystem, we must understand the functionality required by the various control modes of the ADCS system. The following table outlines the control modes and the duration of the attitude adjustments along with the required accuracy and range based on the correspondence with the related subteams: GNC, CCD, POW, and OPL.

ADCS Control Modes					
Control Mode	Duration Requirements	Determination		Control	
		Accuracy	Range	Accuracy	Range
Orbit Insertion	Until in final orbit	2°	3-axis inertial	2°	3-axis inertial
Spin Up	Once after deployment	5°	3-axis inertial	5°	3-axis inertial
Communication	One hour near perigee	N/A	N/A	Passive	Passive
Disturbance Correction	Once per orbit	5°	3-axis inertial	5°	3-axis inertial
Sun Pointing	Once per orbit	5°	3-axis inertial	5°	3-axis inertial
Payload	While in Magnetotail	N/A	N/A	Passive	Passive

Table 54: Various control modes of the ADCS subsystem.

The required orbital insertion accuracy comes from Section 6.7.4 and includes the error of everything from thruster misalignment to disturbance torques. The communication and payload has



no pointing requirement as specified by the OPL and CCD subsystems in Sections 4 & 5. The sun pointing accuracy requirement comes from the $P = P_{max} \cos(\theta)$ power relation. A 5 degree error causes a less than 1 percent decrease in solar panel efficiency, or a 0.38% decrease to be exact. The disturbance correction will also happen during the sun pointing maneuvers, meaning this control mode requires the same accuracy. As disturbance corrections correct spin rate and the direction of the spin axis, the spin up requires the same level of accuracy as well.

Examining the control modes, the main pointing constraint comes for the orbital insertion requirements as it has the finest accuracy requirement. This constraint will dictate the entire subsystem design.

7.7.2 Heritage Mission Precedent

Analysis conducted by: Joseph Norwood (analysis, writing of section).

Formal analysis continued with a trade study of the ADCS systems of magnetotail heritage missions. Using systems similar to those of heritage missions mitigates risk as these systems are already proven to be reliable. The previous magnetosphere missions examined were MMS, THEMIS, Cluster, and TWINS.

Mission	Agency	Launch	Mass/Sat.	Stab.	Spin Rate	Accuracy	Thrusters	Sensors	Algorithm	Refs.
THEMIS	NASA	2007	128 kg	Spin	16 ± 2 rpm	Axis: <1° (K), <3° (C); Phase: <0.1° (K)	4 x 4.4N	MSSS, 2 x IRU, FGM	Ground computed	[20]
MMS	NASA	2015	1,250 kg	Spin	3 rpm	Axis: ±0.1° (K), ±0.5° (C); Phase: ±0.1° (K)	8 x 18N, 4 x 4.5N	ST, 2 DSS, accel., GPS, IRAS	Multiplicative EKF	[112], [5]
Cluster	ESA	2000	1,200 kg	Spin	15 rpm	N/A	1 x 400N, 8 x 10N	N/A	N/A	[21]
TWINS	NASA	2006	N/A	N/A	–	N/A	N/A	N/A	N/A	[113]

Table 55: Trade Study of Magnetosphere Heritage Missions. K = knowledge, C = control.

Some things to note before interpreting the results of our study in Table 55 is that insufficient information was available on the TWINS mission because it was on a classified satellite and on the Cluster mission because of its age and the less public nature of the European Space Agency (ESA). In other words, most of the specific technical information on these missions, which includes specific ADCS information, is not readily available in the public domain. However, the two most relevant missions THEMIS and MMS had sufficient information to carry out this heritage examination. MMS is relevant due to its recency in magnetosphere data collection and THEMIS is relevant to the satellite size. Our satellite currently has a projected mass of 178.6 kg per satellite, which most closely aligns with the satellites of THEMIS.

Examining the results, it immediately becomes clear that spin stabilization is the gold standard for these HEO magnetosphere missions. While spin stabilization is chosen for boom tension [5], the choice of spin stabilization, as opposed to reaction wheels, is primarily due magnetic cleanliness. Reaction wheels spin at high RPMs and induce a magnetic flux around the satellite. This flux causes noise in the readings scientific payload. Using spin stabilization reduces the error of these sensor measurements.



These heritage missions also give us a general idea of satellite thruster placement for satellite attitude control, spin up, and spin down. Given the large size of most of our heritage missions, they have more thrusters than practical for a satellite of our size. Looking at the THEMIS mission, which has a comparable mass to MATRIX, we see that THEMIS has two radial thrusters in opposite corners and two axial thrusters mirrored across the centerline of the bottom of the satellite [20]. THEMIS is designed to move anywhere within the southern hemisphere without attitude precession [20]; however, our mission requires constant sun pointing, as seen in Section 7.7.1, which means we need to reorient our spin axis. This difference in requirements means the thruster orientation must move away from that of the heritage missions.

Finally, the attitude determination sensors used includes sun sensors, the scientific magnetometers and IMUs on the both MMS and THEMIS, and the MMS has a star tracker. These instruments set a primary focus for specific design analysis. Furthermore, given the difference in spin rates between THEMIS (16 rpm) and MMS (3 rpm), the difference in star tracker use probably indicates star trackers aren't viable at higher spin rates.

Based on the precedent set by our magnetosphere heritage mission, MATRIX will also be spin stabilized to increase instrument coverage, decrease ADCS fuel/power consumption, simplify control, and increase magnetic cleanliness. As for the thruster orientation, a unique thruster orientation is required. Instrumentation should consist of IMUs and star trackers with the support of the magnetometer readings. This decision reduces mission risk, as the system was previously tested, and allows us to go forward with finer analysis.

7.7.3 Micro-Torque Disturbances

Analysis conducted by: Ethan Spain (analysis, writing of section), Joseph Norwood (Initial Analysis).

As seen in our initial report, analysis of micro-torques is a crucial part of the ADC subsystem in order to determine how the build-up of these forces can affect attitude control over time. This section intends to build upon the analysis in our previous cycle, providing updated values and new solutions to combat these disturbances. In addition, analysis is now done on a per group basis, giving more information into specific requirements for each orbit. These different disturbances will be discussed below, as well as the effect on desired spin rate.

Gravity-Gradient Torque (GGT)

GGTs build up on our satellites due to slight changes in gravitational forces at different distances, which affect different parts of the structure [4]. This can be estimated with equation (13) below:

$$T_g = \frac{3\mu}{2R^3} |I_z - I_y| \sin(2\theta) \quad (13)$$

Where R is orbital radius, and θ is the worst-case attitude error in radians [4]. For the purposes of this analysis, we assumed a worst-case error of 10° and used perigee radii for R to get an overall worst-case scenario for each orbit. I_z and I_y values were $39.147 \text{ kg} \cdot \text{m}^2$ and $25.969 \text{ kg} \cdot \text{m}^2$ respectively, per analysis from SMM.

Solar Radiation Torques (SRT)



SRT is caused by the Solar Radiation Pressure (SRP) on the satellite, and is the other major factor of torque buildup for our mission. It can be estimated by equation (14):

$$T_{sp} = \frac{F_s}{c} A_s (1 + q) \cos(I) (c_{ps} - c_g) \quad (14)$$

where F_s is the solar constant (1267 W/m^2), c is the speed of light, and A_s is the surface area, which we find to be approximately 1.066 m^2 from CAD analysis done by the SMM team. Additionally, q is the reflectance of the material of the satellite, which for the overestimate will be maximum reflectance. This means all of the solar pressure perfectly reflects off the surface. This means q is 1. Finally we have $c_{ps} - c_g$, which is the difference between the center of pressure of the SRP and the center of mass of the satellite. This is estimated to be around 0.58 m^2 , again per the most up-to-date information from SMM

Disturbance Results

Using the torque estimations calculated above, the required momentum and spin rate values were calculated below. Since this process remained unchanged from our initial report and PDR, the full method is included [A.6.3](#).

Group #	GGT (Nm)	SRT (Nm)	ΔL per Period (Nms)	Required L (Nms)	Spin Rate (rpm)
1	6.99×10^{-6}	8.77×10^{-7}	0.6797	38.946	9.500
2	6.99×10^{-6}	8.77×10^{-7}	0.4532	25.964	6.334
3	6.99×10^{-6}	8.77×10^{-7}	0.339	19.473	4.750

Table 56: Disturbance Values and Required Spin Rates

Table 56 summarizes the calculations of these micro-torques, as well as the angular momentum changes and the spin rate requirements to keep them from impacting attitude control. From this analysis, we can determine that the orbital period is the dominant factor in the size of these disturbances; therefore, extra attention must be paid to the two Group 1 satellites, as they experience the highest torques and require the highest spin rate. We are still targeting a spin rate of 15 rpm, in order to provide margins, and because the ΔV cost of spin-up to that value is within the scope of our current analysis.

Additionally, spin rates have the added benefit of contributing to magnetic cleanliness. For MMS an accuracy of around 0.1 nT was achieved, a direct result of a spin rate of 3 rpm [114]. We anticipate that our spin rate of 15 rpm will have similar effects on magnetic readings and give us cleaner data from the magnetotail.

7.7.4 Nutation & Damper Selection

Analysis conducted by: Ethan Spain (research, trade studies, analysis, writing of section), Joseph Norwood (initial analysis).

Nutation refers to the periodic wobble of our spin axis, causing further attitude errors if not properly taken into account [4]. Nutation frequency can be calculated by equation (16):



$$\omega_n^2 = \frac{(I_z - I_x)(I_z - I_y)}{I_x I_y} \omega_s^2 \quad (15)$$

where ω_s refers to our satellite's spinning rate [115]. Again from the PDR, $I_z = 39.147 \text{ kg} \cdot \text{m}^2$ and $I_y = I_x = 25.969 \text{ kg} \cdot \text{m}^2$ per SMM. Taking our spin rate to be 15 rpm, we get a w_n of 7.612 rpm or 0.797 rad/sec.

As shown in the PDR, we have decided to move forward with a passive fluid nutation damper. In order to properly size this damper, it must fit within the bus of the spacecraft, therefore having an outer radius of less than 0.49 meters in order to be within the apothem of the hexagonal prism with a safe margin of 15 centimeters.

Fluid	η Temperature Coefficient	Density kg/m^3	Radiation Tolerance (rads)
DC-200 100 cSt	0.600	964	High
DC-200 500 cSt	0.600	969	High
DC-200 1000 cSt	0.610	970	High
Krptox GPL 101	0.744	1890	High
Krptox GPL 104	0.860	1930	High
Krptox GPL 107	0.907	1950	High
Mercury	0.184	13534	High

Table 57: Trade Study for Damping Fluids [116] [117]

Table 57 outlines possible choices for the viscous ring damper, coming from common aerospace lubricant choices with high viscosity properties [118]. Historically, mercury has been used for this application, as its viscosity is not greatly affected by temperature changes that would be a constant concern. However, mercury has an extremely high density and extreme toxicity, potentially leading to the degradation of other spacecraft components as well as environmental concerns upon satellite reentry in the atmosphere [119]. To make a clear choice, we must first look at necessary sizing of the ring in based on calculated nutation frequency.

For sizing, a ratio determining the height and thickness of the ring can be determined via the equation below:

$$\frac{h}{r_{thick}} = \frac{K}{\eta \tau} \quad (16)$$

where $K = \frac{I}{\pi \omega_n R^3}$ and $\tau = \frac{t_{1/2}}{\ln 2}$. Here $t_{1/2}$ represents the nutation half-life, or how long it takes for a nutation disturbance to be reduced by half its value [118]. For our mission, we target a half-life of 5 minutes, meaning, for example, a 5° nutation disturbance would be reduced to 2.5° after 5 minutes. We are also targeting a thickness of 5 cm and assuming a radius of 0.49 m to maximize the damper's effectiveness. With these constraints picked, we can see how different fluid choices affect this ratio.

Fluid	Dynamic Viscosity $Pa \cdot s$	Density kg/m^3	h/r ratio
DC-200 100 cSt	0.019	964	16.160
DC-200 500 cSt	0.048	969	6.397
DC-200 1000 cSt	0.097	970	3.165
Krptox GPL 101	0.130	1890	2.362
Krptox GPL 104	0.114	1930	2.693
Krptox GPL 107	0.088	1950	3.487
Mercury	0.0015	13534	204.699

Table 58: Sizing Trade Study for Damping Fluids [116] [117]

Table 58 shows the results of this analysis for our selected fluids. As you can see, a larger dynamic viscosity allows for a more uniform ratio, closer to 1. This makes choices like DC-200 1000 cSt and Krptox GPL 107 more attractive. However, from table 57, we can see that DC-200 1000 cSt has a considerably lower temperature coefficient, meaning its viscous properties are less susceptible to temperature changes. As a result, given its resilience in high-radiation conditions, we are choosing DC-200 1000 cSt as our fluid for our passive nutation damper.

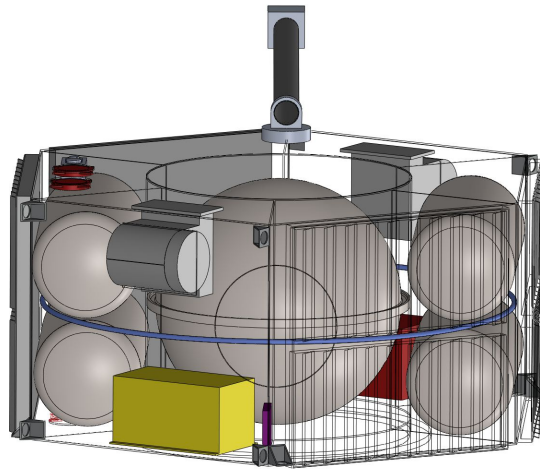


Figure 18: CAD with nutation damper (in blue), front view

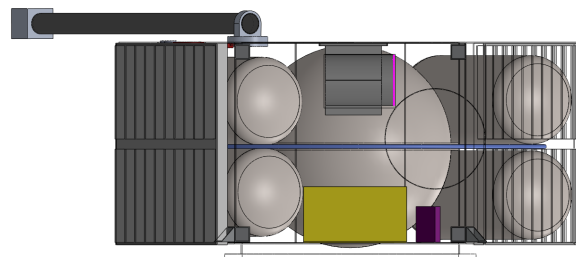


Figure 19: CAD with nutation damper (in blue), side view



Figures 18 and 19 show the sized damper integrated into the CAD model from a front and side view, respectively, verifying it does not interfere with other subsystems. To reiterate, this is assuming an outer radius of 0.49 m, a thickness of 5 mm, and the use of DC-200 1000 cSt fluid.

7.7.5 Thruster Analysis

Analysis conducted by: Joseph Norwood (analysis, writing of section).

Minimum Impulse Bit (MIB)

To narrow down the attitude thrusters' performance requirements, the Minimum Impulse Bit (MIB) required by the ADCS subsystem was calculated. The Minimum Impulse bit is defined by the European Cooperation for Space Standardization (ECSS) as "the smallest impulse delivered by a thruster at a given level of reproducibility, as a result of a given command" [120]. MIB directly determines how accurately you can control the attitude of your spacecraft, meaning a coarser resolution thruster (higher MIB) has worse pointing accuracy. Using the equations in Appendix A.6.1, we can solve for MIB. Since we want a total attitude control accuracy of 2 degrees, which includes the disturbances besides just thruster bit size, we can give a generous 0.5 degree error allowance to thruster bit error, while still ensuring we can achieve the desired attitude orientation. The only other variables we need are the radius of the thrusters to the center of mass and the moment of inertia both of which are found provided by the SMM subteam and can be found in Appendix A.6.2. With this information, we find MIB to be 1.1368 Ns, which is easily feasible MIBs of the ASCENT thrusters that PLV plan to use and analyze in their subteam PDR report. The MIB such thrusters is typically from < 1.6 mNs to < 0.1 Ns [121] meaning they offer significantly higher granularity than the minimum required for our mission's attitude control.

Control Frequency

Next, we can further narrow down the ADCS-related thruster requirements by examining the control frequency, or the minimum and maximum angular control permitted by our satellite's dynamics.

First, the nutation frequency limits the maximum permissible bandwidth before the satellite starts acting as a 3-axis vehicle as opposed to a spin stabilized vehicle [53]. This means we already have our maximum control frequency as it is just the nutation frequency value calculated in Section 7.7.4. This upper control frequency is 7.612 rpm, 0.797 rad/s, as seen in Table 59, and is the same for all the groups of satellites as they all have the same moments of inertia and spin rate.

The lower control frequency bound is provided by the orbit rate [53]. This value tells us how slowly we can control the satellite's attitude before disturbances overpower our control inputs. Calculating this lower bound using the equation:

$$\omega_o = \sqrt{\frac{\mu}{a^3}} \quad (17)$$

The resulting calculations vary for each orbit and are seen in Table 59.

Taking these results, we can then calculate the max torque we can apply on the system with the following equation [53]:



Table 59: Control Frequency Bounds

Parameter	RPM	rad/s
G1 Lower Control Frequency Bound	6.94e-4	7.27e-5
G2 Lower Control Frequency Bound	1.04e-3	1.09e-4
G3 Lower Control Frequency Bound	1.39e-3	1.45e-4
G1, G2, G3 Upper Control Frequency Bound	7.61e0	7.97e-1

$$T_{max} = \omega_n \cdot h = \omega_n \cdot I_z \cdot \omega_s \quad (18)$$

We find that the maximum control torque is 49.014 Nm, which is very large. If all our thrusters are mounted perpendicular to the spin axis that means the max allowable total thrust is 83.899 N. For 4 thrusters firing simultaneously, they can each produce a maximum of 20.975 N.

Orientation

The ADCS constraints for the thruster orientation are that the thruster orientation provides 3-axis control, as specified in Section 7.7.1. Working with the GNC, POW and SMM subteams, the thruster orientation in Figure 48a was chosen, which has 8 100 mN thrusters for attitude control and 4 1 N thrusters for GNC maneuvers. This orientation will allow pure attitude control as every thruster has an opposite canceling out the force on the center of mass. Depending on the desired attitude control or a GNC maneuver is being performed, thrusters will be fired according to Figure 48. Spin up and increasing the spin rate will be done using the thruster firings of 48c and the opposite firing configuration will be done for spin down and decreasing the spin rate. Slewing the spin axis can be performed as shown in Figure 48d using PWM control when the thrusters are aligned with the desired slewing direction. In case of thruster failure, the 1 N thrusters can also act as redundancies as they two can perform axis slewing maneuvers if only two are fired at a time. The choice of 1 N and 100 mN thrusters are well within the required control and this orientation provides the desired 3-axis control. POW is responsible for verifying MIB compliance and SMM for verifying the thruster plume doesn't impinge the satellite's instruments and sensors.

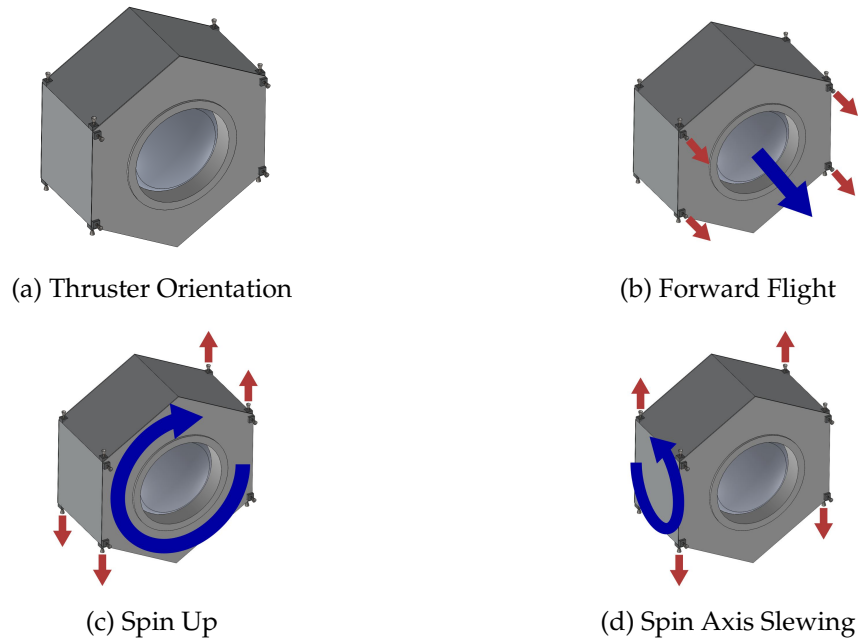


Figure 20: Thruster orientation and different firing maneuvers.

7.7.6 Attitude Delta-V Budget

Analysis conducted by: Joseph Norwood (analysis, writing of section), Ethan Spain (analysis, writing of section), Andrew Simon (analysis).

An important analysis that was omitted from our PDR was an ADCS ΔV budget, which accounted for various maneuvers required for the system (such as pointing and spin-up) as well as losses due to firing misalignment. This section intends to show how these issues were addressed by dividing the analysis into three sections: **spin-up**, **attitude corrections**, and **thruster misalignment**.

Spin-Up Costs

Via SMAD, spin-up costs can be estimated from the equation below

$$\Delta V = \frac{w_s I_s}{m} \quad (19)$$

where I_s is the moment of inertia around the spin axis, w_s is the spin rate and m is our mass [53]. From this equation, to reach our spin rate of 15 rpm, each satellite requires only 0.219 m/s of ΔV .

Attitude Correction

Costs for attitude correction can be determined by summing the costs for both sun-pointing maneuvers and disturbance rejection. From SMAD, these costs can be estimated as J/m , where J is the required impulse for a maneuver [4]. The disturbance rejection impulse was calculated in Section 7.7.3 and the required impulse to keep the spin axis angled at the sun is calculated with the equation

$$J = w_s I_s \alpha \quad (20)$$



where I_s is the moment of inertia around the spin axis, w_s is the spin rate, and α is the angle of rotation of the spin axis [53].

Per orbit, these Δv costs sum to around 0.00626 m/s.

Group #	Delta-V/year (m/s)	Total Delta-v (m/s)
1	2.285	13.711
2	3.428	20.566
3	4.570	27.421

Table 60: Attitude Correction ΔV

As seen in Table 60, these costs over the mission timeline sum up to small amounts, but are pretty consistent across different groups. However, as shown below, most of the ADCS ΔV budget will be allocated to scenarios involving thruster misalignment.

Thrust Axis Misalignment

Misalignment between the thrust axis and the spacecraft's center of mass causes attitude error whenever any thrusters are used. Over the course of the 5-year science mission, the routine orbital maintenance maneuvers will continually affect the attitude of our spacecraft. Changes in angular impulse, ΔH , can be estimated via $\Delta H = m\Delta V L_{cm}$, where L_{cm} is the offset between the center of mass and the thrust axis [4]. This change in impulse, as performed above with the disturbances, can then be used to estimate ΔV costs required to deal with the effects of thruster misalignment.

Group #	Delta-v cost (m/s)
1	124.078
2	160.210
3	224.190

Table 61: Thrust Axis Misalignment ΔV

As expected, table 61 shows that these corrections are more costly for Group 3, since more ΔV is used in that group for station-keeping purposes. Once again, the nominal orbit operations for Group 3 become the limiting factor for our entire mission, thus influencing fuel sizing and orbit maintenance operations.

Total Costs

Table 62 below summarizes the total ADCS ΔV costs estimated for each group. Because of the thruster misalignment calculations, these costs become a significant part of the overall ΔV budget and must be taken into account alongside maneuvers analyzed by GNC.

Group #	Total cost (m/s)
1	138.008
2	180.996
3	251.831

Table 62: Total ADCS ΔV Costs



7.7.7 Instrumentation

Instrumentation analysis aimed to choose the specific instruments that will be used on our spacecraft for attitude determination and started with a general examination. Looking at the possible forms of instrumentation in the NASA State-of-the-Art 2025 Report [122], we find the information in Table 63.

Sensor	Accuracy	Mass (kg)	Power (W)	Cost	Best Orbit	Missions Used
Sun Sensor	Medium/Low	~0.5-1.5	<1.5	Low	Any	Cluster, THEMIS, MMS
Star Tracker	High	~0.5-2	<2	Medium	Any	Cluster, MMS
Magnetometer	Low	<1	<0.5	Low	LEO	Cluster, THEMIS, MMS
GPS	Medium	<0.5	~1	Low/Medium	LEO	MMS
IMU/Gyro	Medium	<0.5	~1	Medium	Any	MMS
Ground Tracking	Low	N/A	N/A	High	Any	Cluster, MMS, THEMIS

Table 63: ADCS Sensor Trade Studies [123]

Examining this table and remembering the heritage mission analysis in Section 7.7.2, we find that GPS and the magnetometer do not work for our highly elliptical orbit and ground tracking is not super accurate for attitude estimation. So, we can continue by performing a more detailed analysis of star trackers, sun sensors, and IMUs.

Star Trackers

Star trackers offer the most accuracy of any attitude determination method making them highly desirable on any spacecraft; however, as previously mentioned in the Section 7.7.2, there were some concerns about the effects of spin stabilization and the corresponding motion blur on this sensors performance. As specific instrumentation was examined, this risk became a reality as cheap star trackers can only function at angular rates of up to 0.3 deg/s, while more expensive star trackers can function at most at 3.0 deg/s [122]. Experimental techniques may be able to further push the bounds of these devices, but our 15 rpm (90 deg/s) spin rate is an order of magnitude higher than the optimistic 3.0 deg/s functionality mentioned in NASA's State-of-the-Art report. This unfortunate reality means star trackers are not a viable option for our satellites.

Sun Sensors

As we don't have star trackers, sun sensors are crucial for maintaining coarse alignment with the sun for our solar panels. Luckily, as seen in the NASA State-of-the-Art report, sun sensors are closing the gap between them and star trackers. A trade study of the accurate sun sensor options in the NASA SOA is seen in Table 64.



Table 64: Sun Sensor Comparison

Manufacturer	Model	Mass (kg)	Peak Power (W)	FOV	Accuracy (3 σ)	Radiation Tolerance (krad)	Ref.
Needronix	Eagle Plus	0.005	< 0.015	128 $\times$ 128 $^\circ$	< $\pm 0.1^\circ$	>20	[124]
Bradford Space	Mini-FSS	0.05	–	128 $\times$ 128 $^\circ$	$\pm 0.2^\circ$	20000	[125]
NewSpace Systems	NFSS-411	0.035	0.15	140 $^\circ$	$\pm 0.1^\circ$	20	[126]
Redwire Space	Digital Sun Sensor ($\pm 64^\circ$)	Sensor: 0.25 Elec: 0.29–1.1	0.5	128 $\times$ 128 $^\circ$	$\pm 0.25^\circ$	100	[127]
Redwire Space	Fine Spinning Sun Sensor ($\pm 64^\circ$)	Sensor: 0.109 Elec: 0.475–0.725	0.5	64 $^\circ$ fan-shaped	$\pm 0.1^\circ$	100	[128]
Redwire Space	Miniature Spinning Sun Sensor ($\pm 87.5^\circ$)	<0.25	0.5	$\pm 87.5^\circ$ (from normal)	$\pm 0.1^\circ$	100	[129]

Table 64 was made primarily taking into account the accuracy and FOV of the sensors, with mass, peak power, and radiation tolerance being secondary considerations. Notably, two of these sensors, the Redwire Space Fine & Miniature Spinning Sun Sensors, were designed specifically for spin-stabilization. This specification along with the highest precision ($\pm 1^\circ$) means the choice becomes between the Fine and Miniature Spinning Sun Sensors. Normally, the larger FOV of the Miniature would make it the clear choice, but because our satellite's spin axis will be sun-facing it is actually better to choose the Fine Spinning Sun Sensor. These two sensors give the sun aspect angle and the sun crossing pulse [128], [129], but the sun-facing spin axis makes it more difficult to get a sun crossing pulse since the sun wouldn't leave the sensor sight. By using a smaller FOV and placing the sensor off-bore, we can get a reliable sun pulse making the best choice of sun sensor the Redwire Space Fine Spinning Sun Sensor.

IMU

IMU will mainly serve to add redundancy to our attitude estimation and help propagate the attitude estimation between the sun sensor signals. To determine the specific IMU to use, the trade study in Table 65 was conducted.

Table 65: IMU Comparison

Manufacturer	Model	Technology	Mass (kg)	Power (W)	Gyro Bias Stability ($^\circ/\text{hr}$)	ARW ($^\circ/\sqrt{\text{hr}}$)	Accel Bias Stability (μg)	VRW ($\text{m/s}/\sqrt{\text{hr}}$)	Ref
Honeywell	HG1700	RLG	0.95	12.5	1	0.125	1000	0.65	[130]
Honeywell	MIMU	RLG	4.73	20	0.05	0.01	100	–	[131]
Silicon Sensing	DMU41	MEMS	<2	<1.5	0.1	0.015	15	0.05	[132]
Safran	STIM320	MEMS	0.057	2.5	0.3	0.13	0.3	0.015	[133]

The driving factors of IMU selection are the Gyro Bias Stability and ARW which determine the accuracy of the attitude estimation. Power must also be noted as some IMU draw a significant amount of Watts. Looking at all the IMU's, the Honeywell MIMU has the least Gyro Bias Stability (0.05 $^\circ/\text{hr}$) but also has a 20W power draw, which is very large for a sensor that will be continually active. The next best option is the Silicon Sensing DMU41 with a Gyro Bias Stability of 0.1 $^\circ/\text{hr}$ and a more reasonable power draw of <1.5 Watts. The Silicon Sensing DMU41 acceleration bias is worse than the Safran STIM320's bias, but the accelerometer will be used only for tracking maneuvers and not for attitude detection. So we choose the Silicon Sensing DMU41 IMU for our subsystem.



7.7.8 Control Algorithms

Analysis conducted by: Josephine Kwok (analysis, writing of section), Joseph Norwood (initial analysis).

Control algorithms used for state estimation is important for controlling and understanding the satellites' attitudes. As discussed in the initial subsystem report, we still plan to use the Multiplicative Extended Kalman Filter (MEKF) and a proportional-derivative (PD) controller.

The decision to use MEKF comes from its use on heritage mission MMS [5]. There are other Kalman filters used for state estimation, the Extended Kalman Filter (EKF) and the Additive Kalman Filter (AKF), but we also chose MEKF because its lower dimension co-variance matrix and therefore, computational complexity [134]. A smoother in addition to the Kalman filters will also not be necessary given our lenient pointing requirements and use of spin stabilization.

For attitude control, the PD controller was selected over the more sophisticated Linear Quadratic Regulator (LQR). Literature consistently identifies the PD controller as the most mature and widely deployed method in spacecraft attitude control [135], and the passive spin stabilization of this mission further reduces the burden placed on active control. LQR offers theoretical optimality by minimizing a quadratic cost function over the linearized system dynamics, but this requires solving the Algebraic Riccati Equation (ARE) at each timestep. For this mission, maintaining LQR optimality would demand periodic ARE re-solves at $\mathcal{O}(n^3)$ cost per cycle. Given the lenient pointing accuracy requirements, this additional computational burden purchases no meaningful performance benefit. The PD controller achieves the required attitude maintenance at $\mathcal{O}(n)$ cost per cycle. Table 66 summarizes the trade off of the discussed control methods.

Property	PD	LQR	EKF	AKF	MEKF
Algorithm Type	Controller	Controller	Estimation	Estimation	Estimation
Covariance Size	-	-	7x7	7x7	6x6
Computational Complexity	$\mathcal{O}(n)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^3)$	$\mathcal{O}(n^3)$	$\mathcal{O}(n^3)$
Storage Complexity	$\mathcal{O}(n)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$
Linearization Cost	Already Linear	$\mathcal{O}(n^3)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$
Heritage	MMS	Limited	Yes	Limited	MMS

Table 66: Control Algorithms Trade Study [134][136]



8 SMM Design

8.1 Subsystem Overview

The Structures and Mechanisms (SMM) subsystem is responsible for designing the spacecraft bus that houses all components across the eight MATRIX satellites. The structure must withstand the vibrational, acoustic, and quasi-static loads imposed by the launch vehicle, as well as the mechanical loads generated during on-orbit thruster firings and spin-stabilized operation.

The SMM subsystem is also responsible for material selection across the spacecraft chassis. Chosen materials must satisfy strength, radiation tolerance, and thermal requirements as defined in the mission-level requirements to ensure the survival of the spacecraft throughout the 6 year-long mission.

In addition, SMM is responsible for the design and reliable actuation of the magnetometer boom deployment mechanism. The boom must achieve the required separation distance and angular position in a single, non-reversible actuation, as a failed or partial deployment would irreversibly compromise magnetic field measurement quality.

Finally, SMM is responsible for the mass and volume budgets across the full team, ensuring each subsystem has sufficient allocation to fulfill its function. SMM also determines the internal configuration of the spacecraft bus, ensuring all components can be accommodated with adequate clearance and correct center-of-mass placement to support spin-stabilization.

The SMM subsystem is mission-critical across all phases, as structural failures are unrecoverable and directly determine whether instruments can operate as intended. Structural integrity, reliable mechanism deployment, and adequate environmental protection are prerequisites for scientific return; without them, no other subsystem can function as intended.

8.2 Subsystem Objectives

The Structures, Mechanisms, and Materials subsystem exists to enable all other subsystems to perform their critical functions. With no mission objective changes since the PDR, the subsystem objectives are to:

1. Provide the physical structure that contains and supports the science payloads and all other subsystems throughout the entire mission duration
2. Allocate mass and volume budgets to all subsystems
3. Integrate the components of all other subsystems into the spacecraft chassis
4. Provide structural integrity to survive launch, flight, and vibration loads
5. Provide protection against the operational environment, including radiation, gravitational, and thermal factors
6. Provide reliable deployment mechanisms for instruments

Providing physical structure, integrating subsystem components, and ensuring structural integrity are prerequisites for reaching operational state. Environmental protection directly supports the



mission's goal of quantifying magnetic field and particle signatures associated with BBF braking, as electronics and instruments must survive six years of radiation exposure to enable reliable long-term measurements. Finally, reliable boom deployment supports both characterizing BBF spatial and temporal structure and quantifying BBF-associated signatures, since full deployment is required to achieve the measurement geometry and magnetic cleanliness distances needed for accurate in-situ data — a failed deployment on any spacecraft directly reduces constellation-wide multi-point measurement capability.

8.3 Subsystem Requirements

The requirements of the SMM subsystem can be broken down into four main categories: functional, performance, constraints, and environmental. No changes have been made to the mission since the PDR, so the SMM subsystem requirements remain unchanged. The derivation of the requirements can be found in the previous report [137].

Table 67 shows some of the fundamental requirements. A list of all requirements, along with trace up requirements and relation with subsystem objectives, can be found in the SMM tab of the "Requirements Spreadsheet" Google Sheet [138].

ID	Requirement	Rationale	Method	Trace	Obj
SMM-F-001	The SMM subsystem shall design a spacecraft bus to house mission components from all other subsystems for the duration of the mission	Consolidate architecture into a bus to house all components.	Analysis	MISS-F-001 to 003	Obj 1
SMM-F-006	The SMM subsystem shall provide deployment and articulation mechanisms for the magnetometer boom as required	The magnetometer boom must be deployed in order to perform its science function.	Analysis and Modelling	MISS-F-001 to 003	Obj 5
SMM-F-007	The SMM subsystem shall ensure the boom deployment mechanism is non-reversible upon successful actuation	A failed or partial deployment is unrecoverable; the mechanism must latch reliably in the deployed state to prevent mission-critical loss of science data.	Mechanism Design	MISS-F-001 to 003	Obj 5
SMM-P-001	The structural materials selected by the SMM subsystem shall have a density of at least 649.2 kg/m^3	Material density determines the total architecture mass.	Analysis	MISS-C-005, 004	Obj 4
SMM-C-001	The structures designed by the SMM subsystem shall have a wet cumulative mass of at most 285 kg	Compliance with mission mass budget.	Mass Budget	MISS-C-005	Obj 6
SMM-E-003	All structures designed by the SMM subsystem must withstand the Launch Vehicle axial load environment of -4 to 7 g during transit from the Earth to the desired orbit	Ensure survival during Launch Vehicle transit [139].	Load Analysis	MISS-E-002	Obj 3

Table 67: SMM Subsystem Requirements

8.4 Subsystem Constraints

The SMM constraints are unchanged from the PDR report, as there haven't been changed in the mission constraints since the PDR. [140].



- **Launch Vehicle Loads and Vibrations:** The structure must survive mechanical loads during liftoff, ascent, staging, and separation, constraining stiffness, strength, interface design, mass, volume, and center-of-mass limits.
- **Propulsion Loads:** On-orbit maneuvers must be transmitted through the bus structure without yielding or fatigue failure.
- **Thermal and Radiation:** Thermal cycling and magnetotail radiation drive material selection for thermal fatigue and cumulative dose tolerance.
- **Mass and Volume:** The structure must comply with launch vehicle dimensional limits while accommodating all subsystem mass and volume requirements.
- **Stiffness and Frequency:** Fundamental frequencies must avoid resonance with the launch vehicle during ascent.
- **Cost:** Structural design must remain within the allocated share of the MIDEX-class \$350M budget.

All constraints trace to mission-level drivers: the \$350M budget bounds material selection and design complexity; mass, volume, stiffness, and frequency requirements flow from launch vehicle selection; and the science orbit defines the radiation and thermal environment.

8.5 Subsystem Drivers

- **Eight-Spacecraft Co-Launch Configuration:** Launching all eight satellites simultaneously on a single launch vehicle imposes strict per-spacecraft mass and volume allocations, driving the structural design toward a compact, lightweight bus while still accommodating all subsystem components and maintaining sufficient stiffness and strength margins.
- **Spin-Stabilized Configuration:** The spacecraft operates in a spin-stabilized attitude, requiring the structural design to maximize the moment of inertia about the spin axis while maintaining precise center of mass placement. This drives the overall mass distribution strategy and the adoption of a central thrust tube as the primary load path, providing the stiffness and structural robustness necessary to sustain spin-stabilized operation throughout the mission.
- **Radiation Environment:** The highly eccentric orbits — with apogees reaching up to $12 R_E$ — result in repeated passes through the inner and outer radiation belts, accumulating significant TID over the mission duration. This drives material selection, wall thickness trades, and the localized Z-graded spot shielding strategy adopted for the most radiation-sensitive components.

8.6 Subsystem Design Approach

The previous design cycle established a consolidated spacecraft bus structure with a preliminary component layout, including instruments fuel and pressurant tanks, and batteries. An updated FEA analysis was performed to simulate expected stresses under launch loads, and an updated mass budget was compiled. A preliminary radiation shielding analysis was also performed, though refinement was required due to mass growth. Finally, a magnetometer boom design was



completed, while deployment mechanism trade studies were conducted and a concept selected, with further analysis required to finalize the design [140]. The main results of the analyses which this design doesn't reiterate on, such as material selection, as well as the ones from the subsystem report can be found in A.7.

This cycle builds upon those designs — with the spacecraft structure unchanged — increasing fidelity by incorporating all remaining subsystem components, specifically the secondary propellant tanks and louvers, with the existing layout refined to accommodate these additions.

Four key tasks were conducted in this design cycle. First, a refined radiation shielding analysis was performed, with spot shielding strategies being the new additions, addressing requirements SMM-P-003 and SMM-E-001.

Second, a structural layout analysis was then conducted to determine final component placement within the spacecraft bus, directly addressing requirements to house all subsystem components (SMM-F-001), provide attachment means (SMM-F-002), and configure their placement (SMM-F-004). Updated mass and volume budgets were compiled to account for any trades conducted between subsystems to comply with allocations (SMM-C-001, SMM-C-002).

Third, updated stress and modal analyses were conducted to verify structural integrity and frequency margins under the expected launch environment, targeting survivability requirements SMM-E-003 through SMM-E-005 and ensuring the spacecraft bus can withstand vibrational, axial, and lateral launch loads.

Fourth, a refined magnetometer boom design was developed following updated OPL specifications, accompanied by a refined deployment mechanism study addressing requirements SMM-F-006 and SMM-F-007.

Together, these tasks advance the subsystem from a preliminary design toward a finalized and verified design, compliant with the mission requirements.

8.7 Formal Analysis

8.7.1 Radiation Shielding Analysis

Analysis conducted by: **Gina Arnau, All categories**

In the PDR report, aluminum was identified as the most suitable material for radiation shielding due to its favorable balance between shielding performance, density, and flight heritage. However, the only mitigation strategy considered at that stage was increasing the spacecraft bus wall thickness, which led to a significant mass penalty, bringing the spacecraft close to its maximum mass constraint.

In this design cycle, alternative lightweight shielding strategies were investigated, including localized spot shielding and multi-layer shielding materials, with the aim of improving radiation protection while minimizing mass growth.

Instead of uniformly increasing the wall thickness of the entire spacecraft bus, shielding is now localized to protect the most radiation-sensitive components, such as onboard electronics and batteries. These components are particularly vulnerable to radiation effects, including performance degradation and single-event effects (SEEs), which can significantly impact mission reliability. The

wall thickness was consequently reverted to the baseline 3 mm established in the first subsystem report, which was previously shown to be sufficiently robust and stiff to survive launch loads, while also offering manufacturing advantages and a favorable mass-to-strength ratio [137].

To verify that the radiation flux does not induce critical mechanical degradation of the spacecraft structure with these thinner walls, an analysis was conducted to evaluate the degradation of the material using SPENVIS. SPENVIS models the displacement damage of the material due to irradiation. The resulting relative material degradation for a 3 mm thick aluminum wall was 0.429 %, which is a small amount.

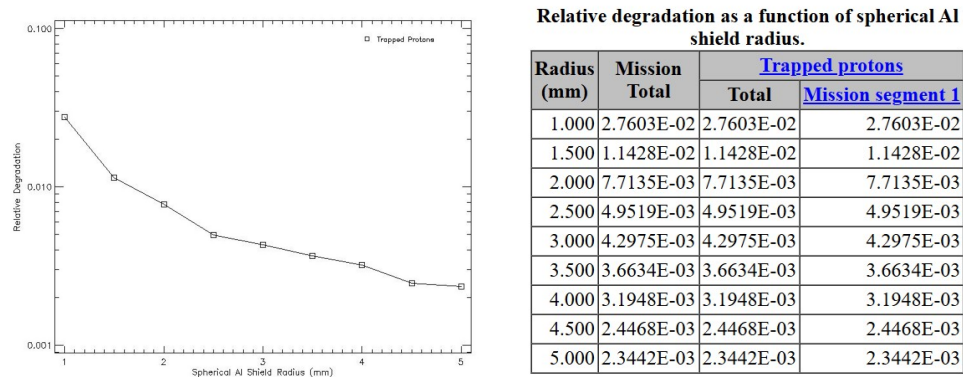


Figure 21: Relative structural degradation due to radiation

However, as per the PDR report, the total ionizing dose (TID) through a 3 mm aluminum wall was found to be 1165 krad, significantly exceeding the mission requirement of 396 krad. Additional shielding measures were thus required for the most sensitive components — namely the batteries and the avionics units.

According to literature, multi-layer configurations outperform single-layer materials by reducing secondary radiation production and improving attenuation across a broader range of particle energies [141]. Of particular interest are Z-graded materials, which consist of layers of materials with varying atomic numbers [142].

These materials are especially attractive for spacecraft applications due to their low density, flexibility, and adaptability to complex geometries. This makes them well-suited for protecting localized, high-risk components where traditional bulk metal shielding is impractical [143]. As a result, combining a reduced baseline aluminum thickness with targeted spot shielding and Z-graded multi-layer materials provides a more mass-efficient and effective radiation protection strategy, as shown below:

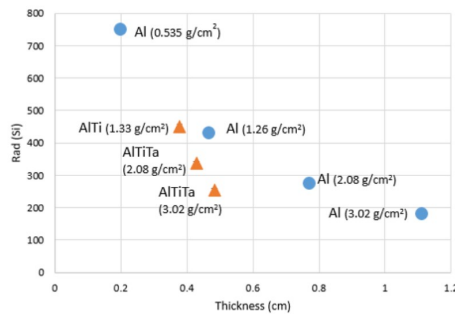


Figure 22: Radiation properties of various Z-graded materials vs Al and thickness (reproduced from [143])

Looking at the figure above, the AlTiTa looks like the most promising material, as it provides a good balance between low density and radiation attenuation. Additionally, this material has been proved to work in NASA's Shields-1 mission [144].

Consequently, the batteries and the BAU will be enclosed in z-grade shielding vaults. The MULASSIS tool in SPENVIS was used to simulate the TID the electronics would receive behind this vault. Following the Shield's 1 mission, a four layer z-grade material was simulated, consisting of Al, Ti, Ta, Ti layers, with a total thickness of 2.0 mm [144].

The shielding stack is ordered to address each radiation type in sequence. The outer aluminum layer intercepts the most energetic particles, followed by a titanium layer — a low-Z material — which attenuates the remaining proton flux. A tantalum layer then targets electron radiation, leveraging its high atomic number for effective attenuation. However, tantalum generates secondary Bremsstrahlung radiation, so a final titanium layer is introduced to attenuate this secondary flux before it reaches the enclosed components. In the simulation, the 3 mm aluminum outer wall was also included in the model to obtain more accurate TID estimates at the innermost layer.

The simulation was ran on both electrons and protons radiation. While the mission did not define how thick each layer was specifically, numbers were estimated to balance out mass growth and radiation shielding. Various configurations of thickness were tested, and the best result was having 0.3 mm of Al, 0.6 mm of Ti, 0.5 mm of Ta, and 0.6 mm of Ti. The resulting TID can be observed below.

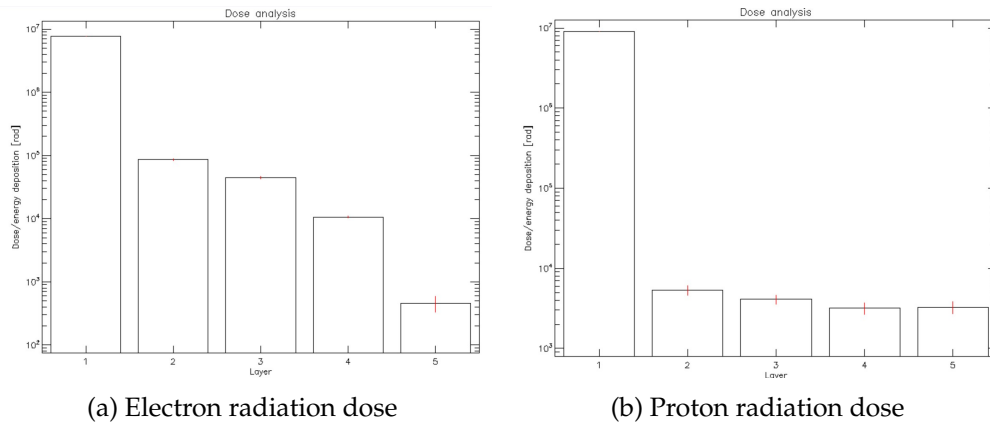


Figure 23: Z-graded shielding TID simulation results for electrons and protons

The MULASSIS results were that the resulting TID ranges from 457 to 3,734 rad, which is significantly below the 396 krad requirement, demonstrating a substantial shielding margin for the enclosed components.

With the 3 mm wall thickness, the structural mass of each spacecraft is 43.57 kg. The BAU and battery spot shielding enclosures, with dimensions of $213 \times 150 \times 154$ mm and $137.9 \times 137.9 \times 260.1$ mm respectively, contribute a combined additional mass of 5.2 kg, bringing the total structural mass to 48.8 kg. Against the SMM subsystem mass allocation of 65.6 kg, which yields a margin of 26%.

8.7.2 Structural Design

Analysis conducted by: **Anthony Matimu, All categories**

At the PDR stage, a number of subsystem components were yet to be accounted for. Since then, the final selection, as well as changes to previous selections, have warranted a refined structural design. The overall structure of the spacecraft remains unchanged with a central load-bearing thrust tube sized to match the standard 24 inch payload interface and radiating wall panels that serve as mounting for components and outer wall panels (Figure 24). The thrust tube is the primary load path during launch, carrying axial and lateral loads from the separation interface through to the spacecraft bus. The wall panel thickness is 3mm as described in Section 8.7.1, selected to meet radiation shielding requirements while minimizing mass. The interface and separation mechanism (24 inch Sierra Space QwkSep system [145] which sets the spacecraft mass limit at 285 kg based on a manufacturer specified mass limit of 300 kg with a margin of 5%) is unchanged from the PDR. It should be noted here that the mission uses a stacked ESPA-derived payload adapter in order to accomplish the deployment of 8 spacecraft in a single launch. As the structure was modified from original specifications to match the LSP launch vehicle interfaces, a brief structural analysis is shown in Appendix A.7.2 to validate the structural design. An overview of the subsystem components and their layout is shown in Figure 25.

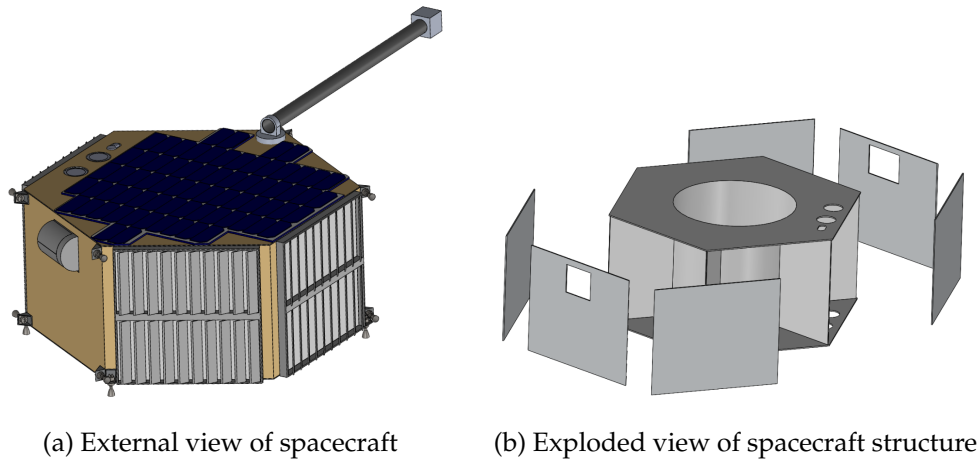


Figure 24: Views of spacecraft structure

Table 68: Spacecraft Structural Parameters

Parameter	Value
Primary Structure Type	Cylindrical thrust tube with hexagonal outer walls
Total Spacecraft Mass	285 kg
Internal Volume	0.439 m ³
Thrust Tube Diameter	24 in (0.6096 m)
Side Length	23 in (0.584 m)
Height	20 in (0.508 m)
Wall Thickness	3 mm
Material	Aluminum 7075-T6

The most notable change from PDR comes from the propellant requirements provided by the PLV subteam. A significant increase in required propellant volume is addressed by the incorporation of 4 custom ellipsoidal head tanks. Tank sizing was conducted by the PLV subteam and shown in Appendix [A.10.3](#). The use of 4 tanks allows the additional propellant mass to be distributed symmetrically about the spin axis without requiring unconventional tank geometries such as a toroidal tank, which would introduce analysis and manufacturing complexity. Symmetric placement is critical for this mission architecture, as mass imbalance about the spin axis directly affects spin stability and pointing performance. The placement of other critical components, as well as rationale, is described below:

- *Instrumentation:* The ESA and EPD are placed on the side opposite the separation mechanism to minimize shock exposure during the separation event. They are oriented such that spacecraft spin sweeps their fields of view, enabling full coverage of the region of interest without requiring active pointing.
- *Thrusters:* The 12 thrusters are arranged to provide full attitude control authority, as described in Section [7.7.6](#). Placement also accounts for plume impingement effects; each thruster is oriented away from spacecraft wall panels, instruments, and sensors to prevent contamination and avoid obscuring fields of view.

- *Pressurized tanks:* The propellant and pressurant tanks are the most massive components of the spacecraft. They are placed symmetrically about the spin axis and as close as possible to the geometric center to minimize the offset between the center of mass and the geometric center. This is critical for spin stabilization and pointing accuracy. The movement of the center of mass as propellant is consumed over the mission lifetime is also taken into account.
- *Sensors and antennas:* Sun sensors are located on the top face of the spacecraft, consistent with the spin-stabilized architecture in which the top face maintains a fixed orientation relative to the sun. Patch antennas are placed on both the top and bottom faces to ensure communication access throughout the orbit and across seasonal variations in geometry, as required by the communication coverage requirement.
- *Solar panels:* Solar panels are mounted on the top surface of the spacecraft, consistent with the sun-pointing mission architecture. Panel area and placement were sized to meet the power budget described in Section 10.7.

The spacecraft also features a deployable boom for the magnetometer instrument. The purpose of this boom is to place the magnetometer at a suitable distance away from the spacecraft to mitigate the effects of spacecraft generated magnetic moments. The boom is sized to be 1 meter from the articulation point to the magnetometer itself. The design of the boom is detailed in Section 8.7.4.

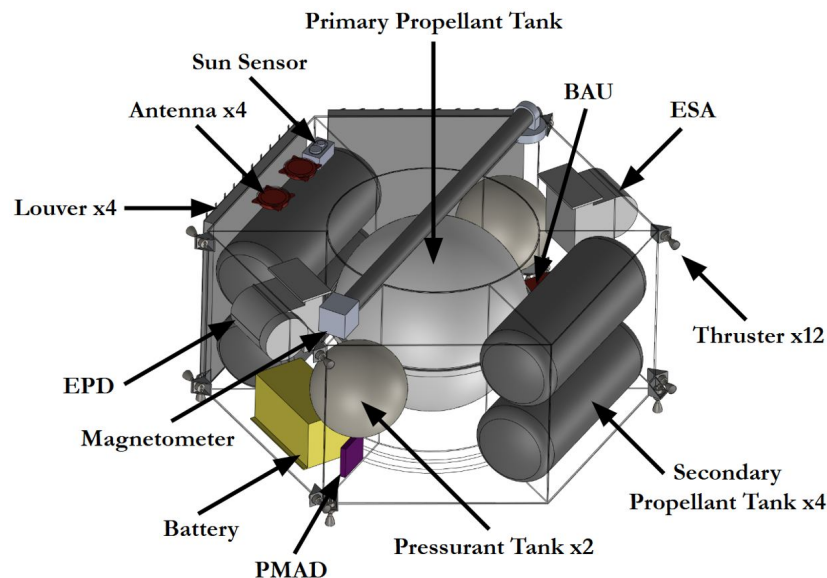


Figure 25: Overview of subsystem components

8.7.3 Static and Modal Analysis

Following the updated structural design, including the addition of the secondary propellant tanks and refinement of component placements, a new set of static and modal analyses was performed using the SolidWorks Simulation environment. All analyses use aluminum 7075 for the primary structure, aluminum 6061 for components (with density overrides to match their actual mass) and

their mounting, and G10 specifically for pressurized tank mounting.

Static Analysis Static analyses were conducted using axial and lateral launch loads derived from the NASA Launch Services Program (LSP) [139]. The spacecraft was constrained with a fixed boundary condition at the separation interface to represent the attachment to the launch vehicle adapter. Launch loads were applied as inertial accelerations of 7g axial and 3g lateral, mapped to the spacecraft body axes in accordance with mission environment requirement MISS-E-002.

A fine mesh was used throughout, with local mesh refinement applied in regions of expected stress concentration such as the thruster brackets and tank mounting brackets. The von Mises stress criterion was used to assess yielding. The results of the static analysis (Figure 26) indicate a maximum stress of 13.2 MPa and a positive margin of safety across all structural members, with no yielding observed under the applied launch loads. The highest stress concentrations are located at the pressurant tank mounting, which is presumably made of G10 (yield strength of 210 MPa). The resulting deformation was found to be 0.012 mm and thus negligible.

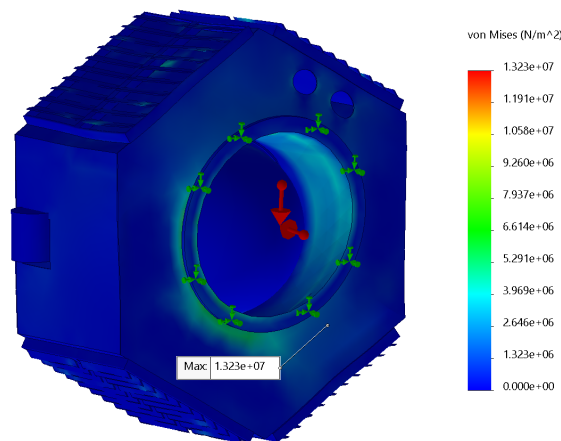


Figure 26: Von Mises stress distribution under combined axial and lateral launch loads

Modal Analysis (Launch Configuration) Modal analysis was performed on a simplified spacecraft model to isolate the primary structural modes of the spacecraft bus. The stowed magnetometer boom was excluded from this analysis, as its significantly lower stiffness relative to the primary structure causes it to dominate the low-frequency modes and obscure the structural response of the bus. The boom is analyzed separately under its own boundary conditions. A fixed boundary condition at the separation interface was maintained to represent launch conditions. The first 40 modes were computed, and modes with effective mass participation greater than 10% were identified as dynamically significant.

Modes 3 and 4 dominate the dynamic response, with significant mass participation in the Y and X directions respectively, as summarized in Table 1. Modes 1 and 2, while present at lower frequencies, have low mass participation and contribute minimally to the overall dynamic response.

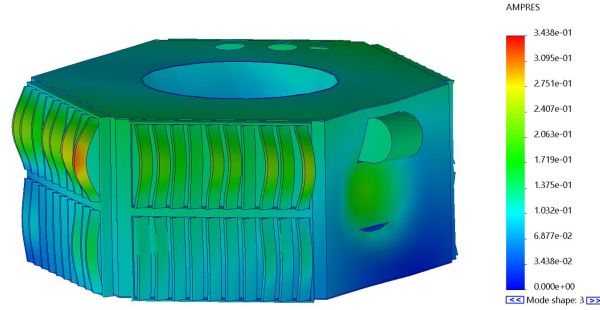


Figure 27: Mode 3 shape showing dominant Y-direction bending at 238 Hz

Mode	Frequency (Hz)	X Participation	Y Participation	Z Participation
1	189.37	0.01936	0.00002	0.00002
2	218.41	0.06651	0.00047	0.00107
3	238.17	0.01038	0.36376	0.00199
4	244.62	0.31416	0.02116	0.01008

Table 69: Effective mass participation factors for Modes 1–4

All modal frequencies exceed 100 Hz, which is well above the typical launch vehicle fundamental frequency envelope of 20–50 Hz. This provides adequate frequency separation from the launch vehicle and reduces the risk of coupled loads excitation at the interface.

Random Vibration Response The equivalent root mean square (RMS) acceleration response was estimated using Miles' equation, which provides a single degree of freedom approximation of the peak response to a random vibration input [146]:

$$G_{\text{RMS}} = \sqrt{\frac{\pi}{2} f_n Q \text{PSD}_{\text{input}}} \quad (21)$$

where f_n is the modal frequency, $Q = \frac{1}{2\zeta}$ is the dynamic amplification factor, and $\text{PSD}_{\text{input}}$ is the input acceleration power spectral density evaluated at f_n . A damping ratio of $\zeta = 0.03$ ($Q \approx 16.7$) was assumed, consistent with welded and bolted structures [147]. The 3σ response represents the statistically expected peak acceleration. A full plot of the random vibration specification is shown in Appendix A.7.1.

For the dominant modes at $f_n = 238$ Hz and $f_n = 244$ Hz, with $\text{PSD}_{\text{input}} \approx 1 \times 10^{-2} \text{ g}^2/\text{Hz}$, the resulting response is:

$$G_{\text{RMS}} \approx 7.93 \text{ g}, \quad G_{3\sigma} \approx 23.8 \text{ g} \quad (22)$$

To account for the fact that each mode contributes only partially to the total response, the RMS acceleration was scaled by the square root of the effective mass participation fraction for each mode. This gives a more representative estimate of the actual dynamic load experienced by the structure at each mode. For Modes 3 and 4, with participation factors of 0.36 (Y-direction) and 0.31 (X-direction) respectively:

$$G_{\text{eff,Mode 3}} \approx 14.3 g, \quad G_{\text{eff,Mode 4}} \approx 13.3 g \quad (23)$$

These effective accelerations are used as inputs to estimate the equivalent dynamic forces and resulting stresses at critical structural interfaces, including tank attachments and thruster brackets.

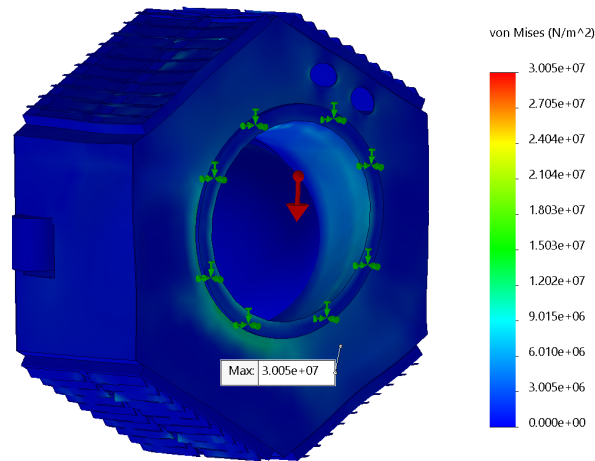


Figure 28: Representative dynamic response under random vibration loading

The result of a static analysis with the random vibrate response input is shown in Figure 28. In this case, a maximum stress of 30 MPa was found in the pressurant tank mounting. This value is still well within the yield strength of the mounting brackets as well as other areas of concern such as the base of the thrust tube, where relatively high levels of stress are also observed.

Modal Analysis (On-Orbit Configuration) A free-free modal analysis was conducted to evaluate the deployed configuration under on-orbit boundary conditions, where no external constraints act on the spacecraft. The first 12 modes were computed. The first 6 modes correspond to rigid-body motion (three translational and three rotational), which is expected for an unconstrained free-free analysis and serves as a check that the model is correctly set up. The remaining modes represent flexible structural deformation of the spacecraft bus and deployed boom.

The fundamental boom bending mode occurs at 11.5 Hz (Mode 7 shown in Figure 29). This is the most important on-orbit structural mode, as boom flexibility can couple with attitude dynamics and disturb pointing. The spacecraft nutation frequency is 0.797 rad/s (0.127 Hz), giving a frequency separation ratio of approximately 90 between the boom bending mode and the nutation frequency. This large separation indicates negligible dynamic coupling between rigid-body attitude motion and the structural response of the boom, and confirms that boom flexibility does not pose a risk to attitude stability during nominal operations.

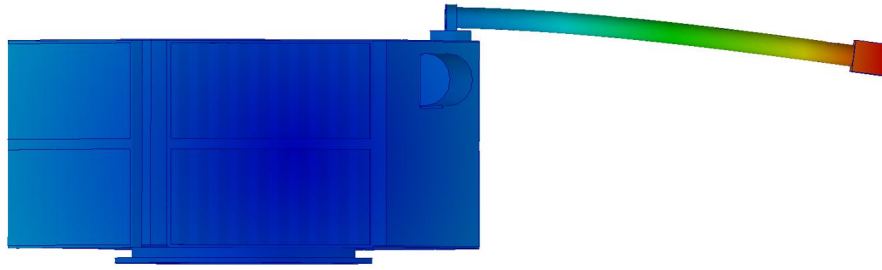


Figure 29: Free-free modal analysis showing deployed boom bending mode at 11.5 Hz

8.7.4 Magnetometer Boom Design and Analysis

Analysis conducted by: **Anthony Matimu:** Literature research, set-up of the analysis, writing of the section **Gina Arnau:** Simulink simulation research, set-up, and analysis. Writing of section

An updated magnetometer boom design was developed for the FDR following new input from the OPL team. The updated design utilizes a 1 meter boom with a damped torsional spring hinge (Figure 31). This mechanism was selected as it provides passive deployment without requiring a motor or external power input. The torsional spring stores energy in the stowed configuration and releases it upon deployment, while the damping element suppresses post-deployment oscillations critical for clean magnetometer measurements. The boom is to be deployed 180°. The damping and spring constants were computed to characterize the dynamic behavior of the hinge and ensure the boom deploys fully and settles within an acceptable threshold for magnetometer operations.

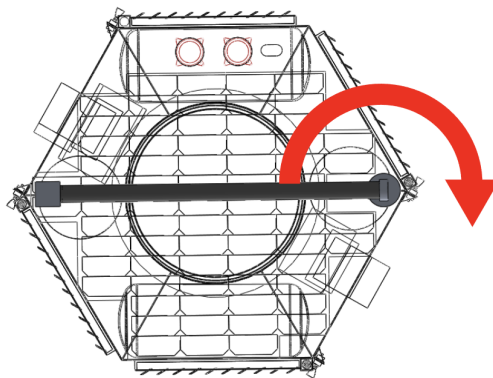


Figure 30: Range of motion for boom deployment

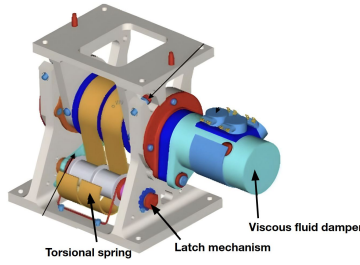


Figure 31: Model of an example HA-90 hinge mechanism [148]

The equation of motion for a damped rotary spring hinge is [149]:

$$I\ddot{\theta} + c\dot{\theta} + k\theta = 0 \quad (24)$$

Dividing through by I :

$$\ddot{\theta} + \frac{c}{I}\dot{\theta} + \frac{k}{I}\theta = 0 \quad (25)$$

Comparing with the standard form of the damped harmonic oscillator:

$$\ddot{\theta} + 2\zeta\omega_n\dot{\theta} + \omega_n^2\theta = 0 \quad (26)$$

the natural frequency and damping ratio are identified by matching coefficients:

$$\omega_n = \sqrt{\frac{k}{I}}, \quad \zeta = \frac{c}{2\sqrt{kI}} \quad (27)$$

For a boom of mass m_b and length L with a tip mass m_t , the total moment of inertia about the hinge is:

$$I = \frac{1}{3}m_bL^2 + m_tL^2 \quad (28)$$

where $\frac{1}{3}m_bL^2$ is the contribution from the distributed mass of the boom (uniform rod rotating about one end) and m_tL^2 is the contribution from the tip mass.

For $m_b = 0.26$ kg, $m_t = 1$ kg, and $L = 1$ m:

$$I = \frac{1}{3}(0.26)(1)^2 + (1)(1)^2 = 0.0867 + 1.0 = 1.0867 \text{ kg}\cdot\text{m}^2 \quad (29)$$

Once the moment of inertia has been computed, the damping ratio can be selected. A preliminary slightly underdamped ratio of 0.8 was chosen due to a fast settling time without harshly overshooting. Using that number as well as a deployment time of 60 seconds for a conservative

estimate, the deployment angular velocity, based on the 2% settling time criterion, can be computed:

$$\omega_n = \frac{4}{\zeta \cdot T} = \frac{4}{0.8 \times 60} = 0.083 \text{ rad/s} \quad (30)$$

Then, the spring constant can be found as:

$$k = \omega_n^2 \cdot I = 0.083^2 \cdot 1.0867 = 0.0075 \text{ N}\cdot\text{m/rad} \quad (31)$$

And the damping constant as:

$$c = 2 \cdot \zeta \cdot \sqrt{k \cdot I} = 2 \cdot 0.8 \cdot \sqrt{0.0075 \cdot 1.0867} = 0.144 \text{ N}\cdot\text{m}\cdot\text{s/rad} \quad (32)$$

A Simulink simulation was performed in MATLAB using the Torsional Spring-Damper model to verify that the computed values resulted in a system compliant with the requirements.

The set-up consisted of the torsional spring-damper from the Simscape Driveline MATLAB add-on. It essentially uses a Torsional Spring-Damper block connected between a fixed reference and an Inertia block representing the boom, with the parameters changed to the computed values. The spring is initialized with a deformation of 180° , representing the stowed position. An Ideal Rotational Motion Sensor measures the angular velocity, which is integrated and converted to degrees to produce the angular displacement.

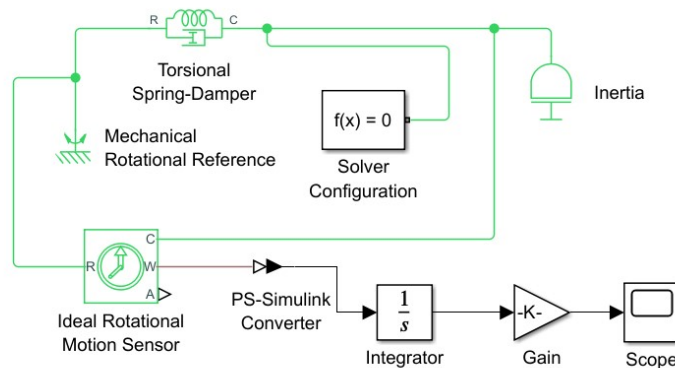


Figure 32: Simulink boom deployment simulation set-up

The simulation was run for 120 seconds, and the result can be seen in figure 33.

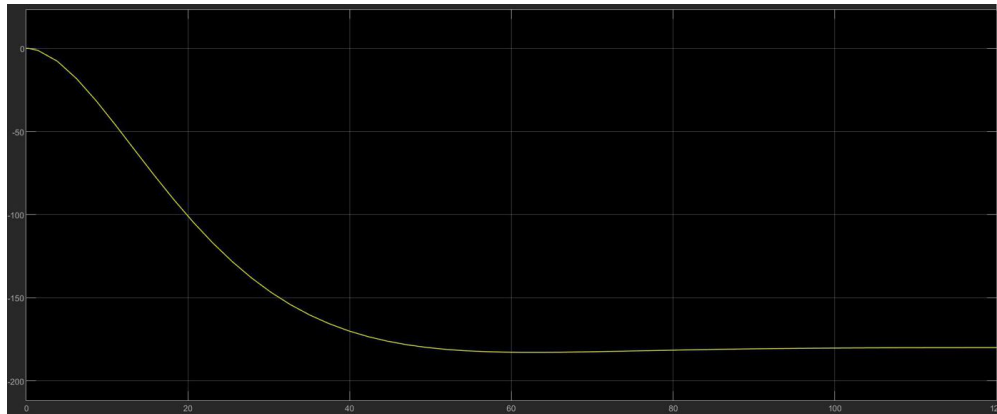


Figure 33: Angular displacement with respect to time

As we can observe, the boom deploys fully from 0° to 180° and settles within 120 seconds, satisfying SMM-P-010. The simulation confirms that post-deployment oscillations decay completely, with the boom settling at -180° with no oscillations at around 80 s, satisfying SMM-P-009.

The deployment torque was verified both through simulation using an Ideal Torque Sensor block and analytically:

$$T_{max} = k \cdot \theta_{max} = 0.0075 \times \pi = 0.0235 \text{ N}\cdot\text{m} \quad (33)$$

The simulated value of 0.0236 N·m shows agreement with the analytical result, validating the model.

Before deployment, the boom is held in the stowed configuration by a frangibolt (Figure 34) which connects the end of the boom to the top surface of the spacecraft. The frangibolt was chosen for its simplicity and reliability, as well as its high load capacity of upwards of 100,000 N depending on its configuration. The activation of the frangibolt releases the boom which articulates according to the previous analysis, and latches into the deployed position.



Figure 34: Frangibolt [150]



9 THM Design

9.1 Subsystem Overview

The Thermal (THM) subsystem is responsible for maintaining and tracking the thermal conditions of all spacecraft hardware across the eight MATRIX satellites. The subsystem estimates the thermal fluxes experienced during all mission phases, designs appropriate thermal insulation, rejection, and heat generation systems to maintain component thermal envelopes, and verifies thermal models through on-orbit and TVAC thermal monitoring.

The highly eccentric orbits, with apogees reaching up to $12 R_E$ and perigees near 1000 km, result in large, continuous thermal cycling as the spacecraft transition between sunlit and eclipse periods and between the near-Earth and deep magnetotail environments. The THM subsystem must ensure all components remain within their allowable thermal ranges throughout these environments across the ≥ 5 -year science phase.

Thermal solutions must adhere to the mass, volume, and power budgets defined by the SMM and POW subsystems. To minimize power requirements and the associated mass and volume implications of greater power storage, passive thermal solutions are preferred where appropriate, with active systems employed only where passive approaches are insufficient.

The THM subsystem is mission-critical across all phases, as components driven outside their allowable thermal ranges may suffer permanent degradation or failure. These outcomes are unrecoverable and directly determine whether instruments and supporting subsystems can operate as intended throughout the science phase.

9.2 Subsystem Objectives

The thermal subsystem aims to control the thermal condition of the spacecraft to ensure the thermal effects of the environment do not compromise overall mission objectives and success. With no mission objective changes since the PDR, the subsystem objectives are to:

1. Characterize thermal environments for the various subsystems and components throughout the key phases
2. Provide heat retention and dissipation capabilities such that components are within their specified operational and survival temperature ranges.
3. Monitor temperature of components and detect unexpected thermal conditions.

Thermal design is mission-inherent rather than objective-driven — every instrument and supporting subsystem depends on operating within its qualified temperature range, making thermal control a prerequisite for any scientific return. Objectives 1 and 3 are therefore critical to achieving all three mission objectives: accurate characterization of BBF spatial and temporal structure, radial evolution, and magnetic field signatures all depend on instruments operating within their qualified temperature ranges, and on anomalous thermal conditions being identified and addressed before they result in unrecoverable hardware degradation. Objective 2 directly supports this by ensuring heat retention and dissipation capabilities maintain components within their specified limits throughout the 6-year mission duration, preserving the continuous multi-point measurement capability required across the 6– $12 R_E$ transition region.



9.3 Subsystem Requirements

The requirements of the THM subsystem can be broken down into four main categories: functional, performance, constraints, and environmental. With no updates to the ConOps since the PDR, the THM subsystem requirements remain unchanged [151]. Table 70 shows some of the fundamental requirements. A list of all requirements, along with trace up requirements and relation with subsystem objectives, can be found in the THM tab of the "Requirements Spreadsheet" Google Sheet [138].

ID	Requirement	Rationale	Method	Trace	Obj
THM-F-002	The thermal subsystem shall provide heat retention and/or dissipation capabilities for the mission architecture.	The mission components must be maintained within the operational temperature range for the mission architecture.	Analysis and thermal system design	MISS-F-012	2
THM-P-003	The thermal subsystem shall maintain structural elements and solar panel surfaces to between -89°C to 89°C for the mission duration	Structural operating temperature range.	Analysis	MISS-E-003, MISS-F-012	2
THM-P-002	The thermal subsystem shall maintain Hydrazine fuel to between 4°C to 60°C for the mission duration	Hydrazine operating temperature range.	Analysis	MISS-E-003, MISS-F-012	2
THM-C-003	The thermal subsystem shall have an average eclipse power consumption of at most 55 W	Compliance with mission architecture power budget.	Analysis	MISS-F-009	2
THM-F-003	The thermal subsystem shall be capable of monitoring the temperature of mission components.	Knowledge of the temperature of mission components is necessary for temperature regulation.	Analysis and thermal system design	MISS-F-012	3

Table 70: Key Thermal Subsystem Requirements

9.4 Subsystem Constraints

With no changes to the mission since the PDR, the THM constraints are shown below:

- **Component Temperature Limits:** All components have manufacturer-defined operational temperature ranges, which drive the sizing of heaters, radiators, and insulation to prevent freezing during eclipse or overheating during peak dissipation.
- **Vacuum Environment:** The absence of convective heat transfer restricts thermal management to radiation and conduction through structural interfaces, significantly constraining design options.
- **Radiation Degradation:** Prolonged exposure to high-energy solar particles degrades thermal materials, requiring selection of materials that maintain sufficient thermal properties over the mission lifetime.
- **Budget Constraints:** The subsystem must remain within its allocated mass, volume, power, and cost budgets, limiting the size and quantity of thermal hardware that can be accommodated.
- **Simulation Constraints:** The subsystem must consider computational, time, and learning



curve constraints in determining how high-fidelity thermal simulations can and need to be. All thermal subsystem constraints flow down from mission-level drivers. The MIDEX \$350M budget cap bounds hardware selection and design complexity. Mass and volume allocations derive from the launch vehicle and spacecraft mass budget. Finally, the magnetotail orbit imposes environmental durability requirements, as thermal control hardware must withstand radiation, vacuum, and thermal cycling for the full mission lifetime.

9.5 Subsystem Drivers

The THM drivers are listed below with no changes since the PDR.

- **Power Budget:** The THM subsystem must remain within its allocated power budget, driving a preference for passive thermal control solutions over active systems, in order to minimize power draw.
- **Environmental Survivability:** The magnetotail orbit exposes the spacecraft to severe thermal cycling, driving the selection of coatings and insulation to maintain integrity and prevent thermally induced failure.
- **Reliability:** A thermal failure risks total mission loss, favoring simple, redundant thermal control architectures to ensure operation throughout the full mission lifetime.
- **Component Temperature Limits:** Each subsystem component has defined operational temperature limits, driving localized thermal management solutions across the spacecraft.

9.6 Subsystem Design Approach

The overall thermal design since the PDR report has involved correcting thermal calculations made, pivoting to a multi-nodal thermal model, and refining temperature sensor placement.

As a refresher, thermal design over the course of the semester has progressed as follows:

1. Characterizing the thermal environment based on ConOps and component-specific thermal envelopes
2. Trade studies on radiative insulation, conductive insulation, heaters
3. Itemizing MATRIX surface optical properties
4. Conducting single-node thermal model
5. Trade study on louvers to dynamically change optical properties to satisfy both worst-case operational cases
6. Conducting multi-node thermal model to refine and lower heater requirements based on component-specific thermal envelopes
7. Identifying placement of temperature sensors

Item 6 was conducted since the PDR. Additionally, corrections were made to items 3 and 4. However, these changes did not alter the broader conclusions derived from the single-node model.



Requirement THM-E-0042 requires the thermal team to accurately characterize the thermal environment MATRIX spacecraft will experience on-orbit. In the PDR, errors in view factors of these thermal fluxes were made. Adjusted view factors are described in this report.

Requirement THM-F-001 and THM-P-001 through THM-P-007 lay out the need to satisfy the thermal envelopes of various temperature-sensitive components on MATRIX. However, with the single-node model conducted in the PDR, meeting the non-structural temperature envelopes across the entire single-node space craft would require too much power. As such, a multi-nodal model is needed.

9.7 Formal Analysis

9.7.1 Thermal Environment Characterization

Analysis conducted by:

Arthur Balva All

On-orbit, MATRIX satellites experience solar flux, earth albedo, earth IR, and radiative loss to space. At various parts of a given orbit, different fluxes are experienced. The balance of these thermal fluxes, in addition to internal heat generation, dictate the temperature of the spacecraft. The equations governing these heat fluxes are listed in table 71.

Solar	$1353 Q_{su} = A_{sc} \cdot s * H_{su}$, Average $1353 \frac{W}{M^2}$
Earth IR	$Q_{et} = F_{et} A_{sc} H_{et}$, Average $216 \frac{W}{M^2}$
Earth Albedo	$Q_{er} = 0.36 F_{er} A_{sc} H_{et}$, Average $216 \frac{W}{M^2}$
Radiative Loss to Space	$Q_s = \epsilon \sigma A (T^4 - T_s^4)$

Table 71: On-orbit Thermal Flux Equations [152]

All MATRIX spacecraft operate in highly-eccentric ecliptic orbits. A task of the thermal subteam is to identify the worst-case high and low thermal fluxes into the spacecraft. These govern the greatest requirements of heat retention/generation and dissipation. The worse cold case occurs when apogee exists at eclipse. At this point, the spacecraft receives no solar flux and Earth albedo and the least amount of Earth IR. The 12 Earth Radii apogee spacecraft experience the longest and coldest eclipse. The worst hot case occurs when perigee exists along the earth-sun vector. Perigee of the different orbits have minuscule difference. For simplicity, we aim for all spacecraft to have the same thermal design. At this point, the spacecraft experience solar flux and maximum earth IR and albedo. Of note, the spacecraft do not do RAAN station-keeping. This means that for the other part of the year, apogee exists in non-eclipse and vice versa. However, these scenarios do not present greater and lower thermal fluxes than already presented.

At these worst case scenarios, the plane of the solar panel and thruster sides of a MATRIX spacecraft are perpendicular to the sun-earth vector. The side panels are parallel. As such, we model view factors from those faces at these points as a plane to a sphere at angle of zero and 90 degrees. For the Magnetometer, using the MMS magnetometer as a geometric reference, a cylinder to planet view factor is used. Tables from a NASA report used to derive view factors are shown in Appendix A.8.1

A compilation of worst-case maximum and minimum incoming thermal fluxes is compiled in table, based on table 71's equations and MATRIX size. Radiative loss to space is a function of



Geometry	Gamma	Altitude	
		1000km	12 Re
Flat Plate	0	0.76	0.013
	90	0.2	0
Cylinder	90	0.88	0.02

Table 72: Radiative View Factors

Hot Case	1365 W
Cold Case	2.85 W

Table 73: Worst-Case incoming thermal fluxes

temperature and is thus highly dependent on to-be-discussed thermal design. These values are not dependent on material absorptivity, rather solely the orbital environment thermal fluxes.

Given the environment, early in the design process, the thermal subteam identified the need to conduct trade studies on the following major components:

- Radiative Insulation
- Conductive Insulation
- High Heat Transfer Components
- Heaters
- Temperature Sensors

9.7.2 Radiative Insulation

Analysis conducted by:

Arthur Balva All

Radiative insulation is needed on the spacecraft to reduce the radiative loss to space and to help thermally isolate components from each other by reducing thermal radiative exchange. This is more-so important in eclipse when, as described in the subsystem drivers, heater power must be kept at a minimum. A trade study was conducted to evaluate various solutions.

MLI blankets are the industry standard for radiative solution, as seen in general literature and broadly on almost all space missions. Table 74 shows that a standard 20-layer MLI blanket with a Kapton outer layer has the lowest IR emissivity. These blankets also satisfy ESD, EMI, and RF isolation requirements. However, MLI can only be applied, generally, on larger surfaces. For small surfaces, such as those on thrusters, we prefer a coating. Acktar's Nano Black™ has a very low solar emissivity and has flight heritage. Acktar Ultra White is a good solution for components that need to reject heat with its high IR emissivity.

9.7.3 Conductive Insulation

Analysis conducted by:

Arthur Balva All

Components on each MATRIX spacecraft have different thermal envelopes. Without conductive



Technique	Solar Absorptivity (α)	IR Emissivity (ϵ)	α/ϵ Ratio
OSR (Optical Solar Reflector)	0.07–0.10 (BOL) 0.16–0.25 (EOL)	0.78–0.85	0.09–0.20
S13G White Paint	0.20 (BOL) 0.44 (EOL)	0.22 (BOL)	0.9
Aluminized Kapton (tape/sheet)	~0.14–0.20	~0.03–0.07 (Al side)	~2–5
MLI Blanket (20-layer typical)	~0.04–0.10 (outer Kapton layer)	~ 0.01 – –0.03	~2–5 (outer layer)
MLI with ITO Outer Layer	~0.10–0.15	ϵ_{eff} ~0.01–0.03	~3–7
Aluminum 6061 Film/Tape	0.11–0.12	0.45–0.73	0.24–0.16
Beryllium Heat Sink	0.31	0.03	10.3
Acktar Nano Black™	0.92	0.03–0.17	5.4
Acktar Ultra White™	0.23	0.85	0.27

Table 74: Radiative Insulation Trade Study

isolation, there will be high heat fluxes between components and temperatures will be more averaged out. As a result, a trade study is conducted to identify conductive isolating materials and compare them to traditional materials.

Technique	Thermal Conductivity (W/mK)	Density (g/cm ³)	Tensile Strength (psi)	Modulus of Elasticity (ksi)
Titanium Structural Mounts	6.7	4.5	31900	16800
Aluminum 6061	167	2.7	45000	10000
Garolite G-10	0.25	1.8	45000	2700

Table 75: Conductive Trade Study [153][154][155]

G-10 is the industry standard in conductive isolation. Compared to aluminum, or the more conductively isolating titanium, G-10 has a much lower thermal conductivity and density. The trade-off is decreased strength. That said, **G-10 appears to be the best conductive isolating solution for MATRIX**. Some examples of use-cases of G-10 on the spacecraft include, but are not limited to:

- Isolating Hydrazine fuel lines from the structure of the bus
- Mounting inner intermediary insulation walls within Hydrazine fuel tank
- Mounting electrical components to limit heater requirements
- Isolating hot propulsion elements like combustion chambers and nozzles
- Generally, isolating subsystem “boxes” that require heaters from spacecraft bus

9.7.4 High Heat Transfer Components

Analysis conducted by:

Sarah Hedley Research, Trade Study, Writing (paragraph 2)

Arthur Balva Writing and editing

Conversely, components may need to have high thermal conductivity between them. This can be done to link high thermal mass components, like the main hydrazine tank, with smaller thermal mass components, like electronics. This can help reduce power consumption and localize



placement of heaters. High thermal conductivity is also useful for linking components with heat rejecting components. This is useful during non-eclipse times when internal components may get too hot. A trade study was conducted to evaluate different solutions to increase and tune thermal conductivity between components.

Heat pipes are a closed liquid-flow cycle that uses an evaporator and condenser to transfer heat without electrical power [156]. Heat pipes are beneficial in that they do not require electrical power to work and are simply constructed without moving parts; however, given that our spacecraft will be operating in temperatures below zero, it is ill advised to use heat pipes as the operating fluid risks freezing. Pumped fluid loops (PFLs) are comprised of a pumping device, a heat exchanger, and a space radiator. PFLs utilize concepts such as Bernoulli's Principle and the Nusselt number to determine its effectiveness at transferring heat in both laminar and turbulent flows and its ideal operating fluid [156]. Because PFLs contain a pump, they consume power to function. Ultimately, pumped fluid loops are ineffective for our mission because our goal is to regulate the drastic thermal gradient between two sides of our spacecrafts in the most mass, volume, and power cost effective way.

Given that thermal straps are the desired solution, another trade study was conducted to evaluate different materials. Different materials can be chosen to tune the conductivity to achieve desired fluxes.

	Density (g/cm^3)	Thermal Conductivity (W/mK)	k/ρ ($m^2/s^3 K$)
Aluminum[157]	2.7	235	0.087
Copper[157]	8.96	400	0.044
Graphite[158]	1.8	1700	0.944

Table 76: Thermal Straps Material Comparison

Generally, in most applications, the thermal subteam wants to maximize heat flux. For these general applications, **graphite is the best solution as it offers the highest thermal conductivity.**

9.7.5 Heaters

Analysis conducted by:

Arthur Balva All Other Categories

Gina Arnau Compilation of Radiative Insulation Optical Properties

Space is cold at 2.7K. Inherently, to meet thermal envelopes when there is minimal to no environmental heat addition requires heaters. There are two main forms of heaters used on spacecraft. The first involves Thin film heaters. These commonly utilize Nichrome wire embedded in a Kapton laminate and are installed over flat surfaces. Another, more powerful solution is cartridge heaters which provide 10-20x more power than thin film heaters.

Method	Cost	Mass Impact	Wattage	Passive?	Controllable?
Thin Film Heaters	Low	Low	3 W/in^2	No	Yes
Cartridge Heaters	High (Custom)	High	55 W/in^2	No	Yes

Table 77: Heater Trade Study[159][160]

The decision on which type of heater to use is dependent on heater requirements and the desire



for redundancy. Many smaller heaters provide easier redundancy. Surface area may be a constraint. If so, and with a high needed power draw, capsule heaters may be the solution. However, as discussed in later thermal simulations, components need, at a maximum 35-40W, with smaller components needing on the order of 5W. As a result, **thin-film heaters will be used**. For redundancy, each component will have full redundancy for its entire heater power needs. Additionally, each heater is budgeted to have 10 percent more heater capacity than projected. Referencing heater requirements in Section 9.3, **six 1.6 in² and two 13 in² heaters are needed**.

9.7.6 Temperature Monitoring

Analysis conducted by:

Gina Arnau Literature research, analysis, and writing of section

Anthony Matimu Literature research, editing and review

In the PDR report, the TE Connectivity RTD Platinum Sensor 20030604-00 as the temperature sensing hardware. In this analysis, sensor placement was determined based on thermal criticality and monitoring objectives, along with sensor integration. The analysis to select this sensor can be found in A.8.3

Sensor placement was driven by three priorities: monitoring components with the tightest operational temperature limits, characterizing the external thermal environment across the spacecraft bus, and detecting anomalous thermal conditions before they result in unrecoverable hardware degradation.

The battery pack is the most thermally sensitive component on the spacecraft. Lithium-ion batteries have a narrow operating temperature range in spacecraft operations; for the MATRIX mission, between, between 0°C to 45°C for charging and -30°C to 60°C for discharge — both of which comfortably satisfy the electronics operating requirement of -5°C to 40°C. It is worth noting that charging will only occur in sunlight, where temperatures will remain above 0°C. Two sensors are assigned to the battery for redundancy, as a sensor failure at this node would leave the most thermally critical component unmonitored. Using the same reasoning, two sensors are placed on the BAU unit, which generates heat internally and must remain within defined limits throughout the mission. Another sensor is placed on each of the fuel tanks, as propellant must remain above its freezing point during the long eclipse periods imposed by the highly eccentric orbits. Three sensors are placed on different places of the solar array to monitor the large temperature swings between sunlit and eclipse phases, supporting thermal model validation. Two sensors are placed on each of the sun-facing and anti-sun-facing panel exteriors, providing a direct measurement of the thermal gradient across the spacecraft structure, as well as on each lateral panel of the bus. Lastly, each instrument (ESA, EPD, and magnetometer) has two dedicated sensors, based on heritage missions [7].

This yields a total of 32 sensors per spacecraft and 256 sensors across the full MATRIX constellation. Despite the large sensor count, the low individual mass of each sensor ensures the THM subsystem remains within its allocated mass budget. Additionally, the number of sensors is still within the same order of magnitude of heritage missions [161].

The TE Connectivity sensor is a flat, 26AWG polyimide insulated twisted pair wires and encapsulated using thermally conductive epoxy resin on a flat aluminum housing (RTD Platinum Sensors



Component	Number of Sensors
Battery pack	2
BAU unit	2
Pressurized tanks	7
Solar array	3
Sun-facing panel	2
Anti-sun-facing panel	2
Lateral bus panels	6
ESA	2
EPD	2
Magnetometer	2
Total	32

Table 78: Temperature sensor allocation per spacecraft

welded to 26AWG polyimide insulated twisted pair wires and encapsulated using thermally conductive epoxy resin on a flat aluminum housing.) Installation is straightforward: each sensor is bonded directly onto the surface of its target component using thermally conductive adhesive, placing it in intimate thermal contact without requiring custom brackets or mechanical modifications to the host structure.

The sensor is a passive device: it behaves simply as a resistor whose resistance varies predictably with temperature. To obtain a temperature reading, the onboard computer supplies a small constant excitation current to the sensor (< 1 mA) and then the resulting voltage across the sensor is measured [162]. These are then converted to temperature measurements, following the sensor's calibration curve.

A source of measurement uncertainty is self-heating. Since the sensors measure temperature by passing a current through a resistive element, the resulting Joule heating elevates the sensor's own temperature above that of the surface being monitored. For a conservative worst-case excitation current of approximately 1 mA through a $100\ \Omega$ element, the dissipated power is:

$$P = I^2 R = (1 \times 10^{-3})^2 \times 100 = 0.0001\text{ W} = 0.1\text{ mW} \quad (34)$$

The resulting temperature rise is

$$\Delta T = P \cdot \theta \quad (35)$$

where θ is the sensor's dissipation coefficient. For thin-film RTD sensors, like the selected one, the dissipation coefficients tend to be of is $2.5\text{ mW}/^\circ\text{C}$ [162].

Consequently:

$$\Delta T = \frac{P}{\theta} = \frac{0.1}{2.5} = 0.04\text{ }^\circ\text{C} \quad (36)$$

The obtained value is small, and so the introduced error by the sensors shouldn't be significant. However, it would be optimal to characterize the self-heating error during ground thermal vacuum testing prior to launch.



9.7.7 Single Node Model

First, the thermal subteam assumed each MATRIX spacecraft to be one thermal node. This is a simplified thermal model but appropriate for a first-order approximation of heater power. This was conducted through the PDR. It was argued that single-node was reasonably appropriate because of the large encompassing structure. Furthermore, the only fully exposed instrument is the magnetometer boom which we assumed to be isolated. The boom is made of carbon fiber, who's thermal conductivity is on the order of magnitude of $5 \frac{W}{MK}$, a very low conductivity. [163]

First, exterior surface area optical properties were compiled. Over the five year mission, optical properties degrade and significantly affect the thermal balance of the spacecraft. This occurs due to radiation, oxidation, and contamination from the launch fairing [156]. Emissivity is not a strongly affected. Absorptivity generally increases due to exposure to UV radiation, the largest factor for MATRIX given its frequent transits of the Van Allen belts, energetic BFFS, and far distance from Earth. Many absorptivity values are publicly available for GEO orbits **vest**. As an arbitrary correction, given MATRIX's more radiation-heavy environment, 50% additional optical degradation is assumed. It is assumed that the thrusters are painted with Acktar Nano Black™. Antennas, are coated, as described by the user manual, in white paint with specified optical properties.

Component	Absorptivity		
		BOL	EOL
Solar Cells	0.5	0.112	0.322
Magnetometer Boom	0.85	0.9	0.945
Thrusters	0.03	0.9	0.945
Antenna	0.88	0.2	0.62
Payloads	0.85	0.9	0.945
ADCS Sensors	0.85	0.9	0.945
MLI	0.01	0.04	0.31

Table 79: Surface Area Component Optical Properties

MATRIX spacecraft station-keep to always be sun-pointing. Thus, we can do a weighted area average of optical properties over the three distinct faces of the spacecraft: sun-facing, thruster-side, and side-panels. Each face experiences unique and isotropic heat fluxes at a moment in time.

Surface	Absorptivity $\times$ Area (m ²)		
		BOL	EOL
Sun-Facing	0.26	0.073	0.28
Thruster-side	0.027	0.061	0.32
Side Panels	0.0027	0.054	0.13

Table 80: Weighted Area Average of Optical Properties

The temperature of the spacecraft node is governed by thermal equilibrium.

$$\sum Q = Q_{su} + Q_{et} + Q_{er} + Q_{heater} + Q_s = 0$$

$$T = \left(\sum_{Sides} \frac{(Q_{su} + Q_{et} + Q_{er})}{\epsilon \sigma A} + Q_{heater} + T_s^4 \right)^{1/4}$$

Figure 35 shows the temperature of the spacecraft as a function of heater power. The green area represents the acceptable structural thermal envelope of -89°C to $+89^{\circ}\text{C}$ from Requirement THM-P-003. Python code that automates this simulation can be found [here](#).

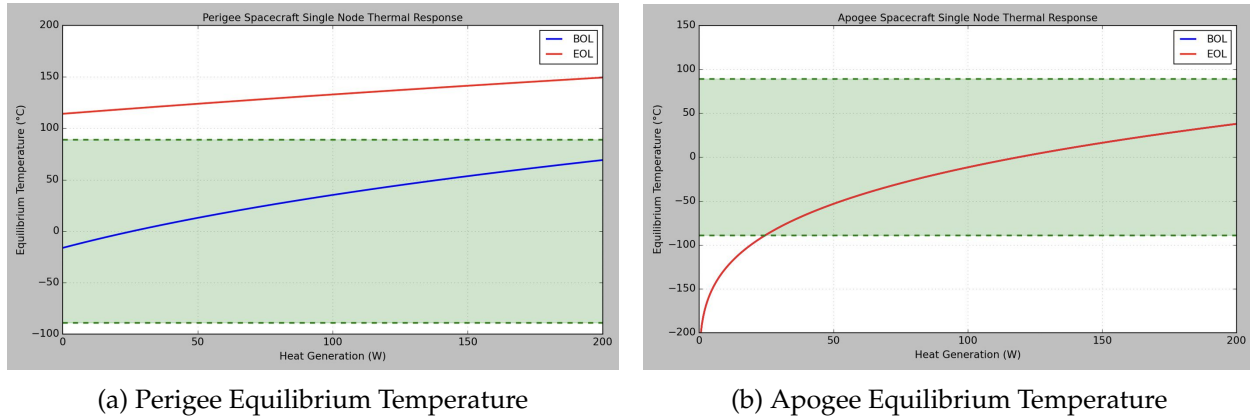


Figure 35: Single Node Thermal Equilibrium Plots

9.7.8 Louvers

As can be seen in Figure 35, the spacecraft is too insulative. At end of life, in non-eclipse, the spacecraft exceeds $+89^{\circ}\text{C}$. However, still at least 25W is needed to maintain the spacecraft temperature above -89°C during eclipse. It is clear that the spacecraft cannot be under 89°C in non-eclipse and still be warm enough in eclipse without overly-burdensome heater power requirements. To avoid additional power draw from radiators, the spacecraft must be able to dynamically change its optical properties.

Louvers are an industry solution to this problem. They have extensive mission heritage, such as on the Juno mission, Cubesats, New Horizons, Parker Solar Probe, and more. Louvers are a passive solution that allows for variable optical properties. They act like venetian blinds and are designed to actuate at a set temperature range. The vanes are actuated by a bimetallic spiral, thermally driven springs. When open, IR radiative loss to space occurs as if the louver nearly did not exist. As the temperature of the louver decreases, the springs variably close the vanes. This lowers the view factor of the radiating surface and thus also IR loss. Coatings are put on the vanes and inner surface to create radiating and insulating surfaces.

Sierra Space is one popular modern supplier of spacecraft Louvers which has extensive mission heritage. Figure 81 provides examples of one of their offerings. Louvers are not intended to be sun-facing [164]. As such, **MATRIX spacecraft will utilize four Sierra Space louvers, each 20"x20". Vanes will be coated with Acktar Nano Black™. The internal radiating surface will be coated with Acktar Ultra White™.** This is a custom size that matches the available area on a side panel.

Life Cycles	Radiating Area	Mass	Thermal Setpoint	Random Vibration	Volume
20,000	248 in ²	1.29 kg	-20 C to 50 C	12.6 Grms	0.013 m ³

Table 81: Sierra Space 20 Panel Louver Technical Specifications [164]

Radiative flux into a louver is modeled by the Stephan-Boltzmann equation in eq. 37 with an effective emissivity. Eq. 38 presents a linearized estimate of the effective emissivity used in practice. Actual effective emissivity depends on the vane angles and associated view factors. Typical open and closed setpoint temperatures are designed 10-18 Celsius apart [156]. For the structure, the setpoints shall be placed at 5°C and 25°C, temperatures that match the lower end of internal component thermal envelopes.

$$\frac{\dot{Q}}{A} = \varepsilon_{\text{eff}} \sigma T^4 \quad (37)$$

$$\varepsilon_{\text{eff}} = \varepsilon_o - \frac{\varepsilon_o - \varepsilon_c}{1 - T_c/T_o} \left(1 - \frac{T}{T_o} \right) \quad (38)$$

The updated single-node thermal models with louvers results in compliance with the structural thermal envelope in non-eclipse and 29W of power in eclipse.

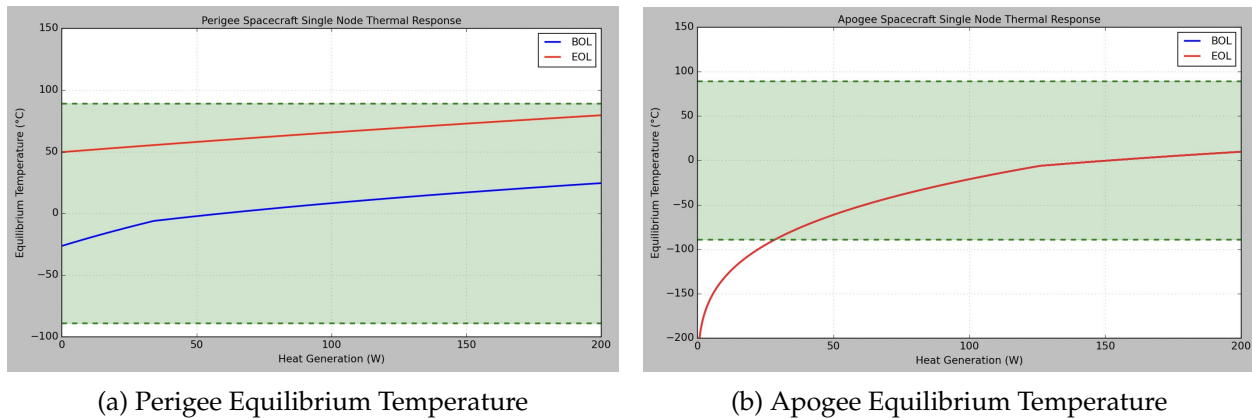


Figure 36: Single Node Thermal Equilibrium Plots with Louvers

9.7.9 Multi-Node Model

In Figure 36, a reasonable power draw is needed to maintain the single-node spacecraft at the structural thermal envelope. However, there are many components with tighter thermal requirements. The most restrictive, on the metric of highest low-bound temperature is the Hydrazine tanks. Hydrazine must be kept at at least 4°C to not freeze. To keep the entire single-node at 4°C in eclipse would require 175W of power, far too much. Thus, a multi-nodal thermal model must be made.

The major components with notable thermal requirements are the structure, tanks, ESA and EPD instruments, magnetometer, and batteries. Their thermal requirements are listed in the requirements spreadsheet [138]. Seven nodes will be used: structure, main tank, side tanks 1, side tanks 2, Instrumentation, magnetometer, and batteries.



A multi-node model requires not just characterization of environmental thermal fluxes but internal thermal fluxes and connections. These come in either conductive or radiative connections. Conductive heat rate is modeled as

$$q_{cond} = kA \frac{\Delta T}{L} \quad (39)$$

We assume a view factor of 1 for all radiative connections for fully internal components. The only radiative connections we model are the structure to components. Since the structure is all that internal components "see", this view factor is reasonable. We assume that the internal spacecraft structure is bare 7075 aluminum and the internal faces of components is MLI. Assuming two parallel equal-area surfaces, radiative heat rate is then modeled as

$$q_{rad} = A\sigma \frac{T_1^4 - T_2^4}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1} \quad (40)$$

Furthermore, to reduce heater power and localize heater placements, we assume graphite thermal straps connect the large, main central Hydrazine tank to small thermal mass components. These are the batteries, instruments, and side tanks. Graphite straps are used. Further tuning of thermal conductivity of these paths is possible and may minimally reduce power draw.

The magnetometer optical properties, orientation, geometry, and thermal envelope are based on the heritage THEMIS mission. Visuals of the magnetometer can be found in Appendix A.8.2. The carbon fiber boom serves as a long thermal conductance path.

The tanks, instruments, and batteries are assumed to be mounted using G-10. This differs from industry practice in which only G-10 spacers are used between mounting structures and components, but this approximation suffices. For the main tank, a 3cm tall, 2cm thick ring is used to mount. Cross-sectional area is assumed to be the mid cross-section. Pocketing is assumed to reduce this area by 4x. Six 45 degree braces, 3 on each hemisphere of the tank are also used. These are assumed to be 3cm in diameter and 0.215m long based on geometry. The side tanks are assumed to be mounted with similar braces, four for each tank. Instrumentation and batteries are assumed to be mounted with 2.5cm thick spacers with cross-sectional area 1/2 of the interface area between the component and structure.

Custom Python code is used to define a spacecraft's nodes, connections, and environment. An optimizer then minimizes the total power draw on the spacecraft while adhering to the thermal requirements of all nodes. Figure 37 provides an example of all thermal heat fluxes for the worst-case cold scenario in eclipse, at beginning of life. This is the maximum power draw throughout the entire mission. A briefer summary of all modes of interest is described in Table 82 Code can be found [here](#).

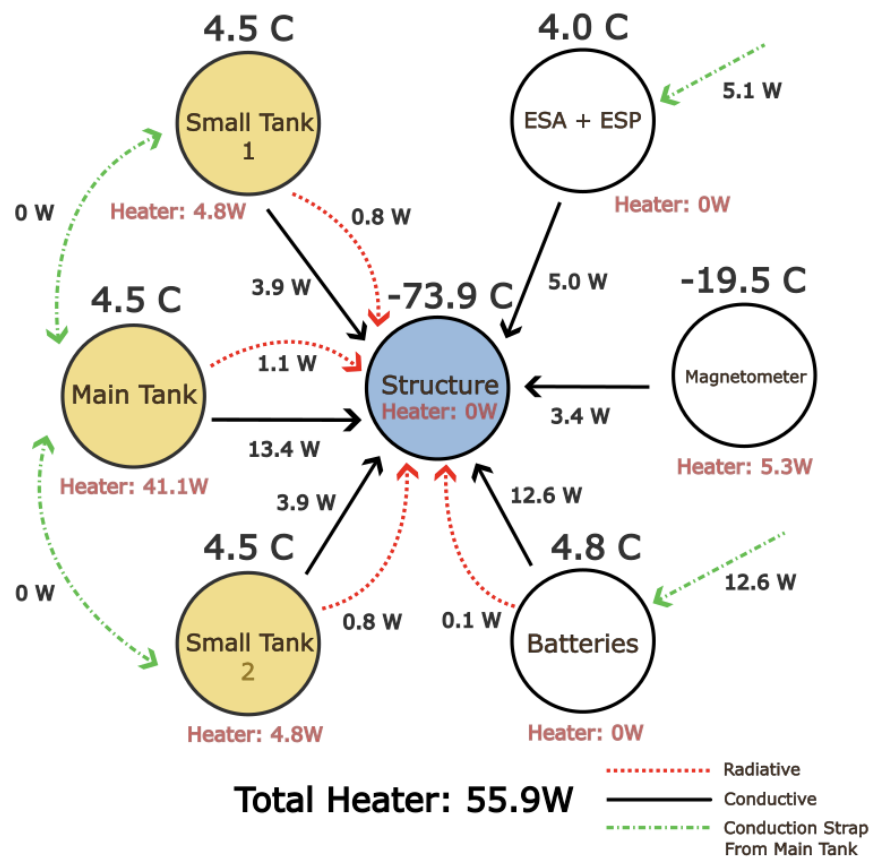


Figure 37: Eclipse, BOL Multi-Node Solution

10 POW Design

10.1 Subsystem Overview

The responsibilities of the POW subsystem are to provide, store, distribute, and control the electrical power on each satellite of the MATRIX mission. The subsystem must be able to satisfy the power requirements of each other subsystem for both the average and peak electrical load during all phases of the mission operation including in times of transit and during eclipse. The goal of the power source portion of the subsystem will be to generate electrical power within the spacecraft to support electrical loads and recharge batteries. The subsystem is also responsible for generating its own power to recharge batteries and provide the power for the electrical load of all the subsystems in the mission. The batteries must always have enough stored energy to supply the necessary peak and average electrical load to other subsystems. The MATRIX mission performs its most significant data collection in the magnetotail transition region while in eclipse mode, therefore it is an essential responsibility for the batteries to be properly sized, so that they are able to provide enough energy for the MATRIX mission to perform all functions during times of eclipse.

Component		Structure	Main Tank	Side Tank 1	Side Tank 2	Instruments	Magnetometer	Batteries	Total
Heater Power (W)	Cold BOL	0	34.35	4.15	4.15	4.6	0	0	47.25
	Cold EOL	0	35.6	4.15	4.15	4.5	0	0	48.4
	Hot BOL	0	8.45	0.1	0.1	0	0	0	8.65
	Hot EOL	0	0	0	0	0	0	0	0

Table 82: Multi-Node Heater Power Summary

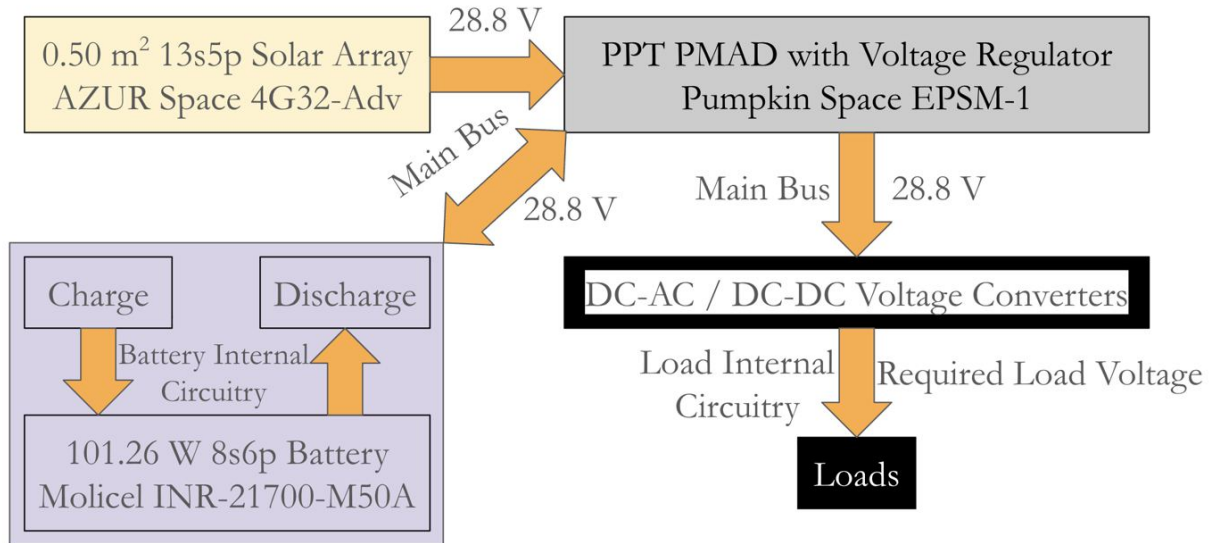


Figure 38: Power & Electrical Design Architecture Block Diagram

Figure 38 shows an overview of the power and electrical subsystem design architecture. The subsystem provides power using 0.50 m² solar array made of AZUR Space 4G32-Adv solar cells. Power is stored inside of a battery pack constructed of MoliceL INR-21700-M50A cells. The subsystem is controlled by Pumpkin Space EPSM-1 power management and distribution system (PMAD) that distributes power to all components and systems in the satellite. The key interfaces between POW and other subsystems are the Power Management and Distribution System (PMAD) and the bus. OPL/CCD, THM, ADCS/GNC, and PLV receive electrical power controlled by the PMAD through the bus. This design allows the subsystem to satisfy the power requirements of each mission subsystem throughout all of the phases of MATRIX mission operations.

10.2 Subsystem Objectives

The MATRIX mission level objectives serve to conduct a targeted and physically grounded investigation into BBF evolution across the 6 – 12 R_E corridor. The three mission objectives are to characterize spatial/temporal structure of BBFs in the region of 6 – 12 R_E , resolve radial evolution/braking in the region of 6.6 – 9 R_E , and quantify magnetic field and particle signatures associated with BBFs in the magnetotail transition region. The POW subsystem objectives directly trace down from these mission level objectives to enable these scientific operations by providing the necessary power to all relevant subsystems.



The primary objective of the Power & Electrical (POW) subsystem is to provide, store, distribute, and control electrical power for all of the satellites of the MATRIX mission:

1. Supply a continuous source of electrical power to satisfy the requirements of each other subsystem for both average and peak electrical loads during all phases of mission operation so that the mission is able to perform all science objectives
2. Generate enough electrical power to recharge the battery and ensure its depth of discharge never exceeds a maximum allowable limit, so that the mission is able to perform all science objectives during eclipse using energy from battery discharge
3. Store enough electrical power to provide power to all subsystems as required by each operational mode, so that the mission is able to perform all science objectives
4. Distribute electrical power to all subsystems as required to perform all necessary functions for each operation mode, so the mission is able to perform all science objectives

10.3 Subsystem Requirements

The most important POW subsystem requirements are listed below in Table 83. They are categorized as functional (F), performance (P), and constraint (C) requirements. The notable requirements listed below are grouped by the specific objectives it satisfies and includes mission-level rationale. The rationale for these requirements provide key sources, calculations, and research done to arrive at these values. A complete and comprehensive list of all requirements, along with trace up requirements and relation with subsystem objectives, can be found in the POW tab of the "Requirements Spreadsheet" Google Sheet: [POW Subsystem Requirements List](#).



ID	Requirement	Rationale	Method	Trace	Obj
POW-F-002	The power subsystem for each of the satellites shall generate at least 143.6 W of power for the mission architecture for the duration of the mission.	Accounts for subsystem power demands plus margins for harness losses, safety factors, and end of life degradation due to radiation.	Analysis	MISS-F-007, MISS-F-016, MISS-P-001	1, 2
POW-P-001	The power subsystem shall maintain enough power for at least 4.5 hours of operation on eclipse mode for all satellites at all times during the mission [53].	Ensures power supply during the maximum calculated eclipse duration of 4.5 hours without solar arrays [12].	Analysis	MISS-F-009, MISS-F-007, MISS-P-001	1, 3
POW-P-003	The power subsystem for each satellite shall store a peak electric power of at least 101.26 W to the mission architecture upon request [53].	Represents the maximum required energy storage during daytime operations.	Analysis	MISS-F-009, MISS-F-007, MISS-P-001	1, 3
POW-P-004	The power subsystem of each satellite shall provide an uninterrupted supply of electric power of at least 143.6 W to the mission architecture .	Represents the energy continuously collected by solar arrays during daytime operations.	Analysis	MISS-F-009, MISS-F-007, MISS-P-001	1, 2
POW-C-001	Each satellite power subsystem shall have a cumulative mass of at most 26.7 kg.	Complies with the mission mass budget based on heritage missions [35] and SMM mass allocation.	Mass Budget	MISS-C-005	-
POW-C-002	The cumulative cost of the power subsystems in the MATRIX mission architecture must not exceed \$30,000,000.	Complies with the allocated cost budget [35] and OPL budget allocation.	Cost Budget	MISS-C-003	-
POW-E-002	The POW subsystem must operate within space thermal conditions, specifically withstanding temperatures of -5°C to 40°C.	Ensures component survival and function against space environment extreme temperatures.	Analysis	MISS-E-003	1,2,3,4

Table 83: POW Subsystem Requirements

10.4 Subsystem Constraints

The most significant constraints in the MATRIX mission are cost, timeline, mass, volume, and environmental durability. The PLV subsystem constraints are derived from the mission level constraints and include the following:

- **Surface Area of Solar Array:** The POW subsystem must generate enough power to support all scientific and operational goals of the mission throughout the mission life time. This requires a solar array to generate this power, and it must fit onto the sun-facing side of each spacecraft. This constraint is linked to **MISS-C-001**, **MISS-C-002**, and **MISS-C-005**. This constraint will heavily influence power mode definition and duty cycling for the POW subsystem.
- **Mass & Volume of Batteries:** The POW subsystem must remain within its mass and volume allocation to meet vehicle and structural constraints. The battery pack consisting of battery cells will take up the most space of any component in the POW system. Linked to **MISS-C-001**, **MISS-C-002**, and **MISS-C-005**. This constraint limits the mass and volume of batteries, which will influence duty cycling in the power budget to ensure that all necessary operations can occur during eclipse times.



- **Eclipse Duration:** The POW subsystem must be able to provide the power for all necessary operations during eclipse times in each cycle. This constrains the amount of power that the system will be able to provide based upon the size of the batteries. This constraint is linked to **MISS-F-001**.
- **Sun Time:** The amount of available sun time constraints the batteries' ability to recharge. The batteries must be able to recharge to full capacity during each cycle. This constraint is linked to **MISS-F-001**.
- **Cost of Power System Components:** The POW subsystem must remain within the mission budget when selecting batteries, solar arrays, etc. Linked to **MISS-C-003**. Power storage or generation may need to be sacrificed in order to remain within budget.
- **Power Budget of Other Subsystems:** The POW subsystem must be able to provide the power required by other subsystems when they require it. Linked to **MISS-C-001** and **MISS-F-009**. This is the main factor constraining the design of the POW subsystem.
- **Thermal Environment:** The power subsystem must operate within the worst case thermal environment of the magnetotail regime at about $65^{\circ}C$. This is especially important as solar cell performance is inversely proportional to temperature. This constraint is linked to **MISS-E-001**, **MISS-E-002**, **MISS-E-003**, **MISS-E-004**, and **MISS-E-005**.
- **Radiation Environment:** The radiation experienced by the solar arrays will provide constraints for their sizing and drive the selection of solar cells from a trade study. This constraint is linked to **MISS-E-001**, **MISS-E-002**, **MISS-E-003**, **MISS-E-004**, and **MISS-E-005**.

10.5 Subsystem Drivers

The POW subsystem drivers are as follows:

- **Supply Power to Meet Each Subsystem Power Budget:** The POW subsystem's greatest driver is providing each MATRIX spacecraft with a reliable and continuous supply of power, so that each subsystem is able to perform its essential tasks. This directly supports all mission level drivers, particularly 2-D Field and Particle Measurements for Mesoscale Science and High Temporal Sampling Rate by providing scientific instrumentation with power for operation, and Data Storage and Communication.
- **Essential Mission Operations during Each Phase/Mode:** The essential operations that each subsystem must perform across different phases and tasks of the MATRIX mission will drive the definition of power system modes and duty cycling. These modes will determine the peak and average power draw necessary at different times for the MATRIX mission.
- **Time in Eclipse:** The maximum time spent by a MATRIX spacecraft in an eclipse drives the power storage design and the sizing of the battery pack.
- **Time in Sun Light:** The minimum time spent by a MATRIX spacecraft in sun light drives the power generation design and the sizing of the solar array.
- **Non-Deployable Solar Arrays:** The available surface area of the the sun-facing side of each MATRIX spacecraft is a significant driver for the POW subsystem design. This driver dictates the allowable surface area of the solar array that will be used to generate power on the

spacecraft.

- **Minimize Cost, Mass, and Volume:** A significant driver for the POW subsystem is to adhere to the cost, mass, and volume budget allocations for the subsystem. This drives the design to place emphasis on the cost, mass, and volumes of all power components selected from trade studies.
- **Environmental Robustness:** The POW subsystem must select components (batteries, solar arrays, etc.) that can survive in the harsh temperatures and radiative environment of the magnetotail. This directly links to all mission level drivers as it ensures the sustained operation of the POW subsystem and its components.
- **End-of-Life (EOL) Performance:** As time passes from the start of the MATRIX mission, the performance of the POW components will decrease from their initial beginning-of-life (BOL) performance. The batteries will experience capacity fade as they experience more charging cycles, and the solar cells will degrade due to their exposure to radiation. Therefore, the EOL performance of these components must drive the POW subsystem design to ensure that the system can meet the power budget requirements of the entire mission through its six year mission life.

10.6 Subsystem Design Approach

The POW Subsystem design approach is demonstrated in the block diagram shown in Figure 39.

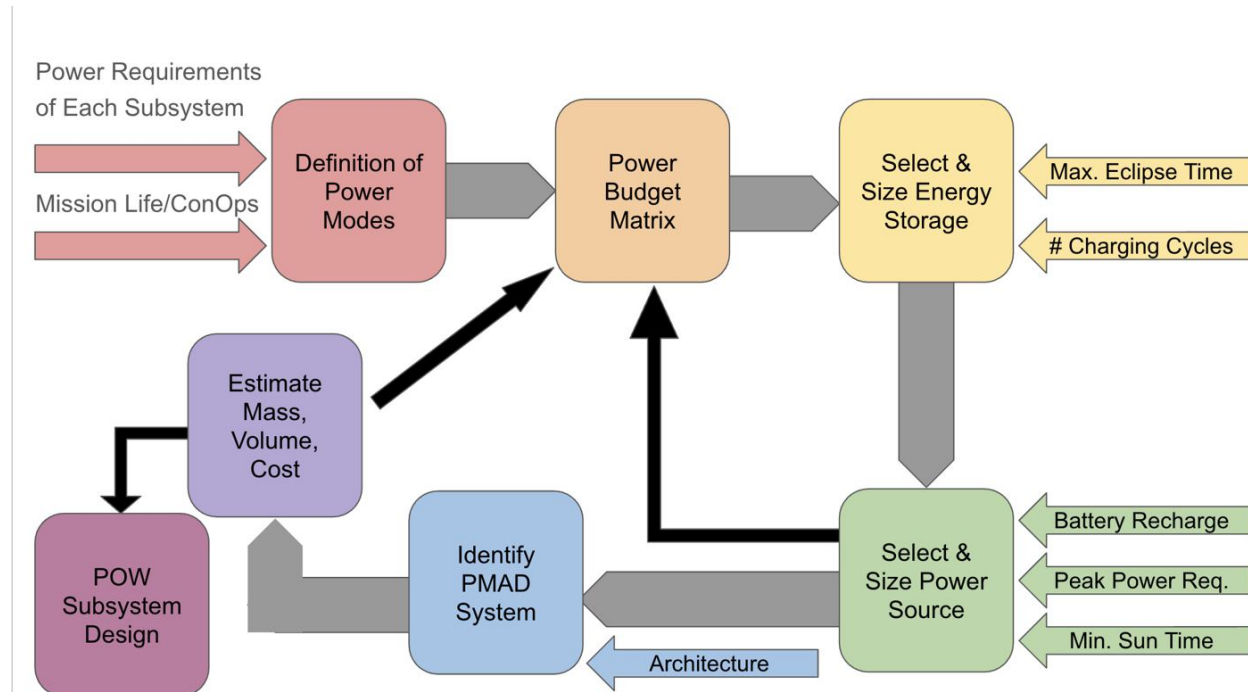


Figure 39: POW Subsystem Design Approach Block Diagram

The first step taken in this analysis design cycle was using eclipse times to perform calculations



for the minimum sun time and maximum eclipse time across all satellite groups. This analysis contributes to satisfying requirement (**POW-P-001**), and is necessary for sizing both the energy storage and power source components of the POW subsystem. Next, operational modes were determined for the POW subsystem (**POW-F-007**). The power requirements of the instruments and components of each other MATRIX mission subsystem were then used to construct a power budget matrix with the input of the power draw for each instrument/equipment from each subsystem. This peak available power draw calculation aims to satisfy (**POW-F-002**). Next a bus voltage was selected (**POW-P-002**). This was used to calculate battery performance and perform trade studies (**POW-F-003**). These trade studies informed the selection of a battery and the design of a solar array. Solar array sizing calculations and trade studies were conducted (**POW-F-003**). The driver of environmental robustness was key to the selection of a solar cell with high radiation hardness. The trade studies that contributed to these design analyses were conducted to satisfy Requirements **POW-F-004**, **POW-F-005**, **POW-F-006**, **POW-F-010**.

In the final design cycle, the most important analysis task was to modify the power budget matrix to ensure POW subsystem compliance to all requirements and constraints. This involved defining a new power mode, adjusting the subsystems that operate in each mode, and modifying the power budget margins, so that the POW system is able to provide the necessary power to complete all scientific and operational objectives during all phases of the mission. This task was driven by the necessity to reduce the sizing of the solar array necessary to generate power for the spacecraft. In the previous PDR design iteration, the solar panels were too large to fit on the sun-facing side of the spacecraft. However, due to mass and volume constraints and the reduced complexity of the structure, it was necessary to reduce the size of the solar array. Therefore, the available surface area on the sun-facing side of the spacecraft became the most important constraint of this final design iteration. In this design iteration, after the definition of a new power mode and the determination of a new power budget matrix for each mode, both the battery pack and solar array needed to be resized for the MATRIX spacecraft. The estimate of the depth of discharge (DOD) for the battery pack was also updated to more accurately reflect standard practices for space missions with a similar number of battery cycles.

These analysis tasks aimed to ensure that the POW subsystem was meeting subsystem objectives 1,2, and 3. The battery and solar panel sizing in this design iteration were also driven by the maximum time of any MATRIX spacecraft in an eclipse and the minimum sun time experienced by any MATRIX spacecraft. A Power Management and Distribution System (PMAD) architecture was also specified for the MATRIX mission. This analysis task served to better achieve subsystem objective 4. All of these analysis tasks serve to complete the MATRIX mission POW subsystem design, while ensuring all power requirements are satisfied with adherence to mass, volume, surface area, and cost budget constraints. The completion of this design supplies power to all of the subsystems whose functions are essential to the accomplishment of all MATRIX mission objectives.

10.7 Formal Analysis

10.7.1 Eclipse Analysis

Analysis conducted by: Tasman Moskowitz, All Components

During the course of the MATRIX mission, each satellite will experience eclipse time when it passes into the shadow of the Earth and sunlight will be blocked during each orbital period. The



MATRIX mission aims to perform measurements in the region of Earth's magnetotail meaning that the mission's most scientifically valuable measurements will occur during eclipse.

The maximum and minimum eclipse times and minimum sun time are shown in Table 84 for each satellite group. This analysis was obtained by the GNC subsystem by propagating each satellite in the mission's orbital trajectory in STK for 1 year. Stationkeeping maneuvers will be performed for each satellite when its apogee most closely aligned with the magnetotail during times of maximum eclipse. While the GNC group only simulated eclipse times over the period of one year, this team chooses to assume that eclipse times in following years will remain similar with proper annual stationkeeping. The total period of the spacecraft is equal to the sum of total sun and eclipse times. Therefore, the minimum sun time will be equal to the difference between the total period and the maximum eclipse time.

Group #	Max Eclipse (hr)	Min Eclipse (hr)	Min Sun Time hr
1	4.356	0.411	19.644
2	3.293	0.382	12.707
3	2.715	0.358	9.285

Table 84: Eclipse Durations per Orbit (1 Year Propagation via STK)

Table 84 shows that the maximum eclipse time of all the satellites is the 4.356 hours experienced by the Group 1 satellites with the highest altitude of apogee. The minimum sun time of 9.285 hours is experienced by satellites in the Group 3 orbit. All of the satellites in the MATRIX mission are designed to be identical to reduce complexity, so each satellite must be designed to perform all necessary functions during this maximum eclipse time. The batteries and solar arrays for the satellites for this condition will be analyzed in subsequent sections. This suggests that all design will be very conservative leading to a high margin of safety in the power budgets.

10.7.2 Mode Determination

Analysis conducted by: Arya Olliviera & Tasman Moskowitz (Definition of Modes), Tasman Moskowitz (Writing of Section), (Refinement Since PDR)

Power requirements differ at various points over the lifetime of the satellite, so it is helpful to define different operational modes for the power system of the satellite. The MATRIX mission will have five necessary modes:

- **Sunlit:** Corresponds to the period of the satellite's operation when it is in direct sunlight, during which power is generated by the solar array to support spacecraft loads and recharge the battery pack.

The satellite performs data processing, attitude sensing, stationkeeping preparation, and communications during this mode. For approximately half the year, each satellite experiences the sunlit mode near perigee while spending most of its orbital period in the magnetotail region near apogee. All components except heaters and thrusters are active in this mode.

- **Eclipse:** Corresponds to the period of the satellites' operation where it is in either the total or partial shadow of the Earth.



As a conservative estimate, no power generation is assumed to occur in the solar panels during this time. This means that batteries are discharged during this mode. The satellite will perform data collection and processing, attitude sensing, heating, and communications during this time. Most scientifically useful data is collected during the time that the satellite reaches apogee in the magnetotail during eclipse times. All components except for thrusters and the antenna power amplifier are operational in this mode.

- **Transit:** Corresponds to the initial period starting when the satellites are first deployed from the launch vehicle and ending when they are in their respective final orbits and about to start its nominal science mode.

This mode requires the use all ADCS/GNC sensors, the flight computer, thermal sensors, and the main propellant thrusters.

- **Stationkeeping:** Corresponds to any period during which stationkeeping burns are executed.

This mode powers all ADCS/GNC sensors, the flight computer, thermal sensors, and the eight 0.8 N stationkeeping thrusters. Science instrumentation and the antenna power amplifier are inactive to allocate available power to propulsion.

- **Safe:** Corresponds to any unusual periods when the satellite experiences an unexpectedly long eclipse due to unforeseen circumstances.

In this mode, only the satellite's most essential functions such as communications, attitude sensing, and heating are permitted to draw power from the batteries.

A summary of the components active in each mode is presented in Table 85. Components listed with non-zero power draw are considered active for the full duration of the mode, while components listed with 0 W are considered powered down. The ADCS/GNC sensors, the flight computer, and the thermal sensors remain active across all modes to preserve attitude knowledge, command processing, and thermal regulation.

10.7.3 Power Budget

Analysis conducted by: Arya Olliviera (power requirements team survey, battery recharge calculations, and power budget allocation), Tasman Moskowitz (battery recharge calculations and budget allocation per mode), Evan Callas, Ted McCormack *Writing of Section:* Arya Olliviera, Tasman Moskowitz, Evan Callas

The power requirements across varying operational modes for the satellites in the MATRIX mission are seen in Table 85. This is a condensed version of the full analysis, which can be found on [this Google Sheet](#).

The major development beyond the PDR stage is the separation of thruster power draws into the In Transit and Stationkeeping power modes. During these newly defined modes, data collection is halted and only necessary instruments receive power. Separating active instruments into these different modes reduced the maximum instantaneous power requirements from ~ 180 W to ~ 90 W.

Each subsystem entered the maximum power draw required by each of its components during



each operational mode. This table is therefore the most conservative estimate of the total power that may be required for each satellite. The maximum power draw for each satellite is $P \sim 90W$, which occurs while the satellite is in stationkeeping mode.

Subsystem/Component	Sunlit Power Draw (W)	Eclipse Power Draw (W)	Transit Power Draw (W)	Stationkeeping Power Draw (W)	Safe Power Draw (W)
ADCS/GNC					
Sun Sensor	0.5	0.25	0.5	0.5	0.5
Sun Sensor	0.5	0.25	0.5	0.5	0.5
IMU	2.5	2.5	2.5	2.5	2.5
Total	3.5	3.0	3.5	3.5	3.5
OPL/CCD					
Antenna Power Amplifier	53.2	0	0	0	0
EPD	2.0	2.0	0	0	0
ESA	1.7	1.7	0	0	0
Magnetometer	0.8	0.8	0	0	0
Flight Computer	7.0	7.0	7.0	7.0	7.0
Total	57.7	11.5	7.0	7.0	7.0
SMM/THM					
Thermal Sensors	1.0	1.0	1.0	1.0	1.0
Structural Heater	0	35.3	0	35.3	35.3
Internal Heating	0	10.0	0	10.0	10.0
Total	1.0	46.3	1.0	46.3	46.3
POW/PLV					
Stationkeeping Thruster	0	0	0	33.0	0
Propelling Thruster (MONARC)	0	0	72.0	0	0
Total	0	0	72.0	33.0	0
Satellite System Total	62.2	60.8	83.5	89.8	56.8

Table 85: Satellite maximum instrument power requirements per subsystem and operational mode.

The table above reflects the maximum potential power requirements per subsystem.

In order to determine the power budget, the power available through solar arrays, batteries, and the PMAD system was determined. Then, power allocated for each section per Mode is directly proportional to the fraction of the total power it requires for that mode.

In the Eclipse, In Transit, and Safe Power Modes, the power will be supplied from the batteries. For Stationkeeping Mode, power may come from the battery or solar arrays depending on position in orbit. In the Sunlit Power Mode, the power will also come from the solar arrays. In this mode, power must be set aside for battery recharge for the worst case Eclipse Mode.

The eclipse-mode power draw with margin is $P_{\text{eclipse}} = 101.26 \text{ W}$ as determined in the next subsections. The eclipse energy and required average battery recharge power for each satellite group are calculated below.

$$E_{\text{eclipse}} = P_{\text{eclipse}} t_{\text{eclipse}}, \quad P_{\text{recharge}} = \frac{E_{\text{eclipse}}}{t_{\text{sun,min}}}$$

Group	t_{eclipse} (hr)	$t_{\text{sun,min}}$ (hr)	E_{eclipse} (Wh)	P_{recharge} (W)
1	4.356	19.644	441.09	22.45
2	3.293	12.707	333.45	26.24
3	2.715	9.285	274.92	29.61

Table 86: Eclipse energy and required average battery recharge power for each satellite group.



The Sunlit Mode power budget will be made using the worst case of about 32.2 W recharge power. The solar array is capable of generating 143.6 W. Therefore, we assume that remaining power after battery recharge is accounted for is about 111.4 W. The power budget allocated for all modes is presented below, and compared against the maximum usage of each subsystem. It can be seen that across all modes, there is significant margin, with the lowest being 11% during Stationkeeping Mode and the highest being 44% during Safe Mode.

Subsystem	Day (W)		Eclipse (W)		Transit (W)		Stationkeeping (W)		Safe Mode (W)	
	Use	Alloc	Use	Alloc	Use	Alloc	Use	Alloc	Use	Alloc
ADCS/GNC	3.50	6.41	3.00	5.00	3.50	4.24	3.50	3.95	3.50	6.24
OPL/CCD	57.70	105.74	11.50	19.15	7.00	8.49	7.00	7.89	7.00	12.48
SMM/THM	1.00	1.83	46.30	77.11	1.00	1.21	46.30	52.21	46.30	82.54
POW/PLV	0.00	0.00	0.00	0.00	72.00	87.31	33.00	37.21	0.00	0.00
Total Subsystem Load	62.20	113.99	60.80	101.26	83.50	101.26	89.80	101.26	56.80	101.26
Battery Recharge	29.61	29.61	–	–	–	–	–	–	–	–
Total Power	91.81	143.60	60.80	101.26	83.50	101.26	89.80	101.26	56.80	101.26
Margin (%)	36%		40%		18%		11%		44%	

Table 87: Comparison of subsystem power usage, allocated power, battery recharge demand, and margin across operational modes.

10.7.4 Power Storage

Analysis conducted by: Evan Callas, All Components (Refinement Since PDR)

10.7.4.1 Power Storage Method

While power is generated from the solar array described in the section above, the spacecraft will demand power draw from an alternate source when the craft enters eclipse mode during every orbital period. When the spacecraft is in eclipse mode, the spacecraft will be in the shadow of the Earth and will not be able to generate power with the AZUR Space solar cells. Because of this, energy storage is an equally important priority while the solar array is operational in Day mode. During Day mode, the solar array is responsible for charging a battery pack capable of energy storage that will power all necessary instruments for the duration of the eclipse. The duration of the eclipse mode varies for each Group of satellites, so Group 1, which had the longest eclipse time, was used to size the battery pack for every spacecraft to ensure it had the capacity to power every instrument throughout any eclipse mode. The design approach for the battery pack sizing was inspired by Joe Troutman's lecture detailing Satellite Electrical Power System Design [165].

In order to design a battery pack that satisfies the requirements of the MATRIX mission, the first design consideration involves selecting the type of battery cell that will make up the composition of the storage module.

The primary step to begin battery selection included trade studies between various battery cells. While analyzing the specifications of cells, some important characteristics that were analyzed include capacity, energy density, discharge rate, charge rate, mass and cycles to 80 % capacity. These specifications allowed the POW subsystem to understand how each cell would perform for the length of the mission duration. Six years is a considerable length of time for the normal performance of these battery cells, yet it is important to analyze the specification such as cycles to 80%



capacity. This is an important specification to analyze for each battery cell because this allowed the power subsystem to make an intelligent decision about the degradation of each battery cell and determine if it would have enough capacity to perform based off its initial beginning of life total battery capacity.

Degradation was a large factor in considering which cell to select for the battery pack that will supply the MATRIX mission with enough power over the duration of the mission. Because over six years there is a 23% drop in capacity based on the frequency that these cells are being used. Throughout each orbital period, the maximum duration that one of the satellites in group one will need to supply power from the battery pack during eclipse mode is approximately 4.356 hours because of this and the fact that the battery cells undergo approximately 12 to 20% calendar aging capacity loss. There would be a large degradation for these battery cells and so the initial beginning of life (BOL) total battery capacity needs to be sufficient in order to provide a end of life (EOL) total battery capacity that alligns with the requirements of the Mission. The complete trade studies for the battery cells can be found Figure 40.

Forge Nano	LG Chem M11	LG Chem M11	Molicel P30B	Molicel P28A	Molicel P28A	Samsung 35E	Samsung30Q	Forge Nano	MolicelP50B	MolicelP45B	Samsung50E
18650	18650	18650	18650	18650	18650	18650	18650	21700	21700	21700	21700
Energy	Power/Energy	Power	Power	Power	Power	Power	Power	Energy	Power/Energy	Power	Power/Energy
3.6	3.5	3.5	3	2.8	2.8	3.35	2.95	5.6	5	4.5	4.9
288	257.14	257.14	229.79	219.13	219.13	241.2	221.25	317.48	253.52	231.43	253.81
1.2	0.7	0.7	0.6	0.56	0.56	8	15	1.68	5	4.5	9.8
7.2	10	10	30	35	35	13	30	5.6	60	45	14.7
1.2	1.75	1.75	3	2.8	2.8	1	1.5	1.68	5	4.5	2.45
3.6	3.5	3.5	9	6	6	1.7	4	5.6	25	13.5	2.45
45	49	49	47	46	46	50	48	63.5	71	70	69.5
500	400	400	300	500	500	350	150	500	500	500	1000

Figure 40: Battery Trade Studies Conducted For Battery Cell Selection

Model	Type	Size	Cell	Cap (Ah)	Wh	CDR (A)	Temp	Vmin	Weight
Molicel	INR-21700-M50A	21700	Energy	4.8	269	15	-	2.5	70

With this in mind, the battery cell that was selected was the **Molicel INR-21700-M50A**. The POW subsystem chose Molicel INR-21700-M50A because of their high discharge rate of 15 A and energy capacity at 4.8 Ah, which was prioritized when considering the cycle life of the battery pack [166]. The power demands of the group on satellites are the highest because of the amount of time in a satellite in eclipse is the greatest of the three groups so the most energy will have to be supplied to group one based off this. This was used as a maximum value that will be provided to every satellite. It is understood that this is an abundant value for both Group 2 in group 3 yet it will suffice and provide the required power for sure based on the calculations from GNC. The power subsystem was able to calculate the maximum number of cycles throughout the six year mission based on the period of group one. Along with this, the power subsystem utilize the power budget made internally based off of the demands of all of the other subsystems based on their instrument selection. The power budget can be referenced in Section 10.7.3

The power budget determined that 89.8 W was the maximum power required by the entire spacecraft and including a margin of 2% for losses throughout the wiring harness and another .5% for losses in the PDM. The total energy that needs to be stored was calculated to be 101.295 W.



$$\text{Max Power Required with Margin} = 89.8 * 1.125 = 101.26 \text{ W} \quad (41)$$

(42)

The power subsystem decided to utilize a voltage bus of 28.8 V for this mission. With this you can divide the total power by the 28.8 V to receive the nominal discharge current of this battery pack.

$$\text{Nominal Discharge of Battery Pack} = 101.26 \text{ W} / 28.8 \text{ V} = 3.516 \text{ A} \quad (43)$$

(44)

With this nominal discharge current, the energy capacity can be calculated. This will be a preliminary value that will lead to the BOL and EOL Energy Capacity. Using the maximum eclipse time from Group 1 and 43, energy capacity is found.

$$\text{Energy Capacity of Battery Pack} = 3.51 \text{ A} * 4.356 \text{ hr} = 15.315 \text{ Ah} \quad (45)$$

(46)

This value is then divided by the depth of discharge (DOD) that the POW subsystem selected for the Molice battery cells. The chosen DOD was 0.6 which was based off of the NASA Space Mission Analysis and Design handbook [53]. An additional source that researched the impact of photovoltaics on battery life span supported the choice of 0.6 as well [167].

The BOL energy capacity can then required by multiplying the discharge current by the time in which you are in the eclipse utilizing this as well as your depth of discharge, which is the amount of the percentage of energy remaining in your cell you find yourself with the end of life, total battery capacity while dividing this by the percentage of health remaining at the end of the mission such as 65% based off of the degradation, we had to determine previously to be approximately 35%. This allowed us to receive the number of 72.75 amp hours for the beginning of life total battery capacity.

With the bus voltage being 28.8 V and the nominal voltage of each Molice being 3.6 V we understand that the number of cells in series in our battery pack will need to be 8. If we return to the BOL total battery capacity and divide this by the capacity of each individual cell, we understand that we will need 6 cells in parallel to achieve this BOL total battery capacity.

10.7.4.2 Total Power Storage Requirement

The battery pack is sized to supply the full spacecraft power demand throughout the longest expected eclipse period, which corresponds to the 4.356-hour maximum eclipse experienced by Group 1 satellites (Table 84). Designing to this worst-case eclipse duration ensures that a single common battery pack design satisfies the storage requirements of all three satellite groups, simplifying manufacturing and reducing program risk.



The end-of-life (EOL) battery capacity is determined from the peak power draw with applied safety factors, the nominal bus voltage of 28.8V, the maximum eclipse duration, and the allowable depth of discharge (DOD) of 60%. The beginning-of-life (BOL) battery capacity is then computed from the EOL capacity by accounting for a 10% capacity fade over the six-year mission lifetime, which is consistent with standard practice for low-cycle-count space missions [165]. Cell aging effects, including both calendar aging and cycle-induced degradation, are absorbed within this capacity fade allowance. The resulting battery pack specifications are summarized in Table ?? . A complete cell-level breakdown, including all intermediate calculations and per-spacecraft sizing parameters, is provided in Appendix A.9.

10.7.4.3 EOL Power Production Capability

The EOL battery capacity is calculated as a function of the peak power draw, the nominal bus voltage, the nominal eclipse time, and the depth of discharge (DOD) using the equations below [165].

$$\text{Current Drawn from Battery (A)} = \frac{\text{Peak Power Draw with Safety Factors (W)}}{\text{Nominal Bus Voltage (V)}} \quad (47)$$

$$\text{EOL Battery Capacity (Ah)} = \frac{\text{Current Drawn from Battery (A)} \times \text{Nominal Eclipse Time (hrs)}}{\text{DOD (\%)}} \quad (48)$$

10.7.4.4 BOL Power Production Capability

The BOL battery capacity is calculated from the EOL battery capacity by accounting for a conservative capacity fade of 10% [165] using the equation below [165].

$$\text{BOL Battery Capacity (Ah)} = \text{EOL Battery Capacity (Ah)} \times (1 + \text{Capacity Fade (\%)}) \quad (49)$$

10.7.4.5 Battery Pack Volume & Weight

The number of series and parallel cells is first determined using the equations below [165]. Each battery is modeled as a cylinder, and the volume of each cell is found accordingly. Therefore, the total volume of the battery pack can be found as the product of the volume per cell and the total number of cells. Similarly, the total mass of the battery pack can be found as the product of the mass per cell and the total number of cells.

$$\# \text{ Series Cells} = \frac{\text{Nominal Bus Voltage (V)}}{\text{Nominal Cell Voltage (V)}} \quad (50)$$

$$\text{Volume per Cell (cm}^3\text{)} = \frac{\pi}{4} \times (\text{Cell Diameter (cm)})^2 \times \text{Cell Height (cm)} \quad (51)$$

$$\# \text{ Parallel Strings} = \frac{\text{BOL Battery Capacity (Ah)}}{\text{Typical Capacity (Ah)}} \quad (52)$$

$$\text{Current Drawn per Parallel String (A)} = \frac{\text{Current Drawn from Battery (A)}}{\# \text{ Parallel Strings}} \quad (53)$$

$$\# \text{ Total Cells in Battery} = \# \text{ Series Cells} \times \# \text{ Parallel Strings} \quad (54)$$

$$\text{Total Mass of Battery (kg)} = \# \text{ Total Cells in Battery} \times \text{Mass per Cell (g)} \times 10^{-3} \quad (55)$$

$$\text{Total Volume of Battery (cm}^3\text{)} = \# \text{ Total Cells in Battery} \times \text{Volume per Cell (cm}^3\text{)} \quad (56)$$

10.7.5 Power Generation

Analysis conducted by: Tasman Moskowitz, All Components (Refinement Since PDR)

In this section, it is first verified that photovoltaic cells are suitable for use in the power generation systems for the MATRIX mission satellites in Section 10.7.5.1. Then, the maximum net power generation requirement, including battery recharge power, is calculated in Section 10.7.5.2. After that, in Section 10.7.5.3, a solar cell is chosen and the maximum power generation per cell is evaluated. The BOL and EOL power production capabilities of this cell are calculated in Sections 10.7.5.4 and 10.7.5.5 respectively. An operating point with power less than the calculated EOL power is chosen in Section 10.7.5.5. This enables the calculation of the number of series, parallel, and total cells in Section 10.7.5.6, so the total solar array area is calculated in Section 10.7.5.6 and the total solar array mass is calculated in Section 10.7.5.7. This follows the procedure outlined in [168].

10.7.5.1 Power Generation Method

Photovoltaic systems in the form of a solar array are commonly used as the power generation method for satellites. This is usually only applicable for missions lasting less than 10 years, as the solar array naturally degrades over time [168]. Since the mission lifetime of MATRIX is six years, a solar array is a good engineering choice for the power generation method. Both Silicon (Si) and Gallium-Arsenide (GaAs) solar cells are available for solar arrays on satellites, where GaAs cells are generally more efficient but also more expensive than Si cells [168]. Multijunction GaAs cells are the standard for most space missions, due to their higher efficiency, radiation hardness, and extensive use in past missions, so choosing a multijunction GaAs solar cell reduces program risk [168].

10.7.5.2 Net Power Generation Requirement P_{sa}

The first step for designing and sizing the solar array is to determine the maximum net power generation requirement for any MATRIX mission satellite. This is given by the sum of the maximum power draw of all components in any mode, and the maximum battery recharge power requirement for any MATRIX mission satellite group, multiplied by an appropriate safety factor.



As calculated in Section 10.7.3, the maximum power draw of all components occurs in the Station-keeping Mode, and is equal to 101.26 W when a safety factor of 10%, a harness drop of 2%, and a further drop in the PMAD system of 0.5% is included. The battery recharge power requirement is different for each satellite group, and is given by Equation 57 [165].

$$\text{Battery Recharge Power (W)} = \frac{\text{BOL Battery Capacity (Ah)} \times \text{DOD (\%)} }{\text{Min. Sunlight Period (hrs)} + 0.08 \text{ hr Margin}} \times \text{Nominal Bus Voltage (V)} \quad (57)$$

The BOL battery capacity is given by Equation 58, the EOL battery capacity is given by Equation 59, and the maximum current draw from the battery is given by Equation 60 [165].

$$\text{BOL Battery Capacity (Ah)} = \text{EOL Battery Capacity (Ah)} \times (1 + \text{Capacity Fade (\%)}) \quad (58)$$

$$\text{EOL Battery Capacity (Ah)} = \text{Max. Current Draw from Battery (A)} \times \frac{\text{Max. Eclipse Time (hrs)}}{\text{DOD (\%)}} \quad (59)$$

$$\text{Max. Current Draw from Battery (A)} = \frac{\text{Max. Power Draw from All Components in Any Mode (W)}}{\text{Nominal Bus Voltage (V)}} \quad (60)$$

Combining Equations 57, 58, and 59, the battery recharge power is found to be proportional to the ratio of the maximum eclipse time to the minimum sunlight period and margin, as seen in Equation 61. This quantity is maximal for Group 3 using the values from the eclipse study table in Section 10.7.1.

$$\text{Battery Recharge Power (W)} \propto \frac{\text{Max. Eclipse Time (hrs)}}{\text{Min. Sunlight Period (hrs)} + 0.08 \text{ hr Margin}} \quad (61)$$

Therefore, the maximum battery recharge power will occur for satellites in group 3. Equations 62-65 outline the battery recharge power calculation for this group. The capacity fade is assumed to be 10%, which is standard for space missions of this length and was used in the design example in [165]. The nominal bus voltage is 28.8 V and the DOD is 60% as stated in the previous section. For Group 3, the maximum eclipse time is 2.715 hours and the minimum sunlight period is 9.285 hours, as outlined in Table 84.

$$\text{Max. Current Draw from Battery (A)} = \frac{101.26 \text{ W}}{28.8 \text{ V}} = 3.52 \text{ A} \quad (62)$$

$$\text{EOL Battery Capacity (Ah)} = 3.52 \text{ A} \times \frac{2.715 \text{ hrs}}{0.6} = 15.93 \text{ Ah} \quad (63)$$

$$\text{BOL Battery Capacity (Ah)} = 15.93 \text{ Ah} \times (1 + 0.1) = 17.52 \text{ Ah} \quad (64)$$

$$\text{Battery Recharge Power (W)} = \frac{17.52 \text{ Ah} \times 0.6}{9.285 \text{ hrs} + 0.08 \text{ hrs}} \times 28.8 \text{ V} = 32.33 \text{ W} \quad (65)$$

The maximum net power generation requirement without safety factors is thus $P_{sa} = 101.26 \text{ W} + 32.33 \text{ W} = 133.6 \text{ W}$. Including an added safety factor of 5%, a harness drop of 2%, and a further drop in the PMAD system of 0.5%, the net power generation requirement used for solar array sizing is $P_{sa} = 143.6 \text{ W}$.

10.7.5.3 Solar Cell Trade Study & Maximum Power Generation per Cell

As outlined in Section 10.7.5.1, multijunction GaAs cells are the standard for most space missions due to their higher efficiency, radiation hardness, and extensive use in past missions, so choosing a multijunction GaAs solar cell reduces program risk [168]. Therefore, a trade study of GaAs solar cells is conducted to determine the optimal cell for the MATRIX mission.

The first step is to compare commercially available options that are recommended for satellite missions. In this trade study the P_{MP} , beginning-of-life (BOL) efficiency, radiation hardness, cost, and flight heritage are compared. The maximum power point is a function of the BOL efficiency and radiation hardness. Solar panel degradation leads to a reduction in the power it can generate over time, which is a quality measured by radiation hardness. A high radiation hardness means less degradation will occur over time. A list of most commercially recommended solar panels is listed in [169] in Table 88.

Parameter	AZUR Space	Rocket Lab	Rocket Lab	Rocket Lab	SpectroLab	Emcore Corp.
Cell Name	4G32-Adv	ZTJ+	Z4J	IMM- α	XTE-SF	BTJ
Technology	Ge-Substrate 4-Junction	Ge-Substrate 3-Junction	Ge-Substrate 4-Junction	Inverted Metamorphic Multijunction	Ge-Substrate 3-Junction	Ge-Substrate 3-Junction
BOL Efficiency, 28°C (%)	31.1	29.4	30.0	32.0	32.2	28.5
P_{MP} at BOL, 28°C (mW/cm ²)	42.0783	39.7782	40.59	43.296	43.5666	38.5605
Radiation Hardness*	0.9	0.87	0.9	0.87	0.84	0.84
Temp. Coefficient**	-118	-107	-100	-93	-98	-71
Cost	Not Available	Not Available	Not Available	Not Available	Not Available	Not Available
Company Flight Heritage	High	High	High	High	High	Medium
Tech Flight Heritage	High	High	High	Minimal	High	High
Source	[170]	[171]	[172]	[173]	[174]	[175]

* EOL Power Remaining Factor at 28°C ** EOL Power Temperature Coefficient ($\mu\text{W}/\text{cm}^2/\text{°C}$)

Table 88: Solar Cell Trade Study

Based upon the results of this trade study, the optimal solar cell to use for the MATRIX mission is the AZUR Space 4G32-Adv. This cell has a high radiation hardness, BOL efficiency, and P_{MP} , and it has a strong flight history as it was used on the EIVE nanosatellite missions in 2023 [176]. Although it has a lower EOL power temperature coefficient than most of the other solar cells studied, that only means it performs more poorly in warm environments. The magnetotail, the ROI of the MATRIX mission, is generally very cold. The other solar cells that have higher efficiencies and P_{MP} s than the AZUR Space solar cell have lower radiation hardness. As surviving the environment of the magnetotail is an essential function for this mission, a high radiation hardness is prioritized above all other criteria.

Therefore, the multijunction GaAs AZUR Space 4G32-Adv solar cell is chosen for constructing the solar arrays for all MATRIX mission satellites. The datasheet for this solar cell [170] states that for a solar constant of $1353 \frac{\text{W}}{\text{m}^2}$, the bare cell has a BOL efficiency of 31.1% if the Sun is normal to the surface of the cells. All values used in this section are calculated for the bare cell. This corresponds to a maximum output power density $P_o^d = 0.311 \times 1353 \frac{\text{W}}{\text{m}^2} = 420.8 \frac{\text{W}}{\text{m}^2}$. As each cell has a surface area of 77.44 cm^2 [170], the maximum output power per cell is $P_o = 420.8 \frac{\text{W}}{\text{m}^2} \times 77.44 \text{ cm}^2 = 3.26 \frac{\text{W}}{\text{cell}}$.



10.7.5.4 BOL Power Production Capability

The actual BOL power production capability is given by Equation 66, where I_d^{BOL} is the inherent degradation at BOL and $\cos \theta$ is the cosine loss due to the solar cells not facing the Sun perfectly normally [168].

$$P_{BOL} = P_o I_d^{BOL} \cos \theta \quad (66)$$

The BOL inherent degradation I_d^{BOL} accounts for degradation due to design and assembly, where substrate area is lost due to the additional area required for assembling and interconnecting all the solar cells together, the array temperature, where the array heats up due to receiving normal incident radiation, and shadowing, where other appendages of the satellite shadow the solar array.

The design and assembly loss can be estimated using the datasheet [170], and is given by the ratio of the surface area of a single cell assembly (78.62 cm^2) to that of a bare cell (77.44 cm^2). This ratio is equal to $\frac{77.44 \text{ cm}^2}{78.62 \text{ cm}^2} = 0.985$.

The degradation due to the array temperature can be found using the temperature coefficients in the datasheet [170]. An engineering choice is made for the array temperature to be 65°C , which is typical for satellite missions at these radii and was used in the design example in [165]. This also aligns within the expected thermal range predicted by the THM group. For the BOL power generation of each solar cell, the datasheet [170] states that the reference maximum power point voltage is $V_{MP}^{25^\circ\text{C}} = 2.97 \text{ V}$ and the reference maximum power point current is $I_{MP}^{25^\circ\text{C}} = 1.097 \text{ A}$, both for a reference temperature of 25°C . The temperature coefficients at BOL [170] are given by $\Delta V_{MP}/\Delta T = -8.6 \frac{\text{mV}}{^\circ\text{C}}$ and $\Delta I_{MP}/\Delta T = 0.078 \frac{\text{mA}}{^\circ\text{C}}$. Therefore, the maximum power point voltage and current at 65°C are $V_{MP}^{65^\circ\text{C}} = 2.626 \text{ V}$ and $I_{MP}^{65^\circ\text{C}} = 1.100 \text{ A}$ respectively, as seen in Equations 67 and 68. This gives a temperature-adjusted BOL power of $P_{BOL}^{65^\circ\text{C}} = V_{MP}^{65^\circ\text{C}} I_{MP}^{65^\circ\text{C}} = 2.89 \text{ W}$. Compared to the P_o , this is a degradation of $\frac{2.89 \text{ W}}{3.26 \text{ W}} = 0.887$.

$$V_{MP}^{65^\circ\text{C}} = V_{MP}^{25^\circ\text{C}} + \frac{\Delta V_{MP}}{\Delta T} \Delta T = 2.97 \text{ V} - 8.6 \frac{\text{mV}}{^\circ\text{C}} \times (65 - 25)^\circ\text{C} = 2.626 \text{ V} \quad (67)$$

$$I_{MP}^{65^\circ\text{C}} = I_{MP}^{25^\circ\text{C}} + \frac{\Delta I_{MP}}{\Delta T} \Delta T = 1.097 \text{ A} + 0.078 \frac{\text{mA}}{^\circ\text{C}} \times (65 - 25)^\circ\text{C} = 1.100 \text{ A} \quad (68)$$

There is assumed to be minimal shadow from other appendages, therefore the degradation due to shadowing is assumed to be 1.00. The total BOL degradation I_d^{BOL} due to all three factors is given by the product of all three degradations, and is equal to $0.985 \times 0.887 \times 1.00 = 0.874$.

A worst-case cosine loss corresponding to 23.5° is assumed to align with the inclination of the satellite as a worst case estimate. This is quite a large value and is equal to the inclination of the MATRIX mission satellites. This is also the worst-case cosine loss angle used in the GEO design example in [168]. This adds a significant margin for power generation. The cosine loss is then $\cos 23.5^\circ = 0.917$.

Equation 66 then gives the actual BOL power production capability per cell as $2.613 \frac{\text{W}}{\text{cell}}$, as shown in Equation 69.

$$P_{BOL} = 3.26 \times 0.874 \times 0.917 = 2.613 \frac{\text{W}}{\text{cell}} \quad (69)$$



10.7.5.5 EOL Power Production Capability

Through an identical approach, the actual EOL power production capability is given by Equation 70, where I_d^{EOL} is the inherent degradation at EOL and $\cos \theta$ is the cosine loss due to the solar cells not facing the Sun perfectly normally [168]. However, note that the output power used is no longer P_o , but rather P_o^{EOL} , which is the maximum output power per cell at EOL. To calculate this degraded value, the total amount of energy passing through a unit cross sectional area over the life span of a solar cell (fluence value) must be estimated. The fluence of a solar cell for a 15-year GEO mission is 1×10^{15} [165]. The MATRIX mission has two satellites with a maximum altitude of apogee at $12R_E$, so this does not perfectly correspond to a GEO mission. However, the duration of the MATRIX mission is 40% that of a 15-year GEO mission. The power also has a logarithmic relationship to fluence value [177], so taking a sufficiently large estimate of fluence at 1×10^{15} will still give a reasonable estimate of the EOL power generated per unit area from the solar cell. From the datasheet [170], the EOL reference maximum power point voltage is 2.744 V and the reference maximum power point current is 1.072 A at a reference temperature of 25°C. Therefore, the corresponding reference maximum power at EOL is $P_o^{EOL} = 2.942 W$.

$$P_{EOL} = P_o^{EOL} I_d^{EOL} \cos \theta \quad (70)$$

The EOL inherent degradation I_d^{EOL} accounts for degradation due to design and assembly, where substrate area is lost due to the additional area required for assembling and interconnecting all the solar cells together, the array temperature, where the array heats up due to receiving normal incident radiation, and shadowing, where other appendages of the satellite shadow the solar array.

In the same manner as the BOL case, the design and assembly loss can be estimated using the datasheet [170], and is equal to $\frac{77.44 \text{ cm}^2}{78.62 \text{ cm}^2} = 0.985$.

The degradation due to the array temperature can be found using the temperature coefficients in the datasheet [170]. As before, the array temperature is assumed to be 65°C. For the EOL power generation of each solar cell, the datasheet [170] states that the reference maximum power point voltage is $V_{MP}^{25^\circ C} = 2.744 V$ and the reference maximum power point current is $I_{MP}^{25^\circ C} = 1.072 A$, both for a reference temperature of 25°C. The temperature coefficients at EOL [170] are given by $\Delta V_{MP}/\Delta T = -9.0 \frac{mV}{^\circ C}$ and $\Delta I_{MP}/\Delta T = 0.179 \frac{mA}{^\circ C}$. Therefore, the maximum power point voltage and current at 65°C are $V_{MP}^{65^\circ C} = 2.384 V$ and $I_{MP}^{65^\circ C} = 1.079 A$ respectively, as seen in Equations 71 and 72. This gives a temperature-adjusted EOL power of $P_{BOL}^{65^\circ C} = V_{MP}^{65^\circ C} I_{MP}^{65^\circ C} = 2.572 W$. Compared to the P_o , this is a degradation of $\frac{2.572 W}{2.942 W} = 0.874$.

$$V_{MP}^{65^\circ C} = V_{MP}^{25^\circ C} + \frac{\Delta V_{MP}}{\Delta T} \Delta T = 2.744 V - 9.0 \frac{mV}{^\circ C} \times (65 - 25)^\circ C = 2.384 V \quad (71)$$

$$I_{MP}^{65^\circ C} = I_{MP}^{25^\circ C} + \frac{\Delta I_{MP}}{\Delta T} \Delta T = 1.072 A + 0.179 \frac{mA}{^\circ C} \times (65 - 25)^\circ C = 1.079 A \quad (72)$$

There is assumed to be minimal shadow from other appendages, therefore the degradation due to shadowing is assumed to be 1.00. The total EOL degradation I_d^{EOL} due to all three factors is given by the product of all three degradations, and is equal to $0.985 \times 0.874 \times 1.00 = 0.861$.

A significant worst-case cosine loss corresponding to 23.5° is again assumed as an engineering choice. The cosine loss is therefore $\cos 23.5^\circ = 0.917$.



Equation 70 then gives the actual EOL power production capability per cell as $2.323 \frac{W}{cell}$, as shown in Equation 73.

$$P_{EOL} = 2.942 \times 0.861 \times 0.917 = 2.323 \frac{W}{cell} \quad (73)$$

As a sanity check, an approximate degradation per year D is back-calculated for the EOL power. The lifetime degradation L_d is given by Equation 74 [168], and is equal to 88.9%. The degradation per year is given by Equation 75 [168] and is equal to 1.94% per year, where the mission lifetime L is 6 years. This is within the expected range for multijunction GaAs cells [168], which suggests this EOL power calculation is correct.

$$L_d = \frac{P_{EOL}}{P_{BOL}} = \frac{2.323}{2.613} = 88.9\% \quad (74)$$

$$D = 1 - L_d^{1/L} = 1 - 0.889^{1/6} = 1.94\% \quad (75)$$

According to [165], a slightly lower power per cell to the left of the maximum power point should be used for calculations, to avoid a large drop in power beyond the maximum power point. This is taken to be $P_{c,EOL} = 2.3 \frac{W}{cell}$, with an operating voltage of 2.3 V and an operating current of 1.0 A. This is only 1% lower than the calculated value of P_{EOL} , however note that the calculation for both P_{BOL} and P_{EOL} was conducted assuming quite a large cosine loss, corresponding to the worst-case of 23.5° . This is very unlikely to be the value of the cosine loss during most of the mission, because it is expected that the spacecraft will continually maneuver to face the Sun. Therefore, the value of $P_{c,EOL} = 2.3 \frac{W}{cell}$ is a valid engineering choice.

10.7.5.6 Solar Array Area

The number of series cells n_s is found by dividing the operating bus voltage by the operating cell voltage, and is equal to $28.8 V / 2.3 V = 13$ [168]. The number of parallel strings n_p is found by dividing the power required by the power generated by each series string, and is equal to $P_{sa} / n_s P_{c,EOL} = 143.6 / (13 \times 2.3) = 5$ [168]. Hence, a total of $n_t = n_s n_p = 65$ cells are required for power generation. This corresponds to a total area of $65 \times 77.44 = 5033.6 \text{ cm}^2 = 0.50 \text{ m}^2$. This total required surface area signifies that the solar arrays will fit onto the sun-facing hexagonal surface of each satellite with an available surface area of about 0.8 m^2 . This signifies that the solar array will not need to be deployable as all of the necessary solar cells will fit onto the sun-facing side of the spacecraft.

Due to the position of the solar arrays mounted on the sun-facing side of the spacecraft, the maximum amount of power that it is possible to produce without deployable solar arrays will be generated as this face is always pointing at the sun. This optimal pointing allows for the solar arrays to generate the most power possible of any orientation if the solar arrays are only positioned on one face and are not deployed off of the body of the spacecraft. In this sun-facing orientation with the solar arrays pointed towards the sun, the spacecraft will provide enough power to support the electrical loads of the system and recharge the battery.

10.7.5.7 Solar Array Mass

As each cell has a mass of 10.4 g [170], the total mass for the solar array, excluding any support for deployment, is $65 \times 10.4 \text{ g} = 676 \text{ g} = 0.68 \text{ kg}$.



10.7.6 Power Management & Distribution

Analysis conducted by: Tasman Moskowitz, All Components (Refinement Since PDR)

The power management and distribution (PMAD) system controls the flow of power to the spacecraft and instruments [169]. It is an essential component in ensuring that power needs of each subsystem are consistently met, and power management is done efficiently between the power generated by the solar panels and the power stored and discharged by the batteries.

The first step taken to select a PMAD system was to choose a power regulation/distribution architecture. The two main PMAD system architectures are peak-power trackers (PPT) and direct-energy-transfer (DET) [53]. A PPT PMAD system is nondissipative, meaning that it will extract the exact amount of power a spacecraft requires up to the array's peak power. In contrast, a DET PMAD system will dissipate power that is not used by the loads in an external bank of shunt resistors.

The batteries in the MATRIX mission were sized off of the maximum eclipse time constraints experienced by the Group 1 satellites, while the solar arrays were sized based off of the minimum sun time constraint experienced by the Group 3 solar arrays. This signifies that solar arrays of the Group 1 & 2 satellites will generate more energy than the peak energy load as they experience greater times in the sun than the Group 3 satellites. This means that this extra power must be regulated using the PMAD system. A PPT architecture is selected to adaptively change the operating point of the solar panel to track the peak power point when energy demand exceeds the peak power [53]. This decision is made to reduce the thermal load due to the power subsystem that would be experienced if power was dissipated in a DET architecture. The PPT PMAD architecture is a DC-DC system in series with the solar panels, which means that if components operate at the same voltage, less mass and volume will be required as no converter is necessary.

To select an optimal PMAD system, commercially available products were researched as listed in Table 89. Mass, volume, peak power output, input and output voltage levels, maximum efficiency, and cost are the parameters compared to select a PMAD system in this trade study.

Parameter	AAC Clyde Space	GomSpace	Pumpkin Space Systems	Pumpkin Space Systems
PMAD System Name	Starbuck Micro	P31U	EPSM 1	AMPS
Mass (kg)	2.45	0.1	0.3	1.3
Volume (cm ³)	3968	127	180	360
Peak Power Output (W)	80.4	30	300	1200
Input Voltages (V_{DC})	28	0-8	4-32	5-32
Output Voltages (V_{DC})	28 / 5	3.3 / 5	3.3-28	-
Max Efficiency (%)	97	96	99	99
Cost (\$)	Not available	Not available	48,000.00	Not available

Table 89: Comparison of Power Management and Distribution (PMAD) Systems [169]

The AAC Clyde Space Starbuck Micro and the GomSpace P31U cannot support the required power output of 184.8 W, so they cannot be selected. Volume is a critical consideration for the PMAD system due to the space constraint within each satellite. The Pumpkin Space Systems

AMPS has twice the volume of the EPSM 1, and literature is not available for its defined output voltage. Therefore, the MATRIX mission POW system selects the Pumpkin Space Systems EPSM 1 PMAD system. This has low mass and volume, meets the power output requirement, has input and output voltage ranges that include the bus voltage of the satellite, and has high efficiency. Generally, these systems are often custom designed to meet specific mission requirements, but well-known producers of PMAD systems also offer OTS solutions recommended by [169].

10.7.7 POW Subsystem Design Summary

Analysis conducted by: Tasman Moskowitz, All Components

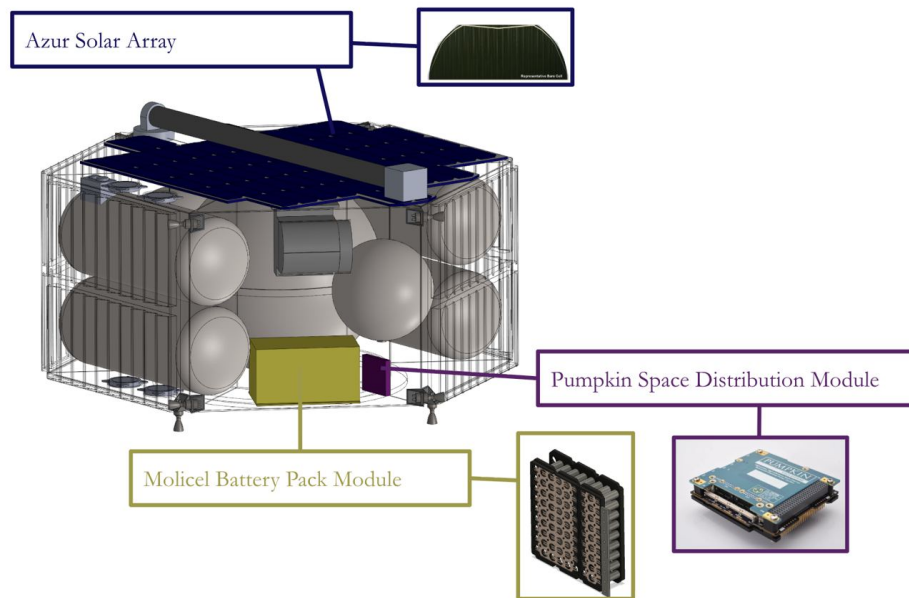


Figure 41: Overview of POW Subsystem Components in MATRIX Spacecraft

Figure 41 shows an illustration of all the significant POW subsystem design components on a MATRIX spacecraft. Table 90 summarizes the design of the power storage and generation components of the POW subsystem. The calculations and design decisions utilized for the sizing of these components are detailed in Appendix A.9. The POW subsystem is controlled by the Pumpkin Space EPSM-1 PMAD system.



1. Common to All Battery Cells: Mission Properties & Requirements	
Parameter	Value
Nominal Eclipse Time (hrs)	4.356
Nominal Bus Voltage (V)	28.8
Nominal Cell Voltage (V)	3.6
Capacity Fade	0.1
DOD	0.6
Peak Power Draw (W)	89.8
Safety Factor	1.12761
Peak Power Draw with Safety Factor (W)	101.3
Current Drawn (A)	3.515950625
EOL Battery Capacity (Ah)	25.5
BOL Battery Capacity (Ah)	28.1
Series Cells	8

3. Common to All Solar Panel Cells: Mission Properties & Requirements	
Parameter	Value
Nominal Sun Time (hrs)	9.285
Nominal Bus Voltage (V)	28.8
Peak Power Draw w/o Batteries (W)	101.3
Sun Time Margin (hrs)	0.08
Battery Recharge Power (W)	32.2
Harness Drop Safety Factor	1.0251
Added Safety Factor	1.05
Total Power Required w/o Safety Factors	133.5
Total Power Required with Safety Factors	143.6

2. Selected Battery Cell: Battery Calculations	
Parameter	Value
Company	Molicel
Battery Product Name	M50A 21700
Cell Type	Li-Ion
Data Source	Here
Typical Capacity (Ah)	4.8
Mass Per Cell (g)	70
Cell Diameter (cm)	2.17
Cell Height (cm)	7.02
Volume Per Cell (cm ³)	26.0
Max Discharge Rate (A)	15
Current Drawn per Parallel String (A)	0.59
Series Cells	8
Parallel Strings	6
Total Cells in Battery	48
Total Mass of Battery (kg)	3.36
Total Volume of Battery (cm ³)	1246
Total Battery Output till Drain (Wh)	6635.52

4. Selected Solar Panel Cell: Solar Panel calculations	
Parameter	Value
Company	AZUR Space
Solar Panel Product Name	4G32-Adv
Data Source	Here
Operating Voltage (V)	2.3
Operating Current (A)	1
Power Generated Per Cell (W)	2.3
Series Cells	13
Parallel Strings	5
Total Cells	65
Area Per Cell (cm ²)	77.44
Total Area Required for Solar Panels (m ²)	0.50
Mass Per Cell (g)	10.4
Total Mass of Solar Panels (kg)	0.68

Table 90: Summary of Battery and Solar Panel Characteristics and Calculations

10.7.7.1 Safety Factors & Margin

A list of the safety factors and margins considered in the design of the POW subsystem are listed below. The high net value of the combined safety factor and margins for the POW subsystem of the MATRIX mission suggest that the POW subsystem will be extremely reliable with a high redundancy. Therefore, the MATRIX POW subsystem will be able to store, generate, and distribute enough power to all of the mission subsystems to support the scientific and operation objectives of the MATRIX mission.

- 10% Margin on peak power draw for any mode
- 2% Harness drop safety factor on peak power draw for any mode
- 0.5% Safety factor assumed due to drop in power across the PMAD system
- Sun time margin of 0.08 *hr*



- 10% Capacity fade over the mission lifetime
- Assumed fluence value of 1×10^{15} over the lifetime of the MATRIX mission
- 5% margin on the power requirement for the solar panels

10.7.7.2 Mass, Volume, and Cost

Table 91 shows the total mass, volume, and cost of the POW subsystem. The POW subsystem remains compliant with the mass, volume, and monetary budget allocated to the subsystem for the MATRIX mission.

Parameter	Per Spacecraft	Total for Mission
Total Mass of Battery Pack (w/o enclosure) [kg]	3.36	26.88
Total Mass of Solar Array [kg]	0.68	5.408
Total Mass of PMAD [kg]	0.3	2.4
Total Volume of Battery Pack (w/o enclosure) [m ³]	0.00125	0.00997
Total Surface Area of Solar Array [m ²]	0.50	4.02688
Total Volume of PMAD [m ³]	0.00018	0.00144
Cost of Battery Pack (\$)	587.76	4702.08
Cost of Solar Array (\$)	29,250.00	234000
Cost of PMAD (\$)	48,000	384000
Total POW Subsystem Mass [kg]	4.34	34.688
Total POW Subsystem Volume [m ³]	0.00143	0.01141
Total POW Subsystem Cost (\$)	77,838	622702.08

Table 91: Power Subsystem Mass, Volume, and Cost Summary

11 PLV Design

11.1 Subsystem Overview

The PLV system is responsible for the launch and orbital insertion of the mission to its parking orbit of $9R_E$. The PLV subsystem then must provide the necessary Δv to perform maneuvers for the satellites to obtain their proper spacing and then perform their apogee raising and lowering burns to their Group 1 & 3 orbital trajectories. Then the subsystem must provide the Δv to support stationkeeping and momentum dumping maneuvers over the course of the mission lifetime, so that the mission is able to stay on its planned orbital trajectory to achieve its science objectives. The final responsibility of the PLV subsystem is to perform the perigee lowering burns that will cause each mission satellite to burn up upon atmospheric reentry for mission disposal. The subsystem also has the goal to comply to the monetary, mass, and volume budgets allocated to the PLV subsystem. The propulsion system on each MATRIX spacecraft also must provide enough Δv for the end-of-life disposal of eight satellites. There is a structural interaction between SMM/THM and PLV when considering propulsion system layout, propellant tank design, and the thermal requirements of the propellant. There are also interactions between GNC/ADCS, POW, CCD, and the propulsion system, so that necessary impulsive maneuvers can be executed.

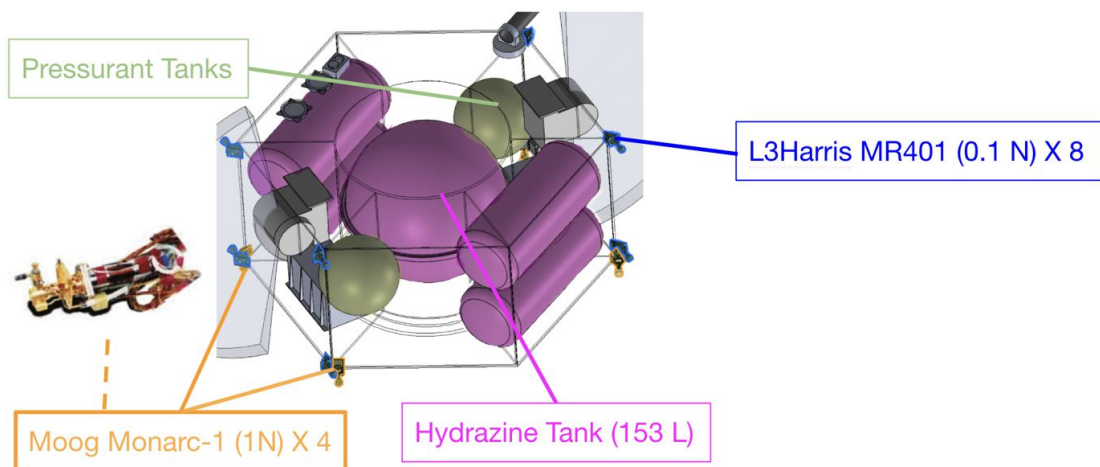


Figure 42: Overview of the PLV Subsystem Architecture Including Hydrazine Propellant Tanks, Pressurant Tanks, and Thrusters

Figure 42 shows an overview of the propulsion subsystem design for each of the MATRIX spacecraft. The design consists of four cylindrical and one spherical hydrazine propellant tanks, two pressurant tanks, and 12 hydrazine thrusters. This propulsion system design aims provide the Δv and maneuver capabilities for each spacecraft to carry out the scientific and operational objectives of the MATRIX mission.

11.2 Subsystem Objectives

The MATRIX mission level objectives serve to conduct a targeted and physically grounded investigation into BBF evolution across the $6 - 12R_E$ corridor. The three mission objectives are



to characterize spatial/temporal structure of BBFs in the region of $6 - 12R_E$, resolve radial evolution/braking in the region of $6.6 - 9R_E$, and quantify magnetic field and particle signatures associated with BBFs in the magnetotail transition region. Each of the seven main PLV objectives is directly traced down from the mission level objectives, because they aim to allow each mission satellite to reach and maintain its proper orbit to fulfill these three mission scientific objectives.

The Propulsion & Launch Vehicle (PLV) subsystem aims to provide the Δv and propulsion related hardware design to allow other mission subsystems to support the overall mission scientific objectives. Therefore the subsystem's main objectives are as follows:

1. Provide Δv capabilities and perform burns to insert all satellites in the MATRIX mission into their parking orbit at an apogee of $9R_E$
2. Provide Δv capabilities and perform burns to allow the eight MATRIX satellites to obtain proper spacing and orbital periods to take measurements at proper spatial scales
3. Provide Δv capabilities and perform burns for the Group 1 and Group 3 satellites to raise/lower their apogees to 7 & $12R_E$ respectively
4. Provide Δv capabilities for GNC and ADCS to perform all necessary stationkeeping maneuvers for each satellite
5. Provide Δv capabilities for ADCS to perform all necessary momentum dumping burns for each satellite
6. Perform all relevant mission operations without impeding the functionality of other subsystems by remaining within the allocated mass, volume, and power budgets.
7. Select a launch vehicle to deliver all mission satellites to their parking orbit with an apogee of $9R_E$ from Earth's surface

11.3 Subsystem Requirements

The primary PLV subsystem requirements are listed below in Table 92. There have been no major changes since the PDR, except for updates to the power, mass, and volume constraint requirements. A complete and comprehensive list of all requirements, along with trace up requirements and relation with subsystem objectives, can be found in the PLV tab of the "Requirements Spreadsheet" Google Sheet: [PLV Subsystem Requirements List](#).



ID	Requirement	Rationale	Method	Trace	Obj
PLV-F-001	The PLV subsystem shall provide a specific impulse (Isp) of at least 200 s to the mission architecture	A specific impulse of at least 200 s is necessary to perform the transfer and stationkeeping maneuvers. This minimum is set through referencing heritage missions utilizing hydrazine.	Analysis	MISS-F-015, MISS-F-001, MISS-F-002, MISS-F-003	1-6
PLV-F-003	The PLV subsystem shall inform and facilitate the Delta-V budget supplied by GNC to perform all required maneuvers	The amount of propellant required, and thus tank sizing, is determined by the Delta-V budget	Analysis	MISS-F-001, MISS-F-002, MISS-F-005, MISS-F-015	1-5
PLV-F-005	The PLV subsystem shall identify and characterize available fuel and thruster options	Different propulsion methods have varying properties (density, storage temperature), informing thruster and tank design	Analysis	MISS-F-015	1-5
PLV-F-010	The launch vehicles selected by the PLV subsystem shall launch all mission components by the end of 2032 at the latest	The timeline for launch is critical for the MATRIX mission to deliver spacecraft on schedule	Analysis	MISS-C-002, MISS-C-003, MISS-F-006	1-5
PLV-F-011	The PLV subsystem shall perform trade studies for launch vehicles based on mass and volume requirements	Launch vehicle choice determines mass/volume budgets; trade studies help identify viable options	Analysis	MISS-C-002, MISS-C-003	7
PLV-F-012	The PLV subsystem shall select launch vehicles to provide proper trajectory insertion	Launch vehicle must insert components into the planned transfer trajectory	Analysis	MISS-F-001, MISS-F-005	7
PLV-F-018	The PLV subsystem shall collaborate with GNC and SMM to determine thruster placement along required axes	To align thrusters with required axes of control	Analysis	MISS-F-015	6
PLV-C-001	The propulsion components shall have a cumulative mass of at most 63 kg (dry) and 157 kg (wet)	Compliance with mission mass budget	Subsystem mass budget	MISS-C-003, MISS-C-001	6
PLV-C-002	The propulsion components shall have a cumulative volume of at most 0.145 m ³	Compliance with mission volume budget	Subsystem volume budget	MISS-C-002, MISS-C-001	6
PLV-C-003	The propulsion components shall have a peak power consumption of at most 87 W	Compliance with mission power budget	Subsystem power budget	MISS-F-016	6

Table 92: PLV Subsystem Selected Requirements

11.4 Subsystem Constraints

The major changes since the PDR include updates to the mass, volume, and power constraints. These new values are reflected in the constraint requirements highlighted in Table 92. The most significant constraints in the MATRIX mission are cost, timeline, mass, volume, and environmental durability. The PLV subsystem constraints are derived from the mission-level constraints and include the following:

- **Mass & Volume of Propellant:** The PLV subsystem must remain within its mass allocation to meet launch vehicle and structural constraints provided by the SMM subsystem. These constraints are linked to **MISS-C-001** and **MISS-C-002**. This limits the mass of propellant available, and therefore heavily informs propellant and thruster selection in order to maximize the allocated mass. Furthermore, propellant selection is limited by volume constraints.
- **Mass & Volume of Propellant Tank:** While the PLV subsystem must adhere to the mass and volume budgets allocated by the SMM subsystem, the propellant tank also must be able to contain the amount of propellant necessary to provide Δv for maneuvers while still



fitting in the spacecraft. These constraints are linked to **MISS-C-001** and **MISS-C-002**. This tank sizing constraint limits the amount of propellant each satellite can carry. It also heavily informs propellant and thruster selection in order to maximize the allocated mass.

- **Cost of Propellant:** The PLV subsystem must remain within the monetary budget allocated to the PLV subsystem, and this cost constraint heavily influences propellant selection. This constraint is linked to **MISS-C-003**.
- **Cost of Thrusters:** The PLV subsystem must remain within the monetary budget allocated to the PLV subsystem, and this cost constraint heavily influences thruster selection. This constraint is linked to **MISS-C-003**. Thruster performance may need to be sacrificed for remaining within budget constraints.
- **Power Draw of Thrusters:** The PLV subsystem must operate within the power budget for safety and preserving power for science, data, and communications. This constraint is linked to **MISS-C-004** and **MISS-F-016**. This constraint is a heavily determining factor for chemical versus electric propulsion, and will heavily influence thruster selection.
- **Propellant Thermal Environment:** The propulsion system must reliably operate within the thermal range necessary for the selected propellant. This will require collaboration with the THM subsystem to ensure that the propulsion system will function in the space thermal environment. This constraint is linked to **MISS-E-001**, **MISS-E-002**, **MISS-E-003**, **MISS-E-004**, and **MISS-E-005**.
- **Time to Perform Burns:** The PLV subsystem design is constrained by the time it takes to perform propulsive maneuvers. It is essential that the propulsion system is able to execute maneuvers as described by the mission and GNC timelines. This constraint is linked to **MISS-F-008**.

11.5 Subsystem Drivers

The PLV subsystem drivers are listed below and remain the same as the previous design iteration:

- **Δv Capability :** The PLV subsystem must provide sufficient total Δv per group to achieve the three required orbit geometries, corrective maneuvers, and perform decommissioning burns. This stems from the mission-level driver Orbit Geometry for Transition Region Access which requires orbit within the magnetotail transition region ($r_a \sim 7 R_E, 9 R_E, 12 R_E$).
- **Stationkeeping and Spacecraft Spacing:** The PLV subsystem must also be able to reliably provide small repeatable thrust impulses to maintain spacecraft spacing within Groups 1, 2 and 3. This comes from the mission driver Accurate Constellation Stationkeeping, which requires precise relative positioning between spacecraft in order to satisfy the other mission driver, 2-D Field and Particle Measurements for Mesoscale Science.
- **High Reliability for Sustained Science Operations:** The PLV subsystem must operate reliably throughout the mission duration. It must be able to maintain its integrity after multiple thruster firings and meet the magnetotail environmental constraints. This subsystem level driver supports the mission level driver 2-D Field and Particle Measurements for Mesoscale Science.



- **Minimized Power Consumption:** The PLV subsystem should operate within the POW budget in order to increase available power for the scientific mission functions. This relates to mission-level driver Data Storage and Communication, which requires significant onboard power for instruments, as well as data handling and communication. By utilizing hydrazine, a monopropellant, electrical load is reduced when compared to electric propulsion or bipropellant systems.
- **Minimized Cost of Thrusters and Propellant:** The PLV subsystem should strive to select thrusters and propellant which minimize cost. This alleviates the budget for other subsystems and allows more funds to go toward other critical areas, including scientific instruments and data storage. This connects to the mission-level drivers 2-D Field and Particle Measurements for Mesoscale Science, and Data Storage and Communication.

These drivers for propulsion and launch vehicle selection follow directly from mission-level requirements. Δv capability is shaped by the multi-spacecraft launch configuration and the science orbit; stationkeeping and spacecraft spacing is dictated measurement requirements for mesoscale science; high reliability supports sustainable mesoscale measurements and reliable downlink; and minimized power consumption and thruster and propellant cost is dictated by the MATRIX power and cost budgets respectively.

11.6 Subsystem Design Approach

The PLV subsystem design approach is to begin with the mission-level maneuver requirements, translate those requirements into propulsion-system sizing constraints, and then use those constraints to select and refine the propellant, storage, pressurization, feed, and thruster architecture. The design process is therefore requirements-first: the maximum required mission Δv , maneuver timing, pointing accuracy, minimum impulse bit, mass allocation, volume envelope, thermal environment, and power availability are used as inputs to the propulsion design rather than checked only after a design has already been selected.

The first step in the approach is to close the mission Δv requirement using a propellant-level trade. Hydrazine and ASCENT are compared using the rocket equation, required propellant mass, required storage volume, thermal behavior, operational risk, heritage, and supplier availability. This establishes the propellant choice before the thruster trade, so the selected propellant becomes a constraint on which catalog thrusters are considered. Once the propellant is selected, the propulsion architecture is sized around the amount of propellant needed to meet the maximum required Δv with margin.

The second step is to fit the propellant and pressurant storage into the spacecraft geometry. Instead of assuming a single idealized tank, the updated approach uses the available SMM packaging envelopes to define a distributed tank architecture. Tank geometries are selected based on pressure-vessel efficiency, available volume, center-of-mass placement, and manufacturability. The tank sizing process includes both volume closure and wall-thickness checks so that the storage architecture is not only volumetrically feasible, but also structurally reasonable.

The third step is to define the feed and pressurization system. The selected architecture treats the distributed propellant tanks as one effective reservoir from the perspective of the thrusters, using a manifolded network of piping, valves, and flow-control elements. Pressurant sizing is performed



using the selected propellant volume and required feed-pressure range, while the plumbing concept is designed to drain symmetric tanks together to reduce center-of-mass migration. This links the feed-system design directly to both propulsion performance and spacecraft dynamics.

The fourth step is to select thrusters and propulsion system architecture and verify subsystem compatibility. Thrusters are evaluated based on compatibility with the selected propellant, required thrust level, feed pressure, minimum impulse bit, burn duration, valve power, and control authority. The thruster trade is therefore not performed independently; it is constrained by the propellant selection, pressurization architecture, power subsystem limits, and ADCS/GNC pointing and control requirements.

The current design meets the subsystem requirements by providing a propulsion architecture that is sized from the required mission Δv , capable of supporting discrete maneuver execution, and compatible with the pointing and minimum impulse bit constraints imposed by ADCS/GNC. It meets the subsystem constraints by using tank geometries that fit within the available spacecraft packaging envelopes, by checking wall thickness and pressure compatibility for the selected storage and pressurization approach, and by accounting for thermal and power impacts associated with propellant storage and thruster operation. It also addresses the subsystem drivers by prioritizing reliable Δv delivery, mature propulsion hardware, manageable integration risk, mass and volume efficiency, center-of-mass control, and compatibility with the rest of the spacecraft. A simplified flow chart of the analysis is shown in Figure 43.

Since the PDR, the PLV subsystem has been updated primarily by replacing preliminary assumptions with finalized design values and architecture decisions. The propellant storage system has been refined from an idealized tank model into a defined five-tank hydrazine architecture consisting of four cylindrical side tanks and one spherical central tank, with the available fuel volume and corresponding propellant mass now explicitly quantified. The pressurization system has also been finalized as a two-tank helium pressurant architecture connected through a common regulated pressurant manifold. In parallel, the propellant and pressurant feed systems have been developed into a defined manifolded plumbing concept, including symmetric secondary-tank depletion followed by central-tank depletion to reduce center-of-mass migration. The delta- v budget has also been finalized, providing a clearer basis for propellant sizing, margin assessment, and thruster selection.

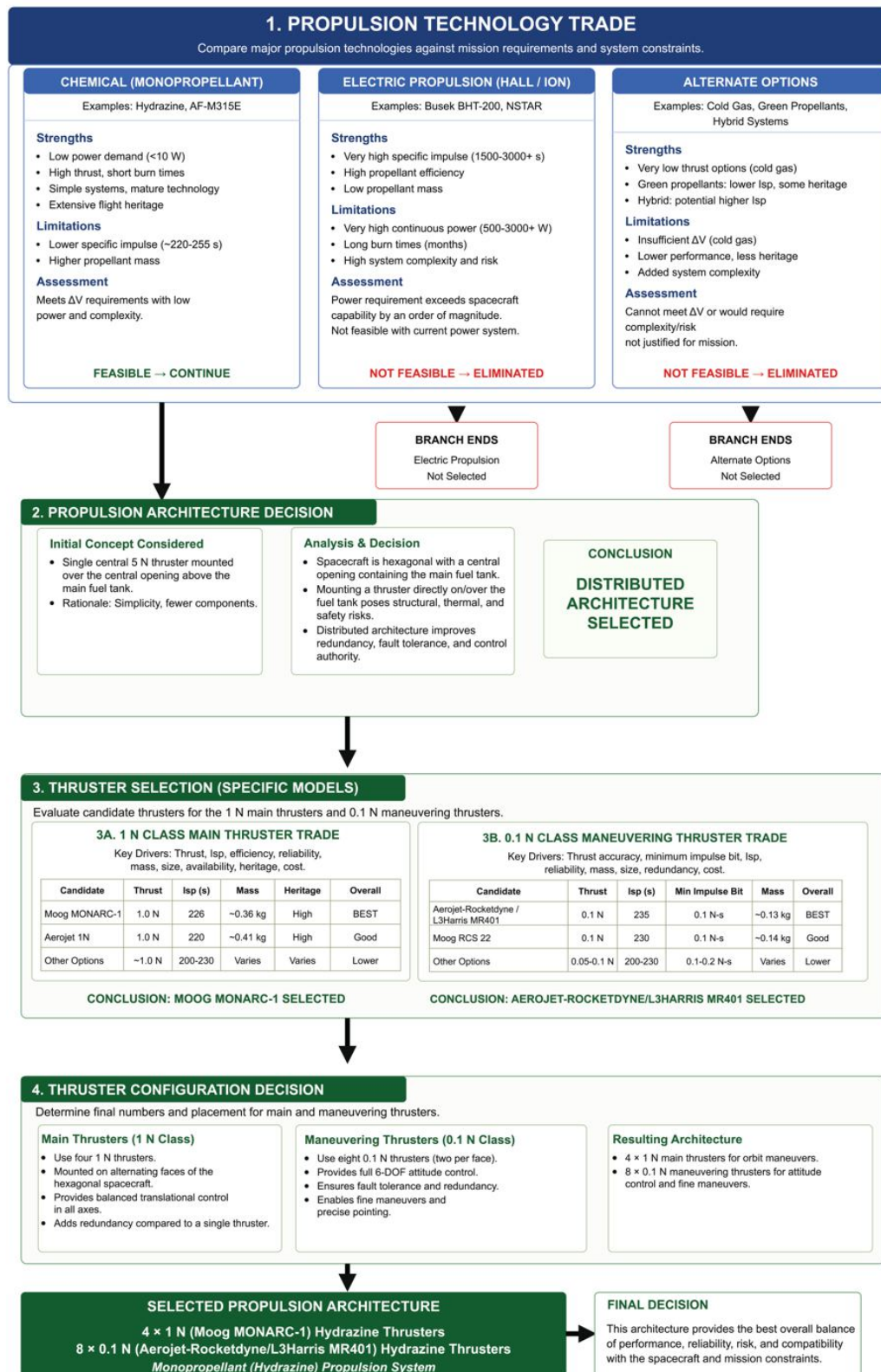


Figure 43: PLV Design Approach Flow Chart



11.7 Formal Analysis

11.7.1 Launch Vehicle Analysis

Analysis conducted by: Tasman Moskowitz, All Categories

For a Medium Explorer (MIDEX) class mission, NASA optionally provides launch services as described in the MIDEX AO Launch Services Program Information Services [139]. The launch service includes the Launch Vehicle (LV) and all associated standard services with mission unique options, all engineering analysis, and a minimum performance standard [139]. The Launch Services Program (LSP) will select a compatible launch service from a set of available vehicles for the mission. In order to utilize LSP, a mission must meet the requirements for using the service. These include mass, volume, payload fairing, stresses, and vibrational loads. The MATRIX requires a launch vehicle to insert all eight satellites into an elliptical orbit with an apogee altitude of $a_a = 9R_E$ and an inclination of $i = 23.5^\circ$. Therefore, it is necessary to determine the C_3 energy of this orbit.

The C_3 energy of an object is defined as twice its specific total energy ϵ , which is the sum of its specific potential energy $-\frac{\mu}{r}$ and specific kinetic energy $\frac{v^2}{2}$, as shown in Equation 76, where μ is the standard gravitational parameter for Earth, r is the instantaneous orbital radius, and v is the instantaneous orbital speed.

$$C_3 = 2\epsilon = 2 \left(\frac{v^2}{2} - \frac{\mu}{r} \right) \quad (76)$$

By the vis-viva equation, the specific total energy ϵ is constant for an object in an elliptical orbit and is equal to $-\frac{\mu}{2a}$, where a is the semi-major axis. The launch vehicle has a parking orbit perigee altitude of $a_p = 1000 \text{ km}$ and an apogee altitude of $a_a = 9R_E$, which corresponds to a semi-major axis of $a = 35,600 \text{ km}$. Therefore, Equation 76 simplifies to Equation 77 and yields $C_3 = -11.2 \frac{\text{km}^2}{\text{s}^2}$ for the launch vehicle parking orbit semi-major axis of $a = 35,600 \text{ km}$.

$$C_3 = 2\epsilon = -\frac{\mu}{a} = -\frac{3.986 \times 10^5 \frac{\text{km}^3}{\text{s}^2}}{35,600 \text{ km}} = -11.2 \frac{\text{km}^2}{\text{s}^2} \quad (77)$$

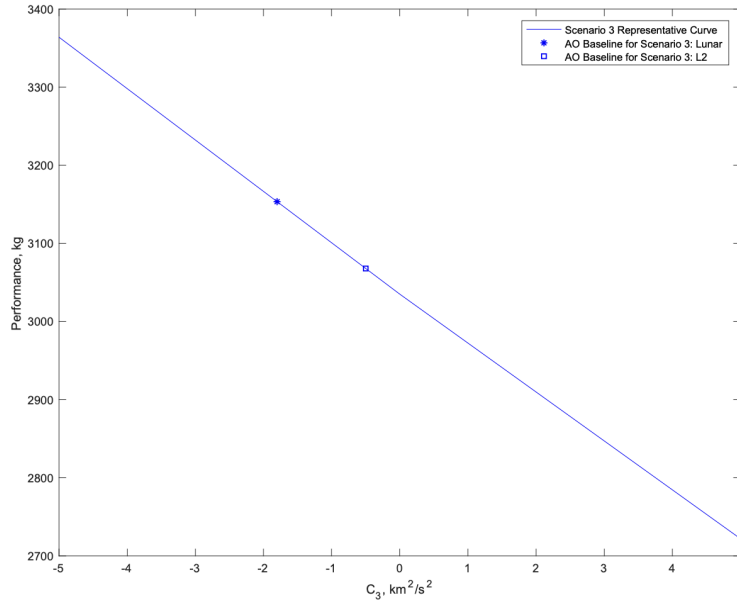


Figure 44: Scenario 3 Representative Performance Capability to High Energy Orbits

The graph of representative performance capacity to high energy orbits in Figure 44 from the MIDEX AO NASA LSP Information Summary [139] demonstrates that the performance varies linearly with C_3 energy. Therefore, a linear relationship can be established between the C_3 energy and the performance P by the linear Equation 78, since the C_3 of the MATRIX mission's required insertion orbit must be extrapolated from the graph. In Equation 78 m denotes a slope and P_0 denotes a P -intercept. Two fixed points are known from Table 3 of [139], where an L_2 orbit with $C_3 = -0.5 \frac{km^2}{s^2}$ has a performance of 3065 kg , and a lunar orbit with $C_3 = -1.8 \frac{km^2}{s^2}$ has a performance of 3150 kg . This allows the slope and intercept to be calculated, as shown in Equations 79 and 80.

$$P = mC_3 + P_0 \quad (78)$$

$$m = \frac{3150 - 3065 \text{ kg}}{-1.8 + 0.5 \frac{km^2}{s^2}} = -65.4 \frac{kg \text{ s}^2}{km^2} \quad (79)$$

$$P_0 = 3065 \text{ kg} - \left(-65.4 \frac{kg \text{ s}^2}{km^2} \right) \left(-0.5 \frac{km^2}{s^2} \right) = 3032.3 \text{ kg} \quad (80)$$

Finally, the calculated values of m and P_0 can be used in conjunction with the calculated value of the C_3 energy of the MATRIX mission launch vehicle parking orbit to yield the maximum allowable mass under Scenario 3. This value is 3764.8 kg , as shown in Equation 81.

$$P = \left(-65.4 \frac{kg \text{ s}^2}{km^2} \right) \left(-11.2 \frac{km^2}{s^2} \right) + 3032.3 \text{ kg} = 3764.8 \text{ kg} \quad (81)$$

This performance agrees with the results of the NASA Launch Services Program Performance Web Site [3] shown in Figure 45.

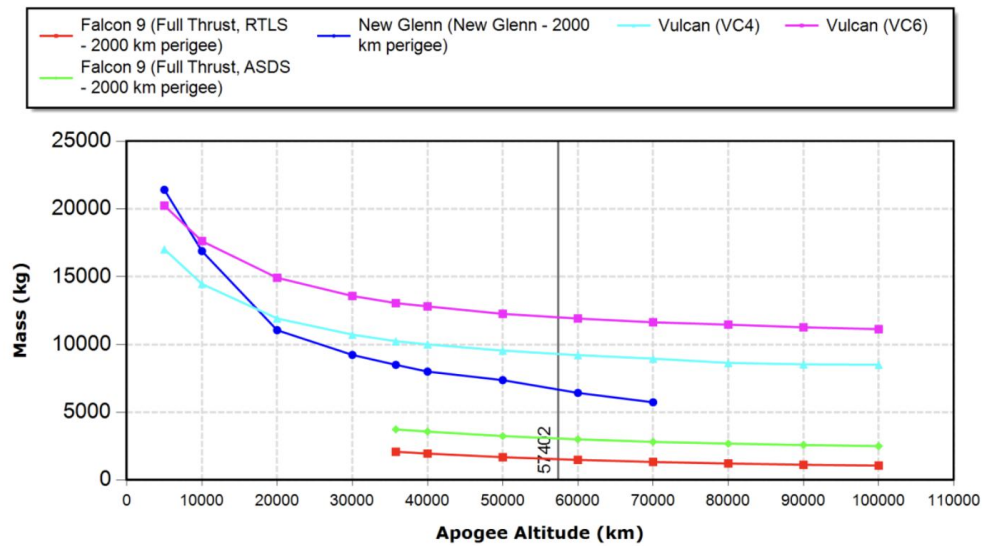


Figure 45: Performance (kg) vs. Orbital Insertion Altitude (km) for NASA LSP Vehicles

Figure 45 shows that as long as the MATRIX mission total payload mass remains under about 12000 kg it will be possible to use the NASA Launch Services Program to launch this MIDEX mission. However, the MATRIX mission is currently compliant with the required performance for the NASA LSP AO MIDEX Scenario 3 from [139] with a mass of under 3764.8 kg. Therefore, it is very likely Scenario 3 corresponds to the Falcon-9 launch vehicle in Figure 45. Therefore, NASA will most likely select the Falcon-9 as the launch vehicle for this mission. Figure 46 shows the dimensions of the Scenario 3 launch vehicle payload fairing envelope [139], and Figure 47 shows a CAD model of the eight MATRIX spacecraft arranged inside the payload fairing [178].

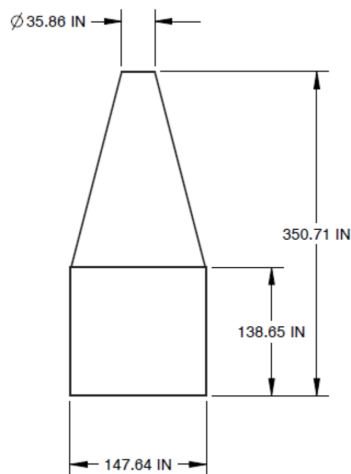


Figure 46: Scenario 3 PLF Static Envelope (inches) [139]

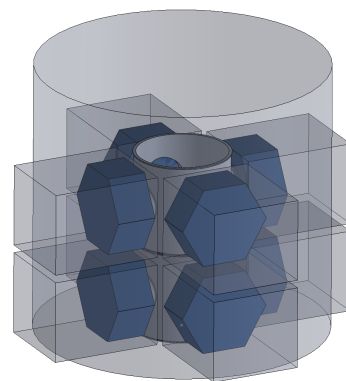


Figure 47: Eight MATRIX spacecraft loaded into the Scenario 3 launch vehicle payload fairing with ring payload adapters [178]

11.7.2 Background Information: Heritage Mission Analysis

Analysis conducted by: Arya Ollivierra (Carried over from PDR report)

Mission	No. of Spacecraft	Spacecraft Mass	Propellant Type	Propellant Mass	No. of Thrusters	Additional Info
CLUSTER II	4	1200 kg	Monomethyl hydrazine (MMH), and mixed oxides of nitrogen (MON1) [179]	650 kg	8 (10 N each)	Six propellant tanks, two pressure tanks, eight thrusters, one main engine [180]. Six spherical fuel tanks made from titanium are attached to the outside of this central cylinder. The fuel they carry accounted for more than half the launch weight of each satellite. Most of this fuel was consumed soon after launch during the maneuvers required to reach operational orbits. Each satellite also carries eight 10 N thrusters for smaller changes of orbit [181].
THEMIS	5	126 kg	Hydrazine	49 kg	4 (4.4 N each)	To control spacecraft attitude each bus hosts a Reaction Control Subsystem (RCS) – a hydrazine blowdown system pressurised with helium [182]. Total ΔV : 940 m/s.
MMS	4	1360 kg	Hydrazine	360 kg	12	Thrusters are used as actuators. The propulsion system is a mono-propellant blowdown system with 12 thrusters sized for both formation maintenance and apogee raise maneuvers. Approximately 360 kg of propellant is contained in four titanium tanks per spacecraft [183].

Table 93: Analysis of Propulsion Systems for Heritage Magnetosphere Missions

The information from these magnetosphere missions was helpful in guiding the PLV design for the magnetotail region. It can be seen that the heritage missions that focused on magnetosphere science typically utilized hydrazine monopropellant propulsion systems. The number of thrusters per spacecraft ranged from 4 to 12, with each being on the order of 4-10 N of thrust. Some included one main engine. Details on the number of propellant tanks and blowdown system was also useful. The wet mass fraction served as a sanity check for the PLV subsystem's calculations for the MATRIX spacecraft.

11.7.3 Background Information: Δv Requirements

Values tabulated by: Arya Ollivierra

The updated Δv requirements since the PDR from the GNC subsystem's STK analysis are presented in the table below:

Group #	Spacing (m/s)	Apogee Change (m/s)	Perigee Change (m/s)	Decommission (m/s)	Stationkeeping (m/s)	ADCS (m/s)	Total (m/s)
1	1.99	147.89	2.49	54.80	347.82	138.01	692.99
2	1.99	0.00	0.00	71.00	449.10	181.00	703.08
3	1.99	132.56	5.11	109.26	628.45	251.83	1129.19

Table 94: Delta V Budget Breakdown by Group

Therefore, the maximum required Δv is required by Group 3 and is about 1129 m/s. It can be seen that the highest stationkeeping and ADCS Δv is Group Three at 880 m/s. The PLV subsystem aims to meet these requirements with sufficient safety margin, discussed in the subsections below.



11.7.4 Propellant Selection and Justification

Analysis conducted by: Ted McCormack

The final propellant selection for the PLV subsystem is hydrazine monopropellant. This selection is based on a combined assessment of mission Δv closure, available tank volume, thermal compatibility, pressurization-system simplicity, component heritage, supplier risk, and integration risk. ASCENT was considered as a higher-performance alternative because it provides higher specific impulse and higher density than hydrazine, reducing the propellant mass and volume required for a given Δv [184], [185]. However, the final design favors hydrazine because it provides lower implementation risk, stronger flight heritage, broader component availability, and a more mature pressure-fed monopropellant architecture [186], [187].

The main quantitative advantage of ASCENT is its improved propellant efficiency. Using representative values of $I_{sp} = 235$ s for hydrazine and $I_{sp} = 257$ s for ASCENT, the propellant mass required for the maximum mission Δv requirement of 1129 m/s is lower for ASCENT. For the updated dry spacecraft mass of 120 kg, the rocket equation gives the required propellant masses shown in Table 95. These results show that ASCENT reduces required propellant mass by approximately 6–9 kg across the relevant Δv range, with the largest required maneuver case needing 75.9 kg of hydrazine or 67.8 kg of ASCENT. The detailed calculation is shown in Appendix A.10.1.

Table 95: Propellant mass and volume required for representative mission Δv cases using a 120 kg dry mass.

Δv Case	Hydrazine Mass	Hydrazine Volume	ASCENT Mass	ASCENT Volume
875 m/s	55.4 kg	55.2 L	49.8 kg	33.9 L
1000 m/s	65.2 kg	64.9 L	58.4 kg	39.8 L
1015 m/s	66.4 kg	66.1 L	59.5 kg	40.5 L
1100 m/s	73.4 kg	73.1 L	65.7 kg	44.7 L
1129 m/s	75.9 kg	75.6 L	67.8 kg	46.1 L
1200 m/s	82.0 kg	81.7 L	73.2 kg	49.8 L

Although ASCENT is more efficient on a mass and volume basis, the finalized hydrazine tank architecture provides enough capacity to close the mission with substantial margin. The current storage configuration consists of four cylindrical side tanks and one spherical central tank. Each side tank holds approximately 20.75 L, and the central tank holds approximately 70 L, for a total hydrazine storage volume of

$$V_{prop,total} = 4(20.75 \text{ L}) + 70 \text{ L} = 153.0 \text{ L}.$$

Using a hydrazine density of approximately 1.004 kg/L, this corresponds to approximately 153.6 kg of hydrazine. With a dry spacecraft mass of 120 kg, the resulting wet mass is

$$m_0 = 120 \text{ kg} + 153.6 \text{ kg} = 273.6 \text{ kg}.$$

This hydrazine load provides an estimated total Δv capability of approximately 1899 m/s. Relative to the maximum required Δv of 1129 m/s, this provides approximately 770 m/s of margin. Therefore, while ASCENT would reduce the propellant volume required, hydrazine is not volume-limited in the finalized tank architecture.



Thermal control is a significant discriminator between the hydrazine and ASCENT propellant options. Hydrazine freezes near 2°C , so the hydrazine storage and feed system is required to remain above 5°C to provide margin against freezing during eclipse and cold-case operation [187]. This requirement increases heater power demand for the propellant tanks, valves, and feed lines, especially because the MATRIX science orbit includes long eclipse periods. ASCENT has a much lower listed freezing point of -80°C , reducing the risk of propellant freezing and relaxing storage-heater requirements relative to hydrazine [185]. However, ASCENT-based propulsion systems generally require higher catalyst-bed preheat and higher-temperature chamber/catalyst materials than hydrazine systems [186]. Hydrazine therefore imposes a larger continuous or survival-heater burden on the propellant storage system, while ASCENT shifts more of the thermal burden toward thruster catalyst preheating and high-temperature thruster materials. For the current design, this reduces but does not eliminate hydrazine's thermal disadvantage: maintaining hydrazine above 5°C costs additional heater power, but hydrazine's lower catalyst preheat requirement and mature catalyst technology partially offset this penalty at the propulsion-system level.

The pressurization system also supports the hydrazine selection. Hydrazine is compatible with a standard pressure-fed monopropellant architecture using inert pressurant gas and a regulated or controlled feed system. For this design, helium is selected as the pressurant because it provides the required pressure-volume capability with significantly lower pressurant mass than nitrogen. A regulated helium architecture also allows the propellant tanks to be maintained near the required feed pressure instead of requiring the propellant tanks to begin at a much higher pressure as in a pure blowdown system. This reduces pressure-vessel burden and supports more stable thrust performance over the mission. The specific feed pressure and valve configuration are then treated as constraints on the following thruster-selection process.

Operational and programmatic constraints further support the hydrazine selection. Hydrazine is a toxic and hazardous propellant that imposes stricter ground-handling requirements than ASCENT, but it has extensive flight heritage and well-established spacecraft handling, storage, and pressurization practices [186], [187]. ASCENT reduces toxicity and improves performance, but the specific ASCENT implementation considered for this mission introduces higher supplier dependence and greater uncertainty in catalog-level component availability, thrust-class maturity, and thermal/preheat integration [184], [188]. Because the finalized hydrazine tank architecture already provides substantial Δv margin, the performance advantage of ASCENT is not mission-enabling for the current design. The trade therefore shifts from determining which propellant closes the Δv budget to determining which propellant closes the mission with lower system risk.

Hydrazine is selected because it satisfies the maximum required mission Δv with substantial margin, fits within the finalized tank volume, is compatible with a straightforward regulated helium pressure-fed architecture, and reduces programmatic risk relative to a less mature single-supplier ASCENT implementation. ASCENT remains a technically attractive alternative for future mass- or volume-constrained designs, but for the current PLV architecture, hydrazine provides the better overall balance of performance, verification confidence, component heritage, and integration risk. This selection establishes the design basis for the following thruster trade, which will evaluate hydrazine-compatible thrusters against the required thrust level, minimum impulse bit, feed-pressure range, burn duration, power draw, and attitude-control constraints.

11.7.5 Thruster Selection and Justification

Analysis conducted by: Arya Olliviera

The thrusters selected have not changed since the PDR. The *NASA Small Spacecraft Technology State of the Art Report* [189] provided data such as thrust, specific impulse, technology readiness levels (TRLs), and power consumption for over 60 thrusters. To identify options which satisfy the GNC ΔV requirements within the mass constraints allocated to the PLV subsystem, the maximum achievable ΔV and required burn time for each thruster were calculated using the equations below. The maximum wet mass constraint was approximated by the PLV subsystem and provided to SMM based on Delta V needs, and then iterated upon during thruster selection.

Max delta V:

$$\Delta V = I_{sp} g_0 \ln \left(\frac{m_0}{m_f} \right)$$

Burn time for ΔV :

$$t = \frac{m_p I_{sp} g_0}{T}$$

Propellant mass:

$$m_p = m_0 \left(1 - e^{-\frac{\Delta V}{I_{sp} g_0}} \right)$$

The full analysis can be found in [this Google Sheet](#).

Based on Delta V ranking, mass of propellant necessary, and burn time, the PLV subsystem selected four of the 1 N Moog MONARC-1 thrusters [190], and eight of the 0.08 N Aerojet Rocketdyne MR-401 Thrusters (L3Harris MR-401 LTR, .02 lbf thruster). The minimum impulse bit (MIB) requested by ADCS is 1.1368 Ns for precise attitude control. From the specs, the MIB of the L3Harris MR-401 thruster is 0.004 Ns [191], which matches this requirement and allows a greater margin of finer control.

The performance capabilities calculated for the ranking process of these two thrusters are highlighted in the tables below. The 1 N thruster is evaluated based on time for the major apogee change burn of 150 m/s, as well as the maximum total Delta V requirement seen in Group 3 (249 m/s). The 0.1 N thruster was analyzed based on a maximum stationkeeping and ADCS requirement of 880 m/s seen in Group 3. The compilation and comparison of all candidate thrusters can be found in the spreadsheet linked above.

Parameter	Moog MONARC-1
Thrust (N)	1
Isp (s)	227
Power (W)	18
Burn Time for 150 m/s ΔV (hr)	11.77
Max ΔV (m/s)	1671.95
Propellant Mass for 249 m/s ΔV (kg)	30.89

Table 96: Moog MONARC-1 Thruster Summary



Parameter	L3Harris MR401
Thrust (N)	0.08
Isp (s)	182
Power (W)	8.25
Max ΔV (m/s)	1340.50
Minimum Impulse Bit (N·s)	0.004
Propellant Mass for 880 m/s ΔV (kg)	113.63

Table 97: L3Harris MR401 Thruster Summary

11.7.5.1 Thruster Nozzle Performance Analysis

Analysis conducted by: Tasman Moskowitz, All Components (Analysis Refined Since PDR Report)

The goal of this section is to characterize the performance of the nozzles for each of the selected thrusters. Parameters such as the thrust coefficient C_F , the characteristic velocity c^* , the exit Mach number M_e , the expansion area ratio ϵ , and the propellant characteristic length L^* are calculated to perform this analysis. This analysis aims to validate the nozzle selection performed in the previous section by demonstrating the strong performance of these thruster nozzles.

The first step in determining such parameters is to determine the radius at the nozzle throat and the exit and the length of the nozzle. This can be achieved by assuming both the Moog MONARC-1 and L3Harris MR-401 thruster nozzle cones are geometrically similar, since both these thrusters operate using hydrazine monopropellant. Therefore, both the thrusters and the reference would share the same nozzle exit cone half angle α and effective momentum ratio $\lambda = 0.5(1 + \cos \alpha)$. From [192], a typical hydrazine monopropellant thruster nozzle has $\alpha \sim 16.4^\circ$, which corresponds to an effective momentum ratio $\lambda = 0.980$. From literature, the Moog MONARC-1 has an exit radius of $R_e = 2.5 \text{ mm}$ and an expansion area ratio of $\epsilon = A_e/A_t = 57 : 1$, where A_e and A_t denote nozzle exit area and throat area respectively [193], while the L3Harris MR-401 has an exit radius of $R_e = 3.5 \text{ mm}$, and a nozzle length of $L_n = 4.8 \text{ mm}$ [194]. It is known from [195] that these parameters are related by,

$$\epsilon = \frac{A_e}{A_t} = \left(\frac{R_e}{R_t} \right)^2 \quad (82)$$

$$L_n = R_t \frac{\sqrt{\epsilon} - 1}{\tan \alpha} = \frac{R_e}{\sqrt{\epsilon}} \times \frac{\sqrt{\epsilon} - 1}{\tan \alpha} \quad (83)$$

which gives $L_n = 7.36 \text{ mm}$ and $R_t = 0.331 \text{ mm}$ for the MONARC-1, and $\epsilon = 2.81 : 1$ and $R_t = 2.1 \text{ mm}$ for the MR-401. Corresponding areas can be calculated by $A = \pi R^2$.

The next step is to calculate the exit Mach number M_e from the following equation presented in [195] using a value for the ratio of specific heats $\gamma = 1.25$ for hydrazine [196]:

$$\epsilon = \frac{1}{M_e} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M_e^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}} \quad (84)$$

which yields $M_e = 4.83$ for the MONARC-1 and $M_e = 2.02$ for the MR-401.

After that, known values of the chamber pressure p_c , which is 16 bar for the MONARC-1 and 17 bar for the MR-401, as well as thrusts $T = 1 \text{ N}$ and $T = 0.08 \text{ N}$ respectively, are used to calculate



the thrust coefficient C_F using the following equation from [195],

$$C_F = \frac{T}{A_t p_c} \quad (85)$$

which gives $C_F = 1.82$ and $C_F = 0.0034$ respectively. Note that the latter C_F is too low, which suggests that the exit flow may not in fact be supersonic and $M_e = 0.337$ (the subsonic solution to Equation 84) for the MR-401 thruster.

The exit pressure p_e can then be solved for by using the following equation from [195] with atmospheric pressure $p_a = 0$ in outer space,

$$C_F = \lambda \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}} \left[1 - \left(\frac{p_e}{p_c}\right)^{\frac{\gamma-1}{\gamma}}\right]} + \frac{p_e - p_a}{p_c} \epsilon \quad (86)$$

This gives a solution of $p_e = 1.68 \text{ kPa}$ for the MONARC-1, while no solution exists for the MR-401, as expected. The throat pressure p_t for MONARC-1 can then be determined to be 9.89 kPa using the following equation from [195]:

$$\frac{p_e}{p_t} = \left(\frac{1 + \frac{\gamma-1}{2}}{1 + \frac{\gamma-1}{2} M_e^2} \right)^{\frac{\gamma}{\gamma-1}} \quad (87)$$

Finally, the given I_{sp} values of 227.5 s for MONARC-1 [193] and 182 s for MR-401 can be used to calculate the characteristic velocity c^* using the following equation from [195]:

$$c^* = \frac{I_{sp} g}{C_F} \quad (88)$$

which yields 1.23 km/s and an unrealistic 525 km/s (due to the unrealistically low C_F), respectively.

Therefore, it appears that using the MONARC-1 nozzle provides satisfactory performance on all counts, as it has a high thrust coefficient and characteristic velocity, small volume, and has a thrust and specific impulse that satisfy the design requirements of the PLV subsystem. However, the L3Harris MR-401 thruster has an extremely low coefficient of thrust. This may occur, because the exhaust gasses are not supersonic at the nozzle exit, which seems reasonable for such a small thruster intended for stationkeeping burns. The consequences of this are that the thrust coefficient C_F , exit pressure p_e , and characteristic velocity c^* are all unrealistic making the nozzle performance analysis inconclusive for the L3Harris MR-401 thruster. However, this thruster will still be utilized in the propulsion design due to its small thrust leading to precise attitude control. A summary of this nozzle performance analysis for the two thrusters is shown in Table 98.



Company	Moog	Aerojet Rocketdyne
Name	MONARC-1	MR-401
Quantity	4	8
Thrust [N]	1	0.08
Specific Impulse I_{sp} [s]	227.5	182
Thrust Coefficient C_F	1.82	0.0034 (unrealistic)
Characteristic Velocity c^* [km/s]	1.23	525 (unrealistic)
Expansion Area Ratio ϵ	57:1	2.81:1 (for precise control)
Exit Pressure P_e (kPa)	1.68	No Solution - likely from low C_F

Table 98: Summary Table of Thruster Performance Characteristics for the Moog MONARC-1 and L3Harris MR-401 Thrusters

11.7.5.2 Thruster Placement & Torque Authority

Analysis conducted by: Evan Callas & Ted McCormack

Placement

Thruster placement is driven by both structural integration and attitude control requirements, and is illustrated in Figures 48a - 48d. After discussion and collaboration with the ADCS, SMM, and GNC subsystems, it was decided that the thrusters are mounted at the corners of the spacecraft bus on external brackets, symmetrically distributed about the spacecraft center of mass [197]. This placement maximizes the available lever arm for generating torque while maintaining unobstructed plume paths. This placement is also significantly more propellant-efficient because it avoids the application of an off-centered thrust.

The configuration consists of multiple thrusters oriented along principal axes, enabling both translational and rotational control. Pure translation is achieved by firing pairs or sets of thrusters in the same direction, as shown in Figure 48b. This allows the spacecraft to perform orbital maneuvers without disturbing its attitude.

The symmetric placement of thrusters also improves system robustness. Because multiple thrusters can contribute to similar control axes, the system retains partial functionality in the event of a single thruster failure.

Torque Authority

Attitude control is achieved through differential thrust, as illustrated in Figures 48c and 48d. If a GNC maneuver is executed, thrusters will be fired according to Figure 48 as well. By firing opposing thrusters, a moment is generated about the spacecraft center of mass without producing

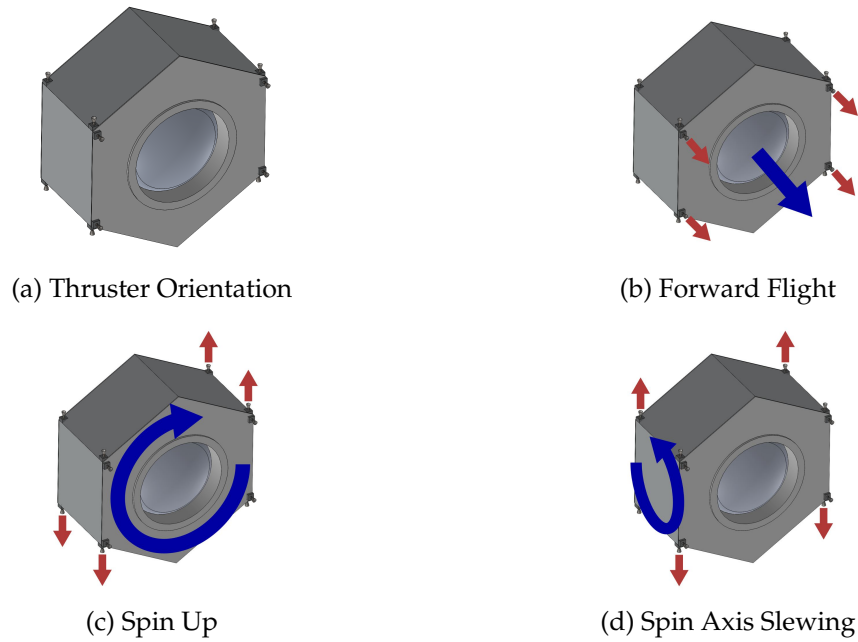


Figure 48: Thruster orientation and different firing maneuvers.

translation. The placement of thrusters at the corners of the structure maximizes the moment arm, increasing torque authority for a given thrust level and improving control responsiveness.

Spin up and increasing the spin rate will be done using the thruster firings of Figure 48c and the opposite firing configuration will be done for spin down and decreasing the spin rate. As shown in Figure 48d, slewing about the spacecraft's spin axis can be executed using PWM control when the thrusters are aligned with the desired slewing direction.

To analyze the resolution and accuracy with which the thrusters can control a small output of propellant, the Minimum Impulse Bit (MIB) was calculated. The Minimum Impulse bit is defined by the European Cooperation for Space Standardization (ECSS) as "the smallest impulse delivered by a thruster at a given level of reproducibility, as a result of a given command" [120]. MIB corresponds to the accuracy that the thrusters have to control the attitude of the spacecraft. A coarser resolution thruster (higher MIB) has worse pointing accuracy. Using the equations in Appendix A.6.1, the MIB can be found for the L3Harris thruster. Since we want a total attitude control accuracy of 3 degrees, which includes the disturbances besides just thruster bit size, we can give a generous 0.5 degree error allowance to thruster bit error, while still ensuring we can achieve the desired attitude orientation. The only other variables we need are the radius of the thrusters to the center of mass and the moment of inertia both of which are provided by the SMM subteam and can be found in Appendix A.6.2.

$$\text{MIB} = \frac{\Delta\theta \cdot I_{\text{spin}} \cdot \omega}{R} \quad (89)$$

With this information, MIB is found to be 0.2556 lb-s or 1.1368 Ns. Based on the MIB of the L3Harris thruster being 0.004 Ns as published in the thruster datasheet [191], the MIB of the PLV



subsystem's selected thrusters is within the limits of the calculated MIB for this mission, so the thruster selection is validated once more. By coming to this conclusion it is acknowledged that the L3Harris thrusters offer significantly higher granularity than the minimum required for our mission's attitude control.

Seeing that the MIB for the L3Harris thruster is 0.004 Ns validates their torque authority and demonstrates how the spacecraft will have precise attitude control. This validation also supports the chosen thruster placement and qualifies the design shown in Figure 48.

11.7.5.3 Pointing Requirements during Burns

Analysis conducted by: Ted McCormack

Pointing accuracy during propulsive maneuvers is a key interface between PLV, ADCS, and GNC because any attitude error changes the direction of the delivered impulse. The spacecraft is required to end each burn pointing within 2° of the intended target direction, which limits both maneuver inefficiency and unintended transverse impulse [198]. For a pointing error θ , the useful impulse component along the desired direction scales as $\cos \theta$, while the transverse error component scales as $\sin \theta$. At the maximum allowable error of 2° , the useful impulse loss is only $1 - \cos(2^\circ) \approx 0.061\%$, while the transverse component is $\sin(2^\circ) \approx 3.49\%$ of the delivered impulse.

The propulsion subsystem is also constrained by a minimum impulse bit of 1.1368 N s, which defines the smallest commanded impulse that can be delivered [198]. For a spacecraft mass of 120 kg, this corresponds to a minimum velocity increment of approximately 9.47×10^{-3} m/s. At a 2° pointing error, the transverse component of this minimum impulse is approximately 3.31×10^{-4} m/s. While this is small for an individual impulse, it could accumulate over repeated correction burns if pointing errors are not controlled. A more in-depth analysis is shown in A.10.4.

11.7.6 P&ID Diagram for Thrusters, Tanks and Fuel Lines

Analysis conducted by: Arya Olliviera

The updates to the P&ID for the PLV subsystem since the PDR include the number of hydrazine tanks increasing from one to five to optimize volume usage, as discussed in earlier sections. The two pressurant tanks feed into a manifold which then feed into the five hydrazine tanks. The hydrazine tanks lead to another manifold which supplies fuel to the thrusters. Multiple valves are utilized to control and direct the hydrazine and pressurant flows, ensuring targeted distribution as well as equal tank drainage to minimize COM shifting.

This diagram does not include all components, such as pressure gauges and thermometers. The pressurant was finalized since the PDR to be helium. This is because helium is most commonly used in heritage missions due to its low molecular weight, and it does not dissolve in the hydrazine to the extent that nitrogen would, which would be unfavorable [199]. There are four 1 N thrusters for propelling and eight 0.1 N thrusters for stationkeeping and smaller maneuvers.

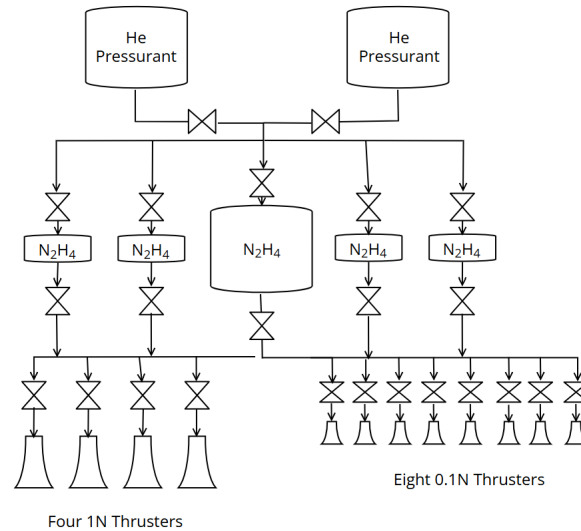


Figure 49: P&ID Diagram for MATRIX PLV System

11.7.6.1 Contamination Zones for Plumes

Analysis conducted by: Arya Olliviera

The exhaust from the nozzles is capable of contaminating the scientific instruments and other external components on the spacecraft. The PLV subsystem could not find research papers on the specific Moog thruster selected, so a geometric estimate for the contamination zone is performed below based on the known specifications of the Moog MONARC-1.

Given the exit radius

$$r_e = 2.5 \times 10^{-3} \text{ m},$$

based on [190], and an estimated nozzle exit cone half-angle

$$\alpha = 16.4^\circ,$$

based on [192].

The contamination zone can be roughly estimated geometrically as a cone by the equation:

$$r(x) = r_e + x \tan \alpha$$

This analysis is based on several assumptions, including that the nozzle exhaust continues along its previous path without expansion into the vacuum of space, which is incorrect. For an accurate estimation of the plume, either experimental data or a CFD analysis factoring temperature of the exhaust, pressure at the nozzle exit as it expands into the vacuum of space, and so on is required.

MATLAB was utilized to plot the plume radius versus distance from the nozzle exit and is unchanged since the PDR [22].

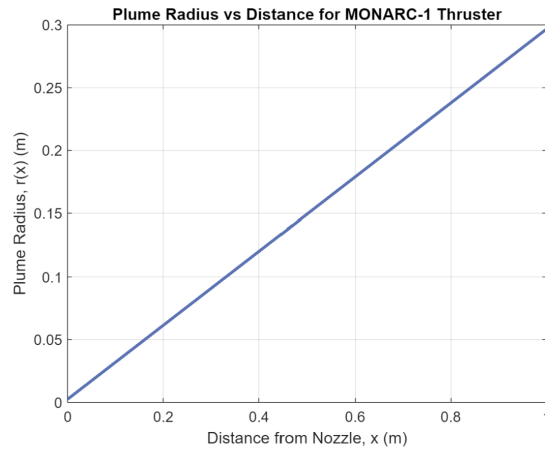


Figure 50: Approximated Contamination Zone for Moog 1N Thruster

The same approach was utilized in analyzing the contamination zones for the eight 0.1 N L3Harris thrusters, with an exit radius $r_e = 3.5 \times 10^{-3}$ m.

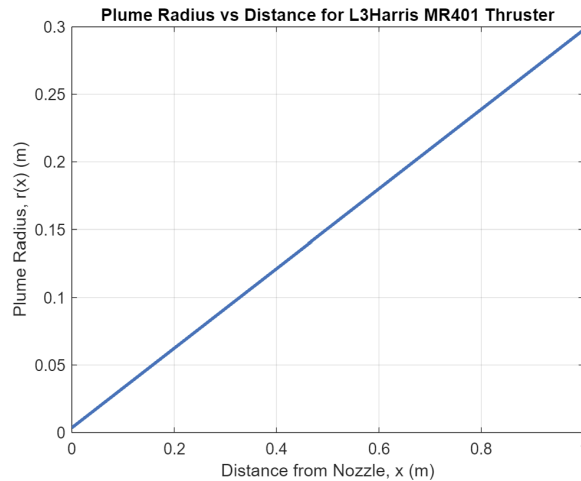


Figure 51: Approximated Contamination Zone for L3Harris 0.1 N Thruster

Based on this geometric estimation, the PLV subsystem collaborated with the SMM subsystem to analyze the potential intersection of the plume with any instruments on the CAD assembly. The figures below show the contamination zones for the four 1 N Moog thrusters and eight 0.1 N thrusters. It can be seen that they stay clear of all instruments and equipment on the spacecraft.

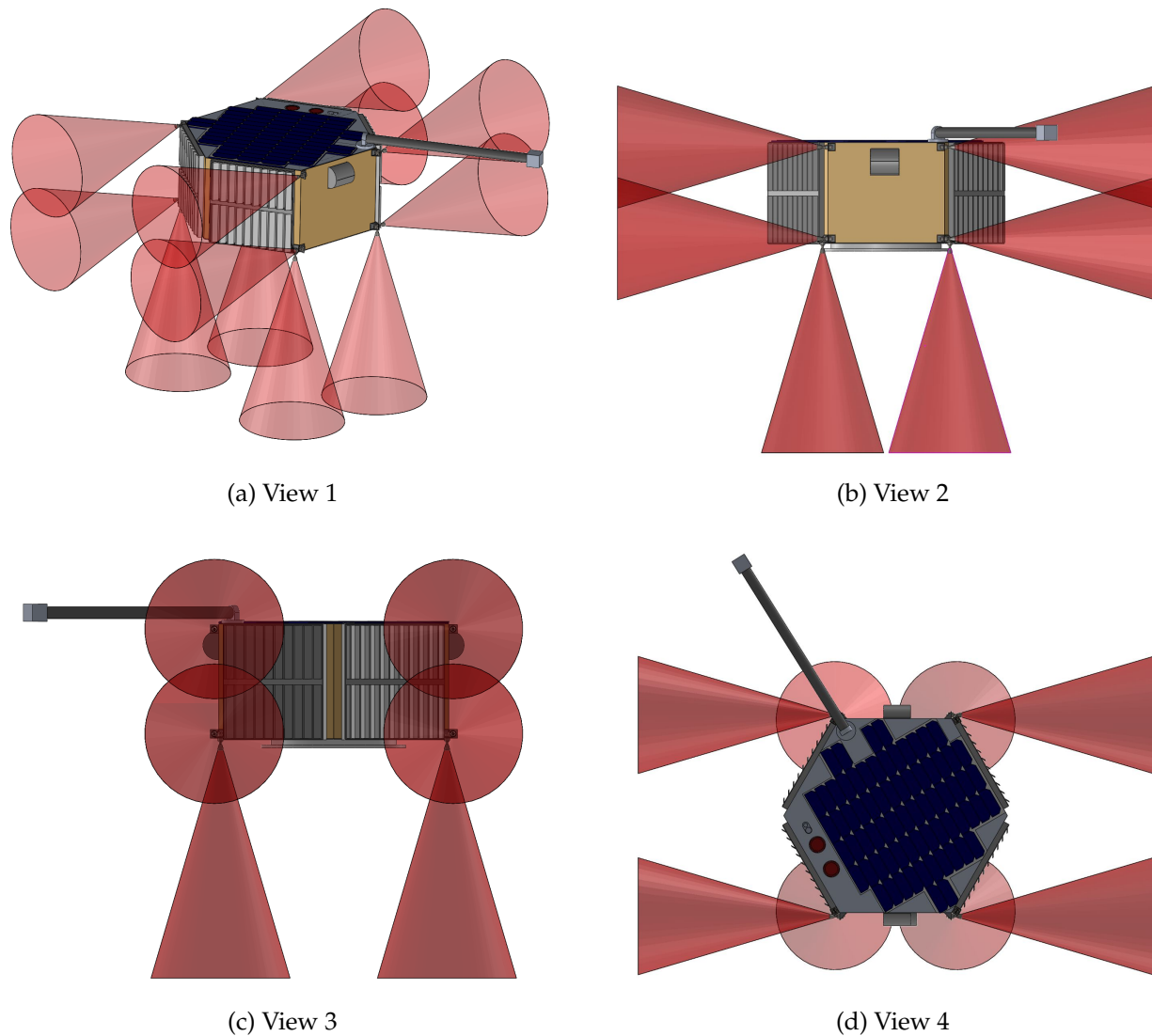


Figure 52: Thruster Contamination Zones for All Thrusters

11.7.6.2 Power Draw during Thruster Firing

Analysis conducted by: Evan Callas, all components

Correspondence between the POW subsystem and the PLV subsystem was required to communicate the expected power that is required to allow effective thruster firing. The PLV subsystem selected four of the 1 N Moog MONARC-1 thrusters [190], and eight of the 0.08 N L3Harris MR-401 Thrusters. The latter was acquired by L3Harris, so the selected thruster is now called the L3Harris MR-401 LTR, .02 lbf thruster (0.07-0.09 N) and the specifications remain approximately the same as seen in the L3Harris Datasheet [191].

The four MONARC-1 thrusters are responsible for boosting the satellite at perigee to maintain its orbit throughout the six year mission whereas the L3Harris thrusters allow each satellite to



maintain translational and rotational control while station-keeping. The MONARC-1 thruster requires 18 W of power in order to actuate the solenoid valve that allows hydrazine propellant to flow into the thruster [193]. For four of these boosting thrusters per satellite, this requires a maximum total power of 72 W in the event that all four are operational at one time. The L3Harris thruster requires 8.25 W of power to heat a catalyst bed to decompose hydrazine based on the literature in the references mentioned above [191]. This power value is required for a maximum of 4 out of 8 thrusters that could possibly be responsible for maneuvering at once, totaling to 33 W required per spacecraft. After duty cycling was taken into consideration for the power budget, the 4 booster thrusters and 4 station keeping thrusters mentioned above are not all active at the same time. Only 72 W is demanded in transit mode and 33 W is demanded in station keeping mode in Table 85.

Based on the power budget derived by the POW subsystem for the MATRIX Mission and seen in Table 87, the PLV subsystem is allocated 87.31 W for transit mode and 37.21 W for station keeping mode. This allows for their to be a margin for the power draw of the thrusters which satisfies all of the subsystem requirements [19].

The total maximum power draw of all mission modes is for the station keeping mode with a total power draw of 89.5 W. This power draw has a built in margin of 12.5% which amounts to 101.26 W of total power being supplied for instrument usage. During transit mode, the total power demand is 83.5 W from the instruments. Due to the instrument demand being less than 89.5 W and accounting for the additional 12.5% margin for power draw, there is a 21% margin built into the instrument draw for power draw during MOOG thruster firing. During station keeping, there is a 12.5% margin for power draw during L3Harris thruster firing.

$$V_{spike} = L * \frac{\Delta i}{\Delta t} \quad (90)$$

The margins mentioned above are critical to consider for power draw during thruster firing because it is possible that current spikes or voltage drops will occur during valve actuation. During thruster firing, high valve actuation switching frequencies can cause there to be power spikes or drops in the main spacecraft bus. Due to this, it is essential that a margin is built into the engineering sizing of the power storage and power generation modules. Equation 90 shows how a voltage spike is induced by a change in current that happens in a short period of time which is often induced if valve actuation occurs too rapidly or there is a rapid switch in actuation direction from the PWM command. Because of this, it is important to account for this in a power margin so that there is room for mechanical and electrical fluctuations. In Equation 90, L = inductance (henries), Δi = change in current (amps), and Δt = time taken for current to cut off (seconds).

In addition to this, the effect of duty cycling was analyzed for the thrusters of the propulsion system. It was determined that MOOG booster thrusters would not be operational at the same time or during the same mode as the L3Harris station keeping thrusters, for the firing occurs at different times throughout a single period. However, it was considered that all four or all eight thrusters could be triggered simultaneously during a maneuver and thus would all be operational at once. This scenario was accounted for during duty cycling as well, and the power allocation reflects these demands. With the margins built into the power storage and generation sizing, there is no concern that power draw during thruster firing will be surpassed during these peak scenarios.



11.7.7 Propellant and Pressurant Feed Systems

Analysis conducted by: Ted McCormack

The final propellant storage architecture consists of four identical side propellant tanks, one spherical central propellant tank, and two spherical helium pressurant tanks. The four side tanks are cylindrical vessels with 2:1 ellipsoidal heads (an obround shape), each constrained by a final outer diameter of 8.78in and a final outer length of 22.77in. The central propellant tank is spherical and sized to hold approximately 70L. The pressurant system uses two identical spherical helium bottles sized for a regulated hydrazine feed. All tanks are modeled as Ti-6Al-4V pressure vessels using thin-shell pressure-vessel theory. NASA pressure-vessel standards are used to justify proof/burst-pressure logic and inspection philosophy, while standard thin-shell and ASME-form equations are used for wall-thickness estimates [200], [201], [202], [203], [204]. Ti-6Al-4V is selected because it provides high specific strength and is commonly used in aerospace pressure-vessel applications; the material density and strength values used for mass and allowable-stress screening are taken from public titanium alloy data sheets and design handbooks [205], [206], [207].

The propellant tanks are sized using a maximum expected operating pressure (MEOP) of 400psi and a burst design pressure of 600psi, corresponding to a 1.5 burst factor. This pressure basis is consistent with the MONARC-1 thruster feed-pressure range of 70–400 psia listed in the Moog monopropellant thruster datasheet [208]. The pressurant tanks are sized for a helium storage pressure of 4000psi, also using a 1.5 burst factor, giving a burst design pressure of 6000psi. The project allowable stress used for sizing is 110400psi. This value is treated as a project-level aerospace design allowable for preliminary wall sizing rather than an ASME code allowable. To avoid relying on best-case seamless construction, the margin table also evaluates each selected thickness with an effective weld/joint efficiency of $E = 0.85$.

For the side propellant tanks, the limiting pressure-vessel stress is the cylindrical hoop stress. Using the ASME-form cylindrical shell relation at 600psi, the required wall thickness is 0.0237in for an ideal seamless shell and 0.0280in using $E = 0.85$. The 2:1 ellipsoidal heads give essentially the same required thickness: 0.0237in for $E = 1$ and 0.0279in for $E = 0.85$. The selected final nominal wall thickness for each side tank is therefore 0.035in. With this wall thickness, each side tank has an internal diameter of 8.71in, an internal length of 22.70in, and an internal volume of 20.75L. The four side tanks provide 82.99L of total propellant volume.

The central propellant tank is spherical, which is the most structurally efficient geometry for a pressure vessel because the membrane stress is equal in all directions and is half the hoop stress of a cylinder with the same radius and wall thickness. A 70L spherical tank has an internal radius of 10.07in, giving an internal diameter of 20.13in. At the 600psi burst design pressure, the spherical wall-thickness requirement is 0.0274in for $E = 1$ and 0.0322in for $E = 0.85$. The selected final nominal wall thickness is 0.040in, giving a final outer diameter of 20.21in. The central tank therefore provides 70L of propellant volume. Combined with the side tanks, the final propellant storage volume is

$$V_{prop,total} = 4(20.75) + 70.0 = 152.99L.$$

The helium pressurant system is sized from the final propellant volume. A regulated feed pressure of 22bar is used to provide margin above a 20bar hydrazine feed requirement while remaining within the MONARC-1 feed-pressure range [208]. With helium stored at 4000psi, or approxi-



mately 276bar, the ideal total bottle volume required to expel 152.99L of hydrazine is 13.25L. A 10% design margin is applied for line volume, regulator residual gas, trapped gas, thermal variation, and non-ideal depletion, giving a final total helium bottle volume of 14.58L. This is split into two identical spherical bottles of 7.29L each. Each bottle has an internal diameter of 9.47in. At the 6000psi burst design pressure, the spherical wall-thickness requirement is 0.129in for $E = 1$ and 0.151in for $E = 0.85$. The selected final nominal wall thickness is 0.160in, giving a final outer diameter of 9.79in for each pressurant tank.

From the perspective of the thrusters, the five propellant tanks are treated as a single effective propellant reservoir. The tanks are connected through a manifolded network of piping, isolation valves, and flow-control elements that controls which tanks feed the propulsion system during each phase of operation. This architecture is intended to preserve both left-right and top-bottom mass balance by draining geometrically matched secondary tanks together before transitioning to the primary central tank. As the secondary tanks deplete, the center of mass may shift slightly depending on small differences in line losses, valve timing, tank fill fraction, or residual propellant; however, paired depletion limits these shifts by keeping the propellant distribution symmetric about the spacecraft centerline. Once the secondary tanks are depleted, the remaining propellant is concentrated in the central tank, which further reduces the moment arm of the remaining liquid mass and limits additional center-of-mass migration. Because the spacecraft is spin-stabilized, any off-axis center-of-mass shift would rotate with the body and could introduce small periodic imbalances or apparent control disturbances if not kept small. Maintaining symmetric tank depletion is therefore important not only for static mass balance, but also for minimizing spin-induced dynamic effects, nutation tendency, and coupling between translational thrusting and attitude motion. A more detailed analysis is shown in Appendix A.10.2.

Table 99 summarizes the final tank dimensions. The shell masses are computed using the Ti-6Al-4V density and the difference between the outer and inner tank volumes; these values are shell-only masses and do not include bosses, weld lands, brackets, propellant management devices, valves, or plumbing [205], [207]. Table 100 summarizes the governing wall-thickness checks. All selected walls satisfy the thin-wall criterion $t/r < 0.1$, with the largest ratio occurring in the helium bottles at $t/r = 0.0338$. Therefore, thin-shell theory is appropriate for this sizing level.

Table 99: Final tank dimensions.

Tank	Qty.	Geometry	Internal volume	Nominal wall	Final OD / dia.
Side propellant tank	4	Obround	20.75L	0.035in	8.78in
Central propellant tank	1	Sphere	70.0L	0.040in	20.21in
Helium pressurant tank	2	Sphere	7.29L	0.160in	9.79in

Table 100: Wall-thickness margins using ASME-form pressure-vessel equations.

Case	Design pressure	$t_{req}, E = 1$	$t_{req}, E = 0.85$	Selected t	MS vs. $E = 0.85$
Side tank barrel	600psi	0.0237in	0.0280in	0.035in	0.25
Side tank 2:1 head	600psi	0.0237in	0.0279in	0.035in	0.26
Central sphere	600psi	0.0274in	0.0322in	0.040in	0.24
Helium sphere	6000psi	0.128in	0.151in	0.160in	0.06

The selected final wall thicknesses are therefore 0.035in for each side propellant tank, 0.040in for



the central spherical propellant tank, and 0.160in for each helium pressurant bottle. These values are governed by burst-pressure membrane stress, then rounded upward to practical nominal gauges that preserve margin after applying a conservative welded-shell efficiency. The final tank set provides 152.99L of propellant storage and 14.58L of total helium bottle storage, with shell-only tank mass totaling approximately 17.43kg. An example of the calculations for this analysis is shown in Appendix [A.10.3](#).



12 Concluding Remarks

This Final Design Report has presented the Magnetotail Systems Laboratory's response to the MAE 342 Request for Proposal: a MIDEX-class, eight-spacecraft constellation called MATRIX (MAGnetotail Transition Region In-situ eXplorer) designed to obtain coordinated, multi-point in-situ measurements of magnetic field and plasma properties across the $6\text{--}12R_E$ magnetotail transition region. The mission addresses a science gap identified by the 2024 NAS Heliophysics Decadal Survey: namely, how energy and mass carried by bursty bulk flows (BBFs) evolve, brake, and deposit energy into the inner magnetosphere, and how this process can be resolved through coordinated mesoscale observations.

The MATRIX architecture distributes eight spacecraft into three coplanar, highly eccentric orbits with apogees of $7R_E$, $9R_E$, and $12R_E$ and perigees near 1,000 km. A buddy pair at $7R_E$ resolves microscale dynamics in the inner transition region; a four-spacecraft train at $9R_E$ resolves the mesoscale braking layer; and a pair at $12R_E$ samples BBFs as they propagate earthward from the mid-tail plasma sheet. This three-group geometry, combined with a continuously operating payload of a fluxgate magnetometer on a deployable boom, an electrostatic analyzer, and a modified energetic ion spectrometer, enables MATRIX to distinguish between the processes governing BBF evolution and the downstream redistribution of energy. To our knowledge, no prior or currently planned mission samples the full transition-region corridor at this combination of spatial scales.

The design closes on every mission-level requirement, constraint, and resource budget. The life-cycle cost rolls up under the \$350M (FY2025) MIDEX cap; the launch wet mass fits within the 2,280 kg allocation and the launch volume within the 14.16 m^3 fairing envelope; the power, thermal, communications, and Δv budgets all close with positive margin against the 6-year minimum operational lifetime requirement. The spacecraft are radiation-tolerant for the 33 krad/year magnetotail environment, and end-of-life disposal is achieved through retrograde apogee burns that lower perigee into the atmosphere.

Several key design decisions defined the final mission. The selection of coplanar orbits, made early in the semester, eliminated prohibitive plane-change Δv that would have consumed the propellant budget reserved for stationkeeping. Direct insertion into the $9R_E$ Group 2 orbit, with subsequent prograde and retrograde perigee maneuvers to raise Group 1 and lower Group 3, minimized total constellation Δv and made the mission feasible on a single NASA Launch Services Program vehicle. Hexagonal-prism spacecraft with body-mounted solar panels, an MLI-and-coatings passive thermal architecture, and a hydrazine monopropellant propulsion system with separate L3Harris 0.1 N attitude-control thrusters and Moog 1 N orbit-change thrusters reflected our preference, articulated repeatedly throughout the semester, for high-heritage hardware and simple, redundant subsystem architectures over higher-performance but lower-TRL alternatives. The Bus Avionics Unit, with its four-module DPM/CIM/PEU/IDPU card stack, consolidates command, data handling, power management, and instrument processing into a single shielded enclosure, simplifying integration and reducing harness mass across the eight-spacecraft fleet.

The mission carries non-trivial risk. The constellation depends on tight time synchronization and on maintaining mesoscale spacing across a 5+ year operational phase, both of which place demanding requirements on GNC, ADCS, and the propellant margin reserved for stationkeeping. Magnetic cleanliness, verified through Helmholtz-coil mapping during integration and test, is critical to the science return of the fluxgate magnetometer and represents a significant pre-launch



verification activity. The radiation environment of the magnetotail, while well below that of the Van Allen belts, imposes a cumulative TID that drives parts selection and shielding mass. Each of these risks is identified in our risk register and mitigated through margin, redundancy, or explicit verification activities, but they remain areas where additional analysis and test would benefit a Phase A study.

Several aspects of the design merit further work. A higher-fidelity thermal model, with full transient orbital cases and detailed component-level temperature predictions, would sharpen the heater sizing and radiator area selection. Detailed FEA of the deployable magnetometer boom under launch loads, and a coupled-loads analysis with the launch vehicle, are natural next steps for SMM. The flight software architecture, currently specified at the cFS framework level with KIPS budgets allocated by SMAD methodology, will require a detailed software design phase. The communications link budget assumes Near Space Network availability that should be confirmed through formal coordination, and the data-handling architecture would benefit from end-to-end testing against realistic BBF burst-mode scenarios. None of these items challenges the feasibility of the mission as proposed; they represent the work that would naturally follow this preliminary design study as MATRIX matures towards the conclusion of a Phase A design towards a Phase B realization.

Beyond the technical content of the mission, the value of this semester for the MATRIX team has been the experience of working as a sixteen-person team on a coupled, multi-subsystem design problem under a fixed cost cap and a fixed timeline. The iterative cycle of mission-level requirements flowing down, subsystem trades feeding back, and cross-subsystem conflicts being resolved at the team level is, in the end, what spacecraft systems engineering actually is. The lessons learned about requirements discipline, margin philosophy, configuration control, and honest communication of risk will outlast any specific technical result reported here. We are grateful to the instructional staff for the opportunity to live through that process on a problem of real scientific importance, and we believe that MATRIX, as designed, has the potential to meaningfully advance the field of magnetospheric physics if flown.



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A Appendix

A.1 OPL

A.1.1 Bill of Materials

The following tables document the component-level Bill of Materials (BOM) used to construct the subsystem cost allocations referenced in the Mission Cost Compliance section. All costs are in FY2025 USD. Quantities reflect the totals required across the full eight-spacecraft MATRIX constellation unless otherwise noted.

A.1.1.1 Operations & Payload (OPL).

The OPL flight hardware BOM consists of the three science payload instruments per spacecraft: a fluxgate magnetometer (FGM), an electrostatic analyzer (ESA), and an energetic particle detector (EPD). Per-instrument first-unit costs were sourced from heritage instrument pricing (THEMIS/MMS analogues, vendor consultations). A 90% learning-curve slope was applied across the eight units per SMAD guidance, with the cumulative recurrence factor computed as $L_N = \sum_{i=1}^N i^b$, where $b = \log(s)/\log(2) = -0.152$ and $s = 0.9$ [4].

Instrument	First-Unit Cost (\$)	Pre-Discount Total (\$)	Post-Recurrence Total (\$)
FGM (Fluxgate Magnetometer)	750,000	6,000,000	4,930,294
ESA (Electrostatic Analyzer)	8,000,000	64,000,000	52,589,807
EPD (Energetic Particle Detector)	2,000,000	16,000,000	13,147,452
Total		86,000,000	70,667,554

Table 101: OPL payload BOM with 90% learning-curve recurrence applied across eight spacecraft.

A.1.1.2 Structures, Materials & Mechanisms (SMM).

Component	Qty.	Unit Cost (\$)	Total (\$)	Notes / Source
Propellant tanks (PTD-96, 96 L)	8	3,000,000	24,000,000	TBC - pending vendor quote
Pressurizing tanks (Sierra Space)	16	750,000	12,000,000	TBC - pending vendor quote
Side panels (24" × 24")	96	25,187	2,417,981	Al 7075, McMaster
Top/bottom panels (48" × 48")	16	50,585	809,365	Al 7075, McMaster
Titanium stock (1" × 1" × 12')	8	30,151	241,207	Thruster bracket material
Sierra Space Quick-Sep 24	8	150,000	1,200,000	TBC - pending vendor quote
Carbon fiber tubing (1/8", 1-3/4" OD)	1	10,798	10,798	McMaster + machining estimate
Spring-damper hinges	16	20,000	320,000	TBC - custom
Frangibolt actuators	16	100	1,600	ARGOS report
Total			24,017,000	

Table 102: SMM Bill of Materials.



A.1.1.3 Thermal (THM).

Component	Qty.	Unit Cost (\$)	Per-S/C Total (\$)	Notes
MLI blankets	2	1,000	2,000	sq m
G-10 isolators	1	5,000	5,000	
RTD sensors	50	500	25,000	
Thin-film heaters	30	500	15,000	
Thermal straps	10	5,000	50,000	
Louvers (custom)	4	25,000	100,000	First-order estimate
G-10 machining	1	5,000	5,000	
Ansys thermal license	1	50,000	50,000	
TVAC testing	1	714,000	714,000	
Misc. assembly + testing	1	100,000	100,000	
Per-spacecraft total			1,066,000	
Constellation total (8 S/C)			8,528,000	

Table 103: THM Bill of Materials.

A.1.1.4 Guidance, Navigation & Control / Attitude Determination & Control (GNC/ADCS).

Hardware	Model	Qty.	Unit Cost (\$)	Mass (kg)	Total (\$)
Sun sensor	Fine Spinning Sun Sensor (64°)	2	15,000	0.725	30,000
IMU	DMU41	1	25,000	0.178	25,000
Damping fluid	Mercury	1	100,000	2.200	100,000
Damper ring	Steel	1	15,000	1.500	15,000
Per-spacecraft total					170,000

Table 104: GNC / ADCS Bill of Materials (per-spacecraft).

A.1.1.5 Power (POW).

Component	Qty.	Unit Cost (\$)	Total (\$)	Source
Molicel INR-21700-M50A cells	64	12.25	783.68	GetFPV
AZUR Space 4G32-Adv solar cells	65	450	29,250	AZUR Space
Pumpkin Space EPSM-1 PCDU	1	48,000	48,000	Pumpkin Space
Per-spacecraft total			78,034	
Constellation total (8 S/C)			624,269	

Table 105: POW Bill of Materials.



A.1.1.6 Propulsion & Launch Vehicle Interface (PLV).

Component	Qty.	Unit Cost (\$)	Per-S/C Cost (\$)	Constellation Total (\$)
1 N thruster	4	150,000	600,000	4,800,000
0.1 N thruster	8	50,000	400,000	3,200,000
Small fuel tanks	4	120,000	480,000	3,840,000
Large fuel tank	1	300,000	300,000	2,400,000
Valves	30	1,000	30,000	240,000
Piping	1	5,000	5,000	40,000
Hydrazine propellant (kg)	153	483.03	73,904	591,229
Loading costs	1	16,875	16,875	135,000
Total				15,246,229

Table 106: PLV Bill of Materials.

A.1.1.7 Communications, Command & Data Handling (CCD).

Component	Qty.	Unit Cost (\$)	Total (\$)	Notes
RH-CF5208 ColdFire processor	8	6,500	52,000	VORAGO Rad-Hard MCU analogue
S-band antenna	32	15,000	480,000	Anywaves (academic discount)
RS-422 interface cable	16	50	800	Markertek
IDPU processor	8	26,000	208,000	Order 10 ⁴ -10 ⁵
NANOLink-boost Gen2 S-band radio	16	17,500	280,000	EnduroSat
NuPower 12B01A power amplifier	16	3,129	50,064	
Bus avionics bundle (PEU+CIM+OBC+SSR)	8	240,000	1,920,000	20× NEPP upper-bound estimate
Total			2,990,864	

Table 107: CCD Bill of Materials.

A.2 CCD

A.2.1 Transmission Bandwidth Allocation Calculations

To justify the selected virtual channel bandwidth allocation, a simple capacity-based analysis was performed using the total available downlink rate.

Let the total downlink capacity be

$$R_{\text{down}} = 3.21 \times 10^5 \text{ bps.} \quad (91)$$

The bandwidth allocated to each virtual channel is given by

$$R_i = f_i R_{\text{down}}, \quad (92)$$

where f_i is the fractional allocation assigned to channel i .

For the selected allocation:

- 5% allocation:

$$R_{5\%} = 0.05 \times (3.21 \times 10^5) = 1.605 \times 10^4 \text{ bps} \quad (93)$$

- 85% allocation:

$$R_{85\%} = 0.85 \times (3.21 \times 10^5) = 2.73 \times 10^5 \text{ bps} \quad (94)$$

Housekeeping telemetry data rates are estimated to be on the order of 10^2 - 10^3 bps. Comparing this to the allocated 5% bandwidth:

$$\frac{R_{5\%}}{R_{HK}} \approx \frac{1.6 \times 10^4}{10^3} \approx 16 \quad (95)$$

This shows that the allocated bandwidth exceeds nominal housekeeping requirements by more than an order of magnitude, providing sufficient margin for protocol overhead, burst telemetry during fault events, and ensuring continuous spacecraft health visibility.

Since engineering and metadata streams together represent a small fraction of the total data volume, the remaining bandwidth is allocated to science data. The 85% allocation ensures that the dominant science data volume can be fully downlinked within each contact window while maintaining guaranteed bandwidth for critical telemetry.

Therefore, the chosen allocation of 5% (critical telemetry), 5% (metadata), 85% (science), and 5% (nominal telemetry) provides a balanced trade between reliability and maximising science return.

A.2.2 Link Budget Equations

	Variable
Frequency (MHz)	f_{MHz}
Slant Range (km)	R_{km}
Transmit Power (dBm)	Ptx_{dBm}
Transmit Antenna Gain (dBi)	Gtx_{dBi}
Receive Antenna Gain (dBi)	Grx_{dBi}
Transmit Antenna Cable/Implementation Losses (dB)	Ltx_{dB}
Receive Antenna Cable/Implementation Losses (dB)	Lrx_{dB}
Pointing/Polarization/Misc Losses (dB)	$Lmisc_{dB}$
Data Rate (bps)	Rb
Receiver Noise Figure	NF_{dB}

Table 108: Link Budget Variables

Effective Isotropic Radiated Power (dBm): $EIRP_{dBm} = Ptx_{dBm} - Ltx_{dB} + Gtx_{dBi}$

Free Space Path Loss (dB): $FSPL_{dB} = 32.45 + 20\log_{10}(R_{km}) + 20\log_{10}(f_{MHz})$

Received Power (dBm): $Prx_{dBm} = EIRP_{dBm} - FSPL_{dB} + Grx_{dBi} - Lrx_{dB} - Lmisc_{dB}$

Thermal Noise Density (dBm/Hz): $N0_{dBm/Hz} = -174$

Effective Thermal Noise Density (dBm/Hz): $N0eff_{dBm/Hz} = N0_{dBm/Hz} + NF_{dB}$

Carrier-to-Noise Density (dB-Hz) [261]: $CN0_{dB-Hz} = Prx_{dBm} - N0eff_{dBm/Hz}$

Normalized Signal-to-Noise Ratio (dB): $EbN0_{dB} = CN0_{dB-Hz} - 10\log_{10}(Rb)$



A.3 Flight Software Throughput Derivation

This appendix derives the 1.49 MIPS DPM processing requirement cited in Section 5.7.3. The derivation follows the SMAD methodology (Tables 16-13 and 16-15 [53]), which builds the throughput estimate from per-function loads, applies a $3\times$ engineering margin to compensate for early-design uncertainty, and then compares the result to the available processor capacity at SMAD's recommended 70% utilisation ceiling.

The flight software runs on the RH5208 ColdFire processor at 16 MHz, providing approximately 16 MIPS of raw computational capacity [55]. The total onboard processing requirement is built up function-by-function as follows.

Housekeeping and command functions. Summing the SMAD Table 16-13 entries that apply to MATRIX (command processing, telemetry processing, attitude sensor processing, attitude determination and control, ephemeris propagation, fault monitoring, power management, and thermal control) yields a total of 269.2 KIPS at nominal load. Applying a $3\times$ engineering margin (per SMAD §16.2.2 for early-phase designs) gives 807.6 KIPS.

cFE Executive. SMAD Table 16-15 estimates the executive throughput as $K_{\text{exec}} = 0.3n$ KIPS, where n is the number of scheduled tasks per second. Assuming $n = 200$ (typical per SMAD),

$$K_{\text{exec}} = 0.3 \times 200 = 60 \text{ KIPS.}$$

cFE I/O Handlers. I/O handler throughput is given as $K_{\text{I/O}} = 0.05m$ KIPS, where m is the number of data words handled per second. Using the MATRIX peak science data rate of $R_{\text{peak}} = 9118.2$ bps observed at the IDPU output, and one 16-bit word per 16 bits,

$$m = \frac{9118.2}{16} = 570 \text{ words/s,} \quad K_{\text{I/O}} = 0.05 \times 570 = 28.5 \text{ KIPS.}$$

Fault monitors and system management. Per SMAD Table 16-13, fault detection monitors and system management routines together contribute approximately 20 KIPS at nominal load.

Science data processing. Although the IDPU performs the bulk of the per-instrument processing (Section 5.7.5), the DPM still performs CCSDS framing, virtual channel routing, and SSR write scheduling for the science stream. Using SMAD's assumption of 100 instructions per byte of input data, and applying it to the 9118.2 bps peak rate, this contributes ~ 114 KIPS at nominal load and 342 KIPS with $3\times$ margin.

cFS framework overhead. The remaining cFS framework overhead (executive, I/O handlers, and fault functions, all carried at the same $3\times$ engineering margin) totals approximately 235.5 KIPS.

The grand total is summarized in Table 109.



Table 109: CCD Processing Budget Summary (TBC)

Function	Load (KIPS)	Source
Housekeeping & command functions ($3\times$ margin)	807.6	SMAD Table 16-13
Science data processing ($3\times$ margin)	342.0	SMAD §16.2.2
cFE Executive ($n = 200$ tasks/s)	60.0	SMAD Table 16-15
cFE I/O handlers ($m = 570$ words/s)	28.5	SMAD Table 16-15
Fault monitors + system management	20.0	SMAD Table 16-13
FSW overhead margin ($3\times$ on OS functions)	235.5	SMAD §16.2.2
Total required	$\approx 1,490$ KIPS = 1.49 MIPS (TBC)	
Available (70% of 16 MIPS)	11,200 KIPS (TBC)	
Margin	$7.5\times$ (TBC)	

Against the RH5208 ColdFire's ~ 16 MIPS raw capacity, the 1.49 MIPS requirement leaves a $7.5\times$ headroom even at SMAD's 70% utilization ceiling ($0.7 \times 16 = 11.2$ MIPS). This satisfies the 50% on-orbit spare reserve recommended for post-launch anomaly correction [53] and confirms that flight software selection is not driven by raw computational performance, hence the same conclusion reached for the hardware trade in Section 5.7.2.

A.4 Flight Computer Mode Configurations

This appendix tabulates the ten operational modes referenced in Section 5.7.3 and shows the corresponding mode-transition diagram. Each row of Table 110 captures the spacecraft-level behaviour during the mode and the corresponding flight computer (cFS) configuration that implements it. ¹⁰

Table 110: Flight Computer Modes & Behavior

Mode	Spacecraft Behavior	Flight Computer Configuration
Start-Up	System boot following LV separation; detumbling via thruster pulses; solar array deployment confirmation from SMM; initial power and comms established via S-Band omnidirectional antenna; spacecraft aliveness checks	cFS ES boots from EEPROM; watchdog timer initialised; cFS SC app executes pre-loaded autonomous deployment sequence; all RS-422 subsystem links polled; IDPU powered and checked; Tier 1 housekeeping only
Commissioning	Sequential subsystem checkout across all 8 spacecraft; orbit-raising and apogee-adjustment burns (Groups 1 and 3); mesoscale constellation phasing for Group 2; instrument calibration and cross-calibration; time synchronisation across constellation	cFS SCH running at reduced cadence; IDPU commanded to instrument checkout mode via IDPU Cmd channel; GNC orbit state published at 1 Hz; science data storage inactive; engineering packets downlinked only

¹⁰This appendix stems directly from the CCD PDR, and is modified slightly to align with the FDR.



Mode	Spacecraft Behavior	Flight Computer Configuration
Nominal (Survey)	Routine science operations in 6-12 $R_{\oplus}$ ROI; spin-stabilised attitude at 15 rpm about Sun-Earth axis; FGM in slow mode (6 bps), ESA and EPD continuously active; periodic NEN downlink at perigee ($\sim 4.2\%$ of orbital period)	cFS SCH: 4 Hz ADCS loop, 1 Hz housekeeping, 0.1 Hz thermal; IDPU in survey mode; cFS DS writing Tier 2 science packets to SSR; GNC ROI flag active; downlink manager armed for perigee window
Burst / High-Rate Science	BBF event detected by IDPU on-board threshold logic ($\Delta B_z \geq 4$ nT in < 8 s); FGM switches to fast mode (3072 bps); all instruments at full rate; prioritised data capture; science data dominates SSR	IDPU publishes <i>BBF-Detected</i> event on cFE Software Bus; DS app switches SSR write queue to Tier 3 (highest priority); SCH increases frame rate; Limit Checker monitors SSR fill level; GNC checks spacecraft remains in ROI
Downlink	Spacecraft at perigee (~ 1000 km altitude); omnidirectional S-Band antenna active; CCSDS TM frames transmitted to NEN; TC frames received and validated; science and housekeeping data downlinked in priority order	POW interlock verified before enabling 40 W transmitter; cFS downlink manager formats CCSDS TM frames by virtual channel (VC0-VC3); up-linked TC commands routed to target subsystem apps via cFE SB; SSR read and cleared post-pass
Maneuver	Thruster burn for orbit raising, apogee adjustment, or stationkeeping; spin-axis precession correction; attitude stabilised before and after burn	ADCS app receives thruster command sequence from DPM at 2 Hz via Thruster Cmd I/F; GNC orbital state updated at 1 Hz; instruments commanded off via IDPU Cmd; SSR writes paused; housekeeping telemetry only during burn
Eclipse	Solar array output drops to zero; spacecraft transitions to battery power; payload sensors powered down; communications temporarily ceased; heaters activated; regular health diagnostics throughout eclipse	cFS LC detects low-power flag from POW; load-shed commands issued to non-critical subsystems; IDPU commanded to standby; SCH suspends science frame; DPM and essential housekeeping remain active; no downlink unless battery SoC permits
Safe Mode	Autonomous fault response triggered by watchdog timeout, EDAC anomaly, or ground command; sun-pointing maintained via spin stabilisation; all non-essential loads powered off; beacon telemetry only	cFS HS app triggers safe mode; all non-critical apps suspended; ES maintains DPM heartbeat and watchdog reset; IDPU powered down; LC monitors power and thermal limits; beacon TM transmitted on every perigee pass pending ground recovery command



Mode	Spacecraft Behavior	Flight Computer Configuration
Standby / Idle	Awaiting scheduled commands or next operational window; no active science collection; spacecraft maintains spin and sun-pointing; minimal power draw	SCH running at minimum cadence (1 Hz HK only); IDPU in low-power standby; DS app idle; GNC orbital state maintained; DPM polls all RS-422 subsystem links at reduced rate
End-of-Life	Minimum- Δv perigee-lowering burn at apogee to initiate atmospheric re-entry; propellant depleted (passivation); all instruments powered off; deorbit sequenced per group (staggered)	cFS SC app executes pre-loaded EOL command sequence; IDPU powered down permanently; SSR wiped; DPM transmits final housekeeping packet; transmitter active until signal lost on re-entry; watchdog timer disabled to allow controlled shutdown

The transitions between these ten modes are summarised in Figure 53. Black arrows denote natural progression through the mission lifecycle (Start-Up → Commissioning → Nominal → End-of-Life), purple arrows denote I/O- and state-driven transitions (e.g. IDPU triggering Burst, POW low-flag triggering Eclipse), and red arrows denote recovery or end-of-life transitions (e.g. HS-triggered entry into Safe Mode from any active state).

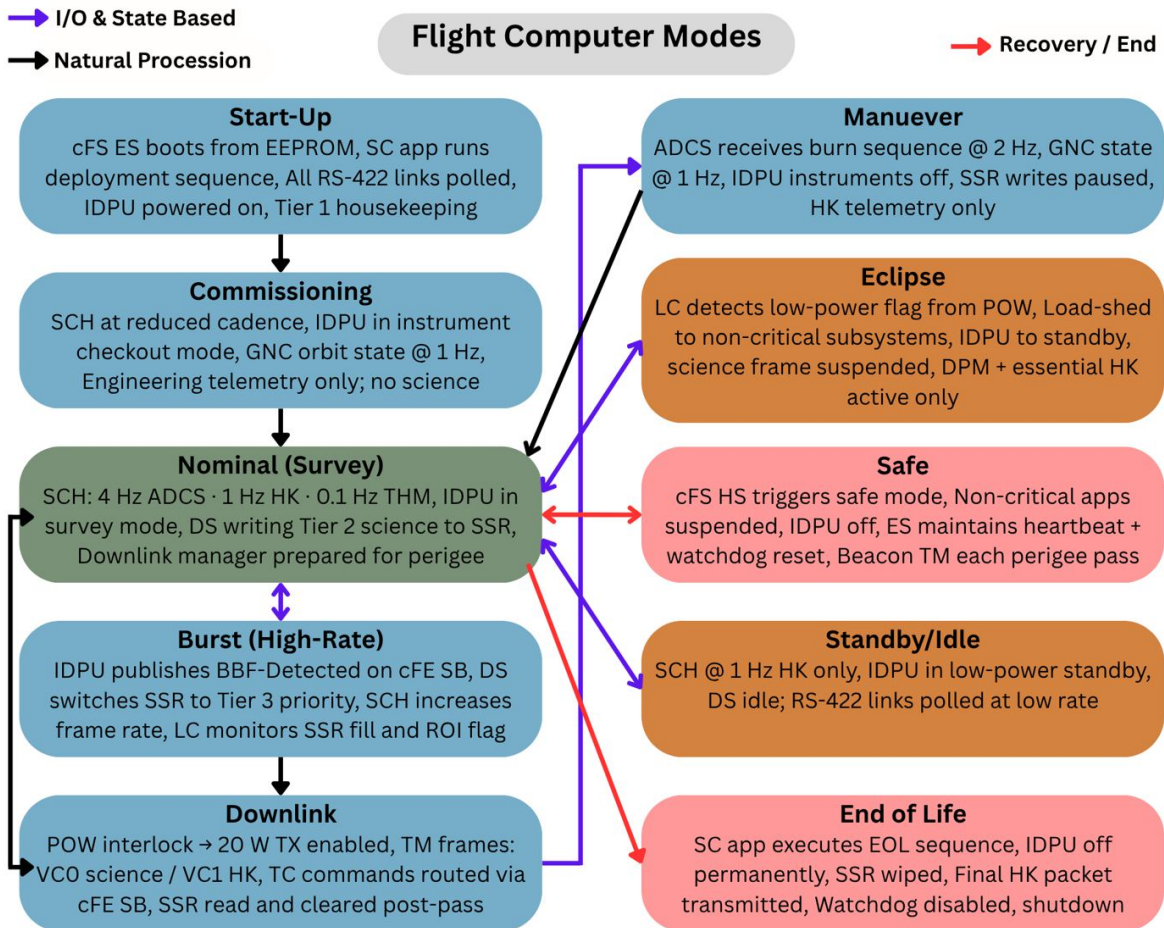


Figure 53: Flight computer mode architecture for the MATRIX spacecraft. The IDPU-initiated Survey ↔ Burst transition (purple) is the primary autonomous mode change in nominal science operations; the HS-triggered Safe Mode entry (red) is the primary fault-response path.

A.5 GNC

A.5.1 Orbital First Principle Calculations

By selecting initial apogee and perigee radii that fit mission specifications, we can then determine semi-major axis length, a , and period T from equations (96) and (97), respectively.

$$a = \frac{r_a + r_p}{2} \quad (96)$$

$$T = 2\pi\sqrt{\frac{a^3}{GM}} \quad (97)$$

From there, we are able to determine the eccentric anomaly, E , at $r = 6 R_E$ via equation (98).



$$r = a(1 - e \cos E) \quad (98)$$

We can then use Kepler's equation, (99) to determine the mean anomaly, M , which tells us what fraction of the period has elapsed from periapsis to a given orbital radius. This allows us to determine the time spent in the ROI beyond $6R_E$ and make more informed decisions on orbital paths.

This was the process in our first design cycle. In this case, we started with exact period values that we wanted and specific perigees, working backward to find apogee radius and determine the percentage of orbit beyond a specific radius.

$$M = E - e \sin E \quad (99)$$

A.5.2 In-Depth ΔV Analysis

This appendix section includes an in-depth analysis of ΔV costs, which was summarized in section 6.7.3. We have broken down these maneuvers into specific sections: **orbit transfer, mission orbit maintenance, and decommission**. Orbit transfer and decommission were analyzed first, as these values will remain relatively constant and will inform how much ΔV is available to use during the main mission component.

Orbit Transfer

Orbit transfer was broken down into two maneuvers: an apogee change and perigee change, designed to be executed at perigee and apogee, respectively [228]. For Group 1, this includes both an apogee and perigee raise, contributing to overall longer period expected for spacing and ROI coverage. For Group 3, this consisted of an apogee and perigee lower. Group 2 has no cost since the launch vehicle will take the satellites directly into Group 2's main orbit.

Group #	Starting $r_p(R_E)$	Starting $r_a(R_E)$	Final $r_p(R_E)$	Final $r_a(R_E)$	ΔV Cost (m/s)	Fuel Used (kg)
1	1.149	8.960	1.157	12.089	150.382	13.255
2	1.149	8.960	1.149	8.960	0	0
3	1.149	8.960	1.141	7.203	137.666	12.160

Table 111: Summary of Transfer ΔV Costs

Table 111 shows a summary of these orbit transfer costs, done through astrogator analysis in STK. From here we can see that so far, both Group 1 and 3 have similar ΔV costs for this initial stage of the mission, a result of the choice to launch directly in Group 2's orbit. This will increase our margin's with satellites in Group 2, allowing us to possibly extend or refit the mission for this specific group.

Mission Orbit Maintenance

Mission orbit maintenance was set to both minimize further costs and ensure completion of mission-level science goals, as well as subsystem requirements. In order to stay within ROI, analysis was done on the fuel, and ΔV costs for controlling both RAAN and inclination (as these are the two defining parameters for the ecliptic plane).



Group #	$\Delta V/\text{year}$ (m/s)	Fuel/year (kg)	Total ΔV (m/s)	Total Fuel Used (kg)
1	69.563	6.401	347.815	32.005
2	89.82	8.684	449.1	43.421
3	125.689	11.374	628.449	56.873

Table 112: Costs of Inclination and Perigee Maintenance

Via STK, we were able to obtain annual ΔV costs for inclination and perigee altitude control, which is shown in table 112 above. This was done by running a sequence that checks inclination and perigee values, and executes a burn at apogee if they are out of specified limits. As you can see, annual stationkeeping becomes more expensive with our lower orbits, and takes up considerable amounts of fuel. While highly expensive, we believe that these maneuvers will be necessary to ensure coverage in the ROI and the fulfillment of mission-level science requirements.

Decommission

ΔV costs related to EOL burns were primarily dependent on apogee altitude for the mission orbit as well as on the desired perigee altitude to facilitate satellite reentry. This altitude desired for reentry is currently modeled at 100 km, based on de-orbit guidelines via NASA and the desire to start the reentry process within a few orbits after the completion of the burn [239]. STK analysis was conducted on a set of perigee lowering burns done at apogee for each group, shown below in table 113.

Group #	Starting Perigee Altitude (km)	Final Perigee Altitude (km)	ΔV Cost (m/s)	Fuel Used (kg)
1	1000	100	54.801	4.664
2	950	100	71.005	6.348
3	900	100	109.261	8.898

Table 113: Summary of Decommission ΔV Costs

The results of this analysis was initially surprising, as Group 3 satellites incur a much higher ΔV cost despite having to lower their perigee altitude by a smaller amount. However, a quick use of the vis-viva equation (100) below,

$$v = \sqrt{\mu \left(\frac{2}{r} - \frac{1}{a} \right)} \quad (100)$$

shows that at apogee, the velocities of Groups 1, 2, and 3 are 950.24, 1259.23, and 1540.44 m/s respectively. This larger velocity at Group 3 makes it more difficult to execute a perigee change, therefore driving up ΔV costs. As a result, through the analysis of these first two sections, we can see that Group 3 will have the least amount of fuel remaining, forcing us to conduct orbit maintenance around this limiting factor.



A.5.3 Effect of Attitude Error during Orbit Raising and Lowering

Table 114: Effect of Attitude Error during Orbit Raising

Error	ΔV (m/s)	Fuel Used (kg)	Δ Fuel Used
0 degrees	147.8883	13.9723	0.000%
0.1 degrees	147.8892	13.9724	0.001%
1 degree	147.9771	13.9805	0.059%
2 degrees	148.2440	14.0050	0.234%
3 degrees	148.6902	14.0461	0.529%
4 degrees	149.3181	14.1040	0.942%
5 degrees	150.1308	14.1788	1.470%
10 degrees in all axes	157.1497	14.8241	6.097%

Table 115: Effect of Attitude Error during Orbit Lowering

Error	ΔV (m/s)	Fuel Used (kg)	Δ Fuel Used
0 degrees	132.4392	12.5453	0.000%
0.1 degrees	132.4400	12.5454	0.001%
1 degree	132.5210	12.5529	0.060%
2 degrees	132.7668	12.5756	0.242%
3 degrees	133.1780	12.6137	0.545%
4 degrees	133.7566	12.6673	0.972%
5 degrees	134.5068	12.7367	1.526%
10 degrees	140.9969	13.3367	6.308%

A.6 ADCS

A.6.1 MIB Equations

These equations are derived from those talked about in the ADCS Lecture Slides and in the Space Vehicle Design textbook [299].

$$\Delta\theta = \frac{\Delta H}{H} \quad (101)$$

$$\Delta H = r \cdot \text{MIB} \quad (102)$$

$$H = I_{\text{spin}} \cdot \omega \quad (103)$$

$$\text{MIB} = \frac{\Delta\theta \cdot I_{\text{spin}} \cdot \omega}{R} \quad (104)$$

A.6.2 Spacecraft Specifications

Table 116: Spacecraft Structural Parameters

Parameter	Value
Primary Structure Type	Cylindrical thrust tube with hexagonal outer walls
Total Spacecraft Mass	285 kg
Internal Volume	0.439 m ³
Thrust Tube Diameter	24 in (0.6096 m)
Side Length	23 in (0.584 m)
Height	20 in (0.508 m)
Wall Thickness	3 mm
Material	Aluminum 7075-T6

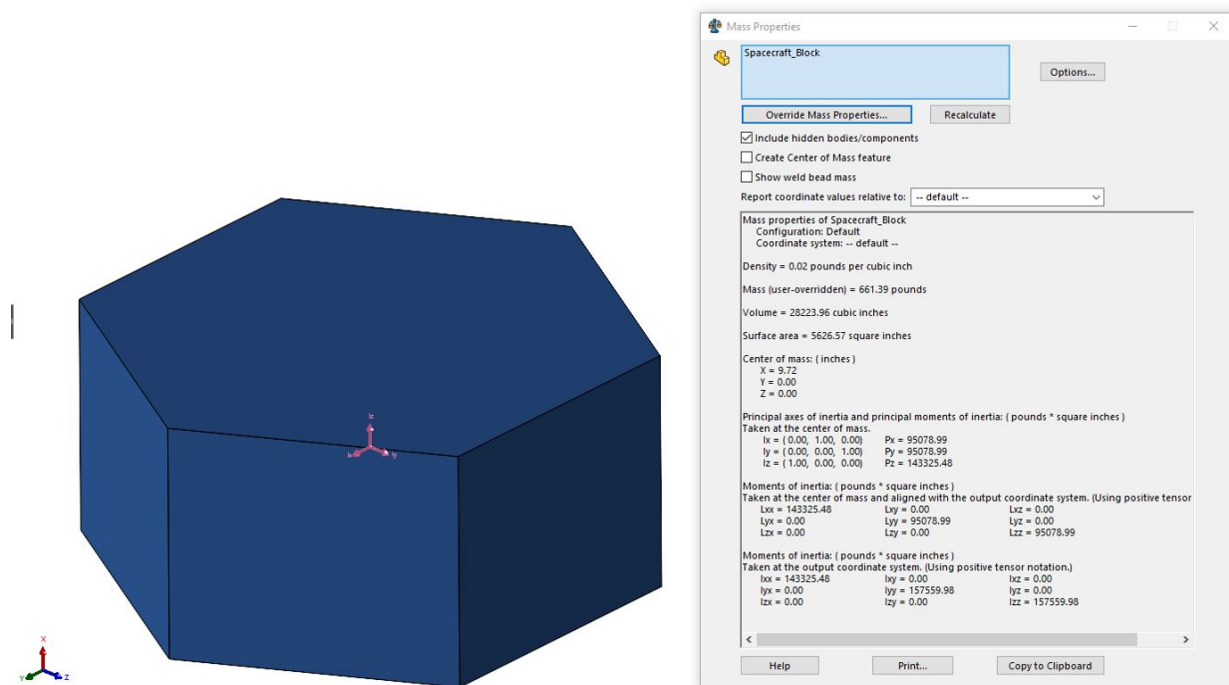


Figure 54: Moments of Inertia of the satellite based on the SMM SolidWorks CAD model.

A.6.3 Spin Rate Calculation

After calculating the disturbance torques, we can combine the worst case GGT and the Solar Radiation Torque overestimate to get a total torque on the satellite by external forces. Retrieving the period of each group orbit, we find then find the net momentum difference per period. While a generous overestimate, this value gives us an upper bound of the momentum the ADC subsystem must be able to manage. With the disturbance torque estimates, we can now find the nominal spin rate required by our satellites. First, we must calculate the required angular momentum using equation (105):

$$L = \frac{\tau \cdot P}{\theta} = \frac{\Delta L}{\theta} \quad (105)$$

where H is the required angular momentum, τ is the total disturbance torque, P is the orbital period, ΔH is the previously calculated momentum change per period and θ is the permissible pointing deviation. Because the science instruments don't require active pointing, the θ is driven by the communication system, which is 1° . The next step is to find the spin rate from this angular momentum using the equation (106)

$$\omega = \frac{L}{I_z} \quad (106)$$

where I_z is the largest inertial axis upon which we are spinning. This process was the same as the first design cycle, it was just done for each of the three groups

A.7 SMM

A.7.1 Random Vibration Environment Specifications

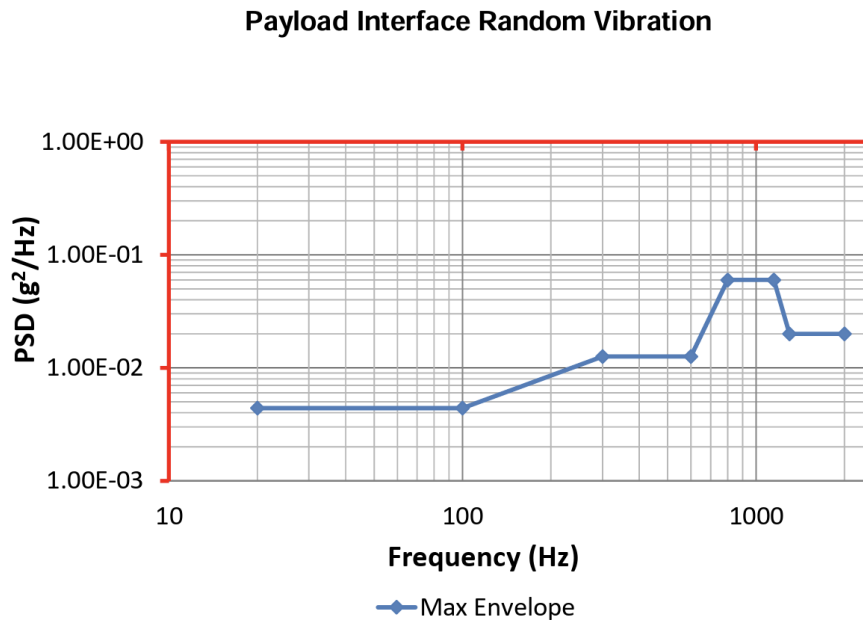


Figure 55: Payload interface random vibration envelope [139]

A.7.2 Secondary Payload Adapter Analysis

To accommodate the launch vehicle conditions, an ESPA-derived secondary payload system was designed. It utilizes the same key parameters such as a wall thickness of 0.5 inches, payload interface diameter of 24 inches, and Aluminum 7050 material [296]. The purpose of this analysis

is to demonstrate that an ESPA ring of a non-standard size would be able to withstand launch loading conditions.

A fixed constraint was applied to the launch vehicle interface and a 7g axial and 3g lateral load was applied to the stacked configuration. Block models of each spacecraft were attached to the payload interfaces as geometry and mass representations.

The results of the analysis showed that the max stress experienced was 50 MPa (Figure 56a) and the total displacement, which notably only occurred at the free end of the spacecraft, was 0.4 mm (Figure 56b). A buckling analysis showed that the load factor was 223, indicating that structure is highly unlikely to experience compressive buckling.

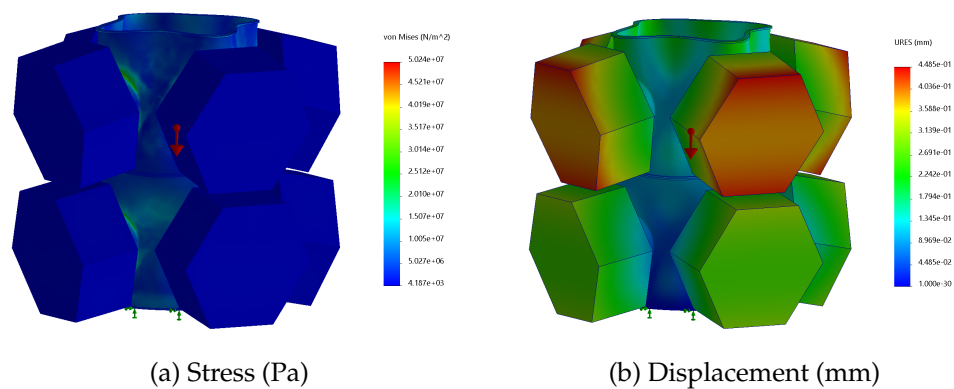


Figure 56: Results of static load FEA analysis

A.7.3 Spacecraft Materials

Selecting appropriate materials for structural components of the spacecraft requires balancing mass, mechanical performance, thermal behavior, and cost, among others. Some of the most common materials for spacecraft, as described by SMAD, are aluminum alloys, steel, titanium, and carbon fiber composites [4]. A trade study was performed to determine the most suitable materials for the spacecraft. Table 117.

Property	Al 7075 AL7075	Al 6061 AL6061	Titanium titanium	Steel Steel, steel 2	Carbon Fiber Composites carbon
Density [g/cm ³]	2.81	2.70	4.43	7.86	1.15 - 2.25
Modulus of Elasticity [GPa]	71.7	68.9	114	196	2.62 - 520
Ultimate Tensile Strength [MPa]	572	310	900	160	0.917 - 4140
Tensile Yield Strength [MPa]	503	276	830	115	4.62 - 3380
CTE [°C ⁻¹]	2.36×10^{-5}	2.36×10^{-5}	8.6×10^{-6}	11.2×10^{-6}	$(-0.700 - 186) \times 10^{-6}$
Thermal Conductivity [W/mK]	130	167	6.7	17.9 - 22.6	0.220 - 400
Electrical Resistivity [Ω·cm]	5.15×10^{-6}	3.99×10^{-6}	1.78×10^{-4}	9.8×10^{-5}	$100 - 1.30 \times 10^{13}$
Relative Cost	Moderate	Low	Expensive	Expensive	Expensive

Table 117: Material properties

Comparing the materials listed above, titanium and steel were eliminated early as candidate materials for the spacecraft due to their high densities and costs, which would impose a significant mass and cost penalties on the mass and cost constraint listed in section 2.2. However, titanium offers a



superior ultimate strength and a low coefficient of thermal expansion, which makes the material a potential candidate to reinforce specific points in the structure that require greater strength and stiffness.

Carbon fiber composites offer an attractive density, ultimate tensile strength, and tensile yield strength ranges. However, their anisotropic nature introduces complexity in structural analysis and manufacturing. The wide variability in their properties also increases design uncertainty. For these reasons, carbon fiber composites were not selected as the primary structural material, though like with titanium, it may be revisited for secondary structural components if needed in future design iterations.

Consequently, the candidate materials for the main structure of the spacecraft are the alloys Al 7075 and Al 6061. Both of these materials are heritage materials in a diversity of NASA missions. As seen in Table 117, the two alloys have similar properties. Al 7075 exhibits greater ultimate tensile and tensile yield strengths than Al 6061, which would allow for thinner-walled structural members that maintain equivalent or superior safety margins under launch and operational loads, resulting in a net reduction in structural mass. Alternatively, Al 6061 exhibits marginally superior thermal conductivity. Due to its greater ultimate tensile and tensile yield strengths, Al 7075 is the preliminary choice of material for the spacecraft bus.

A.7.4 Launch Vehicle Separation System

A payload separation system is required to provide a mechanical interface between the spacecraft and the launch vehicle. The system must restrain the spacecraft during launch and provide a controlled separation once the spacecraft reaches orbit. The separation system must be compatible with the 24-inch payload interface used by the spacecraft thrust tube and must support the spacecraft mass of 280 kg while minimizing additional structural mass.

The preliminary separation system considered for this spacecraft is the Sierra Space QwkSep 24 Low Profile separation system. This system is designed for ESPA-class secondary payloads and provides a clamp band interface compatible with 24-inch payload adapters. The QwkSep system is available in both pyrotechnic and non-pyrotechnic actuation configurations. In order to determine the most suitable configuration, a trade study was performed comparing these two variants as well as an alternative non-explosive actuator separation system.

Parameter	QwkSep Pyrotechnic [145]	QwkSep Non-Pyro [145]	NEA Lightband System
Separation Shock	~1000 g	~100 g	~200 g
Operating Temperature Range	-68 to 104°C	-65 to 75°C	-40 to 85°C
System Mass	9.5 kg	9.5 kg	~12 kg
Payload Capacity	300 kg	300 kg	~400 kg
Actuation Type	NASA Standard Initiator	Mechanical Release	Non-Explosive Actuator
Integration Complexity	Moderate	Moderate	Higher

Table 118: Comparison of payload separation systems

Both Sierra Space QwkSep configurations provide identical structural performance, including a system mass of approximately 9.5 kg and a payload capacity of 300 kg with a maximum payload center-of-mass offset of 508 mm. The primary difference between the two configurations is the actuation mechanism and resulting separation shock environment. The pyrotechnic configuration produces a separation shock of approximately 1000 g near the actuator, whereas the



non-pyrotechnic version reduces this shock level to approximately 100 g. Lower shock levels reduce the risk of disturbance to sensitive avionics and scientific instruments. The pyrotechnic configuration provides a wider operating temperature range (-68 to 104°C) compared with the non-pyrotechnic system (-65 to 75°C). In addition, the pyrotechnic version uses a NASA Standard Initiator, which simplifies integration with launch vehicle systems.

Based on this analysis, the Sierra Space QwkSep 24 Low Profile Pyro separation system was selected as the spacecraft separation mechanism. The pyrotechnic configuration is preferred due to its wider operating temperature range and compatibility with NASA Standard Initiators, which simplifies launch vehicle integration while still maintaining acceptable shock levels for the spacecraft structure and avionics.

A.7.5 Mass and Volume Budget

Subsystem	Component	Unit Mass (kg)	Unit Vol (cm ³)	Quantity	Mass (kg)	Volume (cm ³)	Total Mass (kg)	Total Vol (m ³)
CCD	Anywaves S-Band Antenna	0.139	86	4	0.556	344	1.22	5774.3
	BAU	0	4920.3	1	0	4920.3		
	NANOLink-boost Gen2 Radio	0.248	191	2	0.496	382		
	NuPower™ 12B01A Power Amplifier	0.085	64	2	0.17	128		
OPL	ESA	2.5	3560.65	1	2.5	3560.65	5.70	9056.34
	EPD	2.2	5041	1	2.2	5041		
	MAG	1	454.69	1	1	454.69		
ADCS	Redwire Fine Spinning Sun Sensor	0.725	373.76	2	1.45	747.52	5.33	813.21
	DMU 41 IMU	0.178	5.24	1	0.178	5.24		
	Fluid (Mercury)	2.2	40.3	1	2.2	40.3		
	Ring (Stainless Steel)	1.5	20.15	1	1.5	20.15		
POW	AZUR 4G32-ADV Solar Panels (per cell)	0.0104	21.66	65	0.676	1407.9	4.34	2835.9
	Mollicel INR-21700-M50A	0.07	26	48	3.36	1248		
	Pumpkin ESPM 1 PMAD	0.3	180	1	0.3	180		
PLV	Primary Thruster	0.38	52.86	4	1.52	211.44	22.32	457103.36
	RCS Thruster	0.6	35.24	8	4.8	281.92		
	Primary Propellant Tank	3.7	70000	1	3.7	70000		
	Secondary Propellant Tank	1.74	82990	4	6.96	331960		
	Pressurant Tank	1.07	7290	2	2.14	14580		
	Plumbing (25% of tank mass)	3.2	40070	1	3.2	40070		
SMM	Primary Structure	43.57	15503.95	1	43.57	15503.95	65.63	23668.30368
	Extra Radiation Shielding	5.2	42.1	1	5.2	42.1		
	Secondary Structures (Brackets)	4.357	1550.395	1	4.357	1550.395		
	Sierra Space Qwk-Sep 24	9.5	5448.21	1	9.5	5448.21		
	Boom	3	1106.77	1	3	1106.77		
THM	Frangibolt	0.01	16.87868	1	0.01	16.87868	11.10	26414.2
	20 Layer MLI (per m ²)	0.933	5000	2.98	2.78034	14900		
	TE Connectivity Platinum RTD Sensor	0.02	1	36	0.72	36		
	Thin-Film Heaters	0.03	1	20	0.6	20		
	Thermal Straps	0.2	3	5	1	15		
Totals	Sierra Space Louver	1.5	2860.8	4	6	11443.2	115.63	0.53

Figure 57: MATRIX Mass and Volume Budget Spreadsheet



A.8 THM

A.8.1 Radiation Geometric View Factors

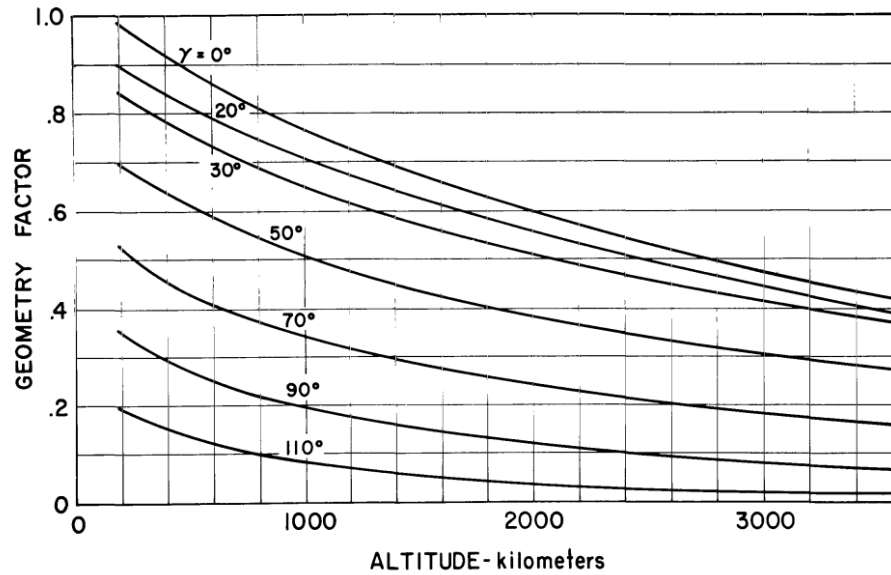


FIGURE 6. GEOMETRY FACTOR VS. ALTITUDE FOR IR TO A FLAT PLATE

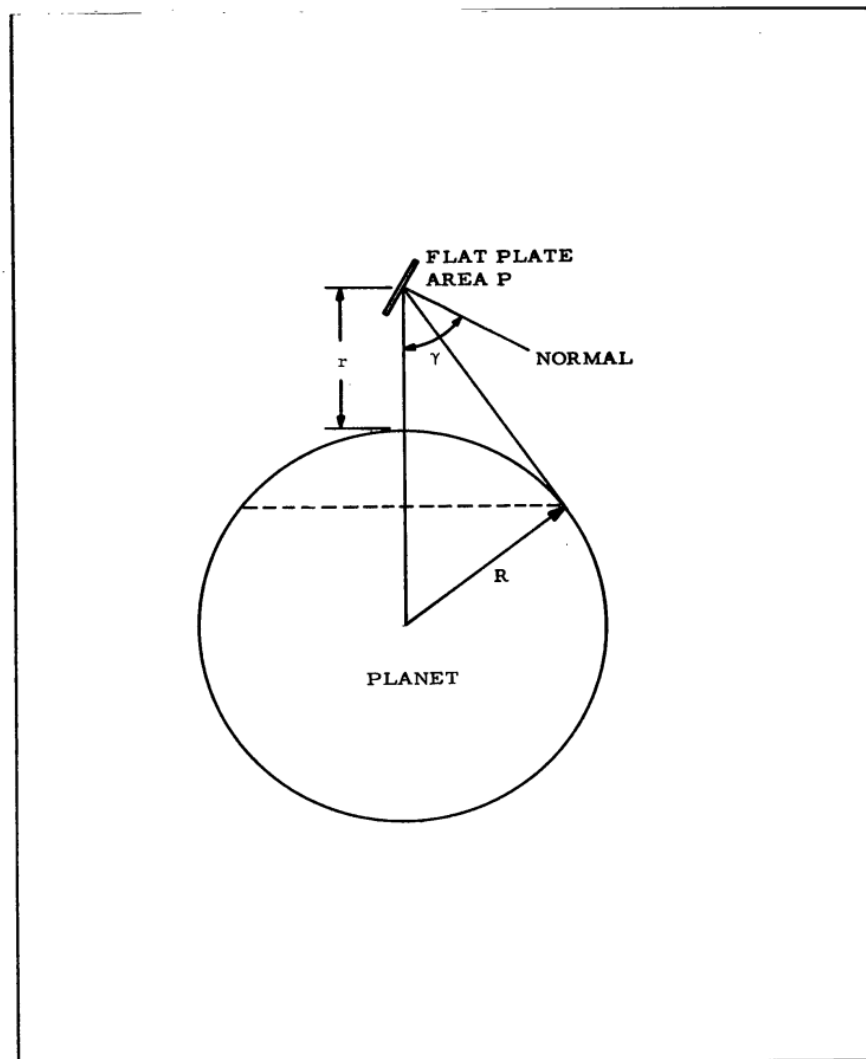


FIGURE 4. GEOMETRY FOR PLANETARY THERMAL RADIATION TO A FLAT PLATE

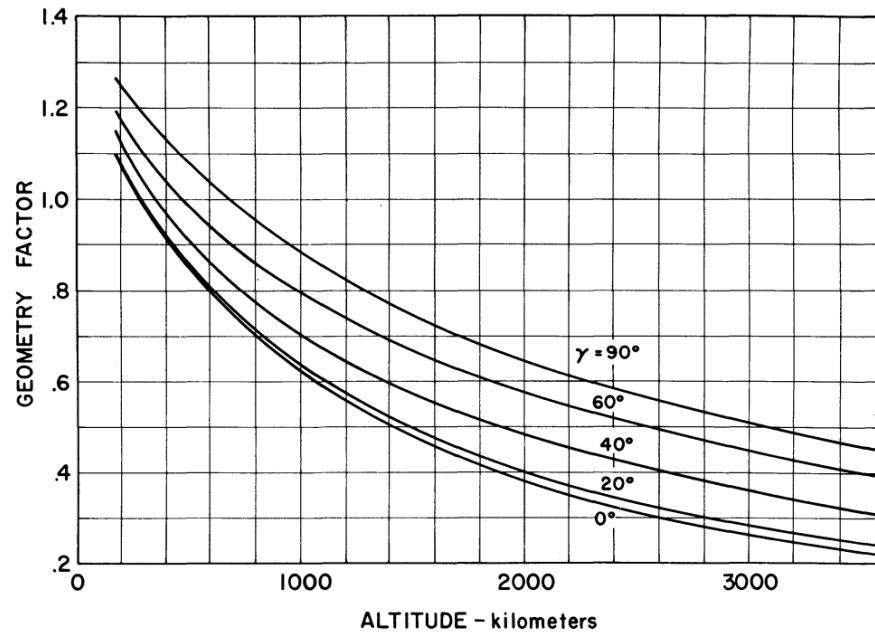


FIGURE 3. GEOMETRY FACTOR VS. ALTITUDE FOR IR TO A CYLINDER

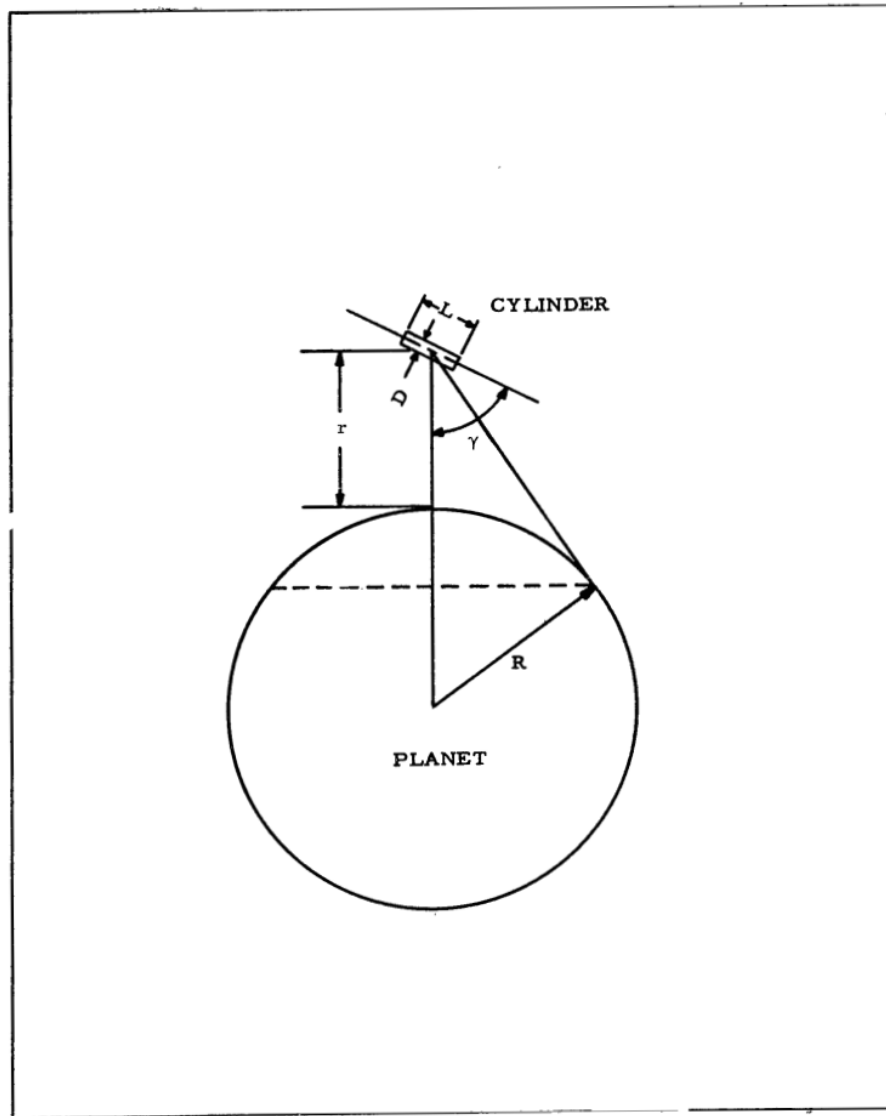
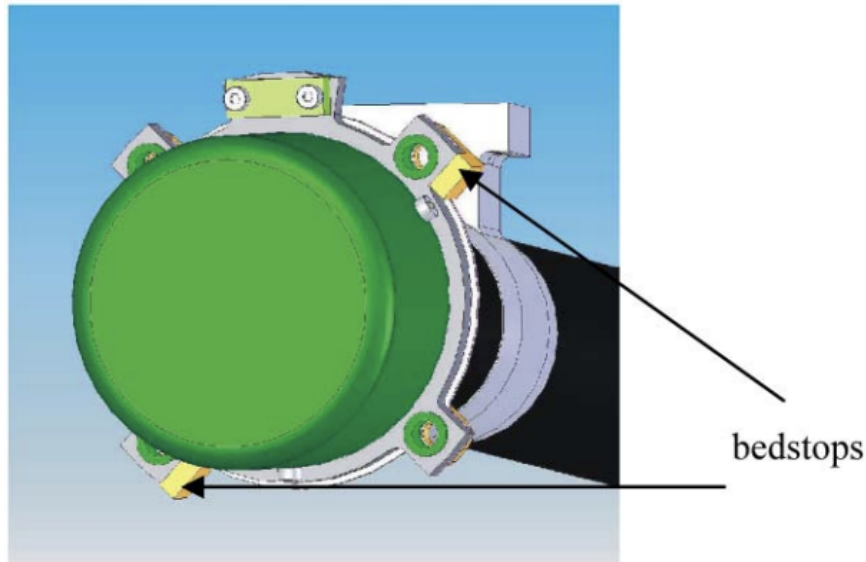


FIGURE 1. GEOMETRY FOR PLANETARY THERMAL RADIATION TO A CYLINDER

A.8.2 THEMIS Magnetometer Geometry

Covering is gold plating with emissivity of 0.02 and absorptivity of 0.06.



A.8.3 Identifying Temperature Monitoring Hardware

As stated in objective 3, one of the critical THM objectives is to monitor the temperature of components and detect unexpected thermal conditions. To fulfill the objective, temperature sensors must be selected. A trade study comparing three of the most commonly used type of temperature sensors for space mission was first conducted, comparing critical criteria to fulfill our subsystem's and mission's requirements such as operational temperature range, power requirement, radiation tolerance, and mass and size [300], [301]. This analysis helped determine which sensor type to use. The results are shown in table 119.

Parameter	Thermistor (NTC) NTC, radiation'NTC	RTD RTD, radiation'RTD	Thermocouple thermocouple, radiation' thermocouple
Temperature Range	-55 to +90 C	~ -125 to +170 C	~ -200 to +1250 C
Accuracy	High	High	Medium
Flight Heritage	Very High - See NASA GSFC S-311-P-18 part	High	Medium - more common in extreme scenarios (rocket nozzles or ground cryogenic)
Power Requirement	Requires small excitation current		No power required
Radiation Tolerance	~ 80 kGy	~ 300 kGy	~ 3000 kGy
Mass & Size	Varies depending on specific commercial sensors, but minimal mass and size		
Electrical Integration	Non-linear response	Linear response	Requires amplifier
Cost	~300/unit	~350/unit	~50/unit

Table 119: Temperature sensor types trade study

The RTD sensor type was selected based on a trade-off analysis of the key mission requirements. Its temperature range fully encompasses all derived operational temperatures for the mission,



while its high accuracy ensures reliable thermal monitoring throughout the mission duration, which is critical for thermal control compliance. Furthermore, its high flight heritage reduces programmatic risk relative to less-proven alternatives. While the thermocouple outperforms the RTD in several areas, including radiation tolerance, cost, power consumption, and temperature range, these advantages do not sufficiently offset its drawbacks for this mission. The thermocouple's medium accuracy introduces greater uncertainty in temperature measurements, which is undesirable for precision thermal control, which is critical to prevent events with the potential to end the mission like thermal runaway. Its requirement for a signal amplifier adds complexity to the electrical integration. Additionally, its flight heritage in standard spacecraft thermal monitoring applications is limited compared to the RTD, increasing risk. The thermistor was similarly eliminated due to its restricted temperature range, which does not fully cover the mission's derived operational temperatures. Therefore, the RTD represents the best balance of accuracy, heritage, temperature coverage, and sufficient radiation tolerance for this mission.

Once the sensor type was selected, a trade study of commercial RTD sensors was conducted, comparing performances through criteria like temperature range, sensitivity, and insulation resistance. The results are shown in Table 120.

Parameter	ANTRIX PTS-84 RTD ANTRIX	TE Connectivity RTD Platinum Sensor 20030604-00 TE platinum	TE Connectivity Glass Wire Wound Sensor GO 2327 SB0871 TE glass	Honeywell Thin Film Platinum RTD HEL- 705 honeywell
Temperature Range [°C]	−200 to +250	−170 to +125	−100 to +250	−70 to +260
Nominal Resistance at 0°C [Ω]	100 ± 0.25	100	100	1000 ± 2
Sensitivity [Ω/°C]	0.385	0.385	0.385	0.375
Thermal Coefficient [ppm/K]	3850	3850	3850	3750
Recommended Excitation Current [mA]	Not stated in datasheet	< 1	1	1
Time Constant [s]	< 3	Not specified in datasheet	25	Not specified in datasheet
Mass [g]	< 100	Not specified in datasheet	Not specified in datasheet	Not specified in datasheet
Electrical Interface	DSM-Circular Electrical Connector	26 AWG polyimide insulated twisted pair wires	Platinum-coated nickel wire leads	28 AWG TFE Teflon insulated nickel-coated stranded copper, 2-wire
Flight Heritage	Yes	Yes, ESA qualified	Not stated	Not stated

Table 120: RTD Sensor Comparison

Upon comparing the results, the Honeywell HEL-705 was discarded first. Despite its higher nominal resistance (and consequently, greater measurement resolution), its lower temperature limit of 70°C does not meet the requirements of the magnetotail environment, where significantly colder temperatures and severe thermal cycling are expected. Between the two TE Connectivity sensors, the surface mount platinum RTD was selected on the basis of its ESA qualification and over 20 years of documented space heritage — decisive factors for a mission operating in the harsh thermal and radiation conditions of the magnetotail. Finally, although the ANTRIX PTS-84 offers a broader temperature range and low time constant, its DSM circular connector and more complex mounting scheme introduce unnecessary integration overhead compared to the TE Connectivity



surface mount design. The TE Connectivity RTD Platinum Sensor was, therefore, selected for this mission.

A.9 POW

The calculations utilized to properly size the battery packs used in the POW subsystem are detailed below. The solar array sizing calculations are detailed in Section 10.7.5.

The first step for determining an operating Depth of Discharge (DOD) is to calculate the maximum number of cycles experienced by the battery over the lifespan of the MATRIX mission using Equation 107 [165]. Since the maximum number of cycles is inversely proportional to the minimum orbital period according to Equation 107, the highest value for the maximum number of cycles occurs for the Group 3 satellites in the MATRIX mission, as they have the shortest orbital period of 12 hours. Therefore, the maximum number of cycles for any group of satellites in the MATRIX mission is 4383, as shown in Equation 108.

$$\text{Max. Number of Cycles} = \frac{\text{Hours Per Day}}{\text{Min. Orbital Period (hrs)}} \times \text{Days Per Year} \times \text{Years of Mission Lifespan} \quad (107)$$

$$\text{Max. Number of Cycles} = \frac{24}{12} \times 365.25 \times 6 = 4383 \text{ cycles} \quad (108)$$

Figure 21-16 of [168] shows that a Depth of Discharge (DOD) of 60% is acceptable for Li-ion batteries that experience less than 10^4 cycles before the end of the mission. Figure 4 of [238] further shows that 4383 cycles corresponds approximately to a DOD of 80% without any safety factor. Therefore, an operating DOD of 60% is chosen. A conservative capacity fade of 10% is assumed over the lifespan of the battery, which is the same value used in [165].



$$\text{Current Drawn from Battery (A)} = \frac{\text{Peak Power Draw with Safety Factors (W)}}{\text{Nominal Bus Voltage (V)}} \quad (109)$$

$$\text{EOL Battery Capacity (Ah)} = \frac{\text{Current Drawn from Battery (A)} \times \text{Nominal Eclipse Time (hrs)}}{\text{DOD (\%)}} \quad (110)$$

$$\text{BOL Battery Capacity (Ah)} = \text{EOL Battery Capacity (Ah)} \times (1 + \text{Capacity Fade (\%)}) \quad (111)$$

$$\# \text{ Series Cells} = \frac{\text{Nominal Bus Voltage (V)}}{\text{Nominal Cell Voltage (V)}} \quad (112)$$

$$\text{Volume per Cell (cm}^3\text{)} = \frac{\pi}{4} \times (\text{Cell Diameter (cm)})^2 \times \text{Cell Height (cm)} \quad (113)$$

$$\# \text{ Parallel Strings} = \frac{\text{BOL Battery Capacity (Ah)}}{\text{Typical Capacity (Ah)}} \quad (114)$$

$$\text{Current Drawn per Parallel String (A)} = \frac{\text{Current Drawn from Battery (A)}}{\# \text{ Parallel Strings}} \quad (115)$$

$$\# \text{ Total Cells in Battery} = \# \text{ Series Cells} \times \# \text{ Parallel Strings} \quad (116)$$

$$\text{Total Mass of Battery (kg)} = \# \text{ Total Cells in Battery} \times \text{Mass per Cell (g)} \times 10^{-3} \quad (117)$$

$$\text{Total Volume of Battery (cm}^3\text{)} = \# \text{ Total Cells in Battery} \times \text{Volume per Cell (cm}^3\text{)} \quad (118)$$

$$\text{Total Battery Output till Drain (Wh)} = \# \text{ Total Cells in Battery} \times \text{Typical Capacity (Ah)} \times \text{Nominal Bus Voltage (V)} \quad (119)$$

A.10 PLV

A.10.1 Propellant Selection Calculations

Rocket Equation Sizing Method

The propellant mass required for a specified mission Δv is computed using the classical rocket equation,

$$\Delta v = I_{sp} g_0 \ln \left(\frac{m_0}{m_f} \right),$$

where I_{sp} is the propellant specific impulse, $g_0 = 9.80665 \text{ m/s}^2$, m_0 is the initial wet mass, and m_f is the final dry mass after propellant depletion. Solving for propellant mass using a fixed dry mass gives

$$m_{prop} = m_f \left(e^{\Delta v / (I_{sp} g_0)} - 1 \right).$$

The updated dry mass basis used in this analysis is

$$m_f = 120 \text{ kg}.$$



Propellant Properties

Representative properties used in the trade study are

$$I_{sp,hyd} = 235 \text{ s}, \quad \rho_{hyd} = 1.004 \text{ kg/L},$$

for hydrazine, and

$$I_{sp,ASCENT} = 257 \text{ s}, \quad \rho_{ASCENT} = 1.47 \text{ kg/L},$$

for ASCENT. The ASCENT performance values are based on NASA GPIM/AF-M315E data [184]. Hydrazine is treated as the heritage monopropellant baseline because of its extensive spacecraft usage and mature component ecosystem [186], [187].

Hydrazine Calculation for the Maximum Required Δv

For hydrazine at $I_{sp} = 235 \text{ s}$ and the maximum required $\Delta v = 1129 \text{ m/s}$,

$$m_{prop,hyd} = 120 \left(e^{1129/(235 \cdot 9.80665)} - 1 \right).$$

Evaluating the exponential term gives

$$m_{prop,hyd} = 75.9 \text{ kg}.$$

The corresponding propellant volume is

$$V_{hyd} = \frac{m_{prop,hyd}}{\rho_{hyd}} = \frac{75.9}{1.004} = 75.6 \text{ L}.$$

ASCENT Calculation for the Maximum Required Δv

For ASCENT at $I_{sp} = 257 \text{ s}$ and $\Delta v = 1129 \text{ m/s}$,

$$m_{prop,ASCENT} = 120 \left(e^{1129/(257 \cdot 9.80665)} - 1 \right).$$

This gives

$$m_{prop,ASCENT} = 67.8 \text{ kg}.$$

The corresponding propellant volume is

$$V_{ASCENT} = \frac{m_{prop,ASCENT}}{\rho_{ASCENT}} = \frac{67.8}{1.47} = 46.1 \text{ L}.$$

Hydrazine Capacity of the Final Tank Architecture



The final hydrazine tank architecture consists of four cylindrical side tanks and one central spherical tank. The total propellant volume is

$$V_{prop,total} = 4(20.75 \text{ L}) + 70 \text{ L} = 153.0 \text{ L}.$$

The corresponding hydrazine mass capacity is

$$m_{prop,cap} = \rho_{hyd} V_{prop,total} = (1.004)(153.0) = 153.6 \text{ kg}.$$

The wet mass corresponding to this propellant load is

$$m_0 = m_f + m_{prop,cap} = 120 + 153.6 = 273.6 \text{ kg}.$$

The estimated Δv capability is therefore

$$\Delta v = I_{sp} g_0 \ln \left(\frac{m_0}{m_f} \right) = 235(9.80665) \ln \left(\frac{273.6}{120} \right).$$

This gives

$$\Delta v \approx 1899 \text{ m/s}.$$

The hydrazine architecture therefore provides approximately

$$1899 - 1129 = 770 \text{ m/s}$$

of margin above the maximum required mission Δv .

A.10.2 Propellant Depletion and Center-of-Mass Shift

The spacecraft center of mass is determined by the mass-weighted average position of all spacecraft components. Considering only the effect of propellant depletion, the instantaneous center of mass can be written as

$$\vec{r}_{CM} = \frac{m_{dry} \vec{r}_{dry} + \sum_{i=1}^N m_i \vec{r}_i}{m_{dry} + \sum_{i=1}^N m_i}$$

where m_{dry} and $\vec{r}_{dry}$ are the dry spacecraft mass and dry center-of-mass location, and m_i and $\vec{r}_i$ are the remaining propellant mass and tank location for the i -th tank. For this propulsion architecture, $N = 5$, corresponding to four secondary side tanks and one primary central tank.

The center-of-mass shift caused by propellant depletion is obtained by comparing the center of mass at different propellant states:



$$\Delta \vec{r}_{CM} = \vec{r}_{CM,2} - \vec{r}_{CM,1}$$

For a symmetric pair of tanks located at equal and opposite positions about the spacecraft centerline, the tank locations can be written as

$$\vec{r}_1 = +\vec{r}_s$$

$$\vec{r}_2 = -\vec{r}_s$$

If the two tanks contain equal propellant mass during depletion,

$$m_1 = m_2$$

then their combined contribution to the center-of-mass numerator is

$$m_1 \vec{r}_1 + m_2 \vec{r}_2 = m_1 \vec{r}_s + m_1 (-\vec{r}_s) = 0$$

Thus, perfectly symmetric depletion produces no first-order lateral center-of-mass shift. This is the reason that geometrically matched secondary tanks should be drained together rather than independently.

If the two tanks deplete unevenly, the residual center-of-mass contribution becomes

$$m_1 \vec{r}_s + m_2 (-\vec{r}_s) = (m_1 - m_2) \vec{r}_s$$

so the resulting center-of-mass shift is approximately

$$\Delta \vec{r}_{CM} \approx \frac{(m_1 - m_2) \vec{r}_s}{m_{tot}}$$

where m_{tot} is the instantaneous total spacecraft mass. This expression shows that the center-of-mass shift scales directly with the propellant mass imbalance and the distance of the tanks from the spacecraft centerline. Therefore, even small depletion imbalances can become important if the tanks are located far from the center of mass.

The same cancellation logic applies to top-bottom tank symmetry. For two tanks located above and below the spacecraft centerline,

$$\vec{r}_{top} = +\vec{r}_v$$

$$\vec{r}_{bottom} = -\vec{r}_v$$



equal depletion gives

$$m_{top}\vec{r}_{top} + m_{bottom}\vec{r}_{bottom} = 0$$

when

$$m_{top} = m_{bottom}$$

Thus, paired depletion preserves both lateral and vertical mass balance.

The primary central tank is located closer to the spacecraft centerline, so its remaining propellant has a smaller moment arm. If the central tank is located at position $\vec{r}_c$, its contribution to center-of-mass shift is

$$\Delta\vec{r}_{CM,c} \approx \frac{\Delta m_c \vec{r}_c}{m_{tot}}$$

Because $|\vec{r}_c|$ is smaller than the side-tank moment arm, propellant depletion from the central tank produces less center-of-mass migration than equivalent depletion from an off-axis tank. This motivates the selected depletion sequence in which the secondary tanks are drained first and the central tank is reserved for later mission phases.

For a spin-stabilized spacecraft, an off-axis center-of-mass offset is also dynamically important. If the center of mass is displaced from the intended spin axis by a distance e , the offset rotates with the spacecraft body. This can introduce periodic imbalance effects and may increase nutation or create small attitude disturbances during thrusting. While a detailed spin-dynamics analysis is deferred to ADCS/GNC, minimizing e through symmetric tank placement and paired depletion reduces the likelihood of spin-coupled disturbances.

The selected manifolded feed architecture therefore treats all five tanks as one effective propellant reservoir from the thruster perspective while controlling depletion order to maintain mass symmetry. The secondary tanks are drained in matched pairs to cancel lateral and vertical center-of-mass shifts, and the primary central tank is used after the secondary tanks have been depleted to keep the remaining propellant mass close to the spacecraft centerline.

A.10.3 Tank Wall-Thickness Calculation Details

Pressure Definitions and Allowables

The propellant tanks are sized to an MEOP of 400psi. Using a burst factor of 1.5 gives

$$P_{burst,prop} = 1.5(400) = 600psi.$$

The helium pressurant tanks are sized to an MEOP of 4000psi. Using the same burst factor gives

$$P_{burst,He} = 1.5(4000) = 6000psi.$$

The stress allowable used in the calculation is

$$S = 110400psi.$$



For the selected Ti-6Al-4V material, this allowable is checked against published titanium alloy strength data and used as a project-level aerospace sizing value rather than a code-stamped ASME allowable [205], [206], [207]. Joint efficiency is included through E , where $E = 1$ represents ideal seamless construction and $E = 0.85$ represents a conservative welded-shell screening case.

Cylindrical Side Tank Volume

Each side tank has a final outer diameter of 8.78in, a final outer length of 22.77in, and a selected final wall thickness of 0.035in. Therefore,

$$D_i = 8.78 - 2(0.035) = 8.71in$$

and

$$L_i = 22.77 - 2(0.035) = 22.70in.$$

For a 2:1 ellipsoidal head, the head depth is

$$h = \frac{D_i}{4} = \frac{8.71}{4} = 2.1775in.$$

The straight cylindrical section length is

$$L_c = L_i - 2h = 22.70 - 2(2.1775) = 18.345in.$$

The internal volume of a cylinder with two 2:1 ellipsoidal heads is

$$V = \frac{\pi D_i^2}{4} L_c + 2 \left(\frac{\pi D_i^3}{24} \right).$$

Substituting the final dimensions gives

$$V = \frac{\pi(8.71)^2}{4}(18.345) + 2 \left(\frac{\pi(8.71)^3}{24} \right) = 1266.05in^3.$$

Using $1L=61.024in^3$, $V = \frac{1266.05}{61.024} = 20.75L$. The four side tanks therefore provide

$$V_{side,total} = 4(20.75) = 82.99L.$$

Side Tank Wall Thickness

For a cylindrical shell under internal pressure, the ASME-form expression used for circumferential stress sizing is [204]

$$t_{cyl} = \frac{PR}{SE - 0.6P}.$$

Using $P = 600psi$, $R = D_i/2 = 4.355in$, $S = 110400psi$, and $E = 1$ gives

$$t_{cyl,E=1} = \frac{600(4.355)}{110400 - 0.6(600)} = 0.0237in.$$

Using $E = 0.85$ gives

$$t_{cyl,E=0.85} = \frac{600(4.355)}{110400(0.85) - 0.6(600)} = 0.0280in.$$



For 2:1 ellipsoidal heads, the ASME-form expression is [204]

$$t_{head} = \frac{PD}{2SE - 0.2P}.$$

Using $D = 8.71in$ and $E = 1$ gives

$$t_{head,E=1} = \frac{600(8.71)}{2(110400) - 0.2(600)} = 0.0237in.$$

Using $E = 0.85$ gives

$$t_{head,E=0.85} = \frac{600(8.71)}{2(110400)(0.85) - 0.2(600)} = 0.0279in.$$

The selected final nominal thickness of 0.035in therefore gives the welded-shell margins

$$MS_{cyl} = \frac{0.035}{0.0280} - 1 = 0.25$$

and

$$MS_{head} = \frac{0.035}{0.0279} - 1 = 0.26.$$

The thin-wall check is

$$\frac{t}{r} = \frac{0.035}{4.355} = 0.0080 < 0.1.$$

Central Spherical Propellant Tank

The central propellant tank is a sphere with an internal volume of 70L. Converting to cubic inches,

$$V = 70(61.024) = 4271.66in^3.$$

The radius of a sphere is obtained from

$$V = \frac{4}{3}\pi r^3,$$

so

$$r_i = \left(\frac{3V}{4\pi}\right)^{1/3} = \left(\frac{3(4271.66)}{4\pi}\right)^{1/3} = 10.07in.$$

The internal diameter is therefore

$$D_i = 2r_i = 20.13in.$$

For a spherical shell, the ASME-form thickness expression is [204]

$$t_{sphere} = \frac{PR}{2SE - 0.2P}.$$

For the propellant tank burst pressure of 600psi, using $E = 1$ gives

$$t_{sphere,E=1} = \frac{600(10.07)}{2(110400) - 0.2(600)} = 0.0274in.$$



Using $E = 0.85$ gives

$$t_{sphere, E=0.85} = \frac{600(10.07)}{2(110400)(0.85) - 0.2(600)} = 0.0322in.$$

The selected final nominal thickness of 0.040in gives

$$MS = \frac{0.040}{0.0322} - 1 = 0.24.$$

The final outer diameter is

$$D_o = D_i + 2t = 20.13 + 2(0.040) = 20.21in.$$

The thin-wall check is

$$\frac{t}{r} = \frac{0.040}{10.07} = 0.0040 < 0.1.$$

Helium Pressurant Bottle Volume

The total final propellant volume is

$$V_{prop, total} = 4(20.75) + 70 = 152.99L.$$

The regulated helium system is sized using

$$V_{b, total} = \frac{P_{reg} V_{prop}}{P_{store} - P_{reg}}.$$

With $P_{reg} = 22bar$ and $P_{store} = 276bar$,

$$V_{b, total} = \frac{22(152.99)}{276 - 22} = 13.25L.$$

Applying 10% design margin gives

$$V_{b, total, design} = 1.10(13.25) = 14.58L.$$

For two equal bottles,

$$V_{b, each} = \frac{14.58}{2} = 7.29L.$$

Helium Pressurant Tank Wall Thickness

Each helium bottle is spherical with internal volume 7.29L. Converting to cubic inches gives

$$V = 7.29(61.024) = 444.7in^3.$$

The internal radius is

$$r_i = \left(\frac{3V}{4\pi} \right)^{1/3} = 4.74in,$$

so the internal diameter is

$$D_i = 9.47in.$$



The helium bottle burst pressure is 6000psi. Using the spherical shell equation with $E = 1$ gives

$$t_{He,E=1} = \frac{6000(4.74)}{2(110400) - 0.2(6000)} = 0.128in.$$

Using $E = 0.85$ gives

$$t_{He,E=0.85} = \frac{6000(4.74)}{2(110400)(0.85) - 0.2(6000)} = 0.151in.$$

The selected final nominal thickness is 0.160in, giving

$$MS = \frac{0.160}{0.151} - 1 = 0.06.$$

The final outer diameter is

$$D_o = 9.47 + 2(0.160) = 9.79in.$$

The thin-wall check is

$$\frac{t}{r} = \frac{0.160}{4.74} = 0.0338 < 0.1.$$

Shell-Only Mass Estimates

Shell masses are estimated using Ti-6Al-4V density, taken as approximately $4.43g/cm^3$, equivalent to about $0.160lbm/in^3$ [206], [207]. For each side tank, the metal volume is estimated by subtracting the internal tank volume from the outer tank volume. This gives a shell-only mass of approximately $1.74kg$ per side tank. For the central spherical tank, the spherical shell volume gives approximately $3.71kg$. For each helium bottle, the shell mass is approximately $3.38kg$. These values do not account for imperfections in construction, joining methods, or external hardware.

A.10.4 Pointing Requirement and Minimum Impulse Bit Analysis

During a propulsive maneuver, any pointing error changes the direction of the delivered impulse. If the total delivered impulse is I and the spacecraft pointing error relative to the desired thrust direction is θ , then the impulse can be decomposed into an intended component and an unintended transverse component:

$$I_{\parallel} = I \cos \theta$$

$$I_{\perp} = I \sin \theta$$

The fractional loss in useful impulse along the intended maneuver direction is therefore

$$\frac{\Delta I}{I} = 1 - \cos \theta$$

For the maximum allowable pointing error,



$$\theta = 2^\circ$$

the fractional useful impulse loss is

$$1 - \cos(2^\circ) = 6.09 \times 10^{-4}$$

or

$$0.0609\%.$$

The unintended transverse impulse fraction is

$$\sin(2^\circ) = 0.0349$$

or

$$3.49\%.$$

The propulsion subsystem is also constrained by a minimum impulse bit of

$$I_{bit} = 1.1368 \text{ N s.}$$

For a spacecraft mass of

$$m = 120 \text{ kg,}$$

the velocity increment associated with the minimum impulse bit is

$$\Delta v_{bit} = \frac{I_{bit}}{m}$$

$$\Delta v_{bit} = \frac{1.1368}{120}$$

$$\Delta v_{bit} = 9.47 \times 10^{-3} \text{ m/s.}$$

The transverse velocity increment caused by a 2° pointing error during a minimum impulse bit is

$$\Delta v_{\perp} = \Delta v_{bit} \sin(2^\circ)$$



$$\Delta v_{\perp} = (9.47 \times 10^{-3})(0.0349)$$

$$\Delta v_{\perp} = 3.31 \times 10^{-4} \text{ m/s.}$$

The useful velocity increment lost due to the pointing error is

$$\Delta v_{loss} = \Delta v_{bit} (1 - \cos(2^{\circ}))$$

$$\Delta v_{loss} = (9.47 \times 10^{-3})(6.09 \times 10^{-4})$$

$$\Delta v_{loss} = 5.77 \times 10^{-6} \text{ m/s.}$$

These results show that the useful impulse loss from a 2° pointing error is negligible for a single minimum-impulse command. The more important effect is the transverse impulse component, which is small for one burn but can accumulate over many burns. Therefore, the burn sequence should require ADCS/GNC pointing convergence before thruster firing, and the thruster selection should support impulse-bit control consistent with this pointing requirement.



B Contributions Table

Team Member	Roles	Sections
Josephine Kwok	Data Curation, Formal Analysis, Writing - Reviewing & Editing	2.1, 6.7.1, 6.7.2, 6.5, 6.2, 7.1, 7.5, 7.4, 7.7.8
Joseph Norwood	Data Curation, Formal Analysis, Writing - Original, Reviewing and Editing	6.7.4, 6.7.5, 7.7.1, 7.7.2, 7.7.5, 7.7.6, 7.7.7, 7.7.3, 7.7.4, 7.7.8, 6.3, 7.6
Andrew Simon	Formal Analysis	7.7.6
Ethan Spain	Data Curation, Formal Analysis, Writing - Reviewing & Editing	6.1, 6.3, 6.4, 6.5, 6.7.1, 6.7.3, 7.7.4, 7.7.3, 7.7.6
Gina Arnau	Data Curation, Formal Analysis, Writing - Original, Reviewing, and Editing	8.1, 8.2, 8.3, 8.4, 8.5, 8.6, 8.7.1, 8.7.4, 9.1, 9.2, 9.3, 9.4, 9.5, 9.7.6
Arthur Balva	Data Curation, Formal Analysis, Writing - Original, Reviewing, and Editing	9.1, 9.2, 9.3, 9.4, 9.5, 9.7.1, 9.7.2, 9.7.3, 9.7.5, 9.7.4, 9.7.7, 9.7.8, 9.7.9
Sarah Hedley	No documented contributions for this report.	-
Anthony Matimu	Conceptualization, Formal Analysis, Visualization, Writing - Original, Reviewing, and Editing	3, 2.2, 2.3, 8.7.2, 8.7.3, 8.7.4, 9.7.6
Evan Callas	Data Curation, Investigation, Formal Analysis, Validation, Writing - Original, Reviewing, and Editing	2.4, 2.9, 10.3, 10.4, 10.5, 10.7.3, 10.7.4.1, 10.7.5, 10.7.6, 11.6, 11.7.5.2, 11.7.6.2
William (Ted) McCormack	Data Curation, Formal Analysis, Writing - Original, Reviewing, and Editing	10.3, 10.4, 11.6, 11.7.4, 11.7.5, 11.7.5.3
Tasman Moskowitz	Data Curation, Formal Analysis, Writing - Original, Reviewing, and Editing	10.1, 10.2, 10.3, 10.4, 10.5, 10.6, 10.7, 11.1, 11.2, 11.3, 11.4, 11.5, 11.7, A.9
Arya Olliviera	Data Curation, Formal Analysis, Writing - Original, Reviewing, and Editing	10.2, 10.7.2, 10.7.3, 11.2, 11.3, 11.4, 11.5, 11.7.2, 11.7.3, 11.7.5, 11.7.6, 11.7.6.1
Maia Abiani	Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Validation, Visualization, Writing - Original Draft and Reviewing & Editing	1.2, 1.1, 4.1, 4.3, 4.6, 4.7, 4.7.9, 4.7.8, 4.7.6, 4.7.7, 5.7.4, 5.7.5
Sonal Bhatia	Project Administration, Conceptualization, Methodology, Data Curation, Formal Analysis, Writing - Original Draft, Reviewing & Editing, Supervision, Software, Visualization, & Validation	1.1, 2.1, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 4.2, 4.3, 4.4, 4.5, 4.7.1, 4.7.2, 4.7.3, 4.7.5, 5.7.3, 5.7.11, 12, A.1.1, A.3, A.4
Sawooly Li	Data Curation, Formal Analysis, Writing - Original, Reviewing, and Editing	1.3, 1.4, 1.5, 4.7.4, 5.3, 5.4, 5.5, 5.7.6, 5.7.2
William Li	Conceptualization, Methodology, Data Curation, Formal Analysis, Writing - Original Draft, Reviewing & Editing, Visualization, & Validation	1.6, 1.7, 4.7.10, 5.3, 5.4, 5.5, 5.7.7, 5.7.8, 5.7.10, 5.7.9



C Contributions Description

1. **Josephine Kwok** : I worked to refine the subsystem requirements to more closely reflect our mission's operating environment. I also ensured that we satisfied all mission-level requirements as reflected in Section 2.1. I found literature to conduct analyses presented in section 6.7.2 in regards to the Group 3 microscale spacing. I also conducted the analyses presented in sections 6.7.1 and 6.7.2 which discuss timing, launch, and satellite spacing which impact operations timeline. Lastly, I also helped to write sections 6.5, 6.2, 7.1, 7.4, 7.5, and 7.7.8.
2. **Joseph Norwood** : I conducted the analysis presented in Sections 6.7.4, 6.7.5, 7.7.1, 7.7.2, 7.7.5, 7.7.6, & 7.7.7 and wrote/edited these sections. I also conducted the preliminary analysis for Sections 7.7.3, 7.7.4, & 7.7.8. I wrote and edited Sections 6.3 & 7.6.
3. **Andrew Simon** : No documented contributions for this report.
4. **Ethan Spain**: I worked to refine subsystem requirements, objectives, and drivers for both ADCS and GNC. This includes sections 6.1, 6.2, 6.5, among others. I conducted an analysis for 6.7.3, 6.7.1, 7.7.4, and 7.7.6, as well as others. This included STK analysis, calculations via SMAD, and heritage mission precedents. Additionally, for 7.7.4, this included a complete analysis for both nutation damper fluid selection and sizing.
5. **Gina Arnau** : I worked to refine the SMM and THM subsystem requirements, constraints, and drivers, finding the literature to change the values for all requirements of the SMM subsystem to reflect our mission's operating environment. I also conducted the analysis presented in Section 8.7.1 - to refine the design approach to protect the spacecraft from the harsh radiation environment - and the one presented in Section 8.7.4 to characterize the dynamic behavior during the deployment of the new magnetometer boom design. For THM, I conducted the analysis in section 9.7.6 to determine the required number of sensors and their placement, as well as their integration. I wrote Sections 8.1 - 8.6 for SMM and Sections 9.1 - 9.5 for THM, as well as section 2.2. Finally, I helped to fix any last-minute latex bugs as well as doing a full review to ensure the document was ready for submission.
6. **Arthur Balva** : I worked on all aspects related to the thermal subsystem design and report writing except the temperature sensor research and trade study. This includes refining requirements, research, trade studies, code-writing, and writing of sections. I also contributed to compiling and ensuring compliance to the mass budget from all sub teams. Early in the design process, I researched launch vehicle interfaces, volumes, and vibrational loads through the Falcon 9 User Guide.
7. **Sarah Hedley** : *Contributions written by group* I conducted the thermal strap trade study and section in the PDR.
8. **Anthony Matimu** : In this report, I described systems integration in Section 3 as well as verifying the mass and volume compliance in Section 2.2 and Section 2.3. I completed all of the design and structural analysis presented in Section 8.7.2 and Section 8.7.3 respectively as well as conducting the literature review and derivation for the equations and simulation for the magnetometer boom in Section 8.7.4. Additionally, I provided heritage mission literature review and contributed to the temperature sensor analysis in Section 9.7.6. I also designed the team and mission logos.



9. **Evan Callas** : I was the lead for the energy storage down for the POW subsystem and assisted with the design and analysis of the power generation system and power distribution and management module for Section 10.7.4.1, 10.7.5.1 and 10.7.6. I also conducted the analysis presented in Section 11.7.5.2 and 11.7.6.2 to refine the thruster placement to ensure that it was in accordance with the demands of other subsystems while providing the torque authority needed for station keeping. I also refined, edited and wrote about the power compliance and pointing compliance in Sections 2.4 and 2.9.
10. **William (Ted) McCormack** : I put together the PLV design approach section, and the accompanying graphic, Figure 43. I worked to refine the analysis, driving the propellant selection and justification. I also conducted the analysis and design behind the propellant and pressurant storage and feed systems presented in Section 11.7.4. I compiled the work for the appendix for the aforementioned analyses, appendix Sections A.10.1, A.10.2, A.10.3, and A.10.4. I also edited the Systems Engineering sections of the POW section.
11. **Tasman Moskowitz** : I brainstormed and originally drafted or updated the subsystem overview, objectives, drivers, and constraints for both the POW & PLV subsystems in the FDR report. I refined the subsystem requirements and wrote the requirements subsection of the FDR report for both POW & PLV, organizing the requirements in a tabular format for the first time. I originally drafted and completed the POW subsystem design approach, and reviewed and edited the PLV system design approach. My entirely individual (all components) analysis tasks for the POW subsystem were the eclipse analysis, power generation, power management and distribution system, power design summary, and the POW appendix. I also collaborated in original analysis and writing of the report to complete the power mode definitions and the construction and refinement of the power budget matrix along with the battery sizing. My individual (all components) tasks for the PLV subsystem in the FDR were the launch vehicle analysis and thruster nozzle performance analysis.
12. **Arya Olliviera** : I worked to refine all PLV subsystem requirements and wrote the PLV requirements section. I reviewed and edited the PLV objectives, constraints, and drivers. I performed a background information compilation and analysis on heritage mission and delta v requirements. I performed the thruster trade study and ranked and selected thrusters based on their specifications and my calculations. I constructed the line diagram (P&ID) illustrating the connections between pressurant and hydrazine tanks, and thrusters. I performed a plume contamination analysis for both thruster types to analyze interference with onboard instruments and equipment. I reviewed the power objectives, and redefined modes for the POW subsystem to have a better representation of maximum power draw. I redefined the power budget to reflect new modes, battery selection, and solar generation capabilities, and compared subsystem allocation to usage.
13. **Maia Abiani** : I was the lead for the payload design and worked on selecting the class of instruments for the payload instrument suite 4.1 by reviewing and analysing the science of the magnetotail transition region itself, as well as past magnetotail missions. I defined all the payload instrumentation requirements 4.3, and completed the trade studies in order to select the magnetometer 4.7.6 and the electrostatic analyser 4.7.9. I also derived the requirements for magnetic cleanliness and estimated the dipole magnetic moment for the MATRIX spacecraft in order to define the required boom length 4.7.8. I also contributed to the overall design of the ConOps based on my review of the literature regarding the magnetotail transition re-



gion in order to decide on the multi-apogee configuration and define the MATRIX mission requirements and objectives 1.2. Additionally, I conducted the analysis in order to define the payload operational modes and confirm OPL's compatibility with the OPL subsystem constraints 4.7.7. On the CCD side, I conducted the design of the modified Bus Avionic Unit, as well as the Instrument Data Processing Unit, and worked to define the IDPU-instrument interface in order to fulfill the key CCD requirements including fault protection, autonomy, and operational mode changes 5.7.4, 5.7.5.

14. **Sonal Bhatia:** I served as Project Manager of the team throughout the semester, with responsibilities reflected across many sections of this report. I drafted and edited the mission overview in Section 1.1, and the concluding remarks in Section 12. I developed and maintained the Gantt Chart and Task Tracker since the Initial Team report, the cost budget and associated sensitivity analyses in Section 4.7.3, and the operations framework spanning all mission modes, procedures, phases, and the prelaunch-through-disposal timeline in Sections 4.7.1-4.7.2. On the CCD side, I led the flight software and onboard subsystem integration effort to the flight computer, including verification of RS-422 interface compatibility across subsystems, documented in Sections 5.7.11 and 5.7.3. I also resolved LaTeX build and formatting issues required to produce the final integrated document. I was also pulled into supporting analyses across subsystems and dedicated time to resolving engineering conflicts (as documented in 4.7.5), thus more of the written contributions and analyses are noted in the table above. Beyond the written deliverables, I facilitated weekly cross-subsystem meetings, coordinated the sixteen-member team across eight subsystems, tracked Directly Responsible Individuals (DRIs) against the project timeline, and reconciled mass, power, data, and cost margins to maintain a coherent mission architecture. This role required sustained effort throughout the semester, including substantial work (and sleepless nights) outside of class hours, and I am proud of the integrated mission concept the team has produced.
15. **Sawooly Li :** I conducted the original and updated analysis of the Onboard Computer Hardware system in Section 5.7.2, as well as the analysis for the transmission protocol and bandwidth allocation in Section 5.7.6. I also constructed the initial risk matrix, enforced key mitigation action items across subsystems, and reassessed risks to develop the final risk matrix in Section 4.7.4. I also wrote and refined the requirements, constraints and drivers for the overall mission level as well as at the CCD subsystem level.
16. **William Li :** I was the Principal Investigator of the MATRIX mission and a member of the OPL/CCD team. I created the ConOps diagrams and descriptions in Section 1.6. For this report I also wrote the Abstract, laying out the big picture of the MATRIX mission. In OPL, I was responsible for conducting a trade study of EPD instruments and I also led proposing modifications to the instrument to make it compatible for MATRIX (Section 4.7.10.) From CCD I did the end-to-end design of the data flow from the BAU to the Near Space Network on Earth (Sections ??.) I used datasheets and properties of Bursty Bulk Flows and the MATRIX orbits to create a data budget that includes all data generated by various MATRIX subsystems. This data budget was then used for almost all CCD hardware selection to ensure that the flight computer and communications hardware was capable of transmitting the data collected.